Debian 10 Server Administration

ARYA PRAMUDIKA

DAFTAR ISI

BAB 1 Pendahuluan

1.1 Pengenalan Linux

Linux adalah keluarga sistem operasi bebas dan sumber terbuka yang dibangun di atas kernel Linux, yaitu sebuah kernel sistem operasi yang pertama kali dikembangkan oleh Linus Torvalds pada 1991.Linux dirilis di bawah Lisensi GNU GPL v2. Secara teknis, Linux dapat merujuk pada kernel-nya itu sendiri. Linux bisa menjadi system operasi yang utuh jika dilengkapi dengan komponen-komponen perangkat lunak lainnya milik GNU, sehingga Linux juga bisa disebut GNU/Linux.

Linux telah lama dikenal penggunaannya sebagai server, tapi sekarang linux banyak digunakan untuk perangkat Desktop dan Smartphone(Android). Linux juga digunakan untuk embedded device seperti Router, Televisi, dan Automobile seperti tesla.

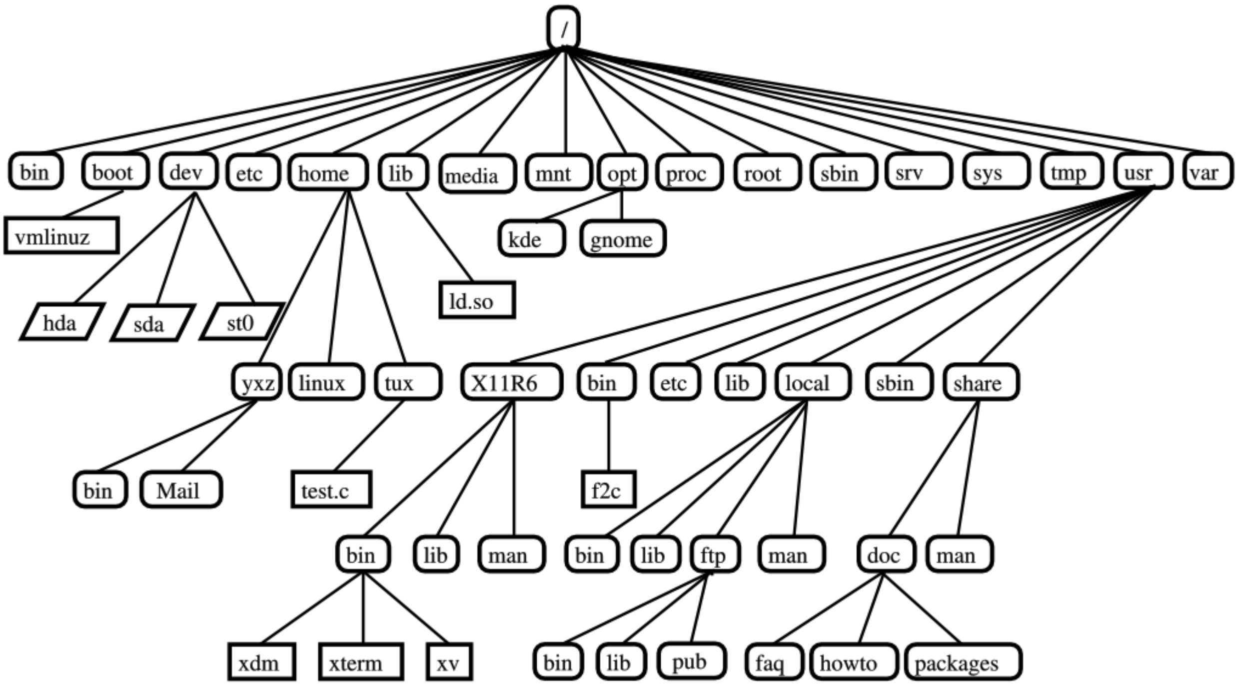
Linux adalah salah satu contoh paling menonjol dari kolaborasi perangkat lunak bebas dan sumber terbuka. Kode sumber dapat digunakan, dimodifikasi dan didistribusikan secara komersial atau non-komersial oleh siapa pun di bawah ketentuan lisensi masing-masing.

Baca selengkapnya di:

https://id.wikipedia.org/wiki/Linux

https://en.wikipedia.org/wiki/Linux

1.2 Struktur Direktori Linux



/	Adalah direktori root atau direktori paling dasar di linux
/bin	Di sinilah file yang dapat dieksekusi berada.
/boot	Berisi file untuk booting system
/dev	Direktori tempat file device, seperti penyimpanan /dev/sda, /dev/hda
/etc	Berisi file konfigurasi system
/home	Berisi direktori user
/lib	Berisi file library yang biasanya berhubungan dengan kernel
/media	Berisi media yang terpasang di komputer. Seperti cdrom, flopy disk, flash disk, hardisk eksternal dsb.
/mnt	Direktori tempat pengaitan sistem sementara
/opt	Berisi paket aplikasi tambahan yang kita install ke dalam system.
/proc	Berisi filesystem virtual dokumentasi kernel dan proces status seperti file text..

/root	Direktori ini merupakan Home-nya user Root. Bukan terletak di /home/root, melainkan folder tersendiri, yaitu di /root.
/sbin	Berisi program biner essensial yang dibutuhkan untuk menjalankan dan memperbaiki sistem. biasanya di eksekusi oleh administrator sistem (root) file-file biner yang ada di /sbin adalah fastboot, fasthalt, fdisk, fsck, fsck.*, getty, halt, iconfig, init, mkfs, mkfs.*, mkfswap, reboot, route, swapon, swaporf, update.
/srv	Direktori ini berisi data untuk semua layanan sistem yang bersangkutan. biasanya nama layanan dituliskan sebagai subdirektori. misalnya /srv/ftp, /srv/www dan sebagainya.
/sys	Direktori special yang memuatkan informasi mengenai hard disk seperti yang dilihat melalui Linux.
/tmp	Tempat untuk menyimpan sementara file-file Linux. Biasanya file yang disimpan dalam direktori ini akan terhapus setiap kalu me-restart komputer.
/usr	Berisi file yang dapat dibagi untuk semua user sistem dan hanya ada akses baca saja (read-only).
/var	Direktori ini berisi file-file variable (spesifik pada mesin bersangkutan). Biasanya berisi informasi seperti log, direktori mail, print dan lain – lain.

Baca selengkapnya di :

https://www.tutorialspoint.com/unix/unix-file-system.htm

https://catatankakimu.wordpress.com/2016/10/25/struktur-hirarki-direktory-direktory-yang-ada-pada-linux

1.3 Distro Linux

Apa itu Distro Linux?

Distro linux adalah sistem operasi yang dibagun dari kernel linux dengan penambahan komponen lain berupa module, aplikasi, service ataupun package lain agar tercipta sistem operasi dengan tujuan yang spesifik yang telah ditentukan oleh pihak pengembang. Distro merupakan singkatan dari Linux **DISTR**ibuti**O**n

Contoh Distro Linux

Debian, Ubuntu, Fedora, Red Hat.

Di Modul ini kita akan membahas tentang Linux Debian 10 (Buster)

Bagaimana cara mendownload file ISO Debian?

Kalian bisa mendownloadnya di situs resminya yaitu di https://www.debian.org/releases/buster/debian-installer

BAB 2 Instalasi dan Konfigurasi Server Debian 10.x (Buster)

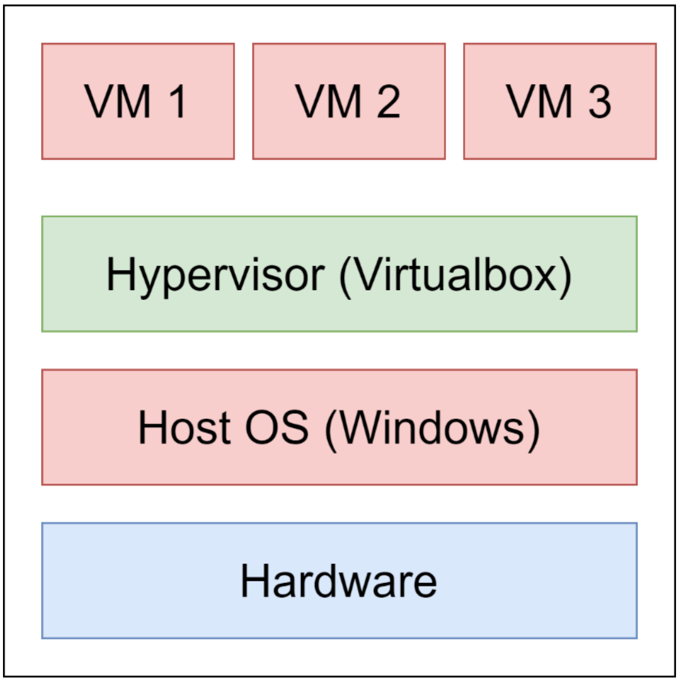
2.1 Topologi yang digunakan untuk praktek

Di modul ini kita praktek menggunakan 3 virtual machine yang berjalan di aplikasi VirtualBox

Apa itu VirtualBox?

VirtualBox adalah perangkat lunak virtualisasi, yang dapat digunakan untuk mengeksekusi sistem operasi "tambahan" di dalam sistem operasi "utama". Sebagai contoh, jika seseorang mempunyai sistem operasi Windows yang terpasang di komputernya, maka seseorang tersebut dapat pula menjalankan sistem operasi lain yang diinginkan di dalam sistem operasi Windows tersebut.

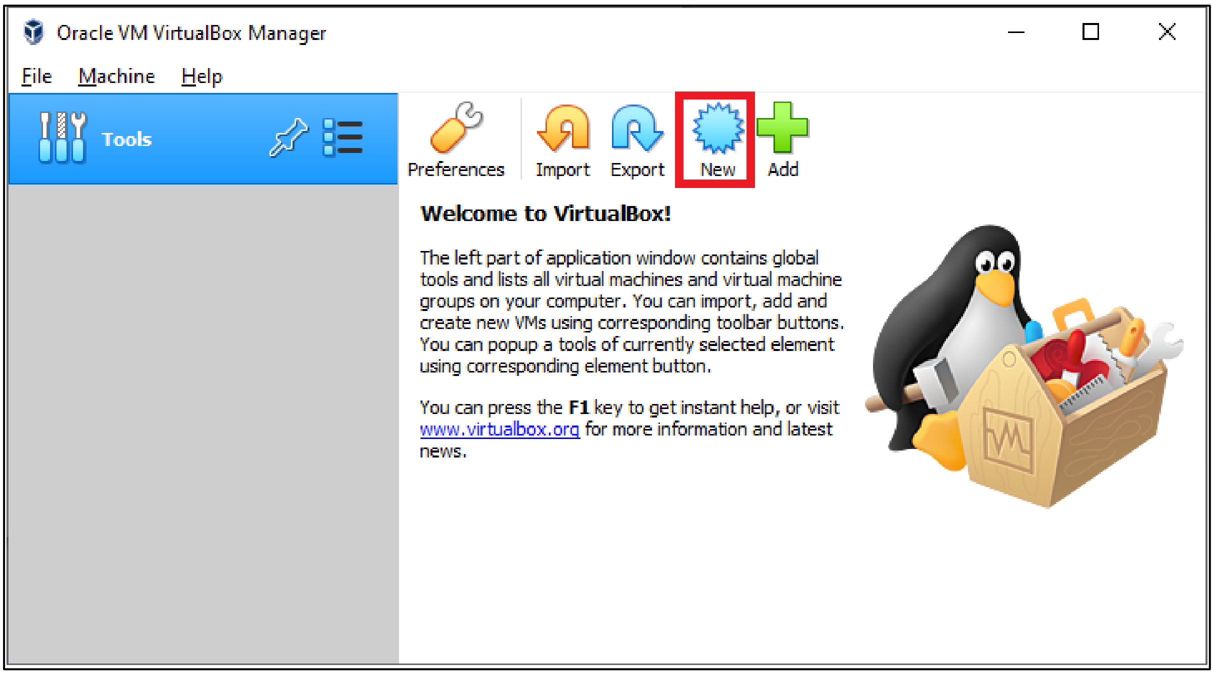
Berikut gambaran virtualisasi yang kita akan buat.



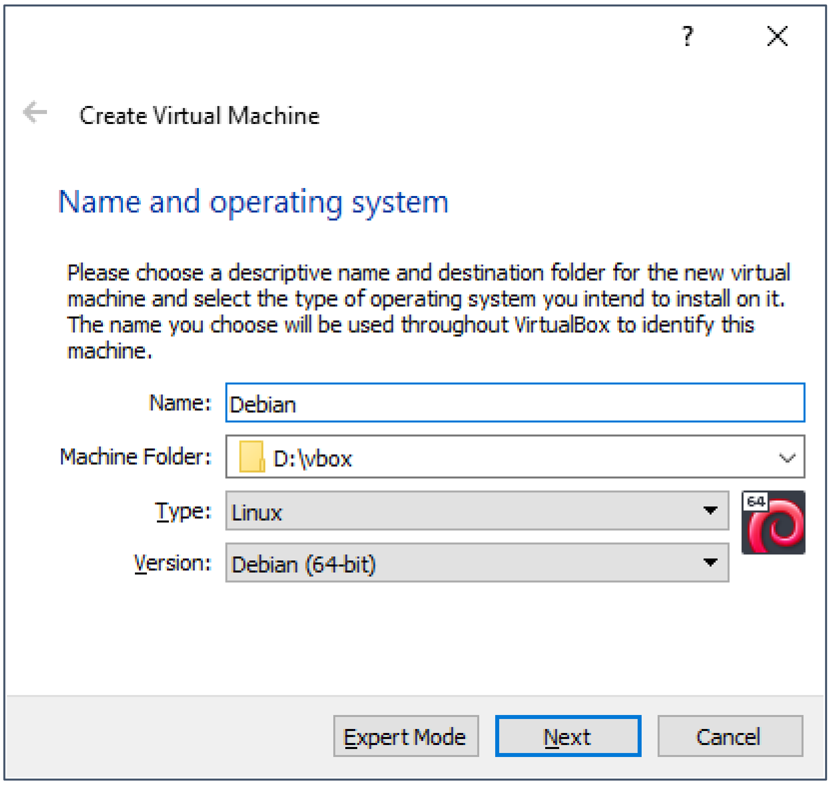
Virtualbox menggunakan RAM Laptop/PC kita, jadi RAM Virtual Machine kita terbatas pada RAM Laptop/PC. Di Modul ini kita membuat 3 VM Debian dengan RAM 512 Mb, maka total RAM yang dipakai yaitu 1,5 GB. Agar berjalan dengan lancar saya sarankan minimal RAM Laptop/PC kalian kurang lebih 4GB.

2.2 Membuat VM dan Instalasi Debian 10 di VirtualBox

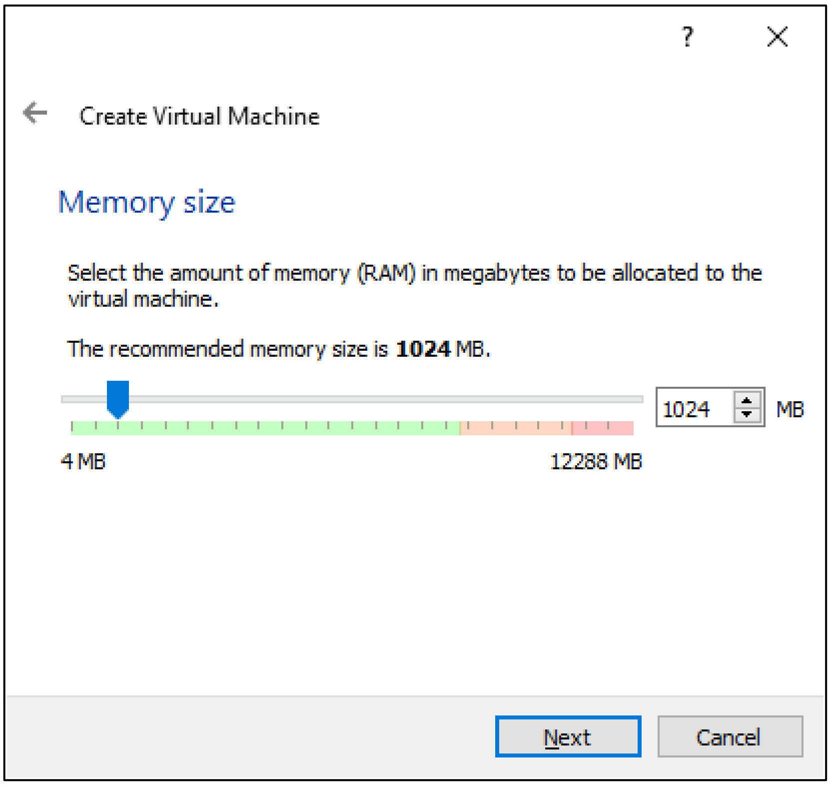
Buka VirtualBox lalu klik New



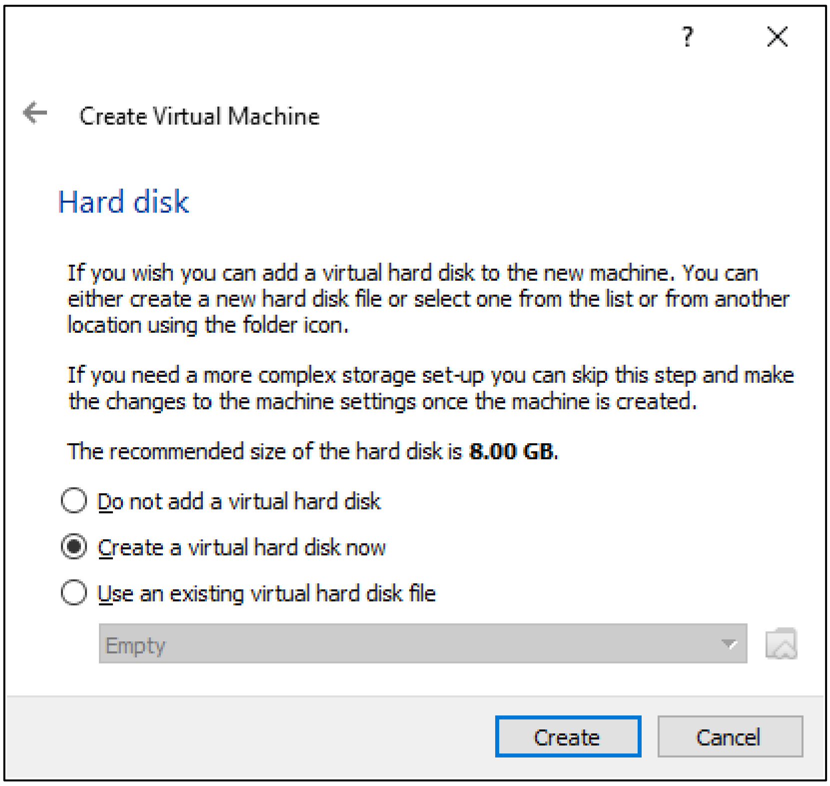
Lalu beri nama sesuai keinginan misal Debian lalu klik next



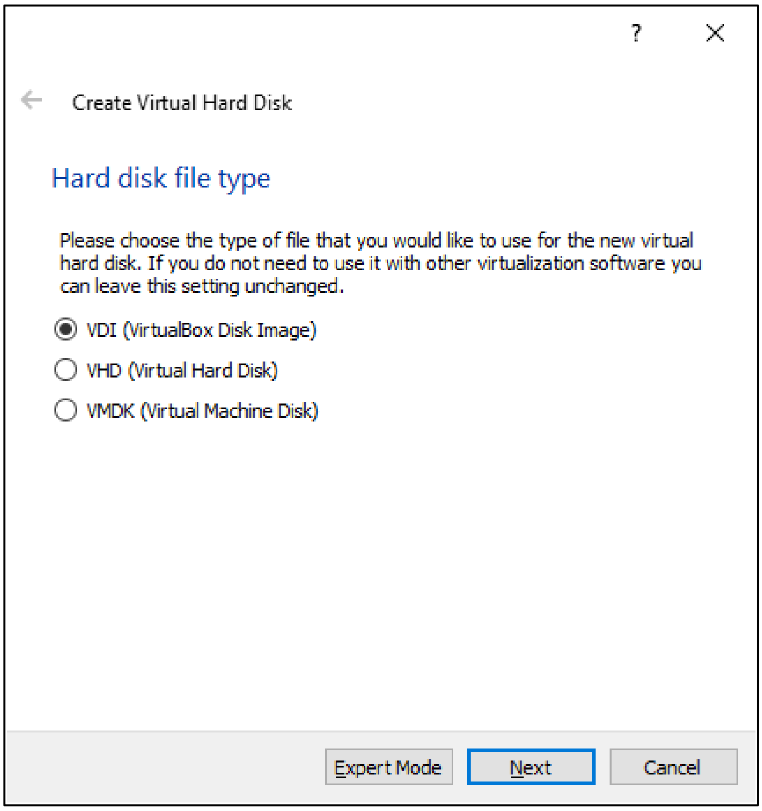
Atur RAM, untuk intallasi kita gunakan RAM 1 GB



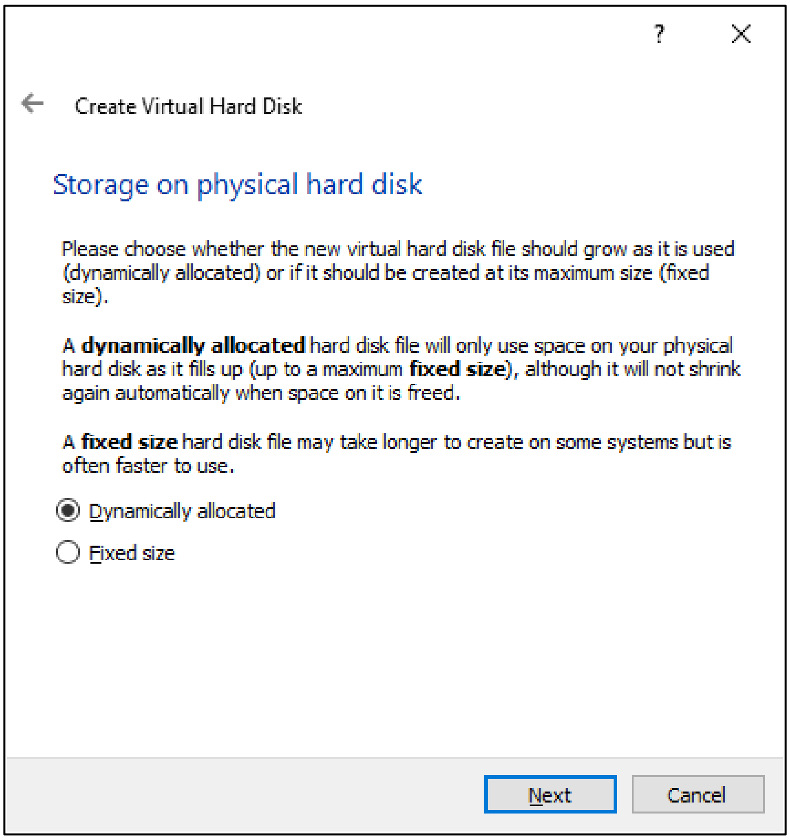
Pilih create a virtual harddisk now, untuk membuat harddisk virtual



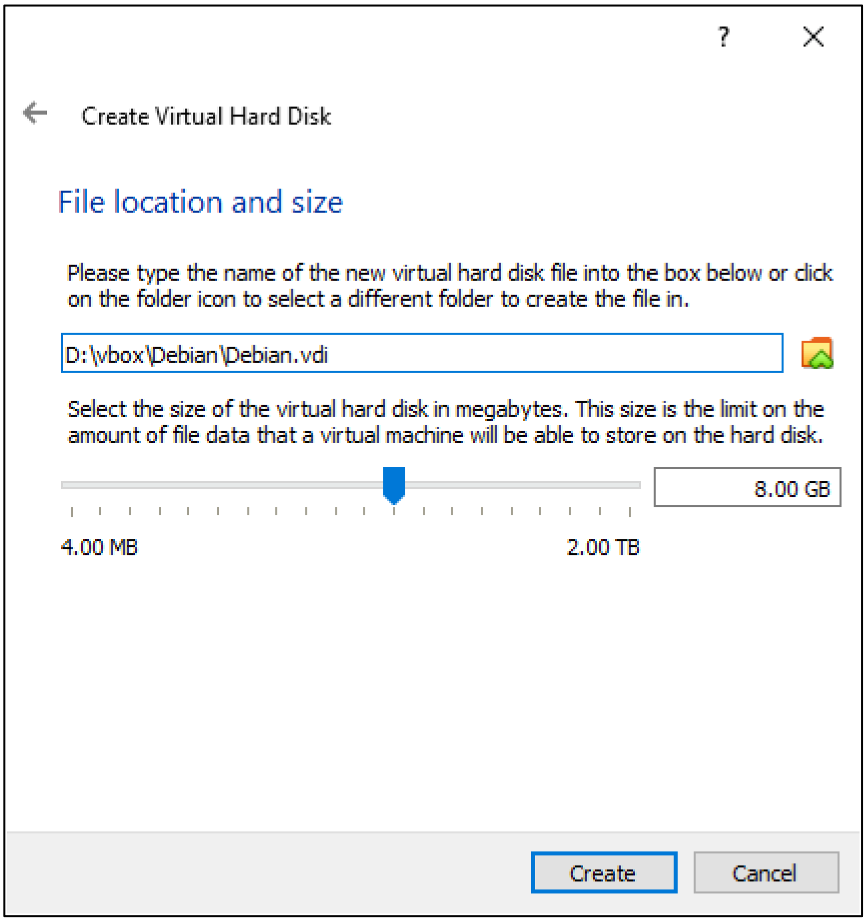
Pilih VDI (Virtual Disk Image)



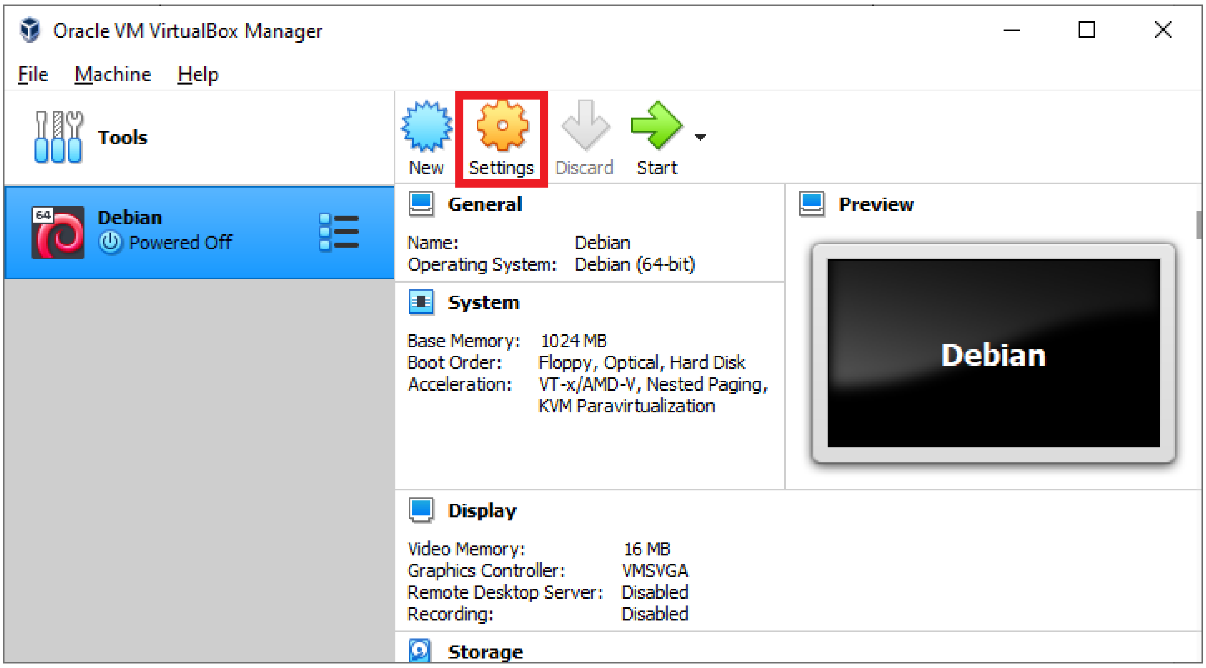
Pilih Dynamic Allocated



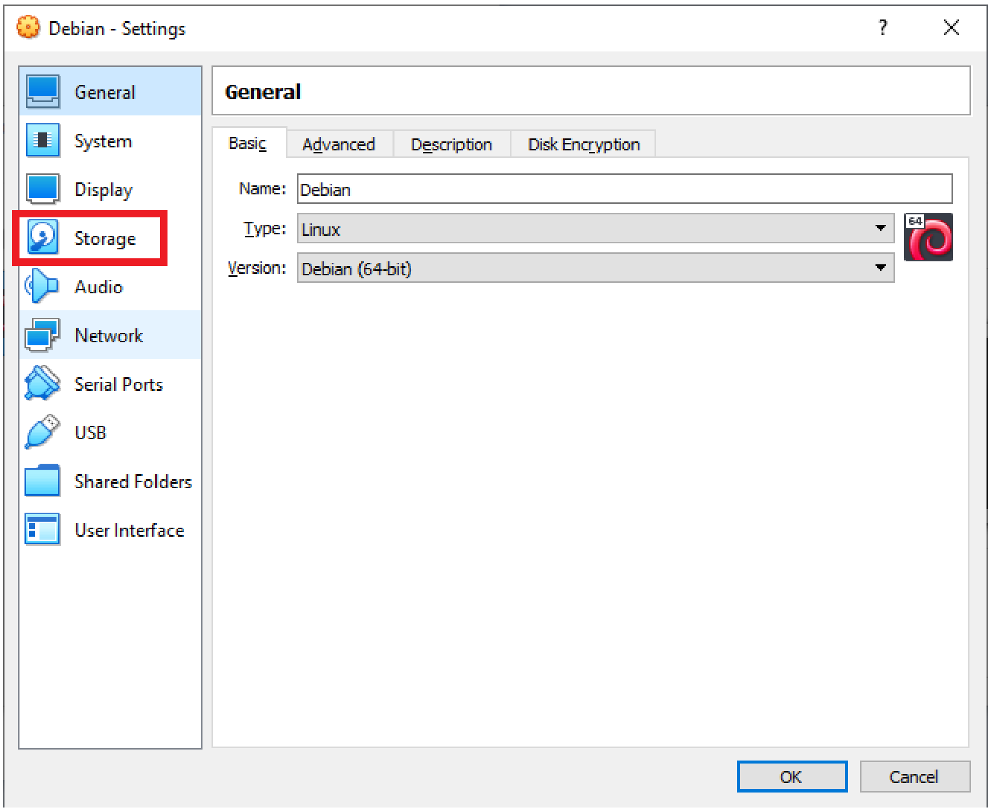
Buat Harddisk sebesar 8GB atau lebih



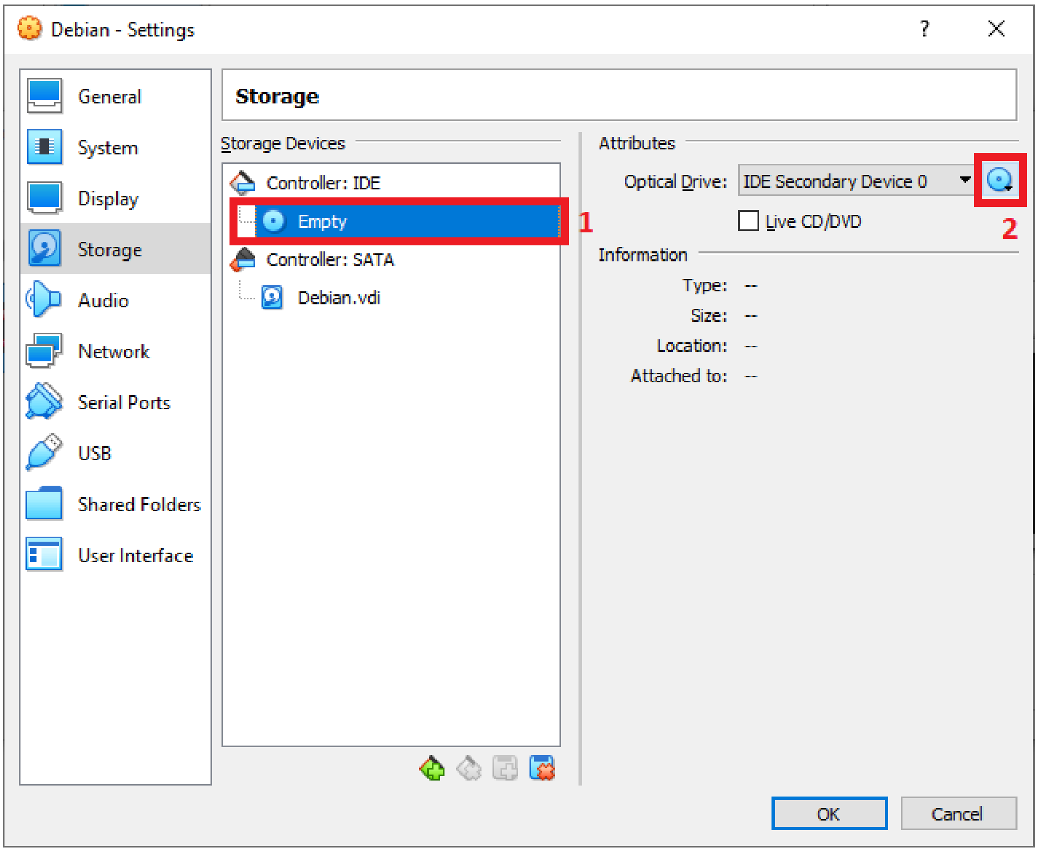
VM Debian berhasil dibuat, selanjutnya klik Settings



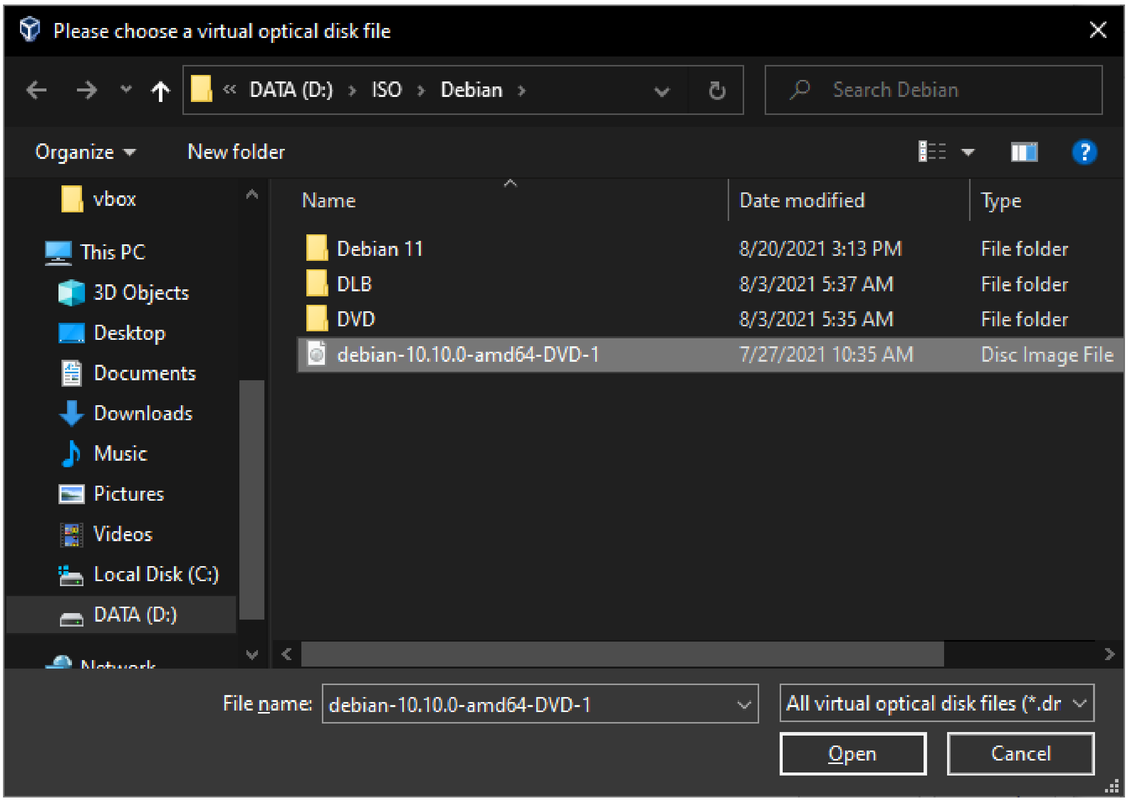
Selanjutnya masuk ke bagian Storage



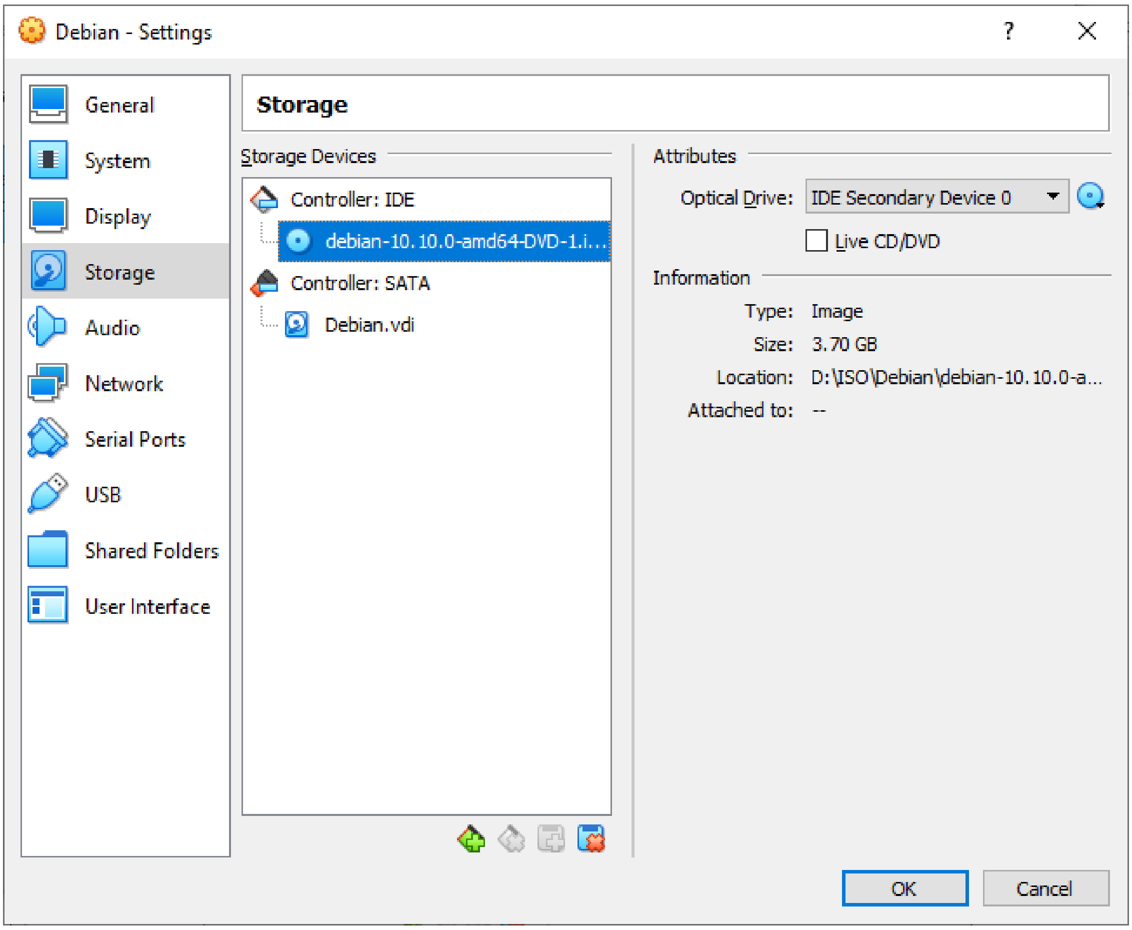
Lalu klik di Bagian Controller:IDE > Empty dan Klik icon disk > Choose a disk file



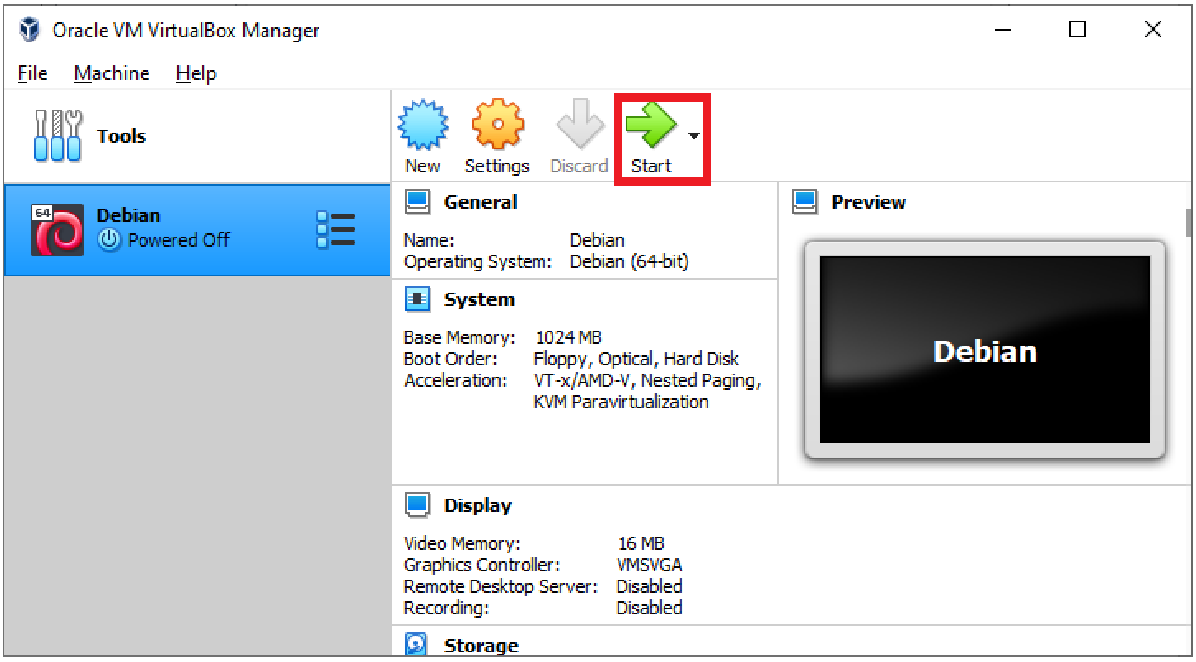
Pilih file ISO Debian



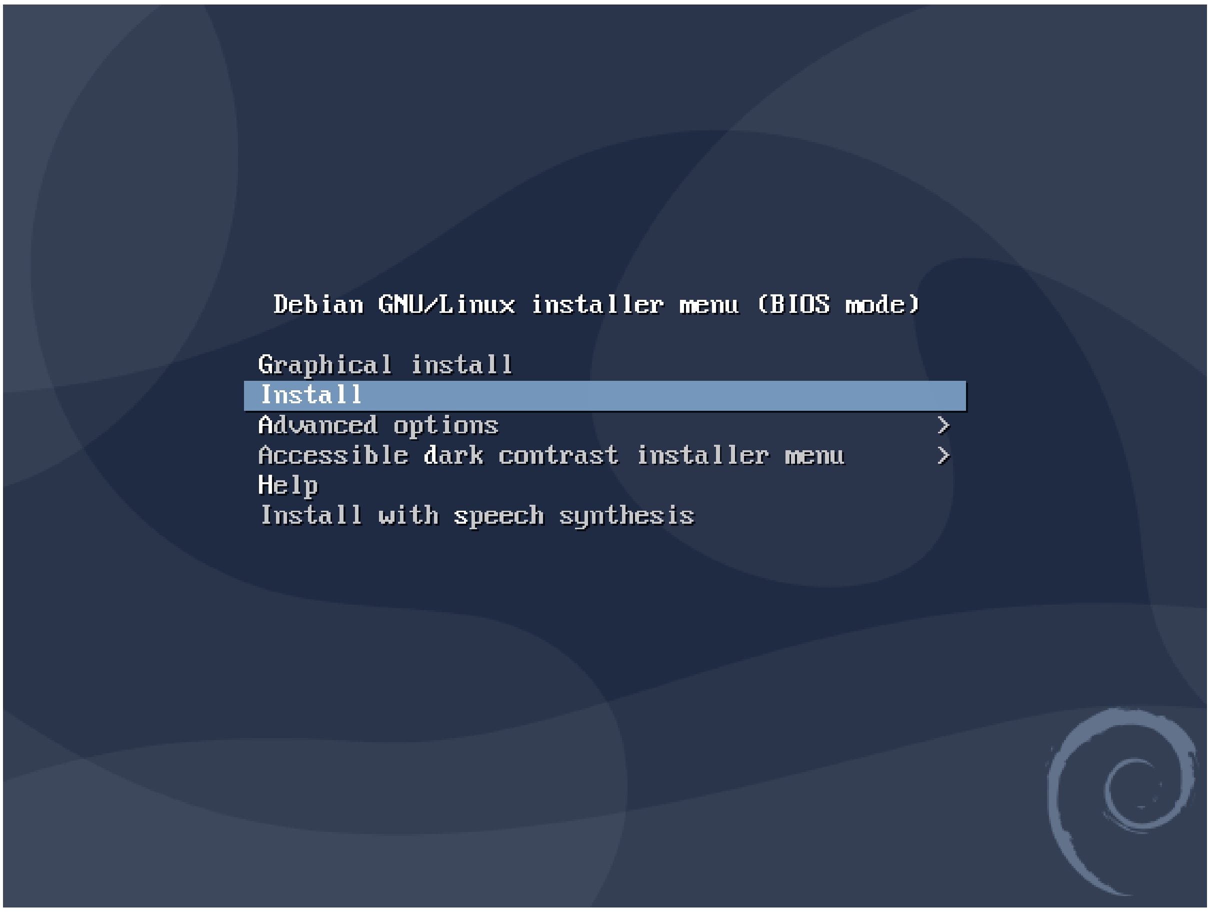
File ISO berhasil ditambahkan, lalu klik OK



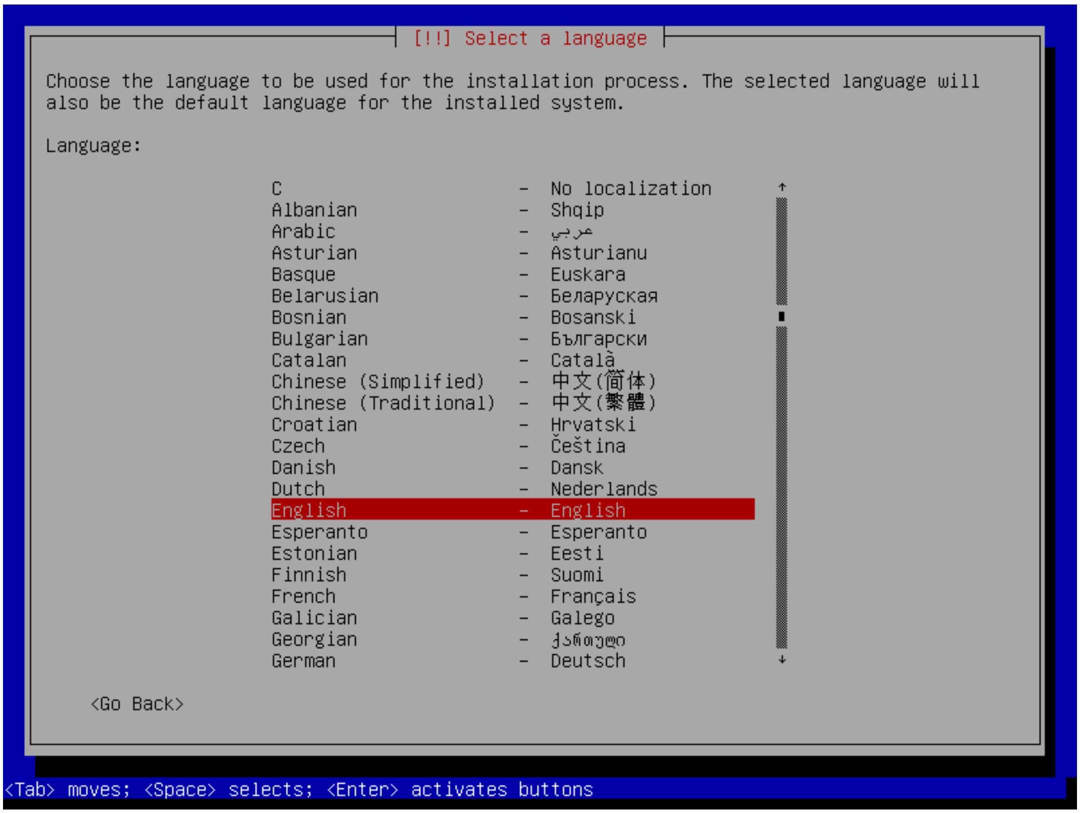
Klik Start untuk menjalankan VM Debian dan Melakukan Instalasi



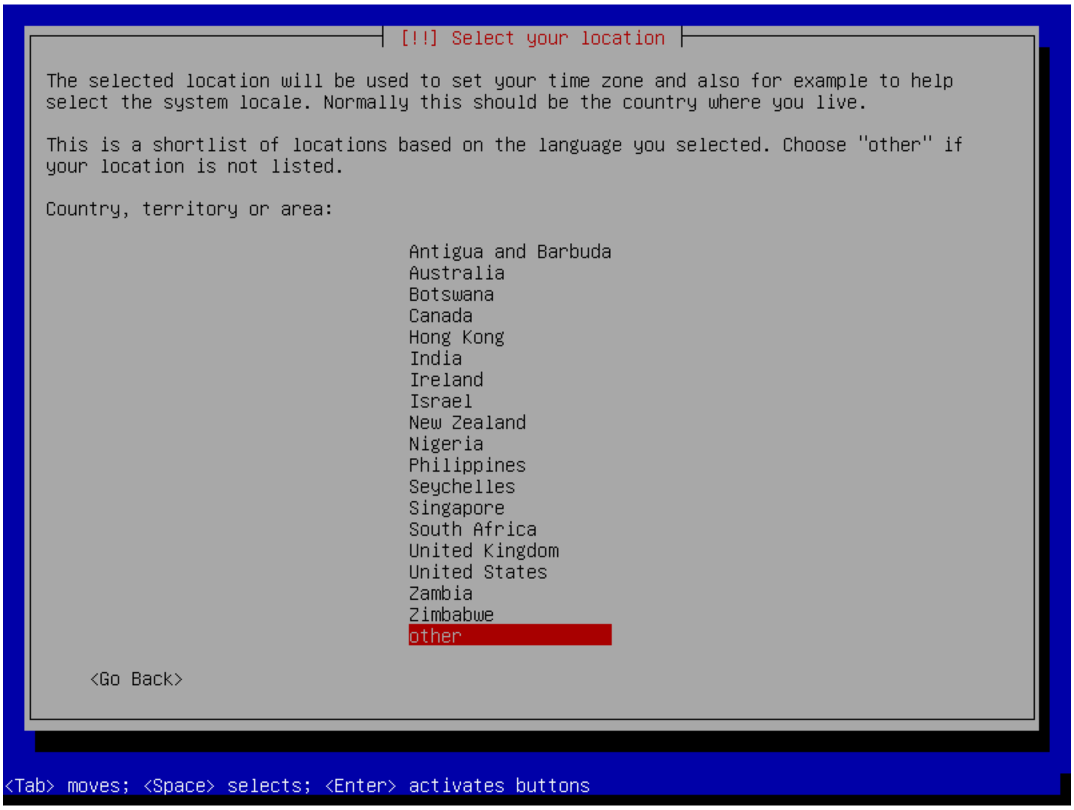
Pilih **Install**



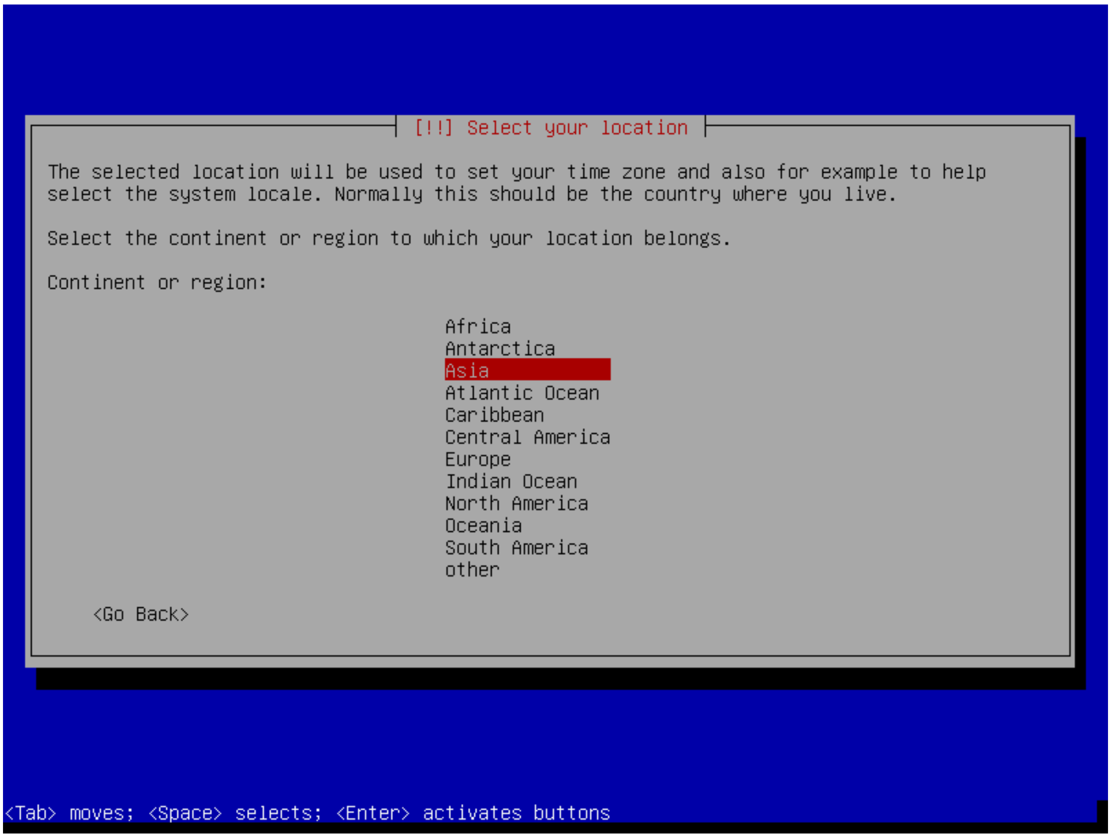
Pilih Bahasa yang digunakan, disini saya memilih **English**



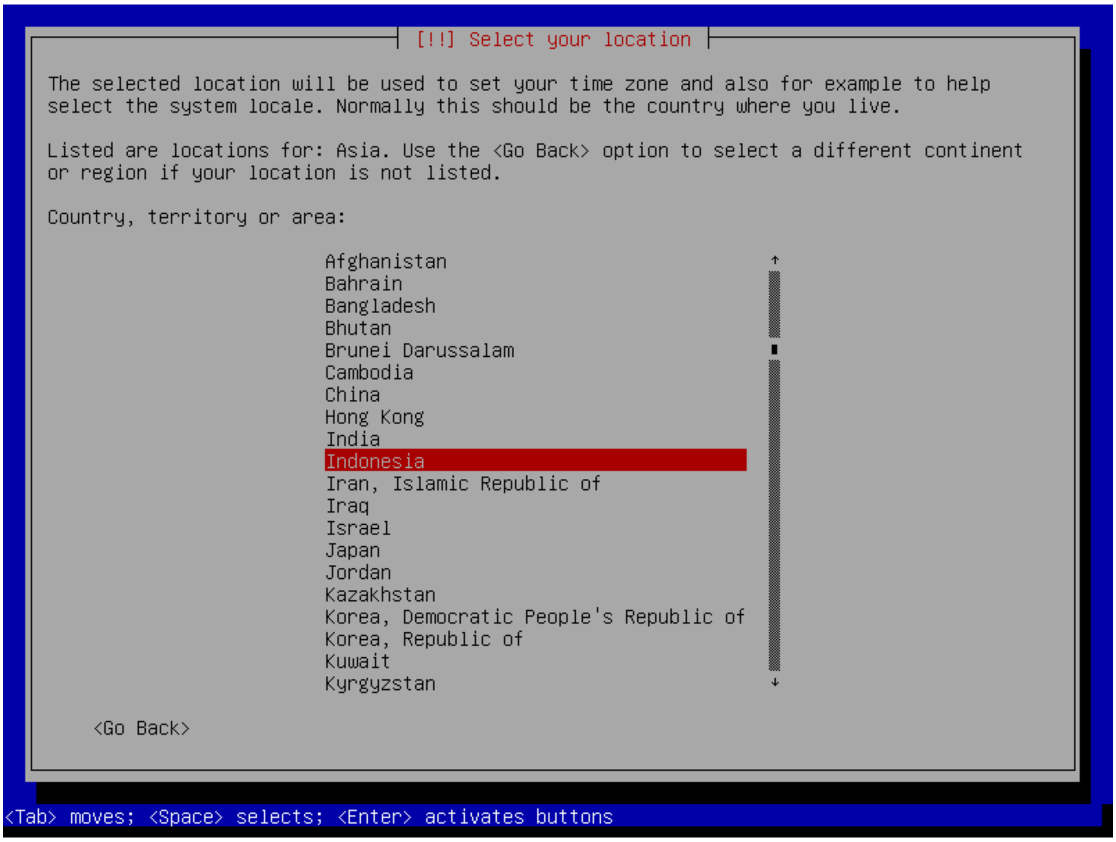
Lalu pilih Country/Negara kita, karena Indonesia tidak ada kita pilih **Other**



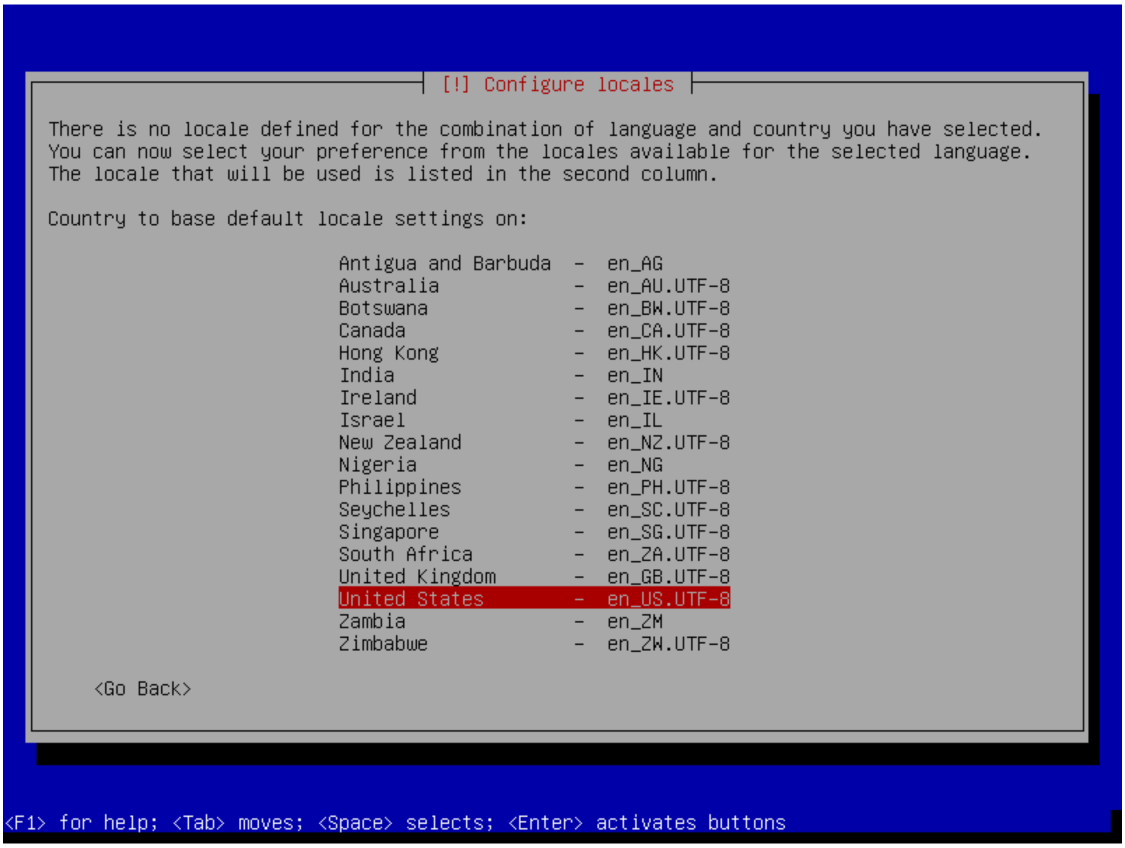
Selanjutnya pilih **Asia**



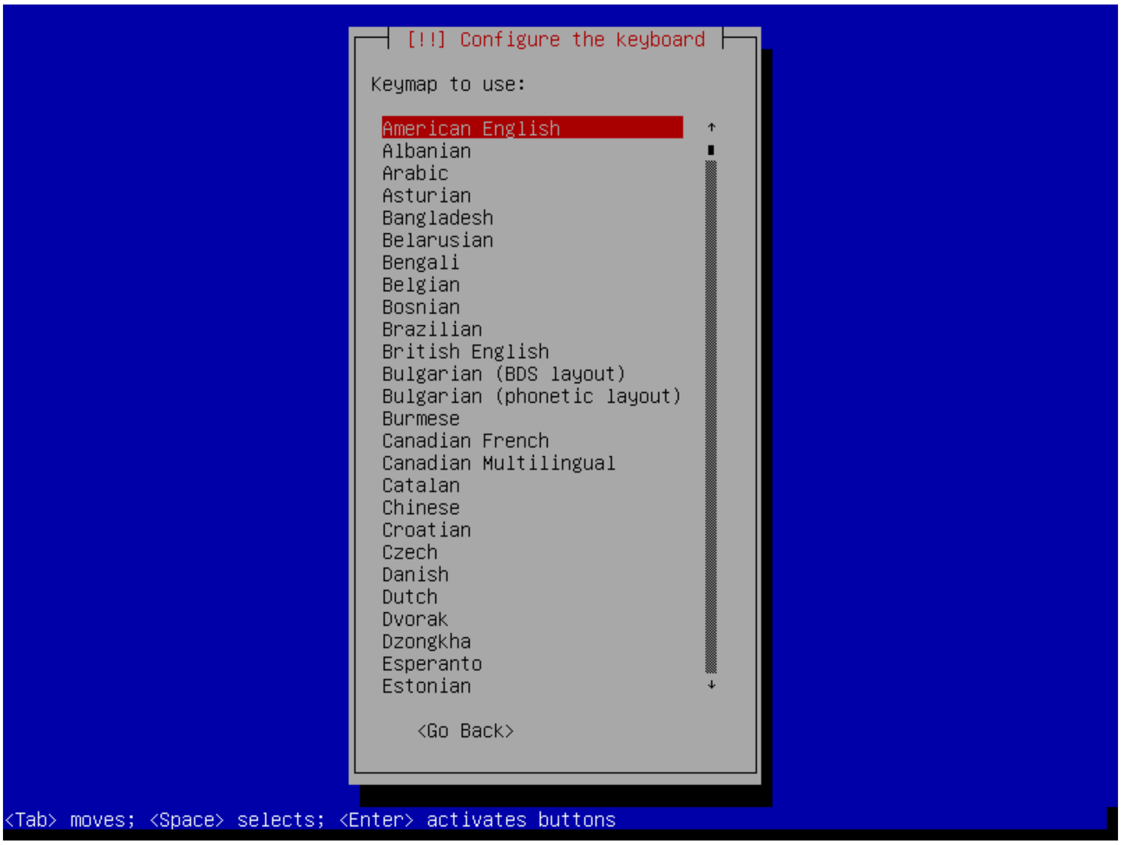
Lalu pilih **Indonesia**



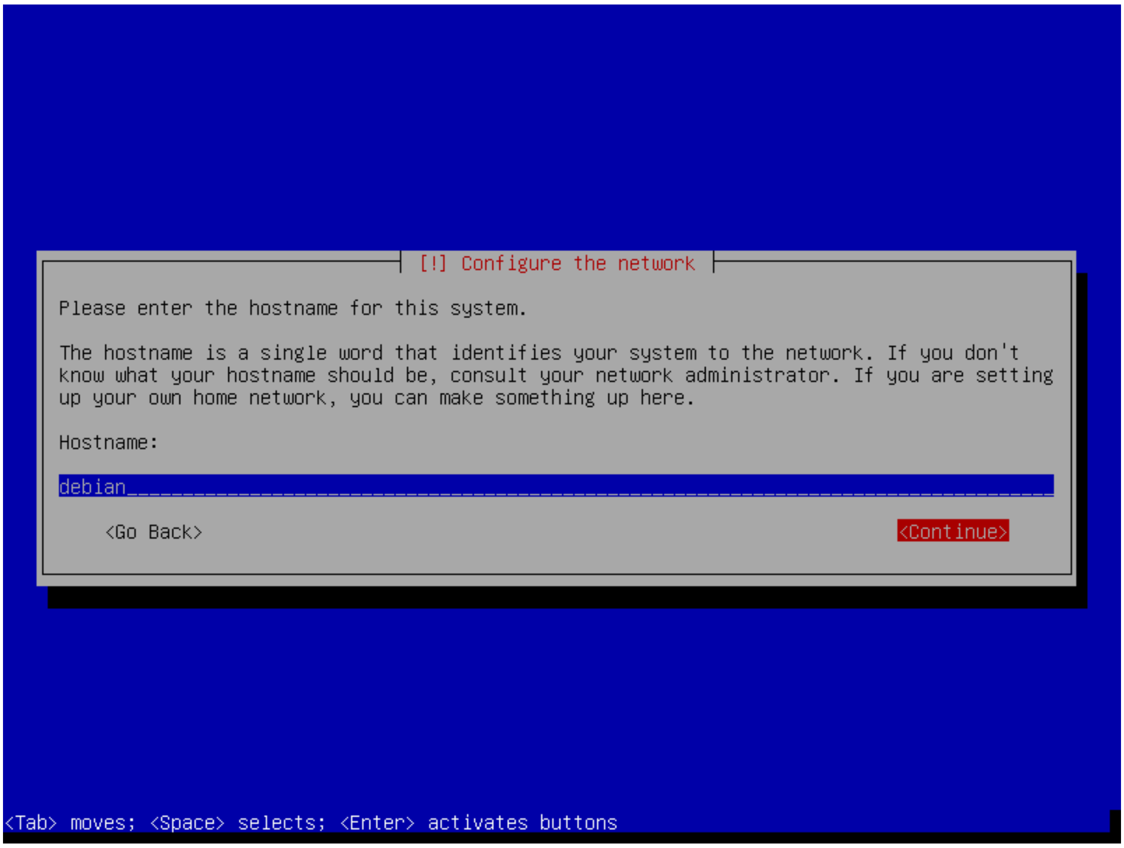
Pilih **United States**



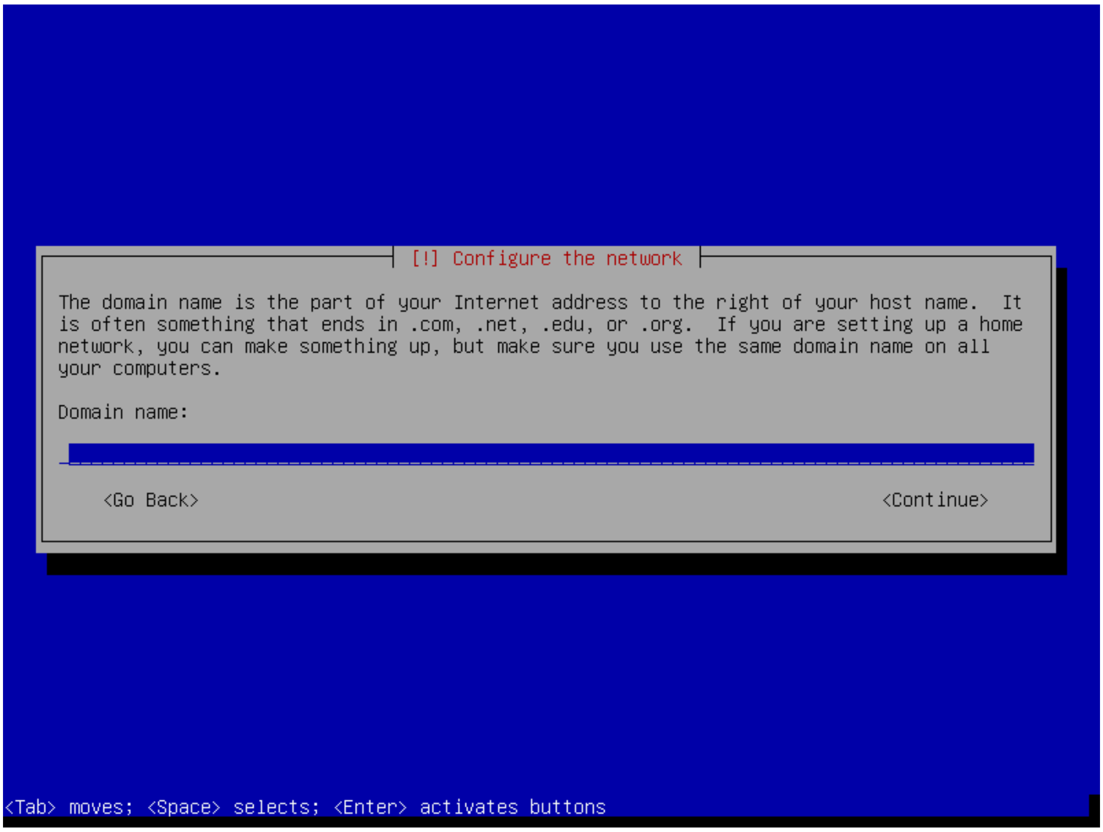
Layout keyboard, pilih **American English**



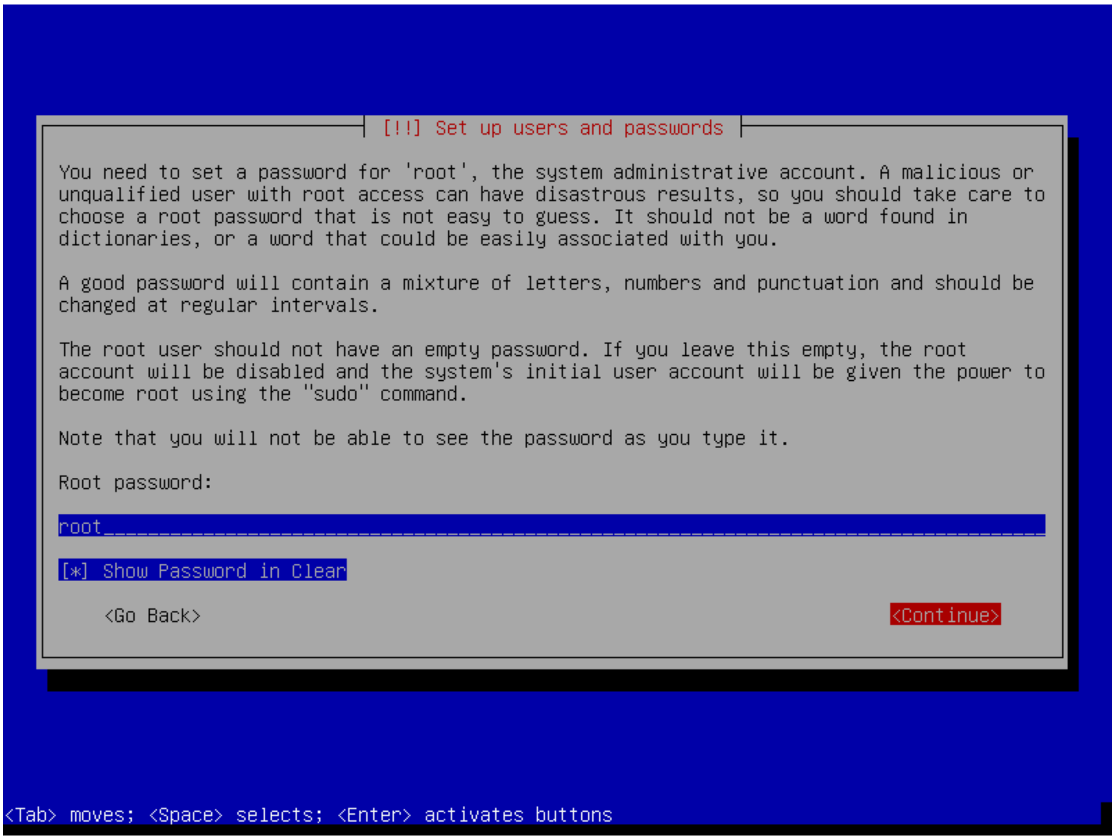
Isikan nama hostname/nama komputer kita, misal **debian**



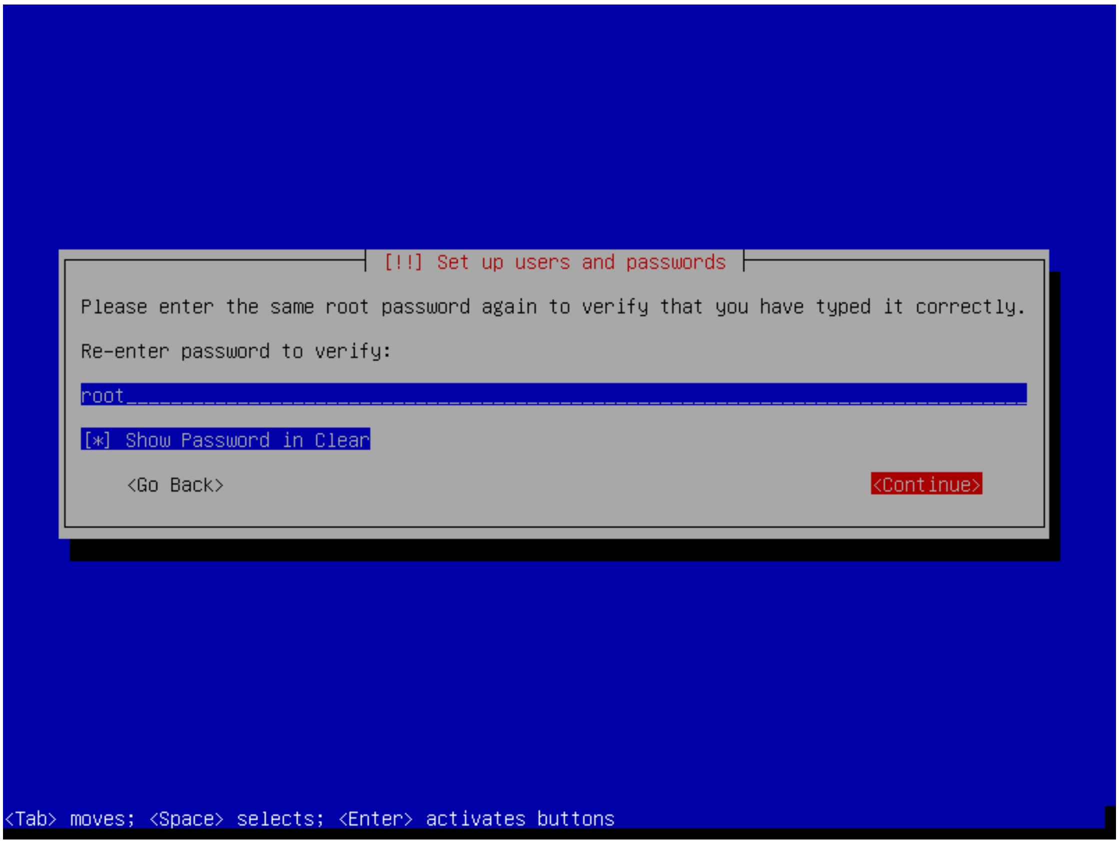
Isikan domain name, atau bisa dikosongkan (di konfigurasi nanti)



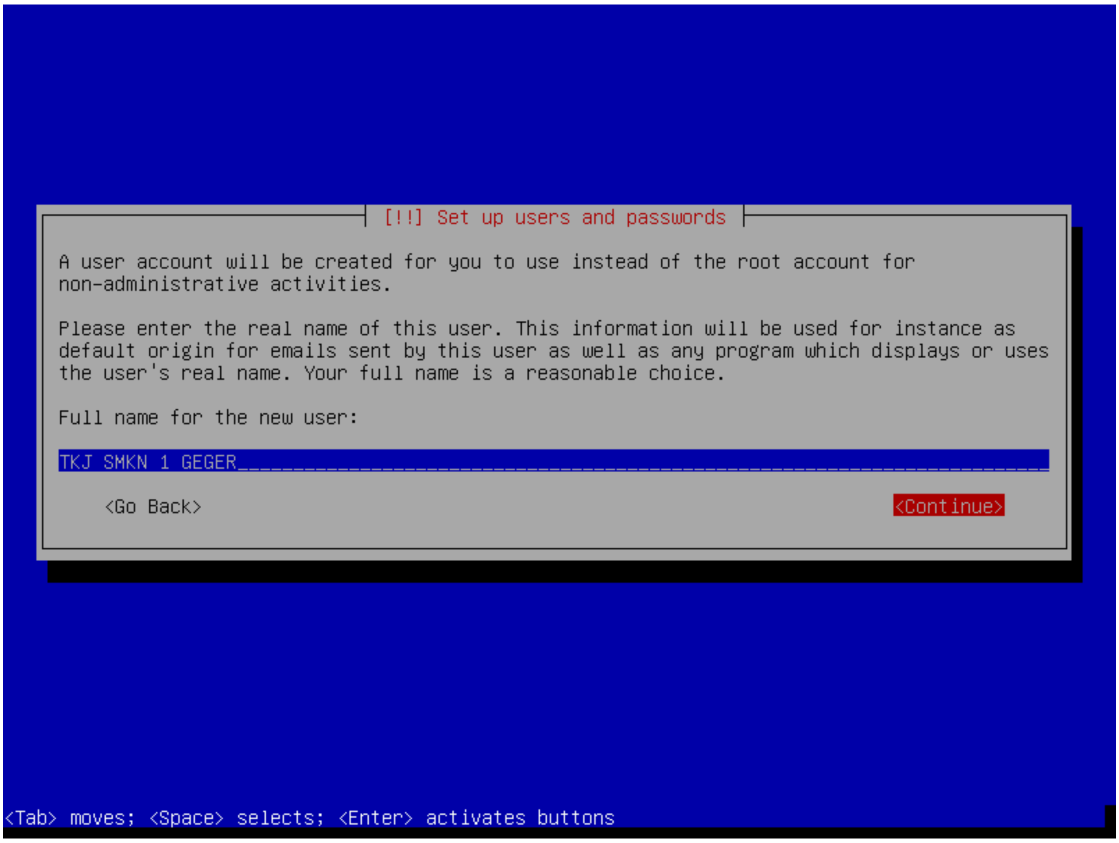
Masukkan password untuk user root, buat sesuai keinginan misal disini saya isi **root**



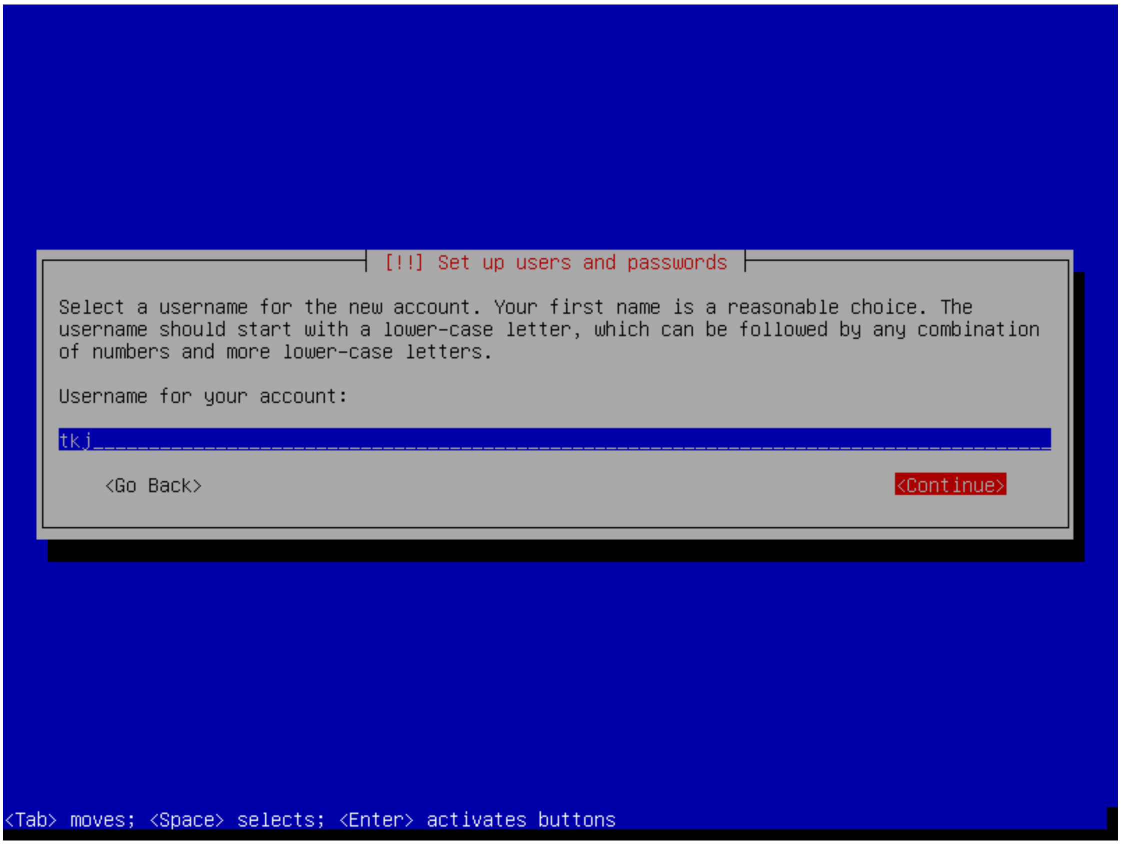
Masukkan password untuk user root sekali lagi



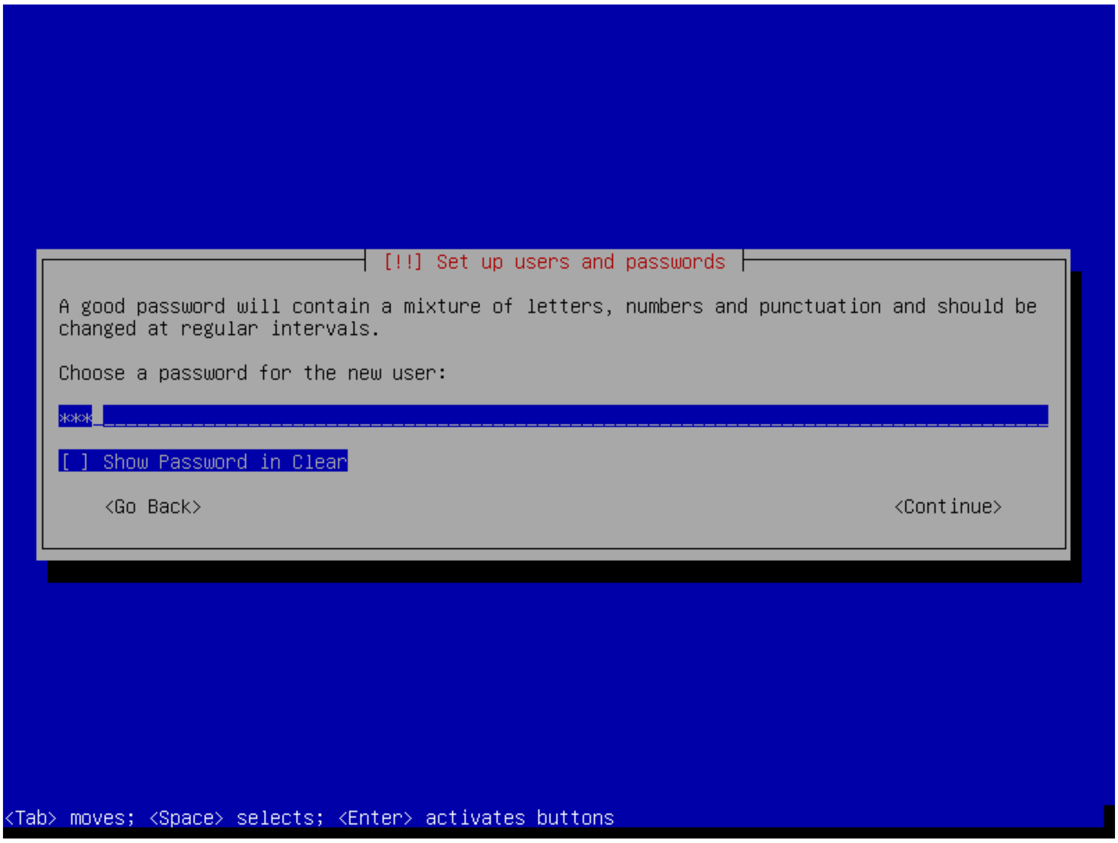
Isikan nama lengkap untuk user baru



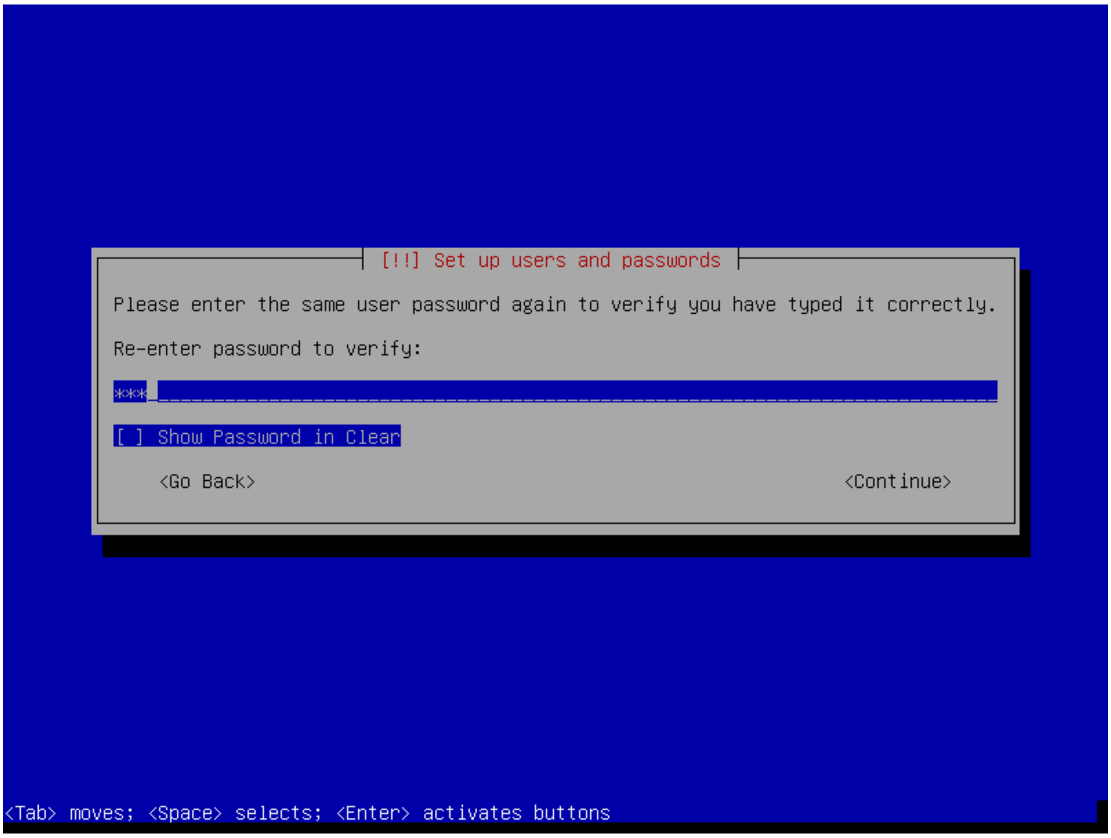
Isikan username untuk user baru



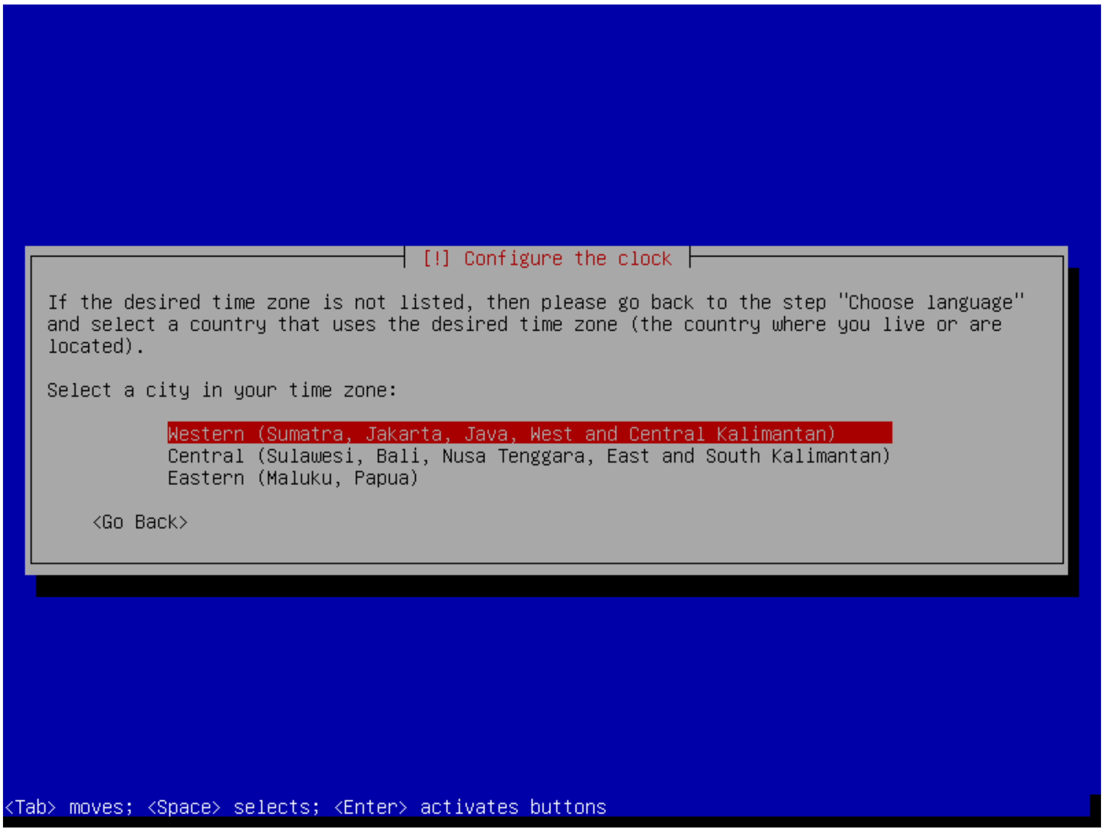
Masukkan password untuk user baru



Masukkan password sekali lagi

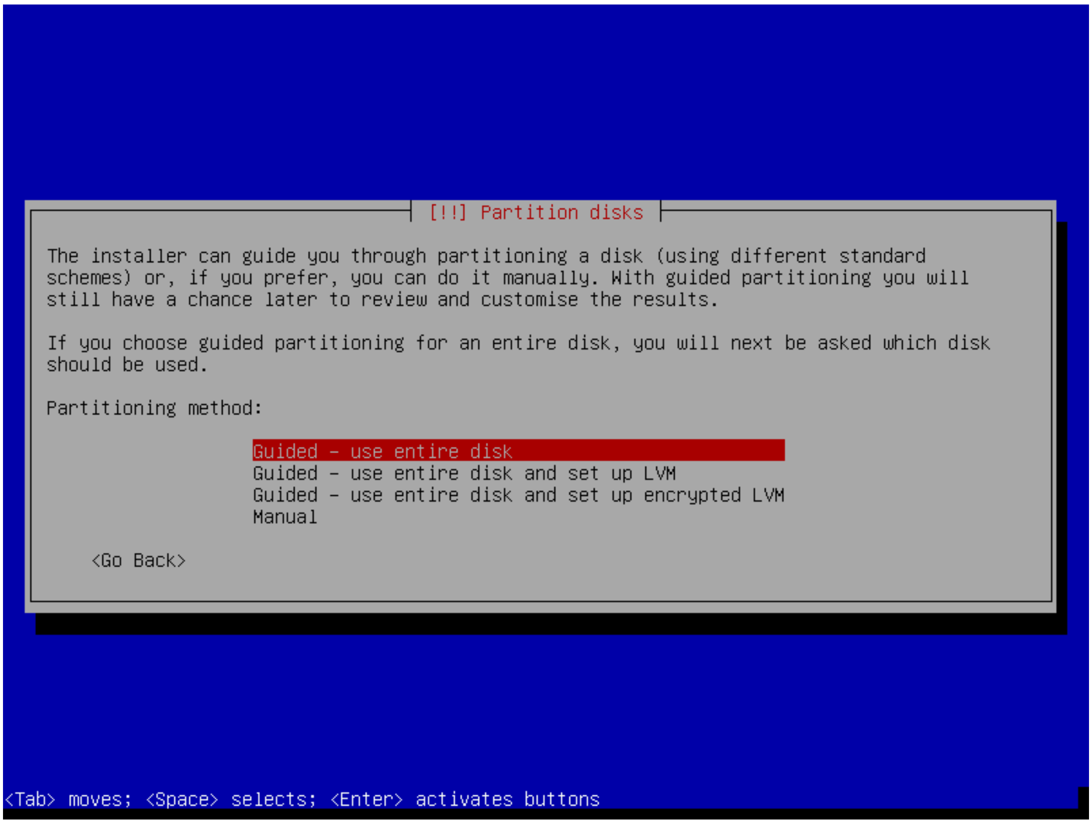


Pilih Zona Waktu, pilih Western untuk WIB

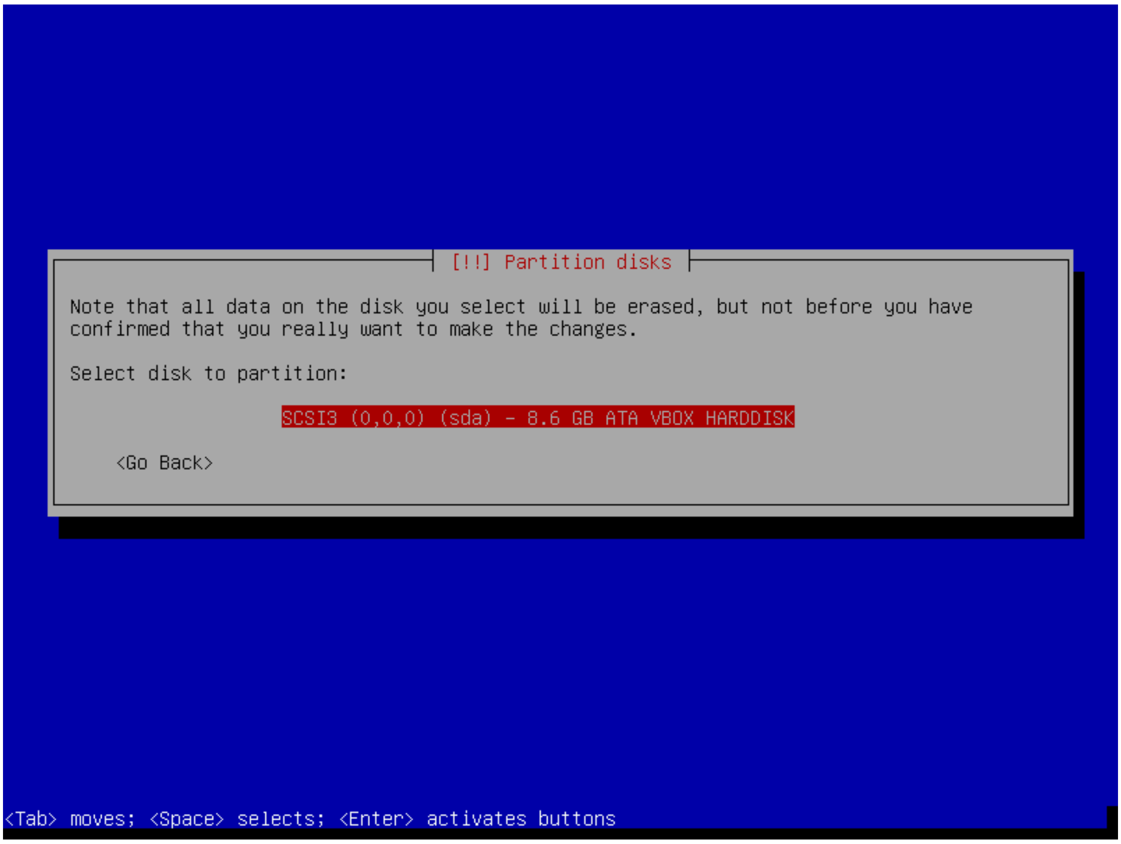


Lalu lakukan partisi, bisa secara otomatis atau manual

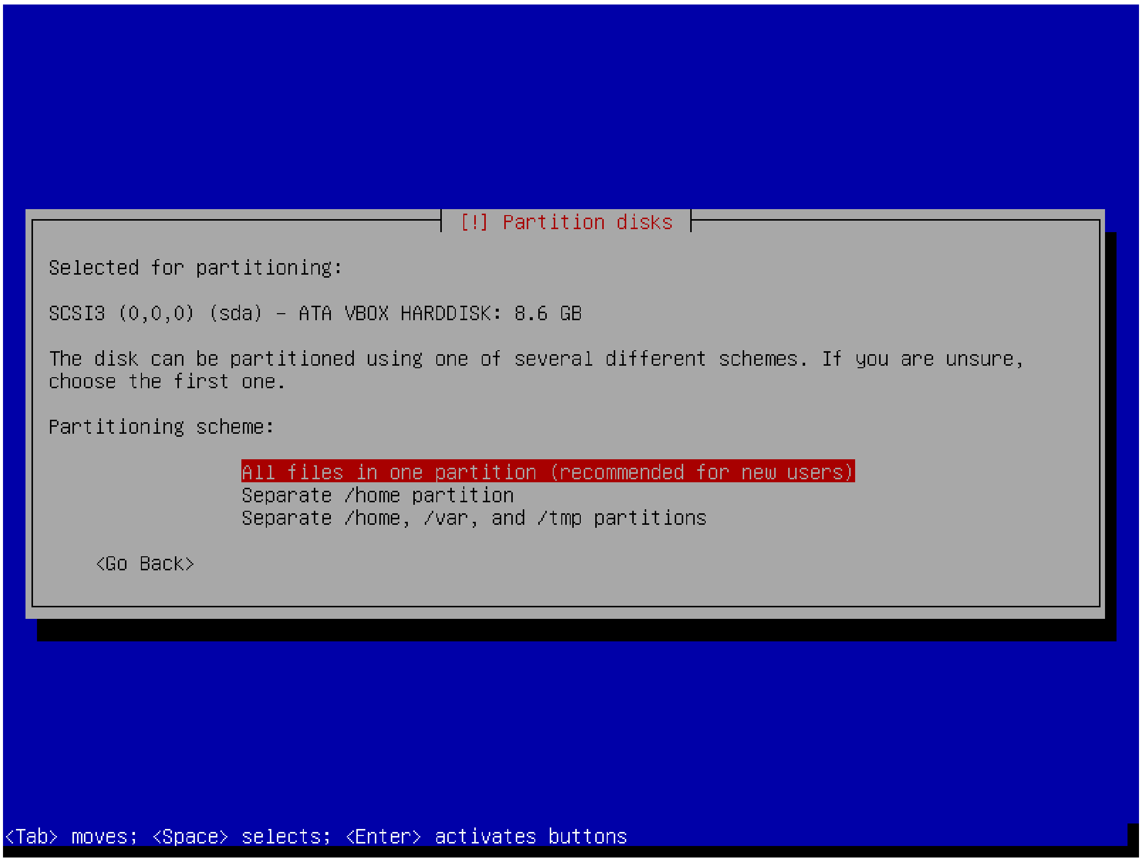
Untu partisi secara otomatis, kita pilih **Guided – use entire disk**



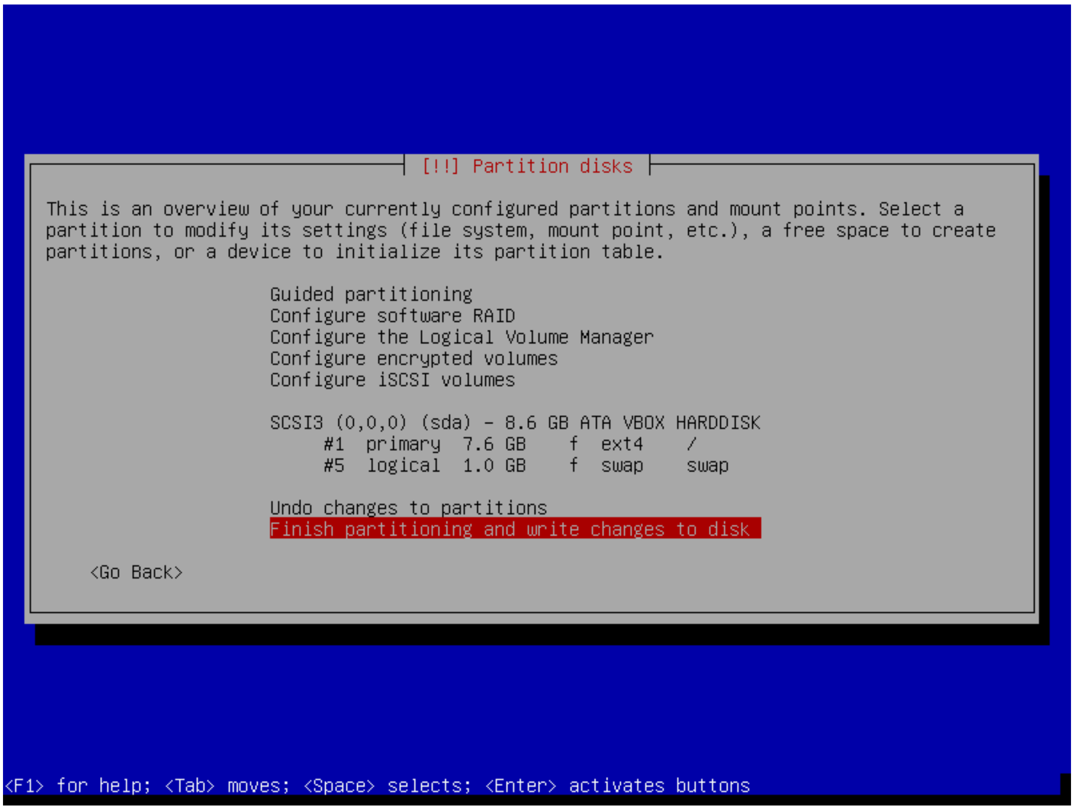
Pilih Harddisk tadi



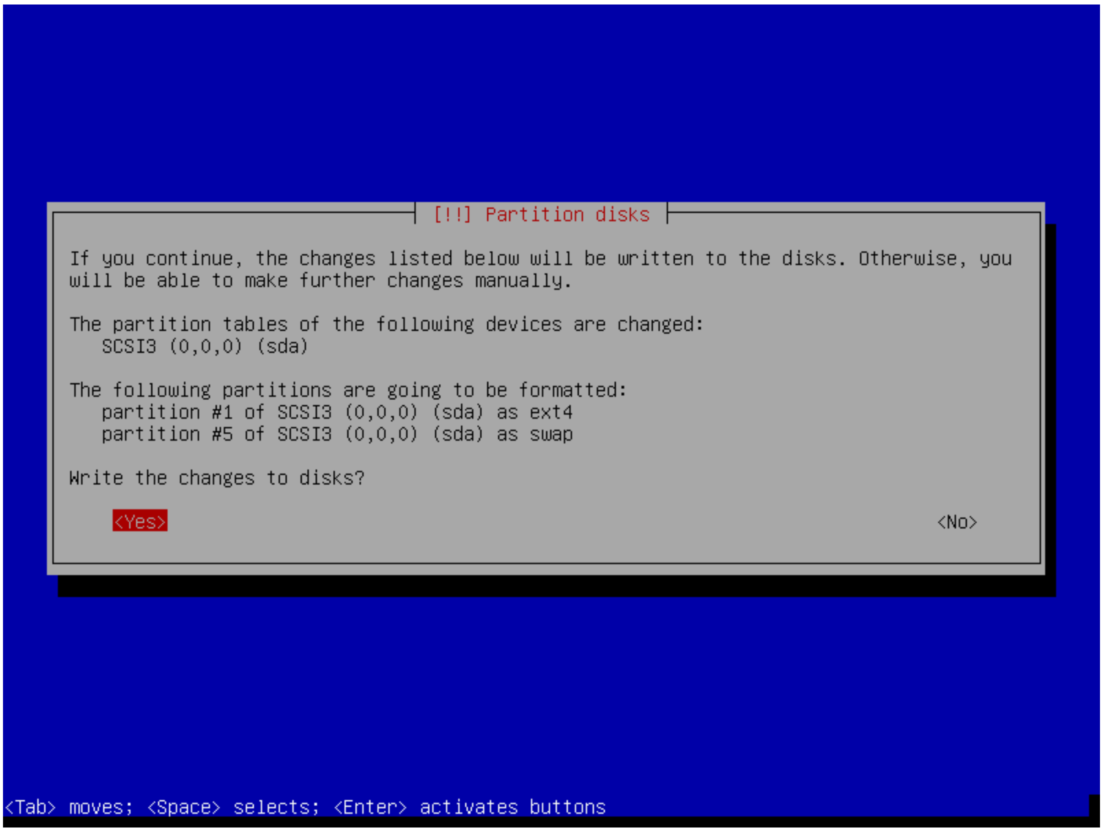
Pilih **All files in one partition (recommended for new users)**



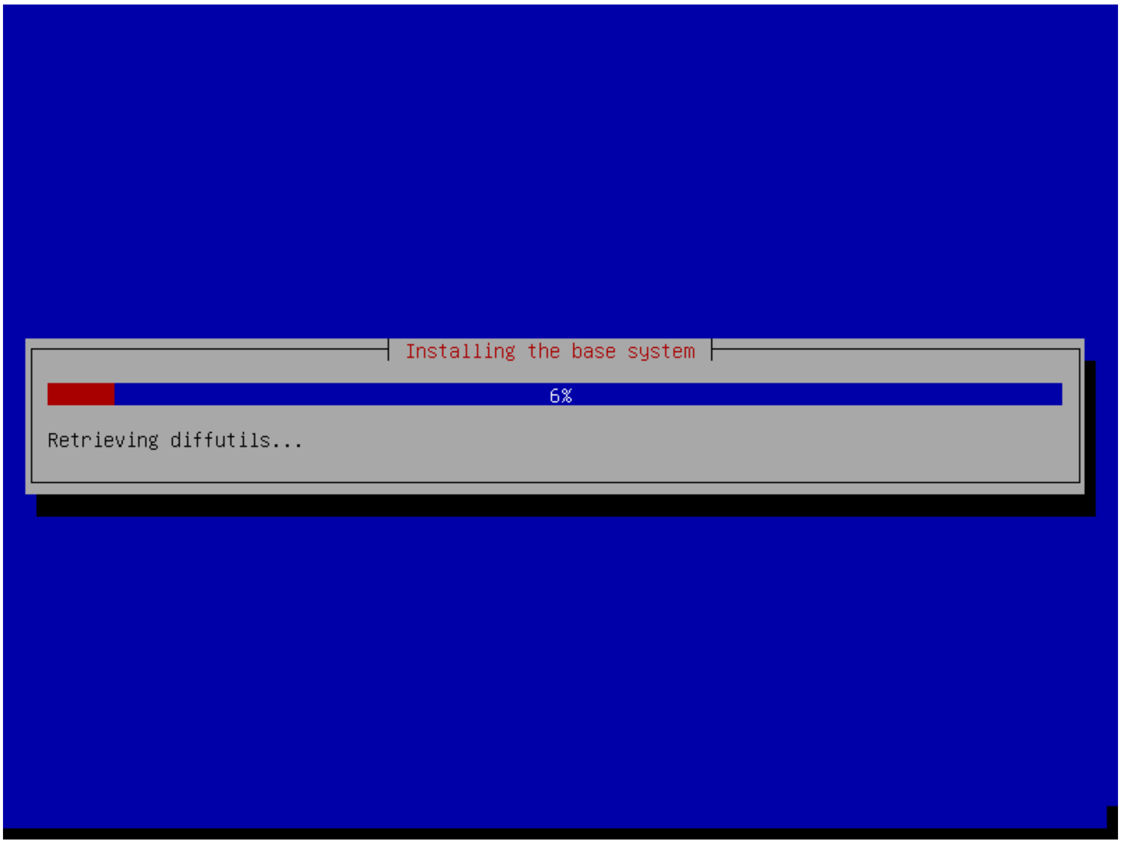
Pilih **Finish partitioning and write changes to disk**



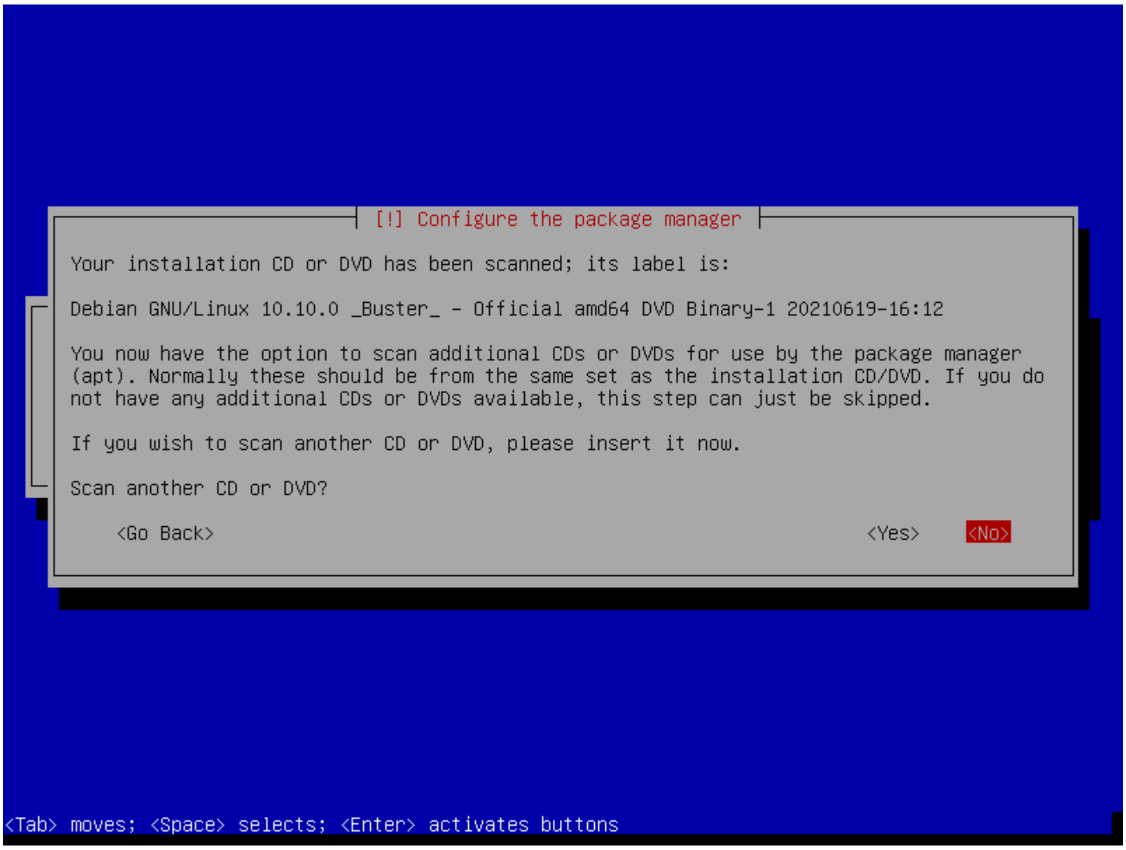
Write changes to disks? Pilih **Yes**



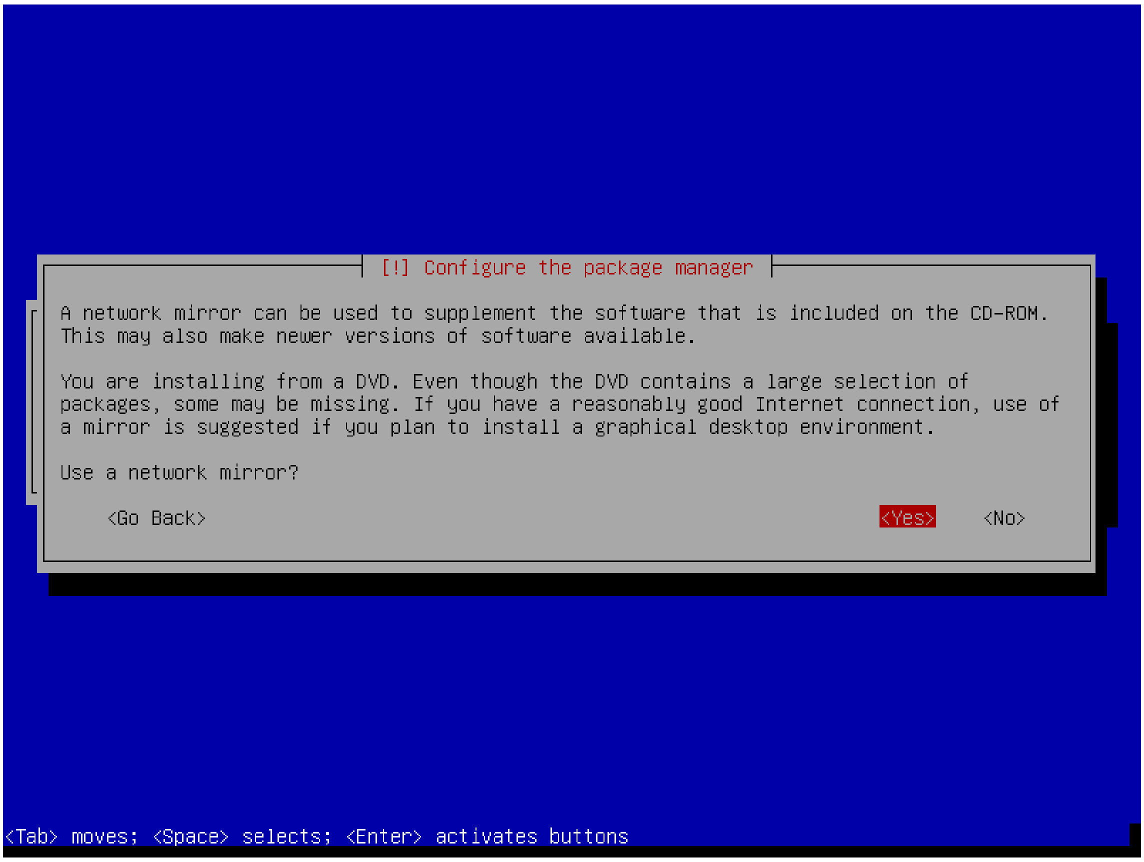
Lalu tunggu sampai install base system selesai



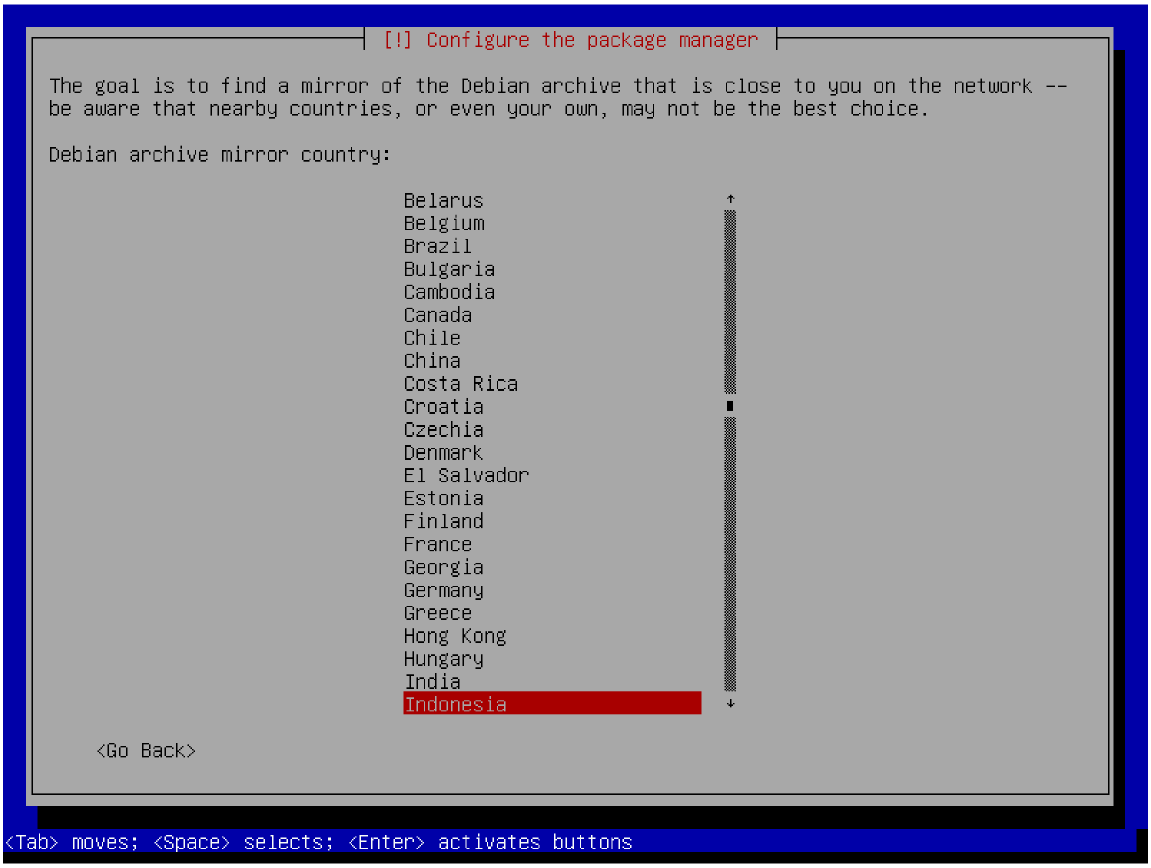
Scan antoher CD or DVD? Kita pilih **No**



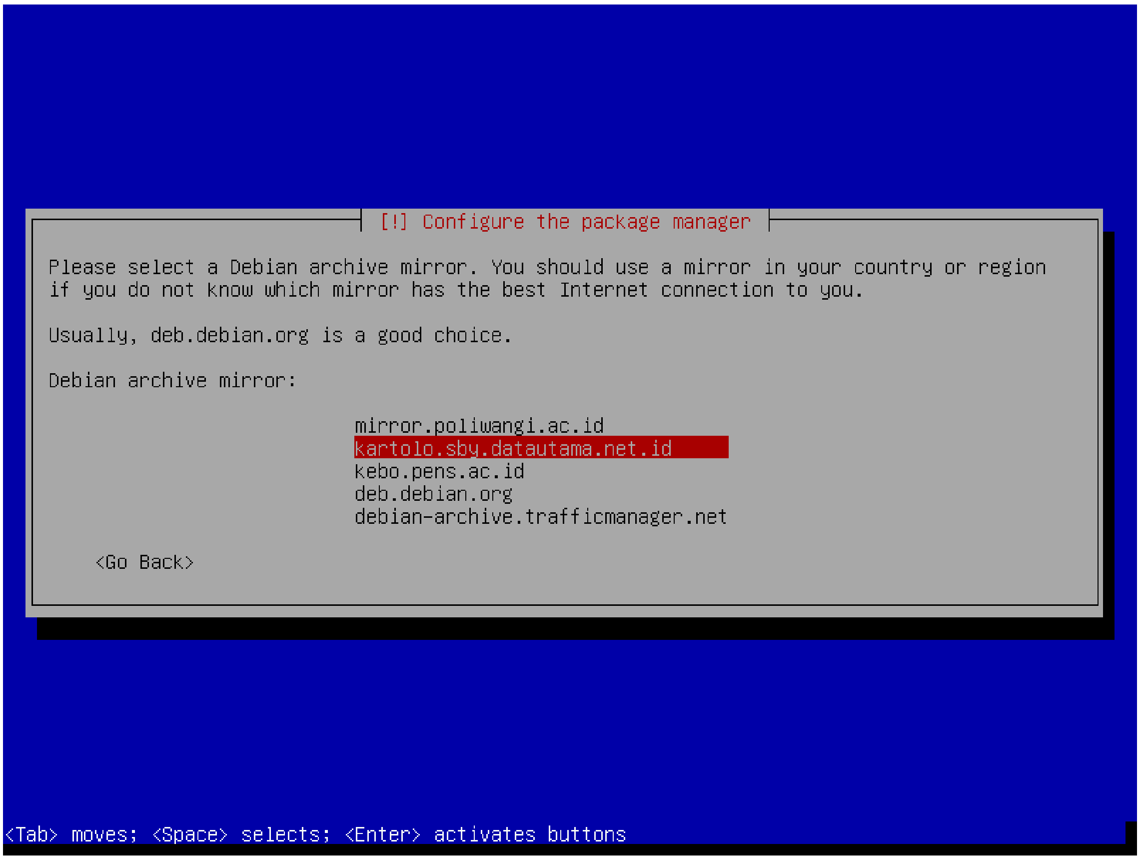
Use a network mirror? Jika ingin menambah repo kita pilih yes



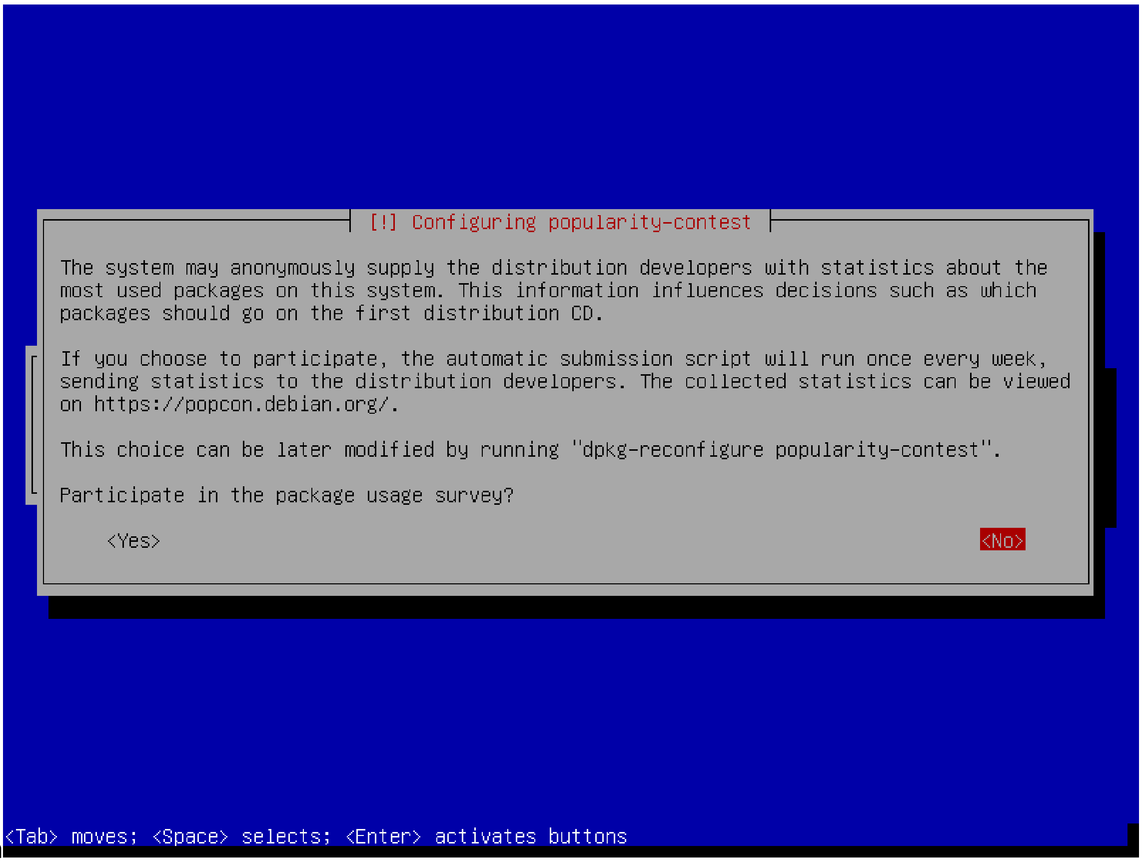
Pilih **Indonesia**



Pilih Repo, misal **kartolo.sby.datautama.net.id**



Participate in the package usage survey? Pilih **No**



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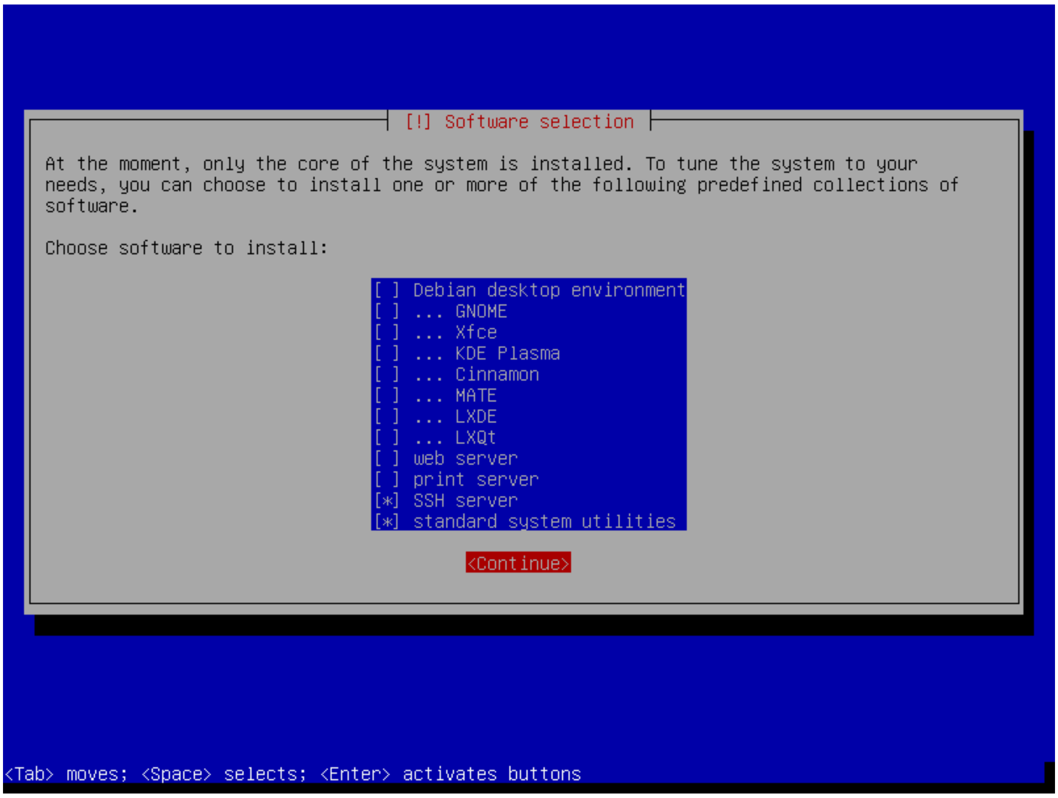
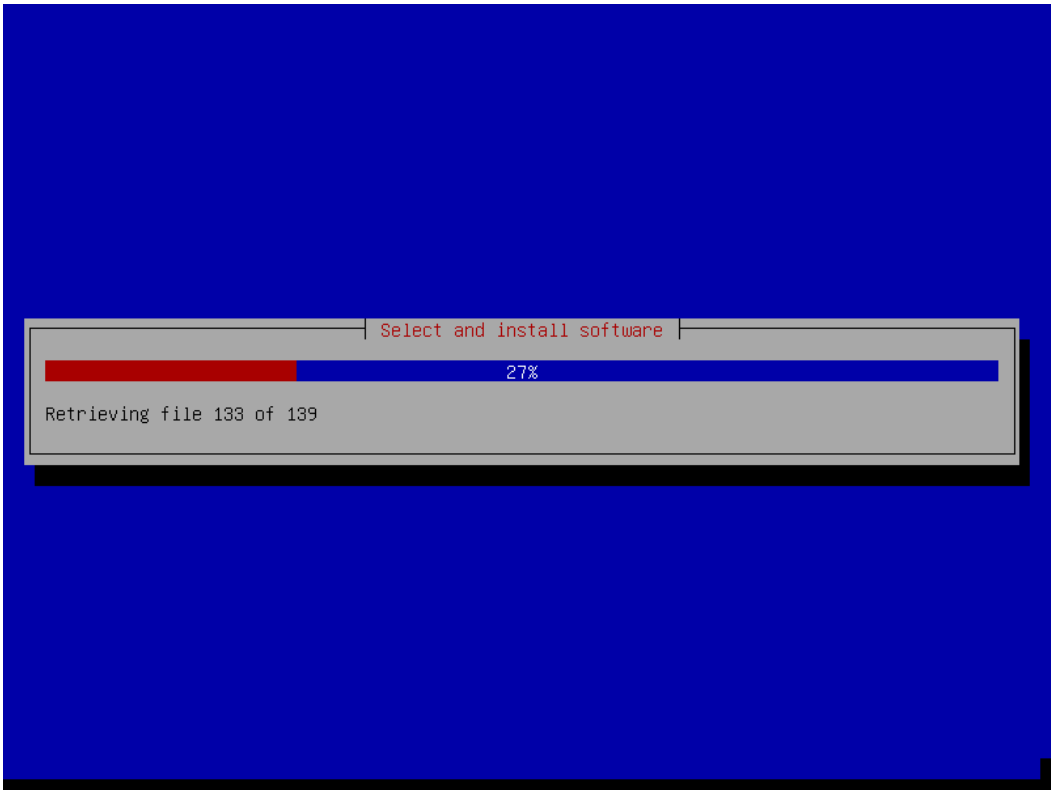
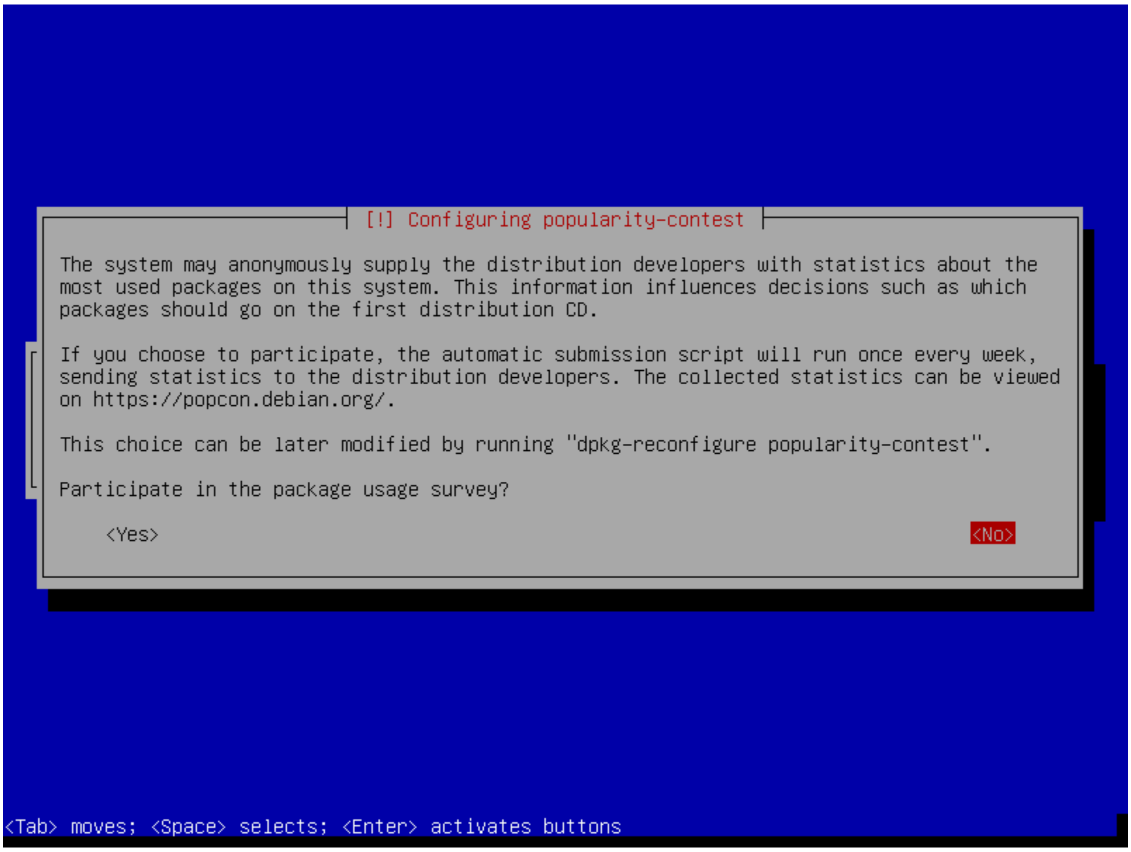
Selanjutnya pilih paket apa saja yang mau di install, untuk server yang berbasis CLI centang **SSH Server** dan **standard system utilities**. Untuk server yang berbasis GUI centang **Debian desktop environment.**



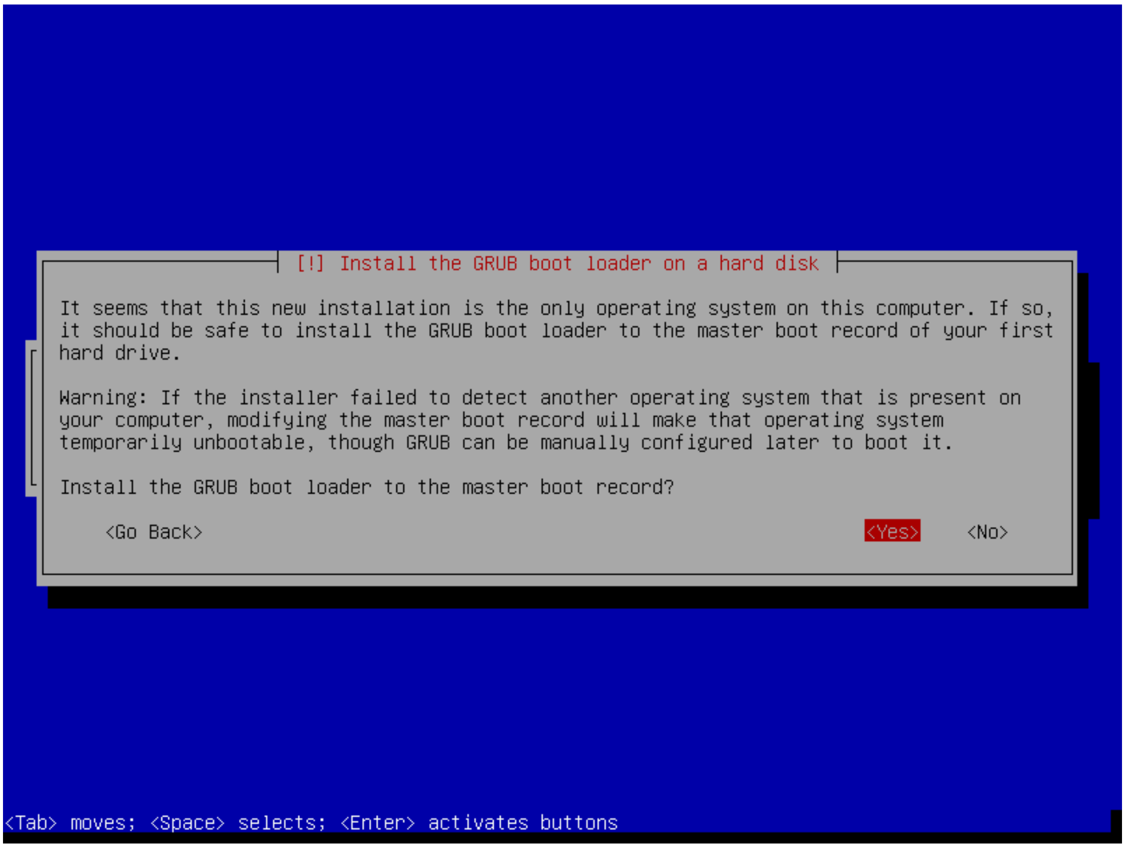
Tunggu Instalasi software sampai selesai



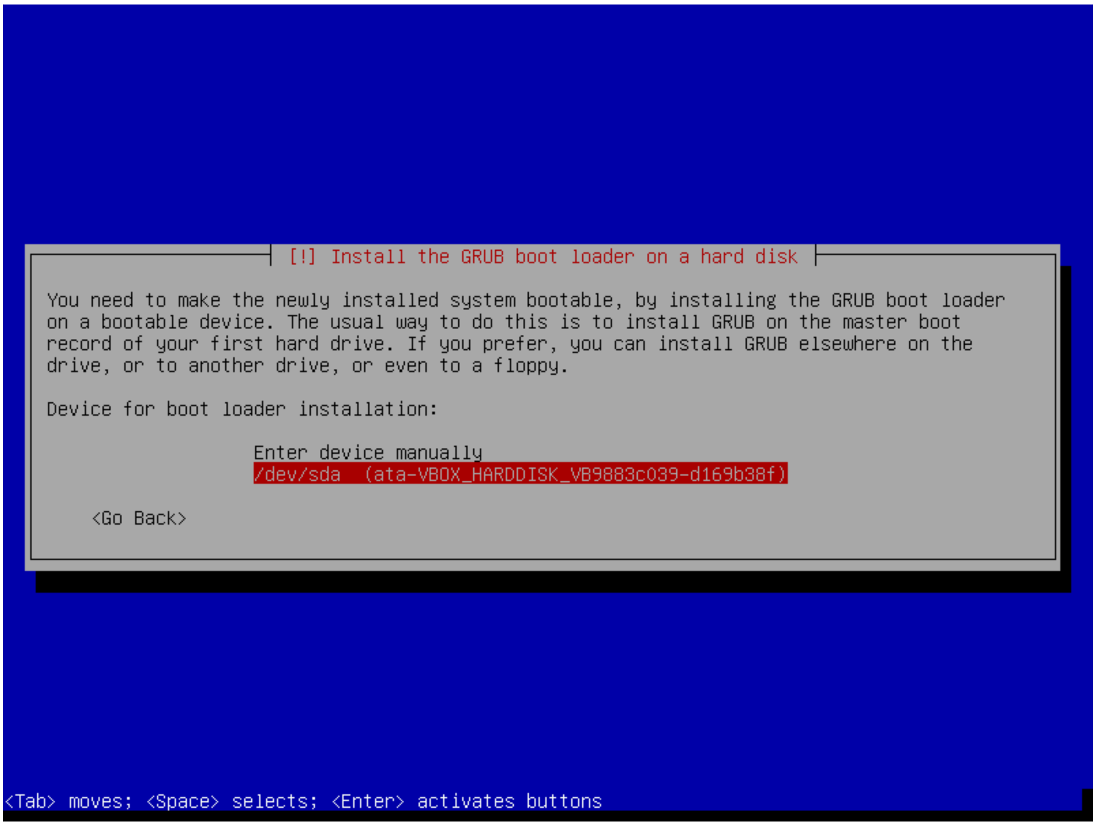
Participate in the package usage survey? Pilih **No**



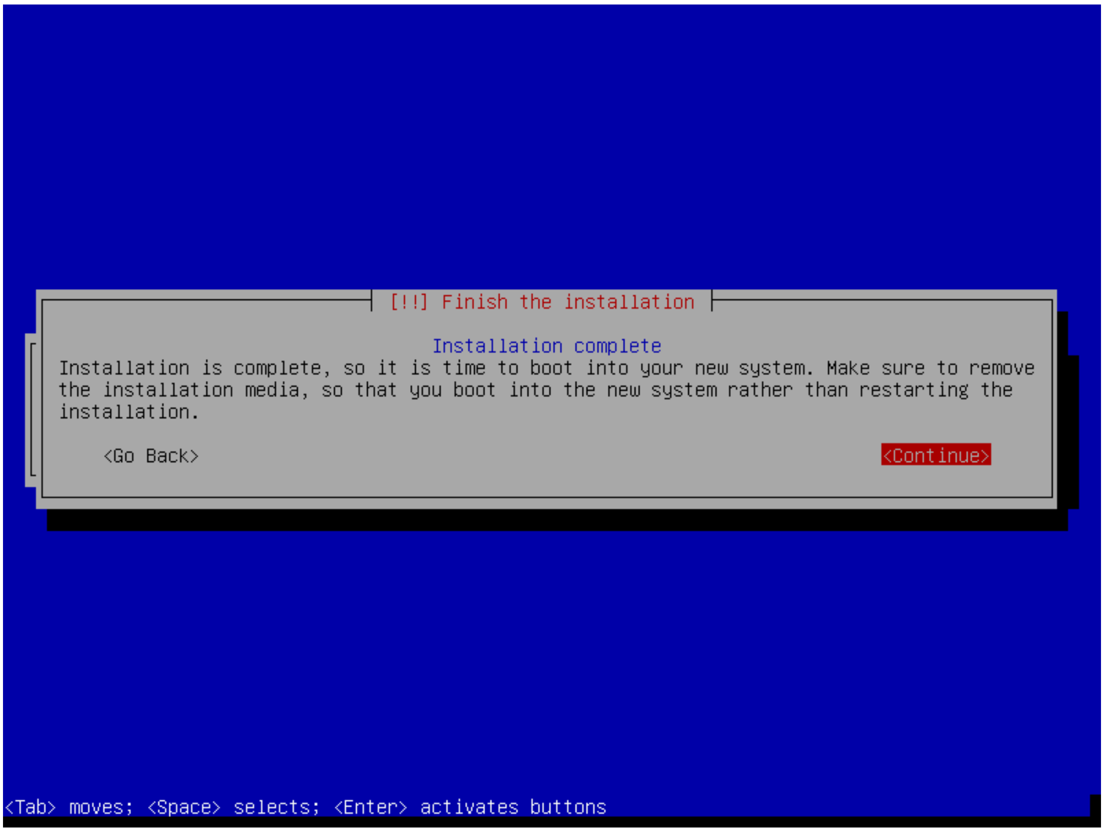
Install the GRUB boot loader to master boot record? Pilih **Yes**



Pilih **/dev/sda** atau harddisk yang di install debian tadi

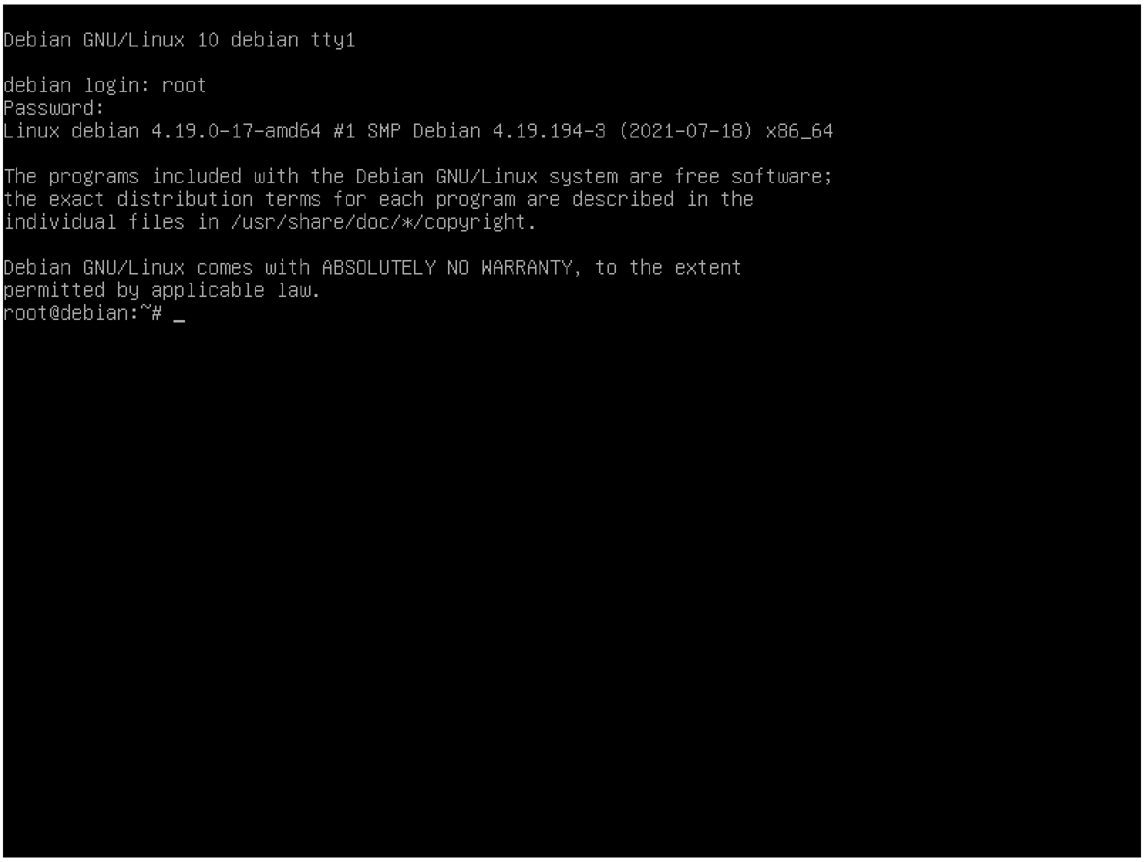


Instalasi berhasil, terakhir ada perintah restart pilih **Continue**

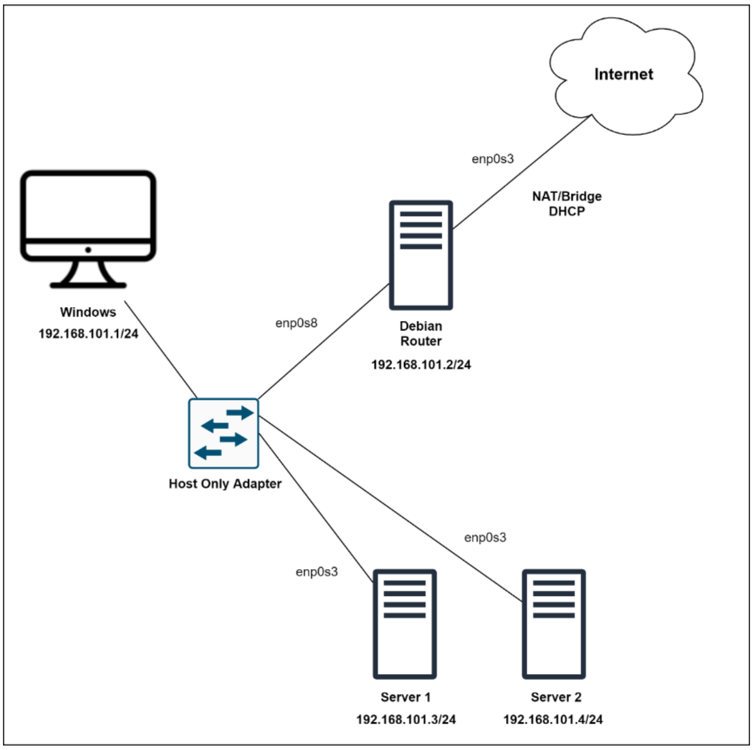


Setelah reboot, maka akan muncul tampilan login debian. Coba login dengan user root/tkj

Berhasil login



2.3 Setup Topology di VirtualBox

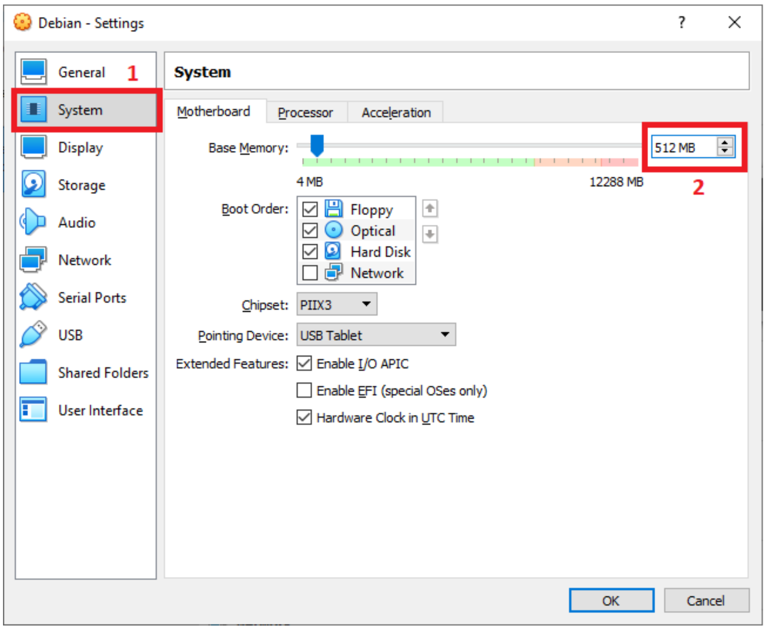


2.4.1 Cloning VM Debian

Apa itu cloning VM? Jadi cloning VM adalah menduplicate VM yang sudah ada menjadi VM yang berbeda, karena di topologi kita butuh 3 VM Debian maka kita lakukan cloning dengan metode linked clone(dengan basis VM Debian yang sudah kita install tadi). Kenapa menggunakan Linked Clone? karena lebih hemat storage.

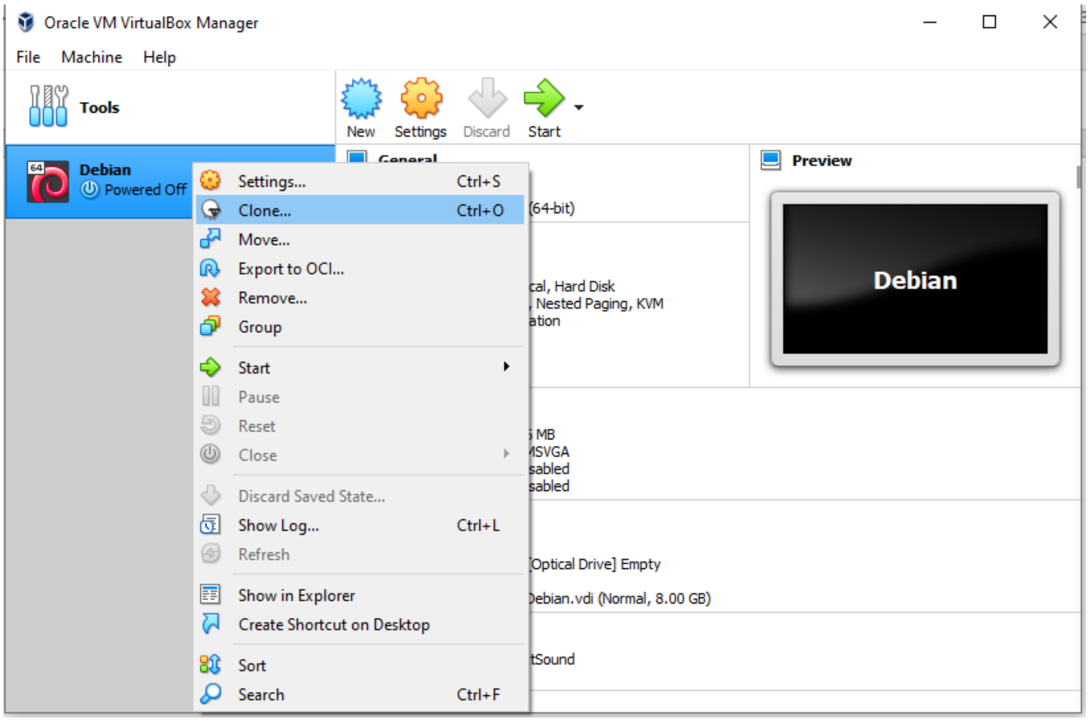
Kita siapkan terlebih dahulu base VM nya

Pilih VM Debian lalu klik Settings > System > Ubah RAM menjadi 512, setelah itu klik OK

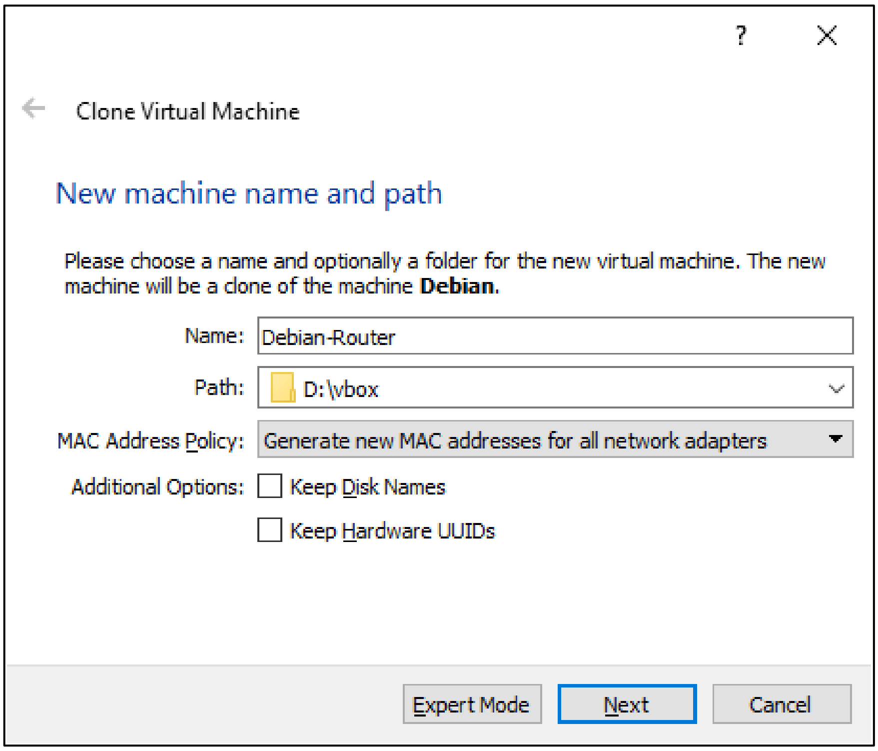


2.4.1.1 Cloning VM Untuk Router Debian

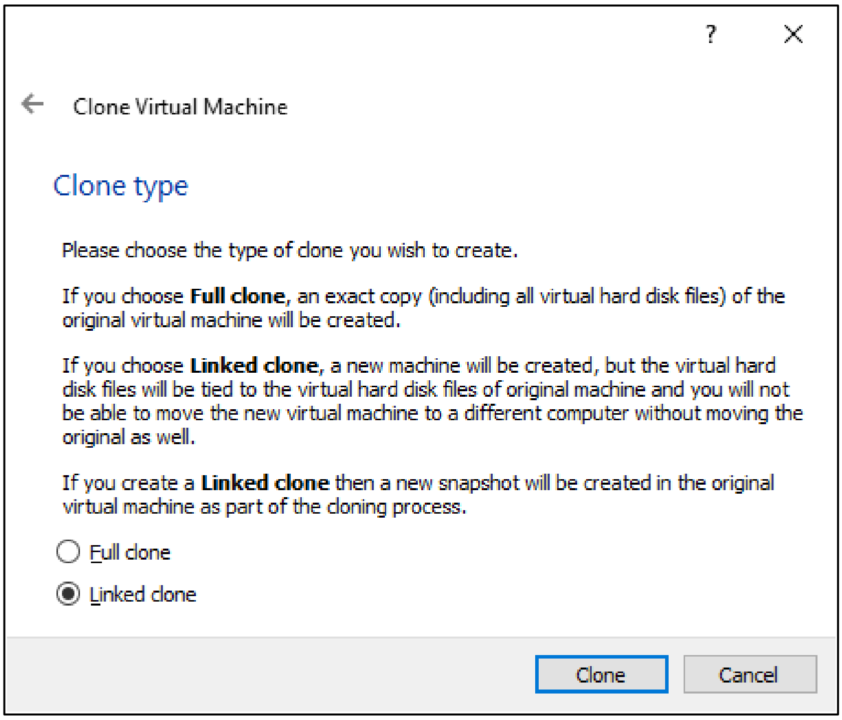
Klik kanan pada VM Debian yang kita buat tadi, lalu klik Clone

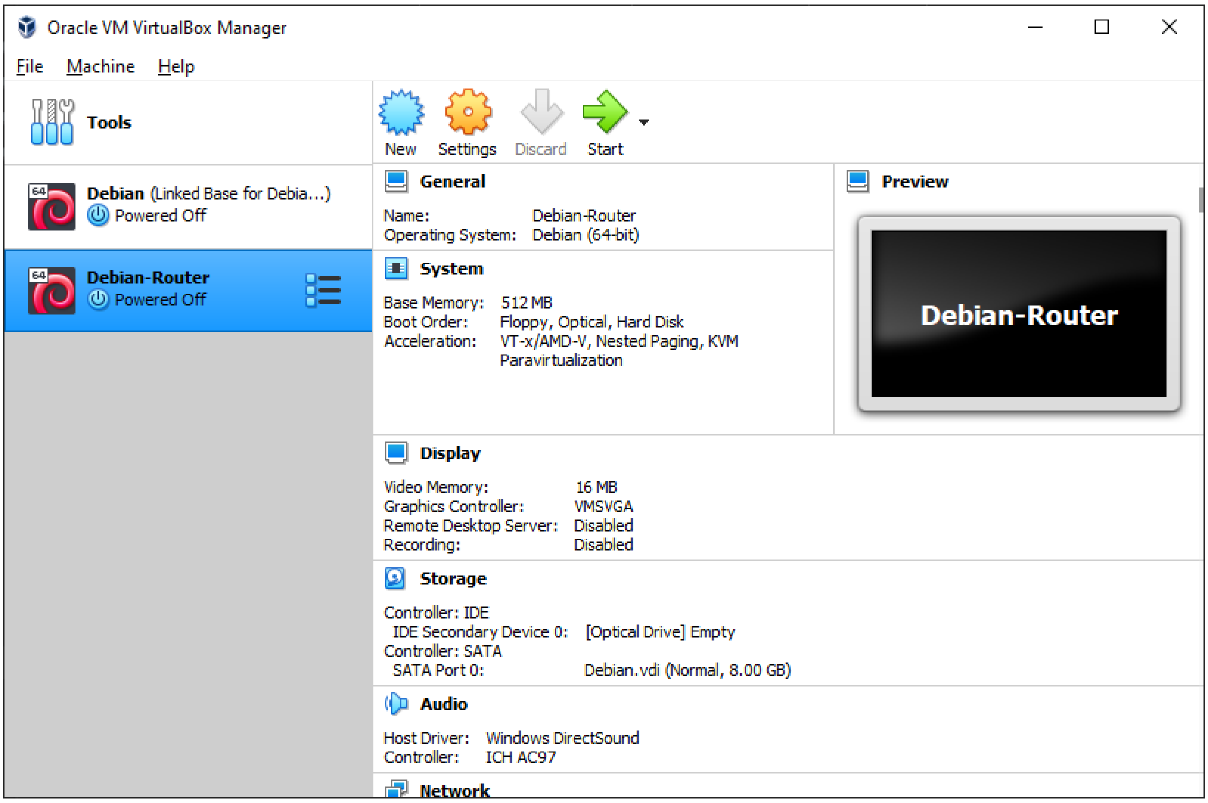


Lalu ubah nama menjadi **Debian-Router**, pada MAC Address Policy kita pilih Generate new **MAC addresses**



Lalu kita pilih linked clone dan klik Clone

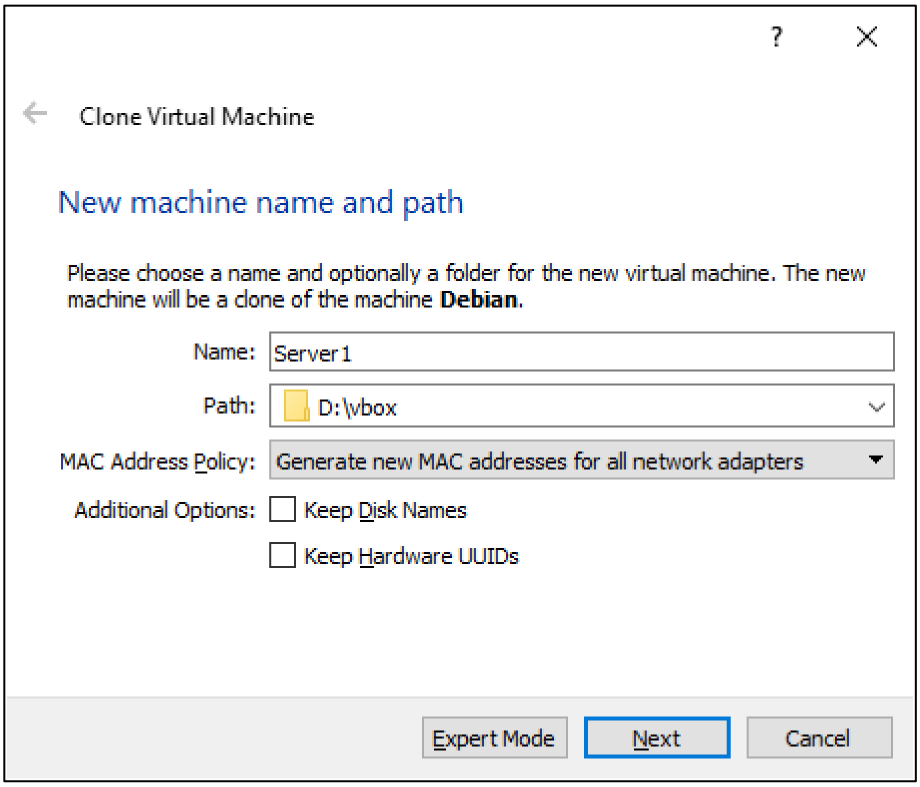




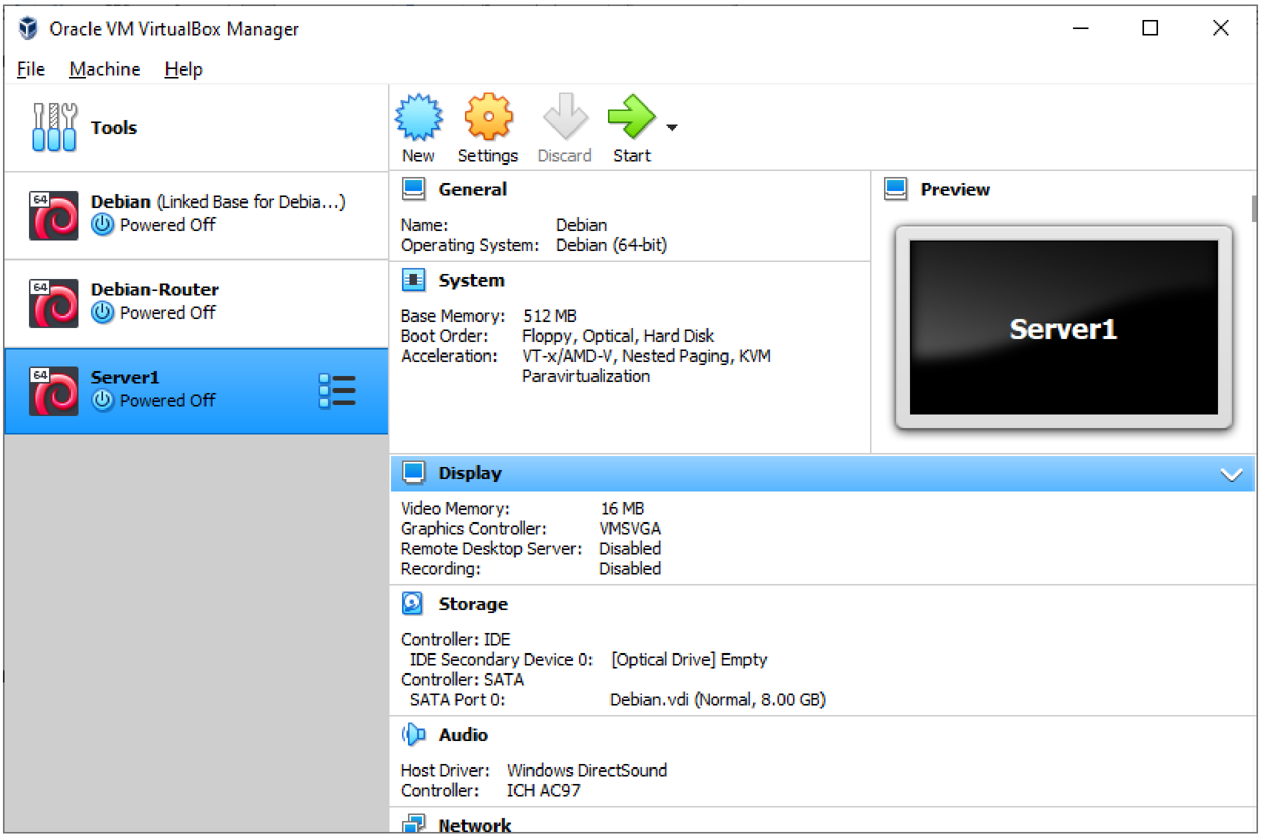
2.4.1.2 Cloning VM untuk Server 1

Sama seperti tadi Klik Kanan VM Debian > Clone

Lalu ubah nama menjadi Server1, MAC Address Policy kita pilih Generate new **MAC addresses**

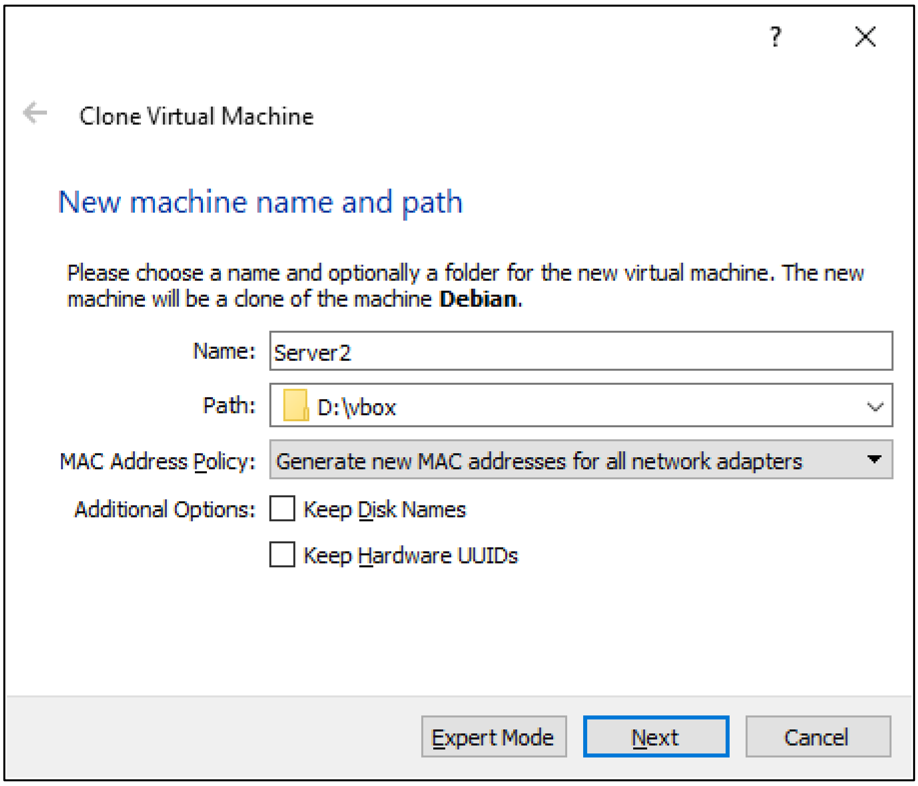


Setelah itu pilih Linked Clone (sama seperti tadi)

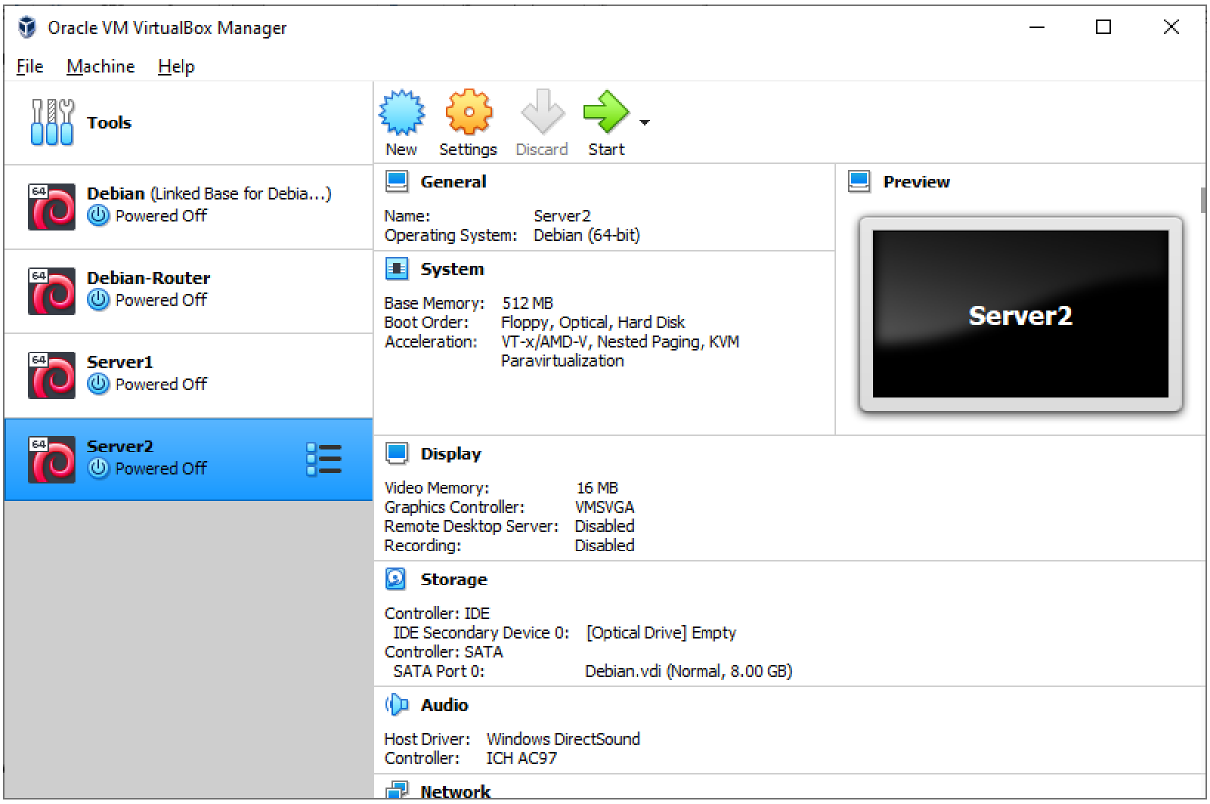


2.4.1.3 Cloning VM untuk Server 2

Ulangi Langkah tadi



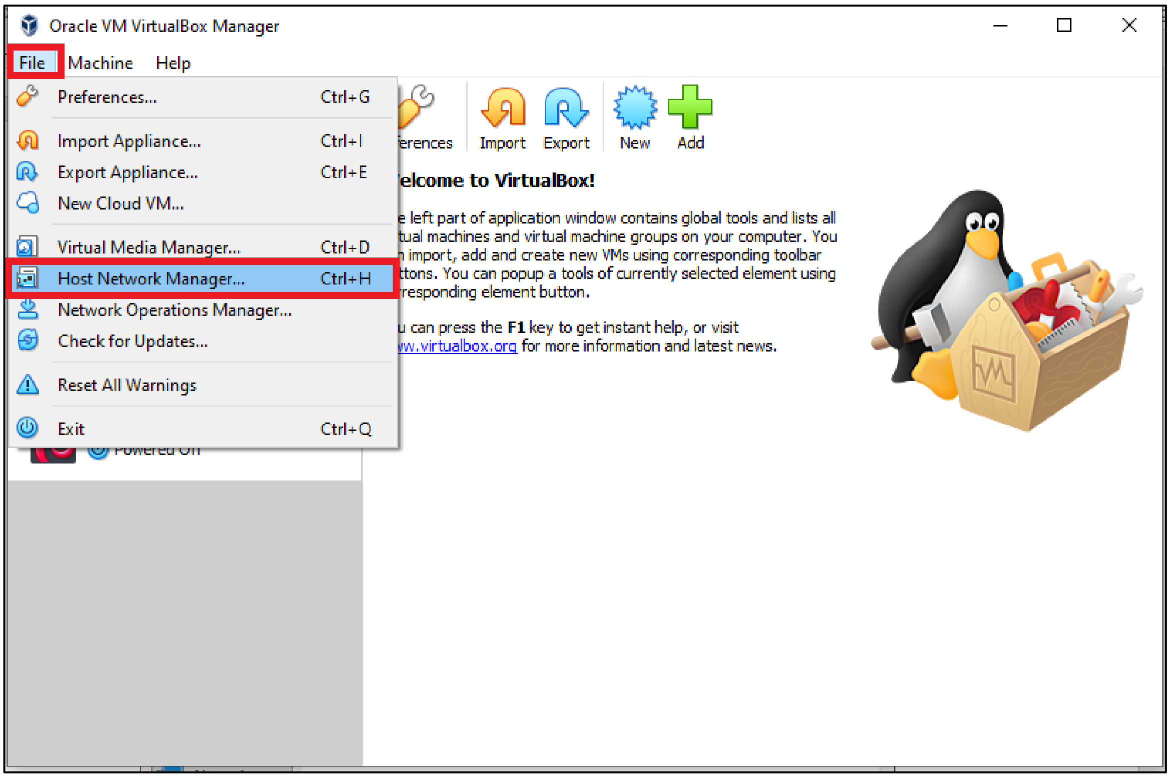
Dan 3 VM pun berhasil kita buat



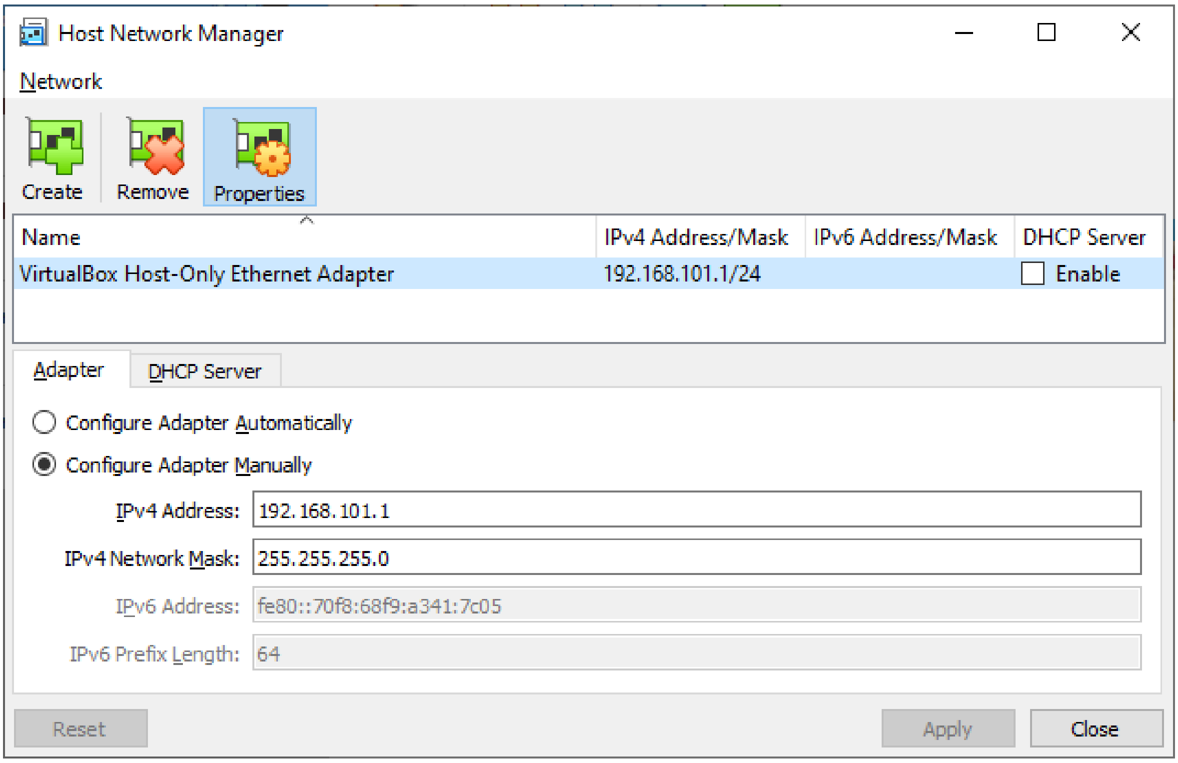
2.4.2 Konfigurasi Network Adapter Debian

Mengubah Alamat IP Host-Only Adapter sesuai topologi

Klik File > Host Network Manager



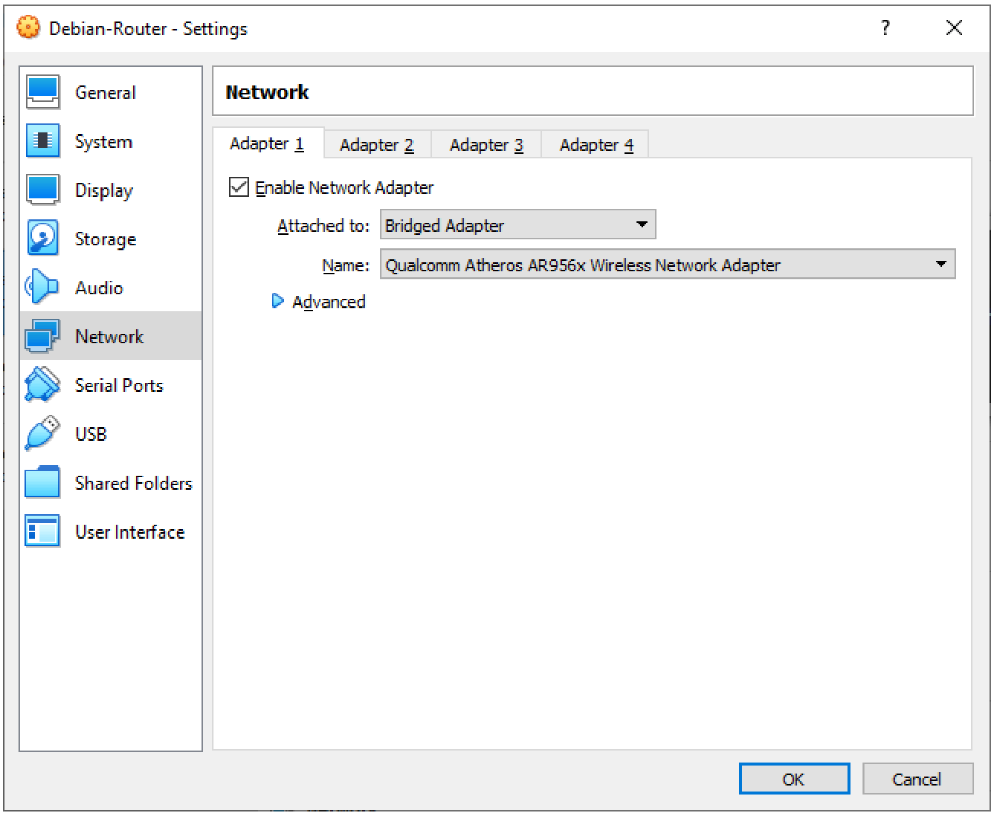
Atur alamat IP dengan topologi, jika IP 192.168.101.x bentok dengan IP LAN maka bisa diganti IP yang lain, jangan lupa uncentang **DHCP Server**

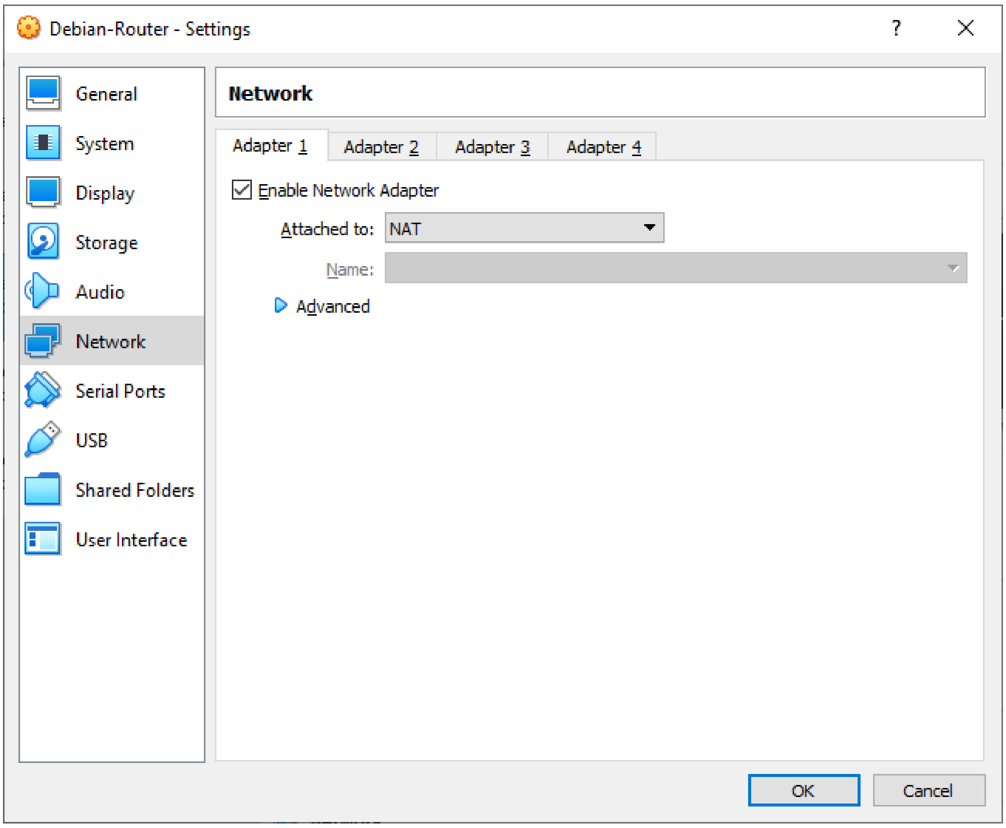


2.4.2.1 Network Adapter untuk Router-Debian

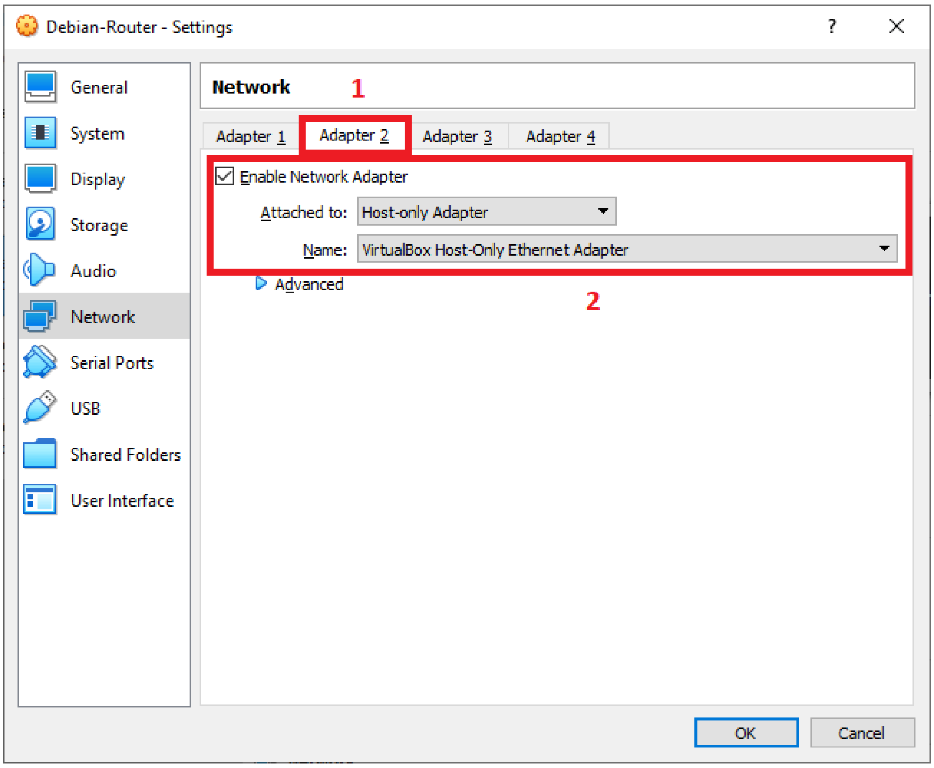
Pilih VM Debian-Router lalu klik Settings > Network > Adapter 1

Pilih Bridged Adapter / NAT (Pilih salah satu)





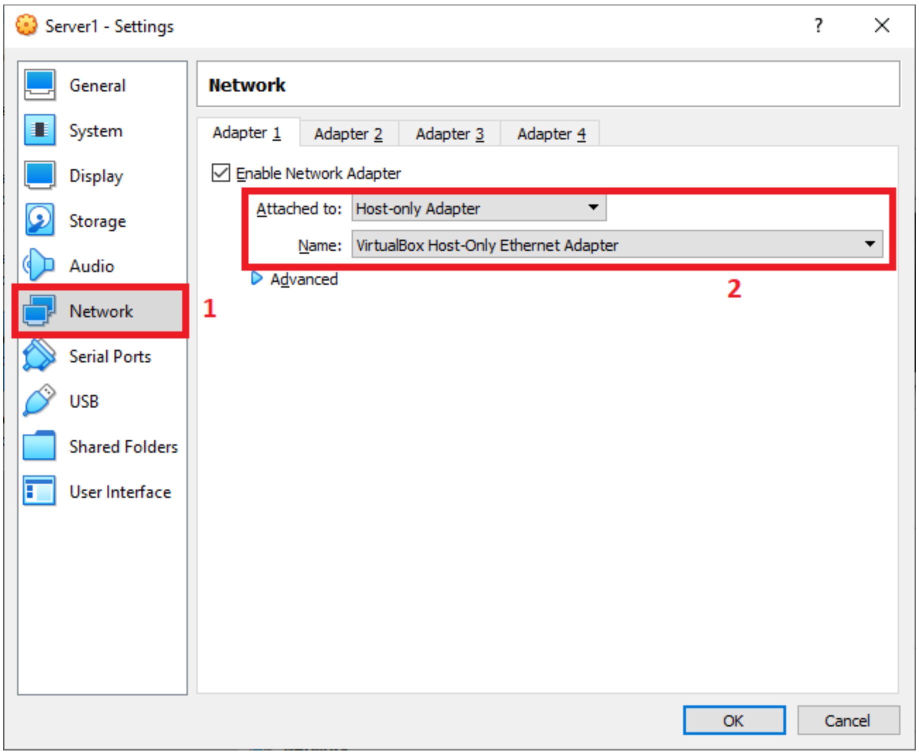
Selanjutnya, klik Adapter 2 lalu centang **Enable Network Adapter** lalu pilih **Host-Only Adapter**



2.4.2.2 Network Adapter untuk Server 1

Pilih VM Server1 lalu klik Settings > Network > Adapter 1

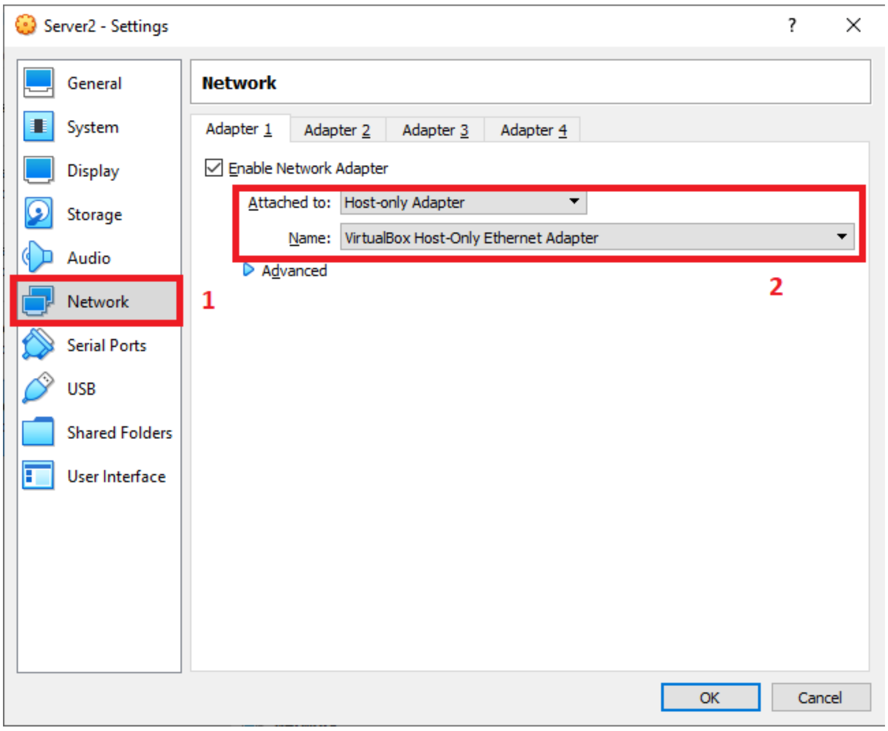
Pilih **Host-only Adapter**



2.4.2.3 Network Adapter untuk Server 2

Pilih VM Server2 lalu klik Settings > Network > Adapter 1

Pilih **Host-only Adapter**



2.5 Konfigurasi Dasar Debian

Jalankan Semua VM Debian yang kita buat dengan cara Klik Start

Sebelum ke konfigurasi dasar kita harus memahami perintah dasar linux di bawah ini

2.5.1 Perintah Linux Dasar

Bentuk linux shell

```
username@hostname: ~ $

root@hostname:~#
```

Simbol $ menandakan kita menggunakan user biasa

Simbol # menandakan kita menggunakan user root

Simbol ~ menandakan kita berada di folder home

username adalah user yang kita pakai, sedangkan hostname adalah nama pc kita

Contoh : user tkj dengan nama pc debian

```
tkj@debian:~$
```

su perintah untuk beralih ke user root dari user biasa

```
tkj@debian:~$ su
```

```
tkj@debian:~$ su
Password:
root@debian:/home/tkj#
```

whoami perintah untuk mengetahui user yang kita pakai

root@debian:~# whoami

```
tkj@debian:~$ whoami
tkj
tkj@debian:~$ _
```

ls perintah untuk mengetahui isi dari folder

Contoh : kita ingin melihat isi folder Downloads

tkj@debian:~$ ls Downloads/

```
tkj@debian:~$ ls Downloads/
file1.txt
tkj@debian:~$
```

ls -l seperti perintah ls namun di buat list dan kita bisa melihat file/folder permission

tkj@debian:~$ ls -l Downloads/

```
tkj@debian:~$ ls -l Downloads/
total 0
-rw-r--r-- 1 tkj tkj 0 Sep  9 13:59 file1.txt
tkj@debian:~$ _
```

ls -a seperti perintah ls namun bisa melihat semua file/folder yang tersembunyi(biasanya dimulai dengan nama .)

tkj@debian:~$ ls -a

```
tkj@debian:~$ ls -a
.  ..  .bash_logout  .bashrc  Downloads  .profile
tkj@debian:~$
```

cd perintah untuk pindah direktori

Contoh: kita ingin pindah ke folder Documents

tkj@debian:~$ cd Documents/

```
tkj@debian:~$ cd Documents/
tkj@debian:~/Documents$
```

pwd perintah untuk melihat kita ada di direktori mana

tkj@debian:~/Documents$ pwd

```
tkj@debian:~/Documents$ pwd
/home/tkj/Documents
```

cp perintah untuk menyalin sebuah file

Contoh : menyalin file1.txt menjadi file2.txt

tkj@debian:~$ cp file1.txt file2.txt

```
tkj@debian:~$ cp file1.txt file2.txt
tkj@debian:~$ _
```

mv perintah untuk memindah sebuah file/folder, bisa juga digunakan untuk merename file/folder

Contoh: memindah file file1.txt ke folder Downloads

```
tkj@debian:~$ mv file1.txt Downloads
```

```
tkj@debian:~$ mv file1.txt Downloads/
tkj@debian:~$ _
```

Contoh: merename file2.txt menjadi file3.txt

```
tkj@debian:~$ mv file2.txt file3.txt
```

```
tkj@debian:~$ ls
Documents  Downloads  file2.txt
tkj@debian:~$ mv file2.txt file3.txt
tkj@debian:~$ ls
Documents  Downloads  file3.txt
tkj@debian:~$
```

rm perintah untuk menghapus file

Contoh: menghapus file3.txt

```
tkj@debian:~$ rm file3.txt
```

```
tkj@debian:~$ rm file3.txt
tkj@debian:~$
```

rm -r dan ***rmdir*** perintah untuk menghapus folder

Contoh: menghapus folder Documents

```
tkj@debian:~$ rmdir Documents
```

```
tkj@debian:~$ ls
Documents  Downloads
tkj@debian:~$ rmdir Documents/
tkj@debian:~$ ls
Downloads
tkj@debian:~$ _
```

touch perintah untuk membuat file

Contoh: membuat file tugas.txt

```
tkj@debian:~$ touch tugas.txt
```

```
tkj@debian:~$ touch tugas.txt
tkj@debian:~$ ls
Downloads  tugas.txt
tkj@debian:~$ _
```

mkdir perintah untuk membuat folder baru

Contoh: membuat folder dengan nama Tugas

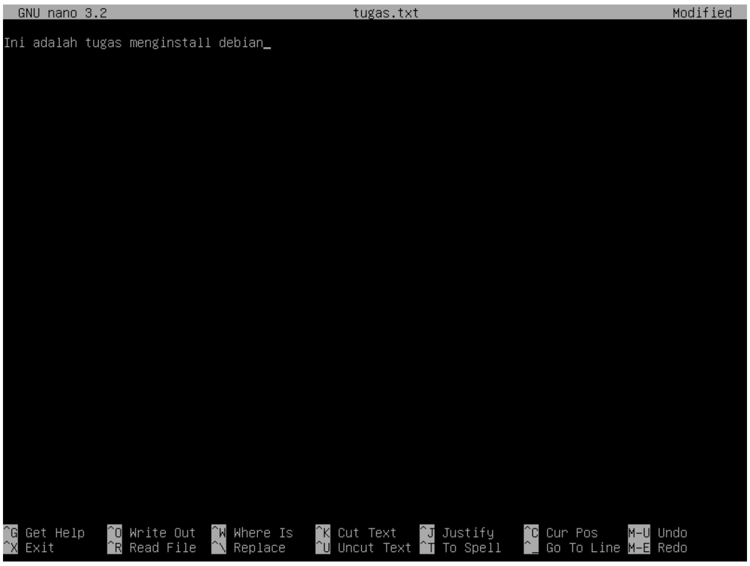
```
tkj@debian:~$ mkdir Tugas
```

```
tkj@debian:~$ mkdir Tugas
tkj@debian:~$ ls
Downloads  Tugas  tugas.txt
tkj@debian:~$ _
```

nano perintah untuk menggunakan teks editor nano

Contoh: kita ingin menambahkan teks ke file tugas.txt

```
tkj@debian:~$ nano tugas.txt
```



Untuk melakukan save CTRL+o untuk exit CTRL+x

cat perintah untuk mengetahui isi file

Contoh: melihat isi file tugas.txt

```
tkj@debian:~$ cat tugas.txt
```

```
tkj@debian:~$ cat tugas.txt
Ini adalah tugas menginstall debian
tkj@debian:~$
```

adduser perintah untuk menambahkan user baru(harus menggunakan user root)

Contoh: menambah user arya

```
root@debian:~# adduser arya
```

```
root@debian:~# adduser arya
Adding user `arya' ...
Adding new group `arya' (1001) ...
Adding new user `arya' (1001) with group `arya' ...
Creating home directory `/home/arya' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for arya
Enter the new value, or press ENTER for the default
        Full Name []: Arya Pramudika
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
root@debian:~# _
```

useradd seperti adduser yaitu perintah untuk menambah user baru

Contoh: menambah user andri

```
root@debian:~# useradd -m andri -s /bin/bash
```

```
root@debian:~# useradd -m andri -s /bin/bash
root@debian:~# _
```

option -m untuk membuat direktori home sedangkan -s mendefinisikan shell yang dipakai oleh user andri

passwd perintah untuk memberi/mengganti password user

Contoh: memberi password pada user andri

```
root@debian:~# passwd andri
```

```
root@debian:~# passwd andri
New password:
Retype new password:
passwd: password updated successfully
root@debian:~#
```

userdel perintah untuk menghapus user

Contoh: menghapus user andri

```
root@debian:~# userdel andri
```

```
root@debian:~# userdel andri
root@debian:~#
```

tar -cvf perintah untuk mengcompress file dengan format .tar.gz

Contoh: Mengcompress folder Documents

```
root@debian:~# tar -cvf file-contoh.tar.gz Documents
```

```
root@debian:~# tar -cvf file-contoh.tar.gz Documents
Documents/
root@debian:~# ls
Documents  file1.sh  file-contoh.tar.gz
root@debian:~#
```

tar -xvf perintah untuk mengekstrak file dengan format .tar.gz root@debian:~# tar -xvf file-contoh.tar.gz

```
root@debian:~# tar -xvf file-contoh.tar.gz
Documents/
root@debian:~#
```

chmod perintah untuk merubah file/folder permission

Contoh: kita buat dulu shell script sederhana yaitu file1.sh dan kita ubah permission nya agar bisa di execute oleh user

root@debian:~$chmod +x file1.sh

```
root@debian:~# chmod +x file1.sh
root@debian:~# ./file1.sh
Thu 09 Sep 2021 02:21:46 PM WIB
root@debian:~# _
```

jadi chmod adalah perintah untuk memberi akses (read, write, executable) kepada group-group tertentu terhadap file

1. Read = r = memberikan kepada group / user untuk hanya sekedar membaca / melihat.

2. Write = w = memberikan kepada group / user untuk bisa menulis file tersebut.

3. Executable = x = memberikan kepada group / user untuk bisa mengeksekusi file tersebut.

yang baru kita lakukan yaitu memberi izin agar file1.sh bisa di execute/eksekusi

kita bisa juga menggunakan angka misal **chmod 777** atau **chmod 775**, dengan keterangan sebagai berikut

0 = —

1 = –x

2 = -w-

3 = -wx

4 = r–

5 = r-x

6 = rw-

7 = rwx

Contoh: kita ubah permission file1.sh ke 777

root@debian:~#chmod 777 file1.sh

```
root@debian:~# ls -l
total 8
drwxr-xr-x 2 tkj  tkj  4096 Sep  9 14:22 Documents
-rwxr-xr-x 1 root root   24 Sep  9 14:21 file1.sh
root@debian:~# chmod 777 file1.sh
root@debian:~# ls -l
total 8
drwxr-xr-x 2 tkj  tkj  4096 Sep  9 14:22 Documents
-rwxrwxrwx 1 root root   24 Sep  9 14:21 file1.sh
root@debian:~# _
```

maka terlihat bahwa kita mengizinkan file1.sh untuk di read,write,execute ke user apapun, group apapun, dan untuk lainnya

7-7-7

rwx-rwx-rwx

bagian pertama yaitu user, kedua group, dan terakhir untuk lainnya

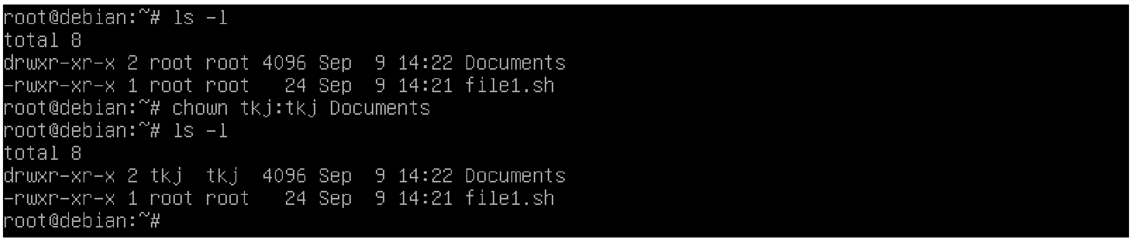
agar lebih paham dan jelas lagi bisa baca di :

https://ss64.com/bash/chmod.html

chown perintah untuk merubah owner dan group dari file/folder

Contoh: ingin merubah owner dan group folder Documents dari root menjadi tkj

```
root@debian:~$chown tkj:tkj Documents
```



```
root@debian:~# ls -l
total 8
drwxr-xr-x 2 root root 4096 Sep  9 14:22 Documents
-rwxr-xr-x 1 root root   24 Sep  9 14:21 file1.sh
root@debian:~# chown tkj:tkj Documents
root@debian:~# ls -l
total 8
drwxr-xr-x 2 tkj  tkj  4096 Sep  9 14:22 Documents
-rwxr-xr-x 1 root root   24 Sep  9 14:21 file1.sh
root@debian:~#
```

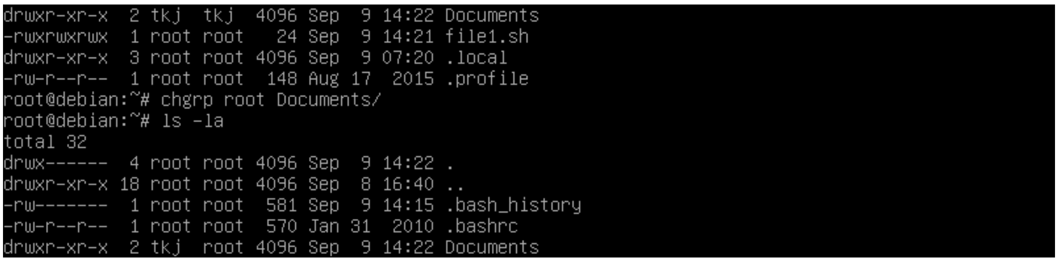
agar lebih paham bisa baca di:

https://ss64.com/bash/chown.html

chgrp mirip seperti chown tetapi hanya group saja yang diubah

Contoh: mengubah group folder Documents yang awalnya tkj menjadi root

```
root@debian:~# chgrp root Documents
```



```
drwxr-xr-x  2 tkj  tkj  4096 Sep  9 14:22 Documents
-rwxrwxrwx  1 root root   24 Sep  9 14:21 file1.sh
drwxr-xr-x  3 root root 4096 Sep  9 07:20 .local
-rw-r--r--  1 root root  148 Aug 17  2015 .profile
root@debian:~# chgrp root Documents/
root@debian:~# ls -la
total 32
drwx------  4 root root 4096 Sep  9 14:22 .
drwxr-xr-x 18 root root 4096 Sep  8 16:40 ..
-rw-------  1 root root  581 Sep  9 14:15 .bash_history
-rw-r--r--  1 root root  570 Jan 31  2010 .bashrc
drwxr-xr-x  2 tkj  root 4096 Sep  9 14:22 Documents
```

Perintah untuk manajemen interface

Untuk melakukan manajemen di Linux Debian kita bisa menggunakan perintah **ip**

Contoh perintah untuk melihat ip address

ip addr bisa disingkat menjadi ***ip a***

```
root@debian:~# ip addr
```

```
root@debian:~# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1
000
    link/ether 08:00:27:14:38:fb brd ff:ff:ff:ff:ff:ff
    inet 192.168.101.30/24 brd 192.168.101.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe14:38fb/64 scope link
       valid_lft forever preferred_lft forever
root@debian:~#
```

Untuk menetapkan IP Address di interface, misal IP Address 192.168.101.31/24 ke interface enp0s3 kita bisa gunakan perintah ip addr add ***ipaddress*** dev ***interface***

Karena sudah ada ip bawaan maka kita tambahkan saja IP Address alias, jadi 1 interface memiliki 2 IP

ip addr add 192.168.101.31/24 dev enp0s3:0

```
root@debian:~# ip addr add 192.168.101.31/24 dev enp0s3:0
```

```
root@debian:~# ip addr add 192.168.101.31/24 dev enp0s3:0
root@debian:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1
000
    link/ether 08:00:27:14:38:fb brd ff:ff:ff:ff:ff:ff
    inet 192.168.101.30/24 brd 192.168.101.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet 192.168.101.31/24 scope global secondary enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe14:38fb/64 scope link
       valid_lft forever preferred_lft forever
root@debian:~# _
```

Perintah untuk menghapus IP Address

```
root@debian:~# ip addr del 192.168.101.31/24 dev enp0s3:0
```

```
root@debian:~# ip addr del 192.168.101.31/24 dev enp0s3:0
root@debian:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1
000
    link/ether 08:00:27:14:38:fb brd ff:ff:ff:ff:ff:ff
    inet 192.168.101.30/24 brd 192.168.101.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe14:38fb/64 scope link
       valid_lft forever preferred_lft forever
root@debian:~#
```

Perintah untuk melihat interface aktif atau mati

ip link

```
root@debian:~# ip link
```

```
root@debian:~# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen
1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP mode DEFAULT group d
efault qlen 1000
    link/ether 08:00:27:14:38:fb brd ff:ff:ff:ff:ff:ff
root@debian:~# _
```

Perintah untuk mengaktifkan dan menonaktifkan interface

```
root@debian:~# ip link set enp0s3 up

root@debian:~# ip link set enp0s3 down
```

```
root@debian:~# ip link set enp0s3 up
root@debian:~# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen
1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP mode DEFAULT group d
efault qlen 1000
    link/ether 08:00:27:14:38:fb brd ff:ff:ff:ff:ff:ff
root@debian:~# ip link set enp0s3 down
root@debian:~# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen
1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST> mtu 1500 qdisc pfifo_fast state DOWN mode DEFAULT group default qle
n 1000
    link/ether 08:00:27:14:38:fb brd ff:ff:ff:ff:ff:ff
root@debian:~# _
```

Perintah untuk melihat tabel routing

```
root@debian:~# ip route
```

```
root@debian:~# ip route
192.168.101.0/24 dev enp0s3 proto kernel scope link src 192.168.101.30
root@debian:~#
```

Perintah untuk melakukan static routing

Menambahkan routing misal untuk network ip 10.10.10.0/24 dengan gateway 192.168.101.3 dan interface enp0s3

```
root@debian:~#ip route add 10.10.10.0/24 via 192.168.101.3 dev enp0s3
```

```
root@debian:~# ip route add 10.10.10.0/24 via 192.168.101.3 dev enp0s3
root@debian:~# ip route
10.10.10.0/24 via 192.168.101.3 dev enp0s3
192.168.101.0/24 dev enp0s3 proto kernel scope link src 192.168.101.30
root@debian:~# _
```

Perintah untuk mengapus static routing

```
root@debian:~# ip route del 10.10.10.0/24
```

```
root@debian:~# ip route del 10.10.10.0/24
root@debian:~# ip route
192.168.101.0/24 dev enp0s3 proto kernel scope link src 192.168.101.30
root@debian:~# _
```

Perintah untuk menambahkan default gateway

```
root@debian:~# ip route add default via 192.168.101.1
```

```
root@debian:~# ip route add default via 192.168.101.1
root@debian:~# ip route
default via 192.168.101.1 dev enp0s3
192.168.101.0/24 dev enp0s3 proto kernel scope link src 192.168.101.30
root@debian:~# _
```

Perintah untuk menghapus default gateway

ip route del default

```
root@debian:~# ip route del default
root@debian:~# ip route
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15
169.254.0.0/16 dev enp0s3 scope link metric 1000
192.168.101.0/24 dev enp0s8 proto kernel scope link src 192.168.101.2
root@debian:~# _
```

2.5.2 Konfigurasi Hostname dan Hosts

Untuk konfigurasi hostname bisa menggunakan perintah **hostnamectl set-hostname** *hostname* atau bisa mengedit file **/etc/hostname**

Untuk Hosts kita mengedit file **/etc/hosts**

Hostname di **Router-Debian**

```
root@debian:~# hostnamectl set-hostname router
```

```
root@debian:~# hostnamectl set-hostname router
root@debian:~#
```

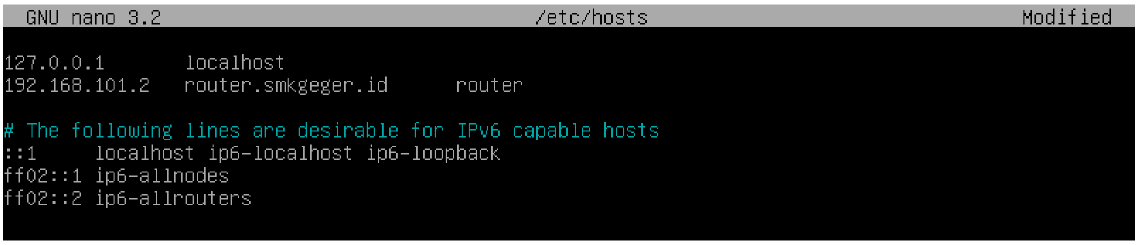
lalu exit dan login kembali, otomatis hostname sudah berubah

```
Debian GNU/Linux 10 router tty1

router login: root
Password:
Last login: Thu Sep  9 19:09:02 WIB 2021 on tty1
Linux router 4.19.0-17-amd64 #1 SMP Debian 4.19.194-3 (2021-07-18) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@router:~# _
```

Hosts di **Router-Debian**

```
root@router1:~# nano /etc/hosts
```

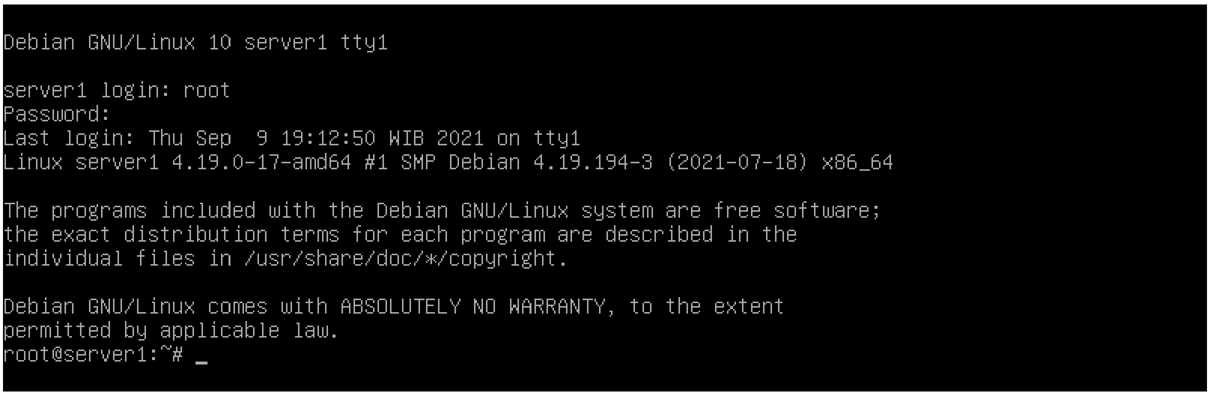
```
127.0.0.1       localhost

192.168.101.2   router.smkgeger.id      router

# The following lines are desirable for IPv6 capable hosts

::1     localhost ip6-localhost ip6-loopback

ff02::1 ip6-allnodes

ff02::2 ip6-allrouters
```



Hostname di **Server 1**

```
root@debian:~# hostnamectl set-hostname server1
```

Lalu exit dan login Kembali



Hosts di **Server 1**

```
root@server1:~# nano /etc/hosts

127.0.0.1       localhost

192.168.101.2   server1.smkgeger.id      server1

# The following lines are desirable for IPv6 capable hosts

::1     localhost ip6-localhost ip6-loopback

ff02::1 ip6-allnodes

ff02::2 ip6-allrouters
```

```
  GNU nano 3.2                              /etc/hosts

127.0.0.1       localhost
192.168.101.3   server1.smkgeger.id     server1

# The following lines are desirable for IPv6 capable hosts
::1     localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

Hostname di **Server 2**

root@debian:~# hostnamectl set-hostname server2

```
root@debian:~# hostnamectl set-hostname server2
root@debian:~#
```

Lalu exit dan login Kembali

```
Debian GNU/Linux 10 server2 tty1

server2 login: root
Password:
Last login: Thu Sep  9 07:00:34 WIB 2021 on tty1
Linux server2 4.19.0-17-amd64 #1 SMP Debian 4.19.194-3 (2021-07-18) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@server2:~#
```

Hosts di **Server 2**

```
root@server2:~# nano /etc/hosts

127.0.0.1       localhost

192.168.101.2   server2.smkgeger.id     server2

# The following lines are desirable for IPv6 capable hosts

::1     localhost ip6-localhost ip6-loopback

ff02::1 ip6-allnodes

ff02::2 ip6-allrouters
```

```
  GNU nano 3.2                              /etc/hosts

127.0.0.1       localhost
192.168.101.4   server2.smkgeger.id     server2

# The following lines are desirable for IPv6 capable hosts
::1     localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

2.5.3 Konfigurasi IP Address

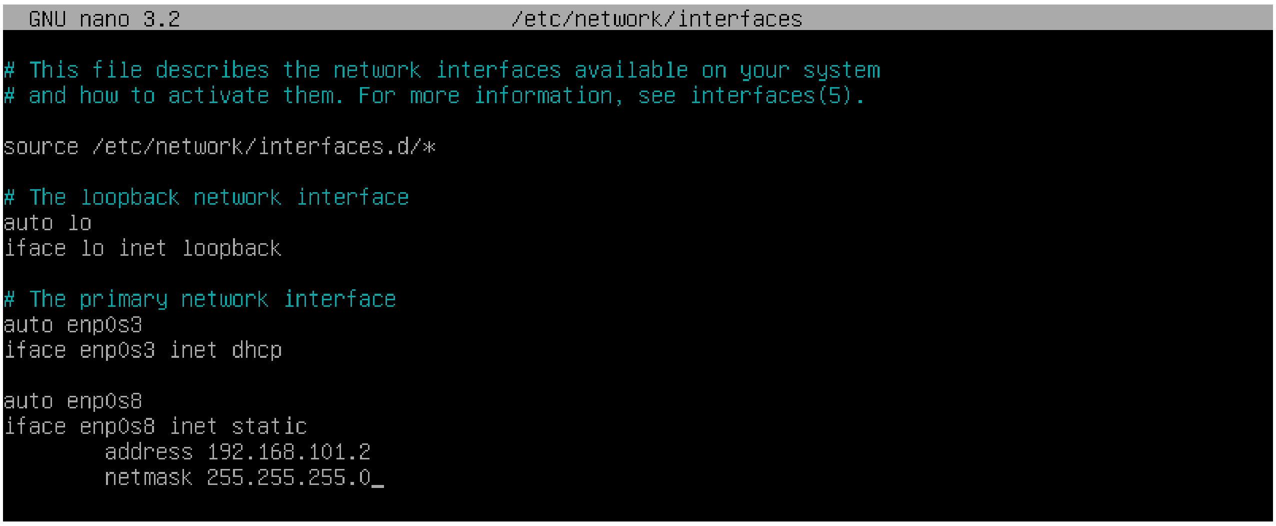
Untuk konfigurasi IP Address kita bisa menggunakan perintah **ip** , tetapi kekurangannya yaitu tidak bisa menyimpan ip secara permanen, ketika komputer reboot maka IP Address hilang oleh karena itu kita konfigurasi IP Address di **/etc/network/interfaces**

Edit file **/etc/network/interfaces** dengan teks editor misal **nano**

nano /etc/network/interfaces

Konfigurasi IP di **Router-Debian**

```
auto enp0s3

iface enp0s3 inet dhcp

auto enp0s8

iface enp0s8 inet static

    address 192.168.101.2

    netmask 255.255.255.0
```



Lalu lakukan restart service networking

```
root@router:~# systemctl restart networking
```

Konfigurasi IP **di Server 1**

```
auto enp0s3

iface enp0s3 inet static

      address 192.168.101.3/24

      netmask 255.255.255.0

      gateway 192.168.101.2
```

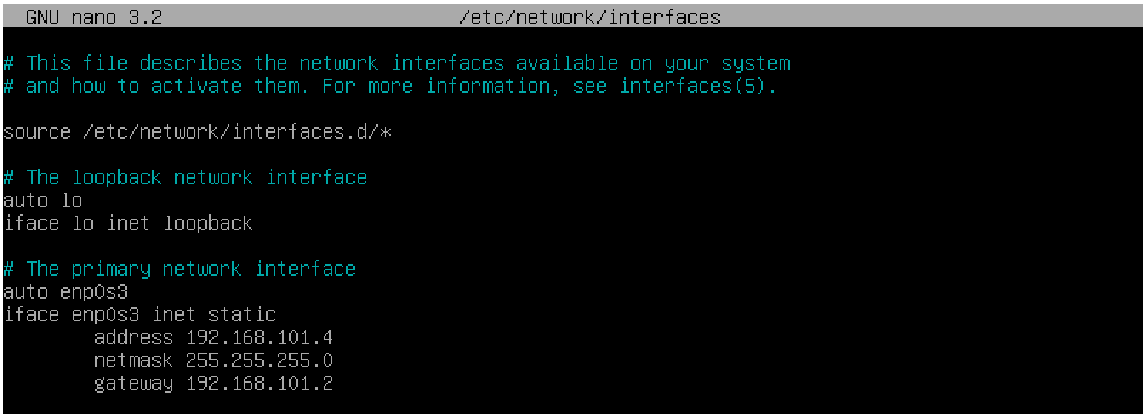
```
  GNU nano 3.2                      /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
auto enp0s3
iface enp0s3 inet static
        address 192.168.101.3/24
        netmask 255.255.255.0
        gateway 192.168.101.2
```

Lalu Restart service networking

```
root@server1:~# systemctl restart networking
```

Konfigurasi IP di **Server 2**

```
auto enp0s3

iface enp0s3 inet static

      address 192.168.101.3/24

      netmask 255.255.255.0

      gateway 192.168.101.2
```



```
GNU nano 3.2                    /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
auto enp0s3
iface enp0s3 inet static
        address 192.168.101.4
        netmask 255.255.255.0
        gateway 192.168.101.2
```

Lalu Restart service networking

root@server2:~# systemctl restart networking

2.5.4 Konfigurasi Repository

Apa itu Repository ?

Repository adalah tempat disimpannya berbagai macam program atau aplikasi di linux, berbeda dengan di windows ketika menginstall aplikasi diharuskan mendownload file berfomat .exe di linux tidak seperti itu (meskipun ada untuk aplikasi tertentu biasanya berformat .deb) akan tetapi secara default kita harus mengkonfigurasi repository bisa berupa alamat url atau DVD.

File untuk konfigurasi repository yaitu file **/etc/apt/sources.list**

2.5.4.1 Repository via URL

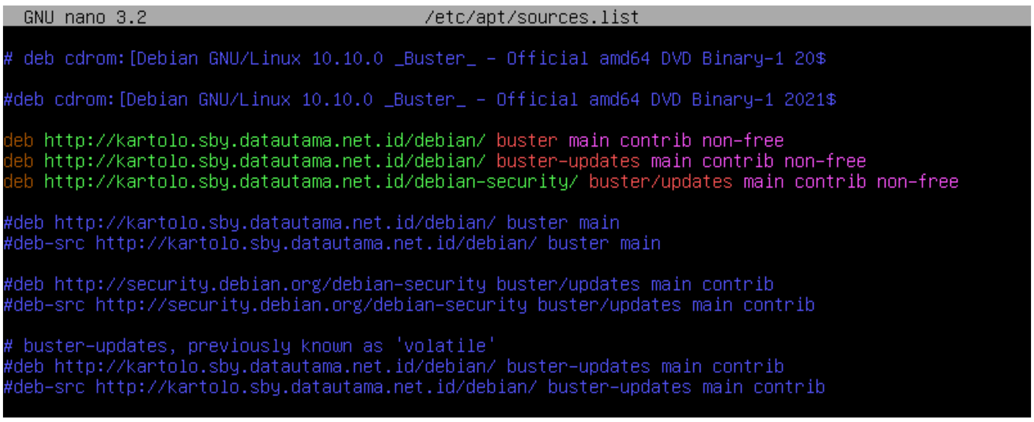
Saat intallasi debian pada bagian **use a network repo?** Ketika kita memilih yes maka otomatis repository url ditambahkan, untuk konfigurasi manual sebagai berikut

Edit file /etc/apt/sources.list dengan teks editor nano

nano /etc/apt/sources.list

Tambahkan url repository misal

```
deb http://kartolo.sby.datautama.net.id/debian/ buster main contrib non-free

deb http://kartolo.sby.datautama.net.id/debian/ buster-updates main contrib non-free

deb http://kartolo.sby.datautama.net.id/debian-security/ buster/updates main contrib non-free
```



```
GNU nano 3.2                    /etc/apt/sources.list

# deb cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 20$

#deb cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 2021$

deb http://kartolo.sby.datautama.net.id/debian/ buster main contrib non-free
deb http://kartolo.sby.datautama.net.id/debian/ buster-updates main contrib non-free
deb http://kartolo.sby.datautama.net.id/debian-security/ buster/updates main contrib non-free

#deb http://kartolo.sby.datautama.net.id/debian/ buster main
#deb-src http://kartolo.sby.datautama.net.id/debian/ buster main

#deb http://security.debian.org/debian-security buster/updates main contrib
#deb-src http://security.debian.org/debian-security buster/updates main contrib

# buster-updates, previously known as 'volatile'
#deb http://kartolo.sby.datautama.net.id/debian/ buster-updates main contrib
#deb-src http://kartolo.sby.datautama.net.id/debian/ buster-updates main contrib
```

Kita bisa menggunakan repository lokal maupun dari web debian langsung, untuk daftar repository lokal dan dunia bisa kunjungi web di bawah ini

https://www.debian.org/mirror/list.id.html

https://www.linuxsec.org/2019/08/repository-lokal-debian-buster.html

Setelah itu lakukan update repository

```
root@router:~# apt -y update
```

2.5.4.2 Repository Dengan DVD

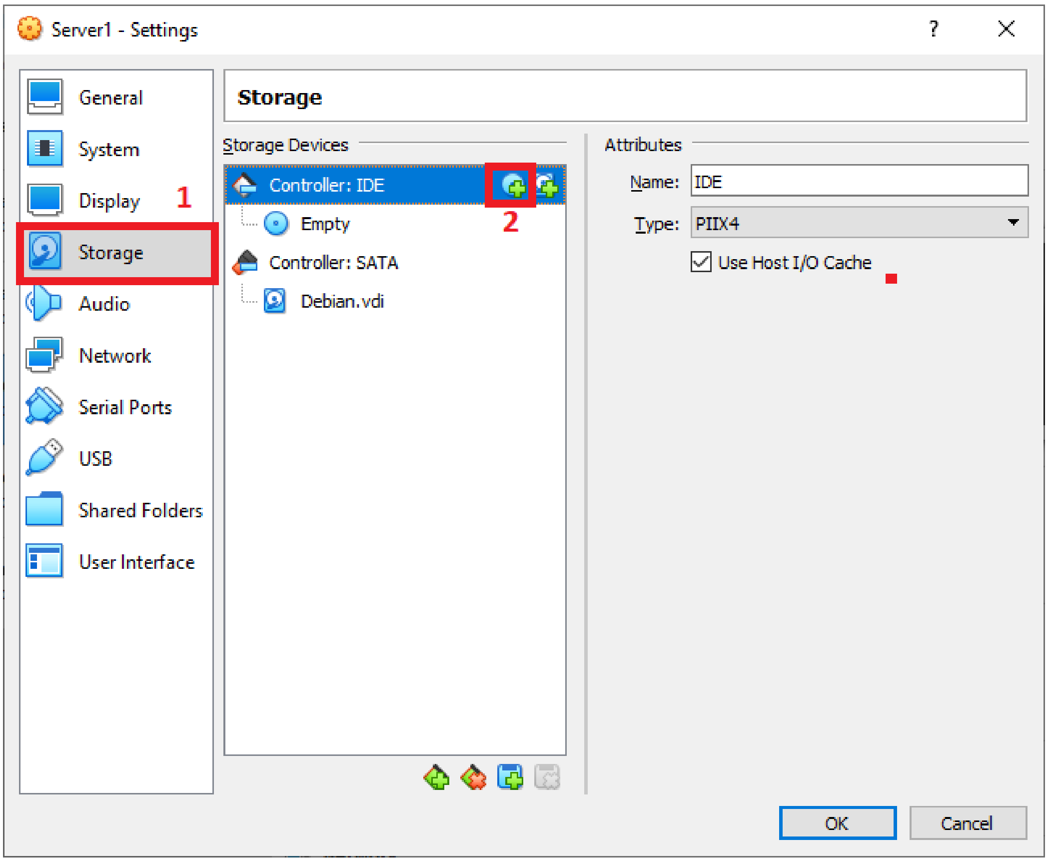
Kita bisa menggunakan DVD Debian 1,2,3 sebagai repository dengan cara menambahkan ke /etc/apt/sources.list

Keuntungan menggunakan DVD yaitu kita bisa menginstall aplikasi tanpa koneksi internet, tapi repotnya harus memasukkan DVD satu-persatu jika aplikasi yang di install terletak di DVD yang berbeda.

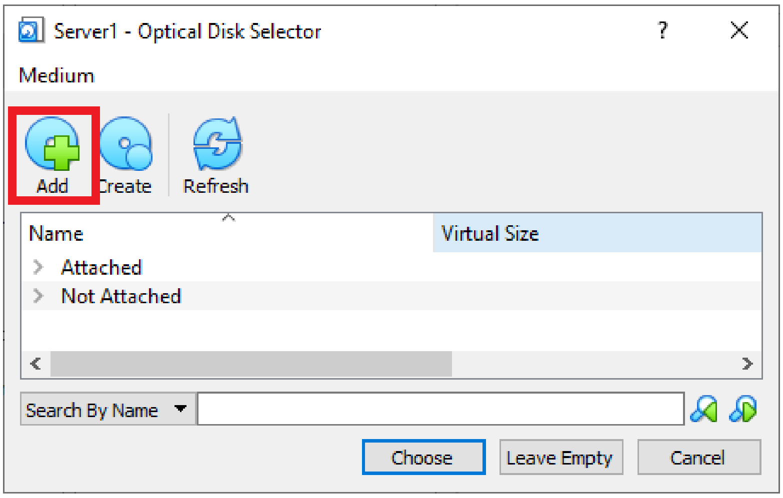
Untuk mendownload file DVD Debian 1,2,3 bisa di link di bawah ini

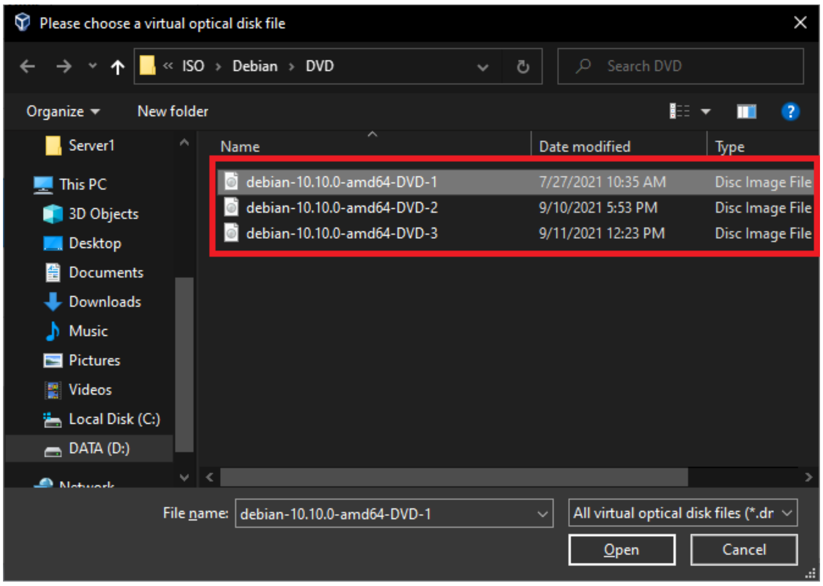
https://cdimage.debian.org/cdimage/archive/10.10.0/amd64/iso-dvd/

Pertama, pastikan VM Debian dalam keadaan mati lalu klik Settings > Storage > Klik logo CD

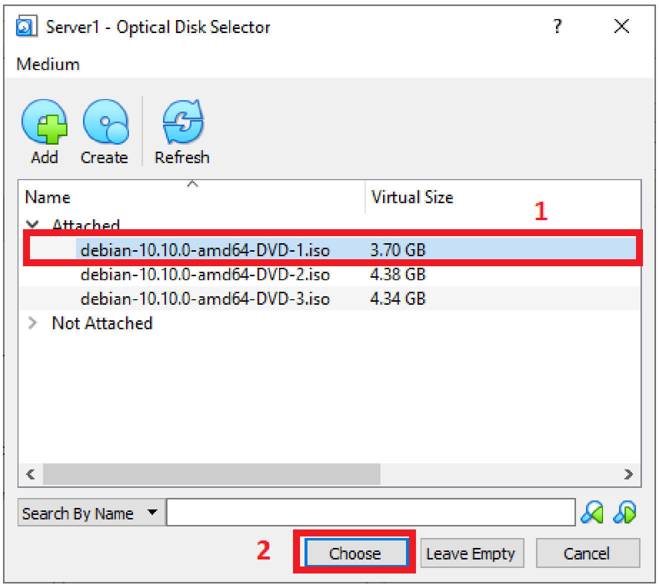


Lalu klik Add dan tambahkan DVD satu persatu

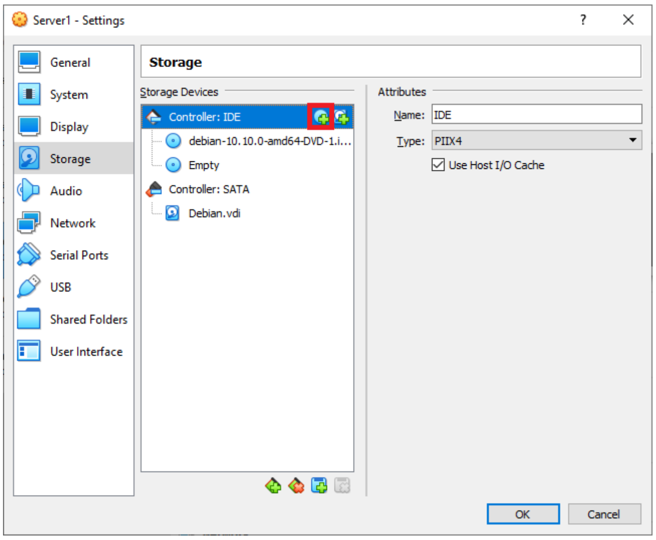


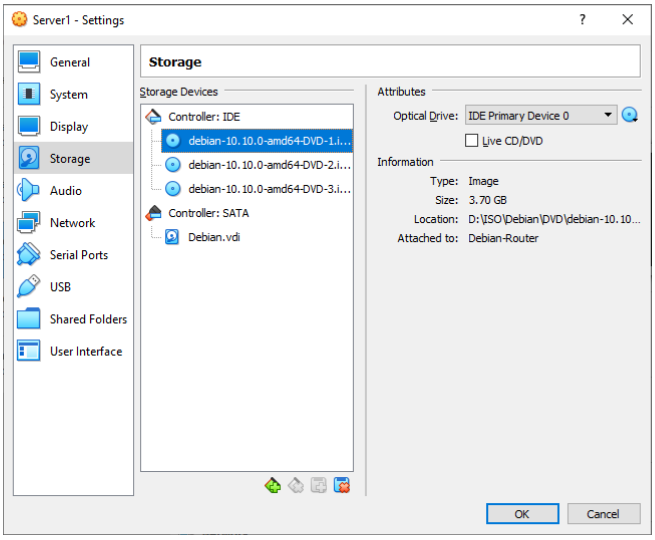


Tambahkan DVD 1

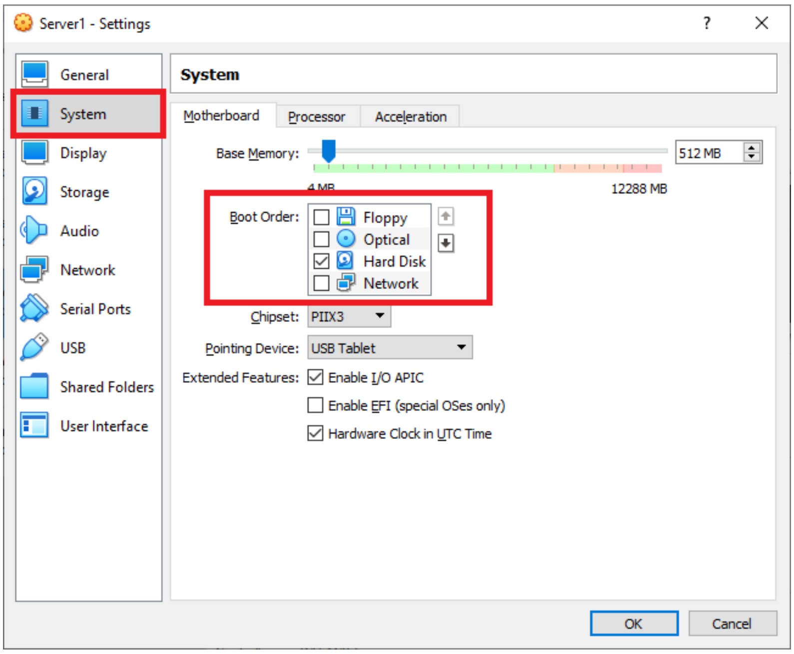


Lalu tambah lagi DVD 2 dan DVD 3





Setelah itu pada pindah ke System > Uncentang Floppy & Optical lalu klik OK



Lalu jalankan VM dan gunakan perintah

root@router:~# apt-cdrom add

Lalu edit file sources.list, beri tanda komentar (#) ke semua repo http dan tambahkan option [trusted=yes]

```
root@router:~# nano /etc/apt/sources.list

deb [trusted=yes] cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-3 2021$

deb [trusted=yes] cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-2 2021$

deb [trusted=yes] cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 2021$
```

```
# deb cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 20210619-16:12]/ buste$

#deb cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 20210619-16:12]/ buster$

#deb http://kartolo.sby.datautama.net.id/debian/ buster main
#deb-src http://kartolo.sby.datautama.net.id/debian/ buster main

deb [trusted=yes] cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-3 20210619-1$
deb [trusted=yes] cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-2 20210619-1$
deb [trusted=yes] cdrom:[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 20210619-1$
```

Lalu lakukan update repository

```
root@router:~# apt -y update
```

```
Ign:11 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-3 20210619-16:12] bust
er/main all Packages
Ign:14 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-3 20210619-16:12] bust
er/main Translation-en_US
Ign:15 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-2 20210619-16:12] bust
er/contrib all Packages
Ign:17 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-2 20210619-16:12] bust
er/contrib Translation-en_US
Ign:20 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-2 20210619-16:12] bust
er/main all Packages
Ign:21 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-2 20210619-16:12] bust
er/main Translation-en_US
Ign:24 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 20210619-16:12] bust
er/contrib all Packages
Ign:26 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 20210619-16:12] bust
er/contrib Translation-en_US
Ign:28 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 20210619-16:12] bust
er/main all Packages
Ign:29 cdrom://[Debian GNU/Linux 10.10.0 _Buster_ - Official amd64 DVD Binary-1 20210619-16:12] bust
er/main Translation-en_US
Reading package lists... Done
Building dependency tree
Reading state information... Done
All packages are up to date.
root@server1:~# _
```

2.5.5 Konfigurasi Sudo

Apa itu Sudo? sudo adalah singkatan dari superuser do dalam bahasa Indonesia berarti “apa yang dilakukan oleh superuser” dengan menggunakan sudo, user biasa bisa mengeksekusi perintah yang hanya bisa dilakukan oleh superuser (user root).

Contoh : Kita ingin mendisable interface enp0s8

```
tkj@router:~$ ip link set enp0s8 down
```

```
tkj@router:~$ ip link set enp0s8 down
RTNETLINK answers: Operation not permitted
tkj@router:~$ _
```

Perhatikan muncul error Operation not permitted, berarti user biasa tidak bisa melakukan enable/disable interface. Contoh dengan sudo

tkj@router:~$ sudo ip link set enp0s8 down

```
tkj@router:~$ sudo ip link set enp0s8 down
tkj@router:~$ ip l
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen
1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP mode DEFAULT group d
efault qlen 1000
    link/ether 08:00:27:da:e2:8c brd ff:ff:ff:ff:ff:ff
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc pfifo_fast state DOWN mode DEFAULT group default qle
n 1000
    link/ether 08:00:27:04:be:2f brd ff:ff:ff:ff:ff:ff
tkj@router:~$
```

Sudah tau kan perbedaannya? Oke selanjutnya cara konfigurasi sudo

Install Package sudo (pindah ke user root terlebih dahulu)

root@router:~# apt -y install sudo

```
root@router:~# apt -y install sudo
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  sudo
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,244 kB of archives.
After this operation, 3,882 kB of additional disk space will be used.
Get:1 http://kartolo.sby.datautama.net.id/debian buster/main amd64 sudo amd64 1.8.27-1+deb10u3 [1,24
4 kB]
Get:1 http://kartolo.sby.datautama.net.id/debian buster/main amd64 sudo amd64 1.8.27-1+deb10u3 [1,24
4 kB]
Fetched 1,150 kB in 2min 7s (9,070 B/s)
Selecting previously unselected package sudo.
(Reading database ... 28375 files and directories currently installed.)
Preparing to unpack .../sudo_1.8.27-1+deb10u3_amd64.deb ...
Unpacking sudo (1.8.27-1+deb10u3) ...
Setting up sudo (1.8.27-1+deb10u3) ...
Processing triggers for man-db (2.8.5-2) ...
Processing triggers for systemd (241-7~deb10u8) ...
root@router:~# _
```

Lalu konfigurasi sudo dengan mengedit file /etc/sudoers

```
nano /etc/sudoers

tkj     ALL=(ALL:ALL) ALL
```

```
# Host alias specification

# User alias specification
tkj     ALL=(ALL:ALL) ALL
# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
```

2.5.6 Konfigurasi Remote Login SSH

Apa fungsi remote login? Dengan remote login kita bisa mengakses server dari jarak jauh (fungsi secara umum) dan juga bisa melakukan copy paste di terminal kita, ini sangat penting karena ketika ada trouble tertentu yang jawabannya ada di google kita bisa langsung copy paste.

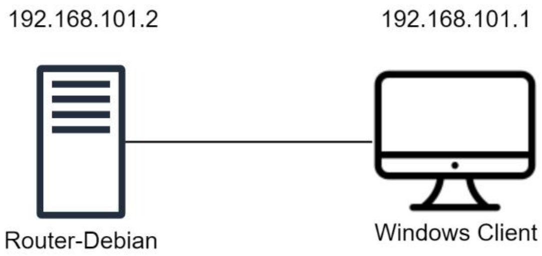
SSH Server

SSH Server di konfigurasi di server yang mau di remote, saat Instalasi kita sudah menambahkan paket SSH Server jadi server kita sudah bisa di remote

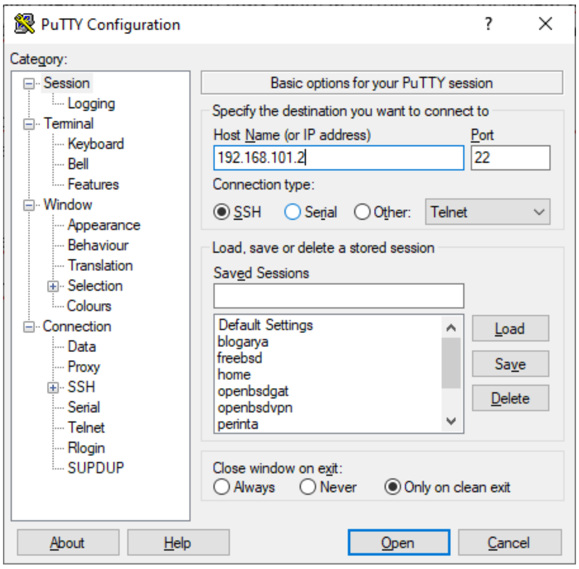
SSH Client

SSH Client, sebagai contoh saya menggunakan aplikasi Putty

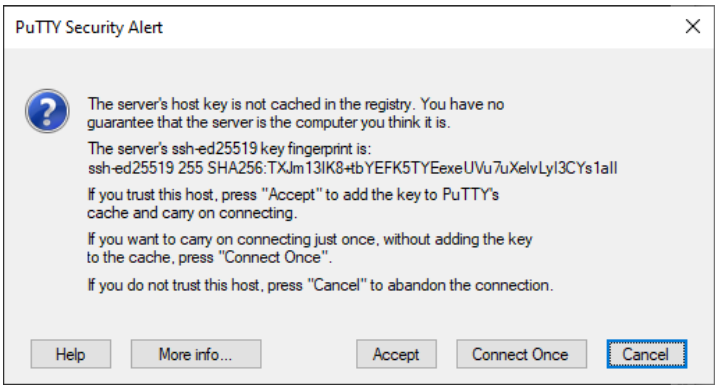
Contoh: Meremote Debian-Router dari Windows



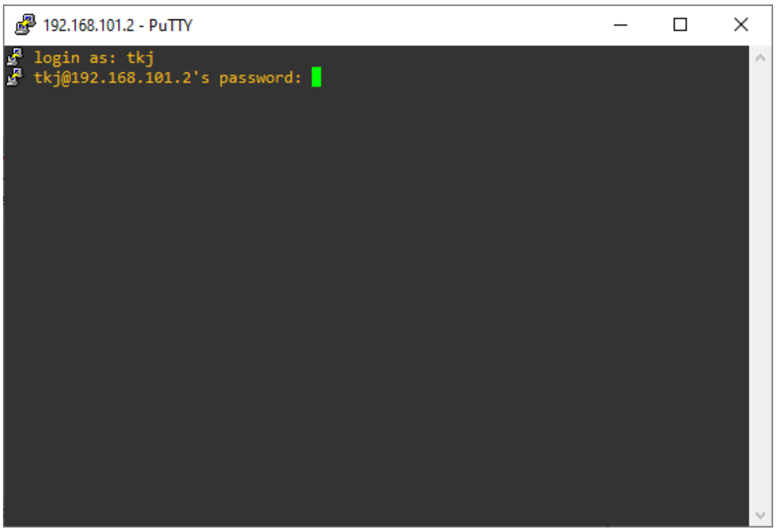
Buka Aplikasi Putty, Masukkan IP Router-Debian dan Port 22



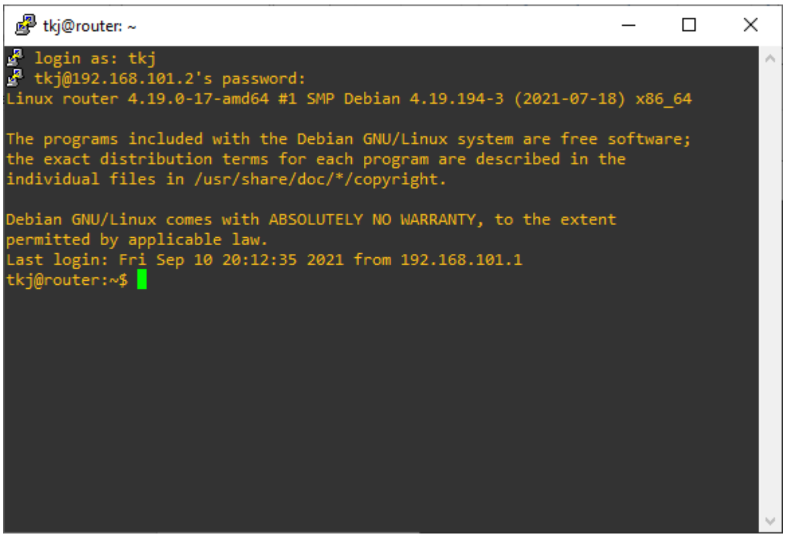
Lalu klik Open, Muncul Dialog pilih Accept



Isikan username dan password



Dan berhasil login



Mengganti Port SSH

Fungsi mengganti port ssh yaitu agar koneksi lebih aman dan terhindar dari brute force

Edit file konfigurasi SSH yaitu file **/etc/ssh/sshd_config**

```
root@router:~# nano /etc/ssh/sshd_config
```

```
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

Hilangkan tanda komentar (#) dan ubah port misal menjadi 2233, Lalu simpan

```
Port 2233
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

Disable Root Access SSH

```
root@router~# nano /etc/ssh/sshd_config
```

```
# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

Restart Service SSH

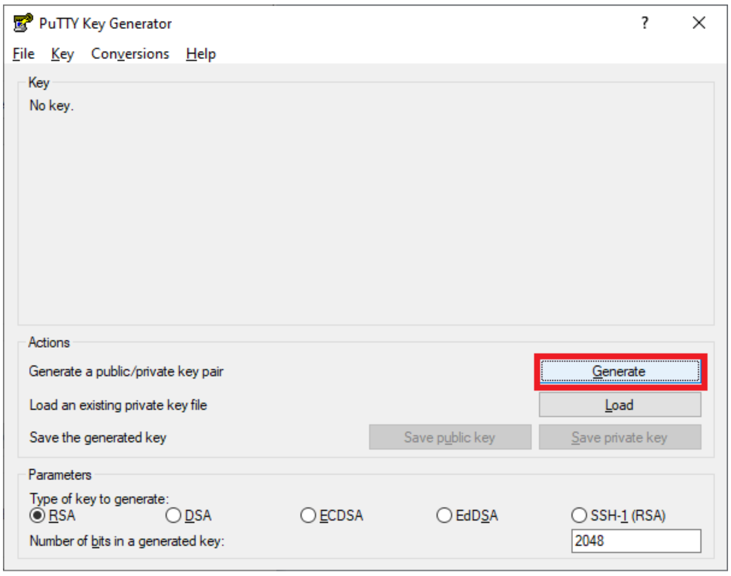
```
root@router:~# systemctl restart sshd
```

Autentikasi SSH dengan Private dan Public Key

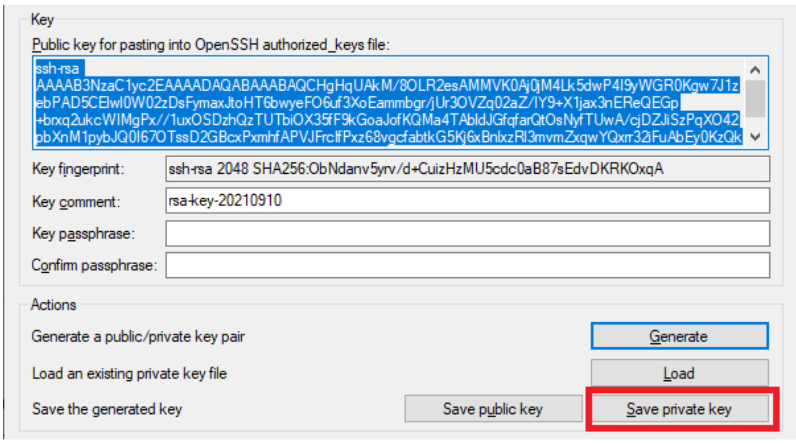
Berfungsi agar hanya Client yang memiliki Private key yang dapat meremote server

Membuat Private & Public Key di Puttygen

Buka PuttyGen > Klik Generate

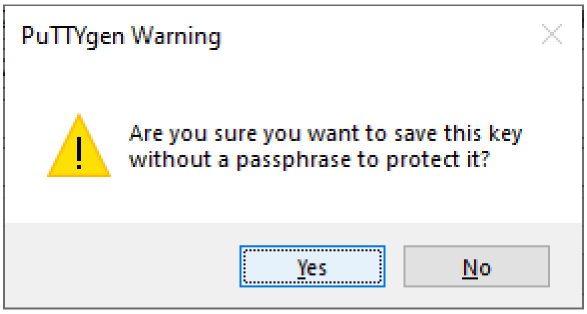


Lalu klik Save Private Key

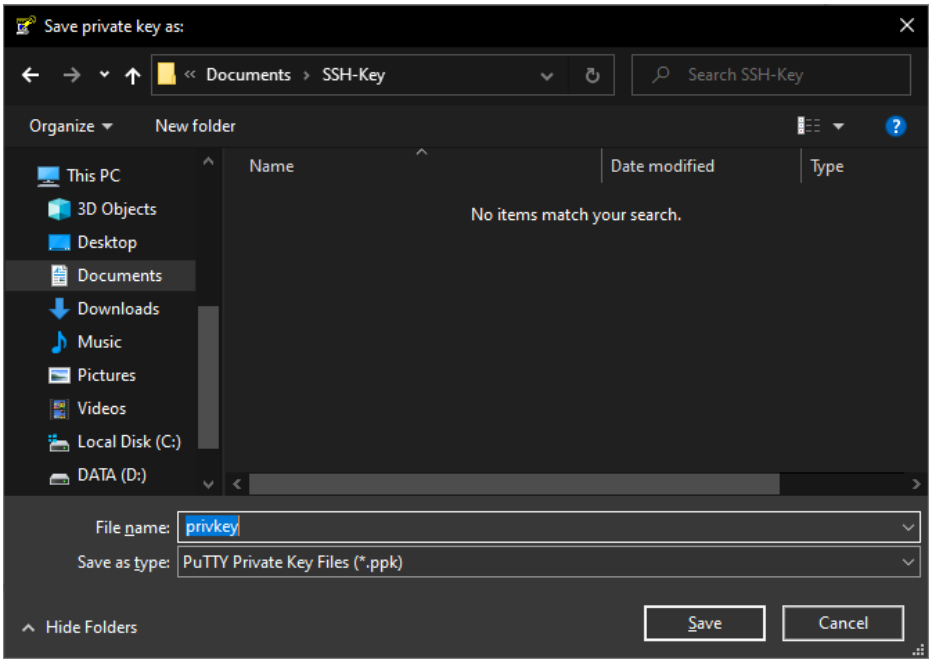


Are you sure you want to save this key without a passphrase to protect it?

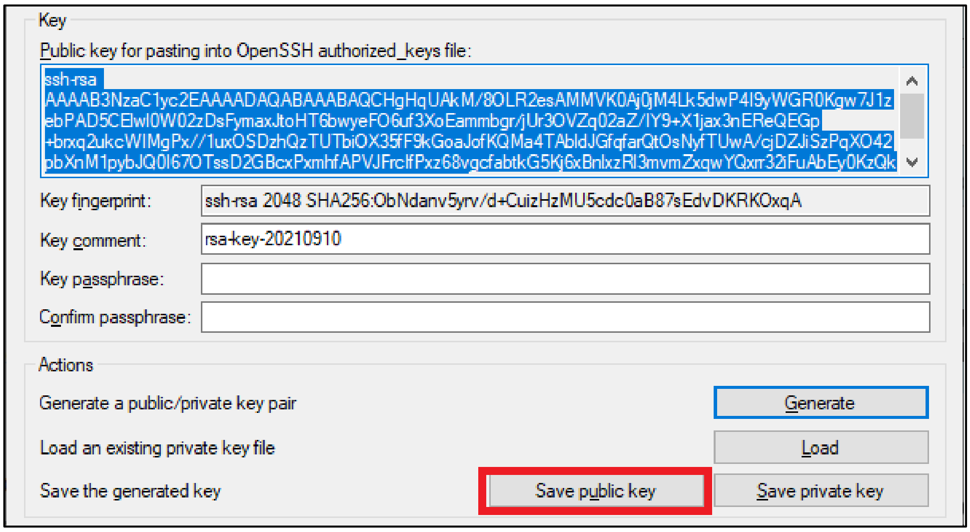
Pilih **Yes**



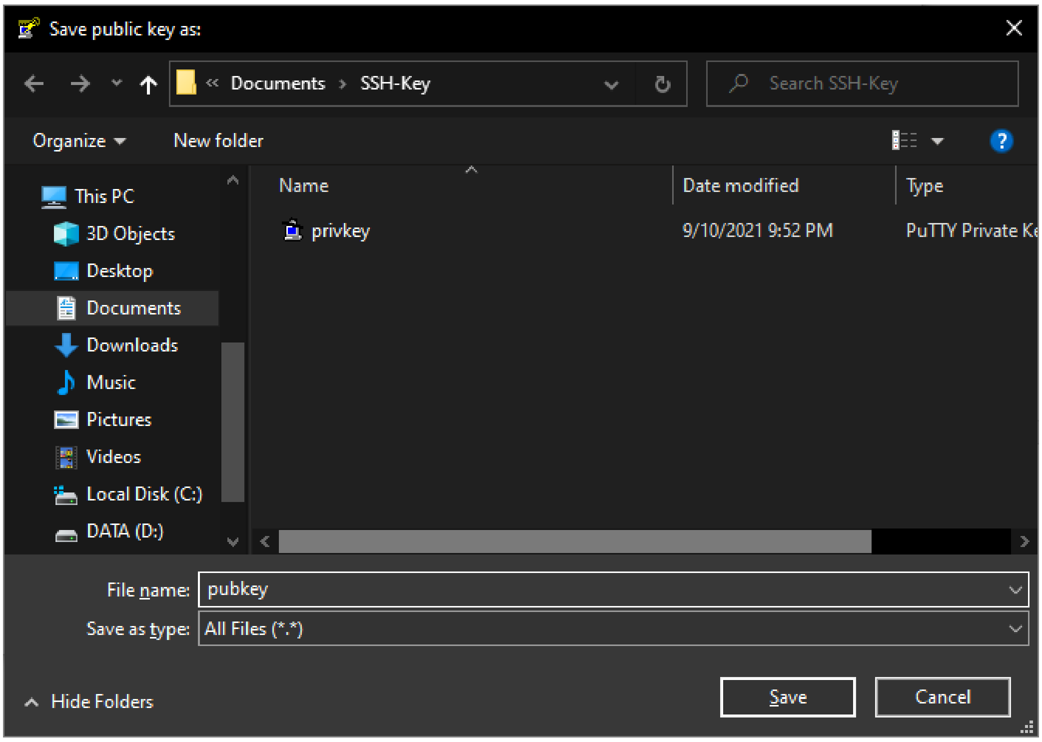
Lalu simpan file



Selanjutnya klik Save Public Key



Lalu simpan file



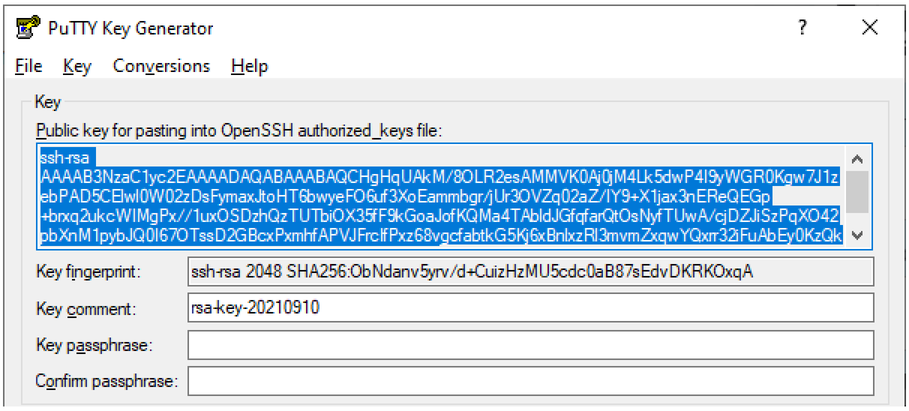
Ubah konfigurasi SSH ke PublickeyAuthentication

```
root@router:~# nano /etc/ssh/sshd_config

PubKeyAuthentication yes

AuthorizedKeysFile        .ssh/authorized_keys
```

Restart Service SSH

```
root@router:~# systemctl restart sshd
```

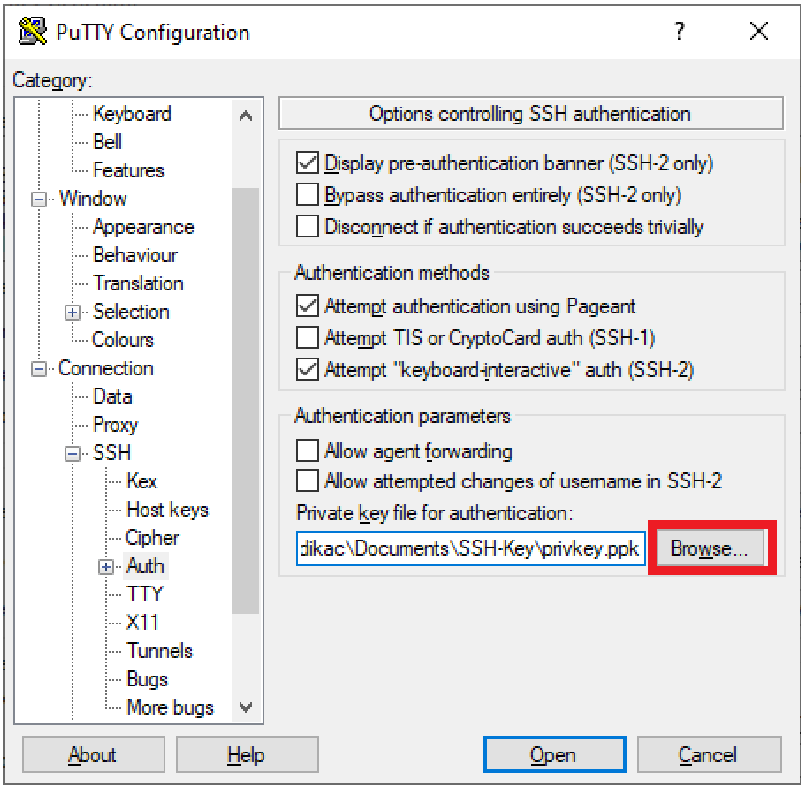
Selanjutnya tambahkan public key ke server, lalu copy public key dan paste ke file di home direktori user .ssh/authorized_keys



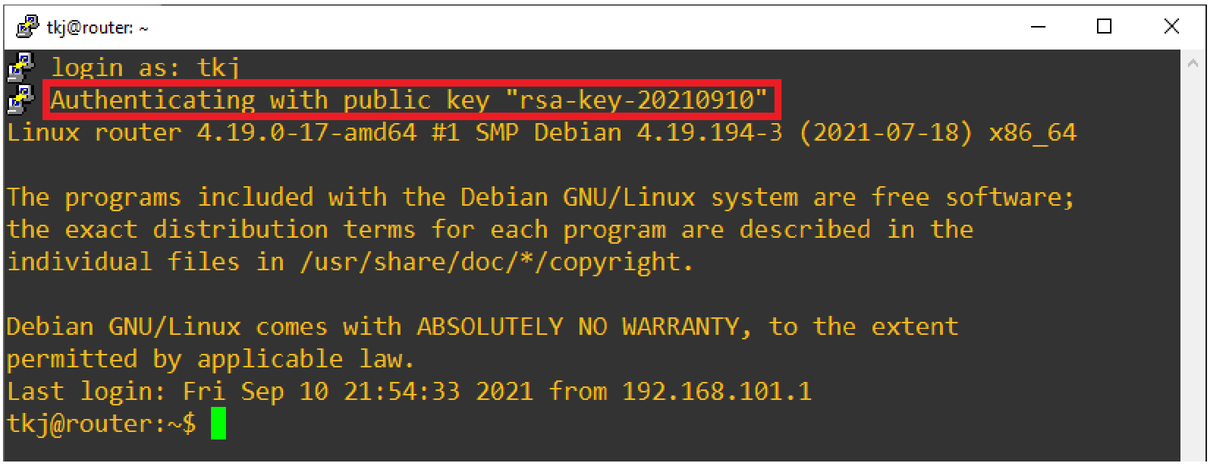
```
tkj@router:~$ mkdir .ssh

tkj@router:~$ nano .ssh/authorized_keys

ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQCHgHqUAkM/8OLR2esAMMVK0Aj0jM4Lk5dwP4I9yWGR0K
gw7J1zebPAD5CElwl0W02zDsFymaxJtoHT6bwyeFO6uf3XoEammbgr/jUr3OVZq02aZ/IY9+X1jax3nERe
QEGp+brxq2ukcWIMgPx//1uxOSDzhQzTUTbiOX35fF9kGoaJofKQMa4TabldJGfqfarQtOsNyfTUwA/cjD
ZJiSzPqXO42pbXnM1pybJQ0I67OtssD2GBcxPxmhfAPVJFrclfPxz68vgcfabtkG5Kj6xBnlxzRl3mvmZxqwY
Qxrr32iFuAbEy0KzQkYjni9soY62pgDrUVL+zvolZDp6Uxfx7 rsa-key-20210910
```

Koneksi dari Putty

Buka Putty > Connection > SSH > Auth lalu klik browse untuk memasukkan file private key



Lalu coba koneksikan



2.6 Konfigurasi Debian-Router

Agar Debian bisa menjadi sebuah router kita hanya perlu mengaktifkan IP Forward dan NAT

Di Modul ini kita tidak perlu DHCP Server karena semua IP Address untuk client di setting static

Enable IP Forward

```
root@router:~# nano /etc/sysctl.conf

net.ipv4.ip_forward=1
```

```
# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1_
```

Lalu aktifkan perubahan pada **/etc/sysctl.conf**

```
root@router:~# sysctl -p /etc/sysctl.conf

net.ipv4.ip_forward=1
```

```
root@router:~# sysctl -p /etc/sysctl.conf
net.ipv4.ip_forward = 1
root@router:~# _
```

Konfigurasi NAT

```
root@router:~# iptables -t nat -A POSTROUTING -o enp0s3 -j MASQUERADE
```

Lalu kita lihat list NAT dengan perintah ***iptables -t nat -nvL***

```
root@router:~# iptables -t nat -A POSTROUTING -o enp0s3 -j MASQUERADE
root@router:~# iptables -t nat -nvL
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 MASQUERADE  all  --  *      enp0s3  0.0.0.0/0            0.0.0.0/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
root@router:~# _
```

Simpan rules iptables secara permanen

Install paket iptables-persistent

```
root@router:~# apt -y install iptables-persistent
```

Nanti akan ada dialog, Save current IPv4 rules pilih **Yes**. Untuk IPv6 pilih **No**

```
Configuring iptables-persistent

Current iptables rules can be saved to the configuration file /etc/iptables/rules.v4. These
rules will then be loaded automatically during system startup.

Rules are only saved automatically during package installation. See the manual page of
iptables-save(8) for instructions on keeping the rules file up-to-date.

Save current IPv4 rules?

                  <Yes>                                    <No>
```

Simpan secara manual

```
root@router:~# iptables-save  > /etc/iptables/rules.v4
```

Tes ping ke internet dari Server1

```
root@server1:~# ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=108 time=31.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=108 time=33.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=108 time=33.6 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 4ms
rtt min/avg/max/mdev = 31.703/33.045/33.877/0.957 ms
root@server1:~#
```

Coba kita traceroute

```
root@server1:~# traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  192.168.101.2 (192.168.101.2)  0.341 ms  0.400 ms  0.384 ms
 2  10.0.2.2 (10.0.2.2)  0.806 ms  0.673 ms  0.808 ms
```

Terlihat bahwa paket melewati IP Router-Debian

2.7 Konfigurasi DNS Server

DNS Server adalah protocol yang digunakan internet untuk memetakan Alamat IP menjadi Sebuah Domain atau sebaliknya.

Misalnya, nama domain smkn1geger.sch.id diterjemahkan ke alamat *103.233.102.32* (IPv4)

Ada beberapa Tipe DNS:

1. DNS Caching
2. DNS Forwarding
3. DNS Authoritative

2.7.1 Konfigurasi DNS Caching & Forwarding di Debian-Router

DNS Caching berfungsi sebagai resolver, misal ketika client menanyakan alamat IP dari domain smkn1geger.sch.id maka DNS Caching bertanya ke DNS Server kemudian memberi jawaban dan menyimpan jawaban tersebut untuk beberapa saat, nanti ketika ada client yang menanyakan alamat IP dari smkn1geger.sch.id lagi, maka DNS Caching langsung memberi jawaban tanpa harus bertanya ke DNS Server.

Tahap pertama kita install aplikasi bind9

```
root@router:~# apt install bind9 dnsutils
```

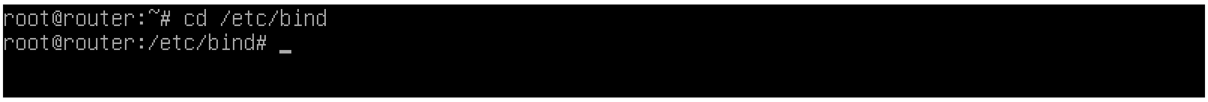
```
root@router:~# apt install bind9 dnsutils
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  bind9utils dns-root-data libirs161 python3-ply
Suggested packages:
  bind9-doc resolvconf ufw rblcheck python-ply-doc
The following NEW packages will be installed:
  bind9 bind9utils dns-root-data dnsutils libirs161 python3-ply
0 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,740 kB of archives.
After this operation, 5,148 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Ubah default konfigurasi bind9

```
root@router:~# nano /etc/default/bind9
```

File Konfigurasi bind berada di direktori **/etc/bind**, pindah ke direktori tersebut

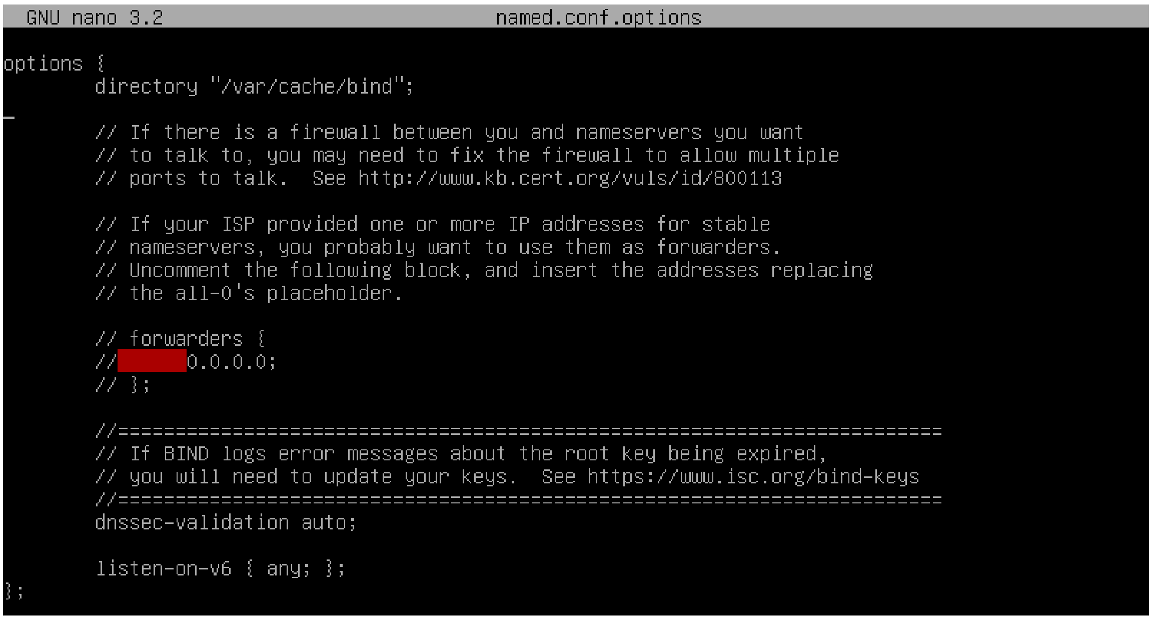
```
root@router:~# cd /etc/bind
```



```
root@router:~# cd /etc/bind
root@router:/etc/bind# _
```

Di dalam direktori /etc/bind terdapat banyak file, kali ini kita hanya fokus ke file **named.conf.options**

```
root@router:/etc/bind# nano named.conf.options

options {

        directory "/var/cache/bind";

       dnssec-validation auto;

        listen-on-v6 { any; };

};
```



```
  GNU nano 3.2                          named.conf.options

options {
        directory "/var/cache/bind";

        // If there is a firewall between you and nameservers you want
        // to talk to, you may need to fix the firewall to allow multiple
        // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

        // If your ISP provided one or more IP addresses for stable
        // nameservers, you probably want to use them as forwarders.
        // Uncomment the following block, and insert the addresses replacing
        // the all-0's placeholder.

        // forwarders {
        //      0.0.0.0;
        // };

        //========================================================================
        // If BIND logs error messages about the root key being expired,
        // you will need to update your keys.  See https://www.isc.org/bind-keys
        //========================================================================
        dnssec-validation auto;

        listen-on-v6 { any; };
};
```

Tanda // merupakan komentar jadi konfigurasi tersebut tidak akan di jalankan

Diatas konfigurasi options kita definisikan IP Client kita dengan acl

```
acl client {

        localhost;

        192.168.101.0/24;

};
```

```
  GNU nano 3.2                              named.conf.options

acl client {
        localhost;
        192.168.101.0/24;
};
options {
        directory "/var/cache/bind";
```

Setelah itu tambahkan konfigurasi di options

```
recursion yes;

allow-recursion { client; };
```

```
options {
        directory "/var/cache/bind";

        // If there is a firewall between you and nameservers you want
        // to talk to, you may need to fix the firewall to allow multiple
        // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

        // If your ISP provided one or more IP addresses for stable
        // nameservers, you probably want to use them as forwarders.
        // Uncomment the following block, and insert the addresses replacing
        // the all-0's placeholder.

        // forwarders {
        //        0.0.0.0;
        // };

        //========================================================================
        // If BIND logs error messages about the root key being expired,
        // you will need to update your keys.  See https://www.isc.org/bind-keys
        //========================================================================
        recursion yes;
        allow-recursion { client; };
        dnssec-validation auto;

        listen-on-v6 { any; };
};
```

Lalu save dan restart service bind9

```
root@router:/etc/bind# systemctl restart bind9
```

```
root@router:/etc/bind# systemctl restart bind9
root@router:/etc/bind#
```

Selanjutnya kita test di Server1

Ubah DNS Resolver ke 192.168.101.2 (IP Debian-Router)

```
root@server1:~# nano /etc/resolv.conf

nameserver 192.168.101.2
```

```
  GNU nano 3.2                              /etc/resolv.conf                              Modified

nameserver 192.168.101.2
```

Kita coba dengan cara perintah dig ke suatu web, misal youtube.com

```
root@server1:~# dig youtube.com
```

atau

```
root@server1:~# dig +noall +answer +stat youtube.com
```

```
root@server1:~# dig +noall +answer +stat youtube.com
youtube.com.            30      IN      A       74.125.24.190
youtube.com.            30      IN      A       74.125.24.91
youtube.com.            30      IN      A       74.125.24.136
youtube.com.            30      IN      A       74.125.24.93
;; Query time: 1544 msec
;; SERVER: 192.168.101.2#53(192.168.101.2)
;; WHEN: Sat Sep 11 17:17:14 WIB 2021
;; MSG SIZE  rcvd: 915
```

Query Time pertama

```
root@server1:~# dig +noall +answer +stat youtube.com
youtube.com.            28      IN      A       74.125.24.190
youtube.com.            28      IN      A       74.125.24.91
youtube.com.            28      IN      A       74.125.24.93
youtube.com.            28      IN      A       74.125.24.136
;; Query time: 0 msec
;; SERVER: 192.168.101.2#53(192.168.101.2)
;; WHEN: Sat Sep 11 17:17:15 WIB 2021
;; MSG SIZE  rcvd: 915

root@server1:~#
```

Query Time kedua

Terlihat bahwa query time pertama yaitu 1544 msec, setelah itu menjadi 0 msec berarti Konfigurasi DNS Caching kita berhasil.

Selanjutnya yaitu konfigurasi DNS Forwarding, Apa itu DNS Forwarding?

DNS Forwarding hampir mirip dengan DNS Caching perbedaannya yaitu hanya pada client yang melalukan request, misal: Server1 melakukan Request ke google.com maka DNS Forwarding hanya meneruskan request tersebut ke DNS Server misal (1.1.1.1) setelah mendapat jawaban , DNS Forwarding mengembalikan jawaban ke Server1.

Berikut kira-kira perbedaannya

1. DNS Forwarding berfungsi meneruskan request dari client ke DNS Server kemudian mengembalikan jawaban ke client.
2. DNS Caching berfungsi menanyakan request dari client kemudian memberi jawaban ke client.

Konfigurasi tetap di file **named.conf.options**

```
root@router:/etc/bind# nano named.conf.options
```

Hilangkan tanda // forwarders di dalam options, isi alamat DNS Server misal 1.1.1.1

```
options {
        directory "/var/cache/bind";

        // If there is a firewall between you and nameservers you want
        // to talk to, you may need to fix the firewall to allow multiple
        // ports to talk.  See http://www.kb.cert.org/vuls/id/800113
        // If your ISP provided one or more IP addresses for stable
        // nameservers, you probably want to use them as forwarders.
        // Uncomment the following block, and insert the addresses replacing
        // the all-0's placeholder.

        forwarders {
                1.1.1.1;
        };
        //========================================================================
        // If BIND logs error messages about the root key being expired,
        // you will need to update your keys.  See https://www.isc.org/bind-keys
        //========================================================================
        recursion yes;
        allow-recursion { client; };
        dnssec-validation auto;
```

Lalu save, dan restart service bind9

```
root@router:/etc/bind# systemctl restart bind9
```

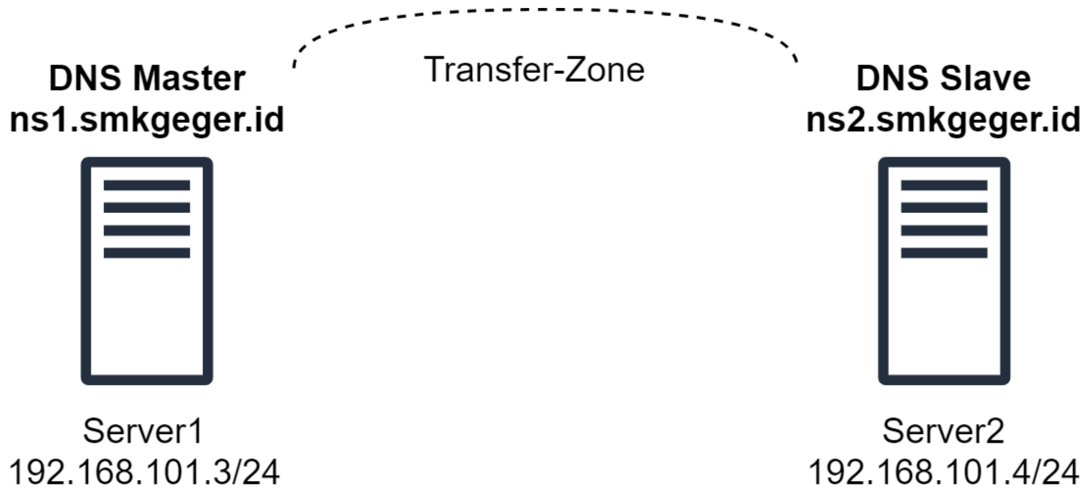
Konfigurasi Lengkap DNS Caching+Forwarding

```
acl client{
	192.168.101.0/24;
};
options {
	directory "/var/cache/bind";
	forwarders {
		1.1.1.1;
	};
	recursion yes;
	allow-recursion { client; };
	dnssec-validation auto;
	listen-on-v6 { any; };
};
```

2.7.2 Konfigurasi DNS Authoritative

DNS Authoritative adalah DNS Server yang berisi record Alamat IP dan Domain, jadi DNS Server ini yang mempunyai kepemilikan suatu domain, sebagai contoh kita gunakan domain smkgeger.id

Berikut topologi DNS Server yang kita praktekkan



DNS Master yaitu DNS Server utama sedangkan DNS Slave yaitu DNS Server sebagai backup jika DNS Utama Mati/Down

2.7.2.1 *Konfigurasi DNS Master(Primary DNS) di Server1*

Install BIND9

```
root@server1:~# apt install bind9 dnsutils
```

```
root@server1:~# apt install bind9 dnsutils
Reading package lists... Done
Building dependency tree
Reading state information... Done
dnsutils is already the newest version (1:9.11.5.P4+dfsg-5.1+deb10u5).
The following additional packages will be installed:
  bind9utils dns-root-data net-tools python3-ply
Suggested packages:
  bind9-doc resolvconf ufw python-ply-doc
The following NEW packages will be installed:
  bind9 bind9utils dns-root-data net-tools python3-ply
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/1,386 kB of archives.
After this operation, 5,123 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

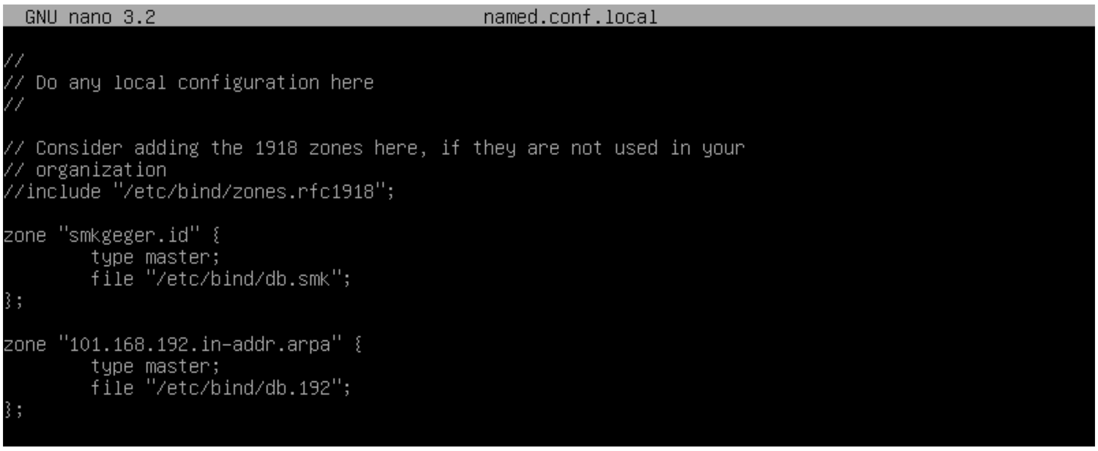
Pindah ke direktori bind

```
root@server1:~# cd /etc/bind
```

```
root@server1:~# cd /etc/bind
root@server1:/etc/bind#
```

Konfigurasi Zone File

```
root@server1:~# nano named.conf.local

zone "smkgeger.id" {
        type master;
        file "/etc/bind/db.smk";
       allow-transfer { 192.168.101.4; };
       also-notify { 192.168.101.4; };
};
zone "101.168.192.in-addr.arpa" {
        type master;
        allow-transfer { 192.168.101.4; };
       also-notify { 192.168.101.4; };
        file "/etc/bind/db.192";
};
```



```
  GNU nano 3.2                    named.conf.local

//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "smkgeger.id" {
        type master;
        file "/etc/bind/db.smk";
};

zone "101.168.192.in-addr.arpa" {
        type master;
        file "/etc/bind/db.192";
};
```

Ada dua zone yang kita buat, pertama yaitu zone forward yang berisi sebuah pemetaan Nama Domain ke Alamat IP, yang kedua yaitu zone reverse kebalikan dari zone forward (pemetaan Alamat IP ke Domain Name).

Penjelasan syntax

zone “smkgeger.id” {	Baris ini menunjukkan bahwa kita membuat sebuah domain dengan nama smkgeger.id
type master;	Baris ini menunjukkan bahwa zone yang kita buat sebagai dns utama.
file “/etc/bind/db.smk”;	Baris ini menunjukkan bahwa file record untuk domain smkgeger.id terletak di /etc/bind/db.smk
allow-transfer { 192.168.101.4; };	Baris ini menunjukkan bahwa kita mengizinkan mentransfer zona ke Alamat IP 192.168.101.4 (Slave DNS)
also-notify { 192.168.101.4; };	Baris ini menunjukkan kita selalu memberi notifikasi perubahan ke Alamat IP 192.168.101.4 (Slave DNS)
};	Penutup konfigurasi forward zone
zone “101.168.192.in-addr.arpa”{	Baris ini menunjukkan bahwa kita membuat sebuah reverse zone dengan Alamat IP 192.168.101.x dengan format terbalik yaitu 101.168.192
type master;	Sama seperti forward zone tadi, ini menunjukkan bahwa zone yang kita buat yaitu sebagai master (dns utama).
file “/etc/bind/db.192”;	Sama seperti forward zone tadi, ini menunjukkan bahwa file record dari zone 101.168.192.in-addr.arpa terletak di /etc/bind/db.192
allow-transfer { 192.168.101.4; };	Baris ini menunjukkan bahwa kita mengizinkan mentransfer zona ke Alamat IP 192.168.101.4 (Slave DNS)
also-notify { 192.168.101.4; };	Baris ini menunjukkan kita selalu memberi notifikasi perubahan ke Alamat IP 192.168.101.4 (Slave DNS)
};	Penutup konfigurasi reverse zone

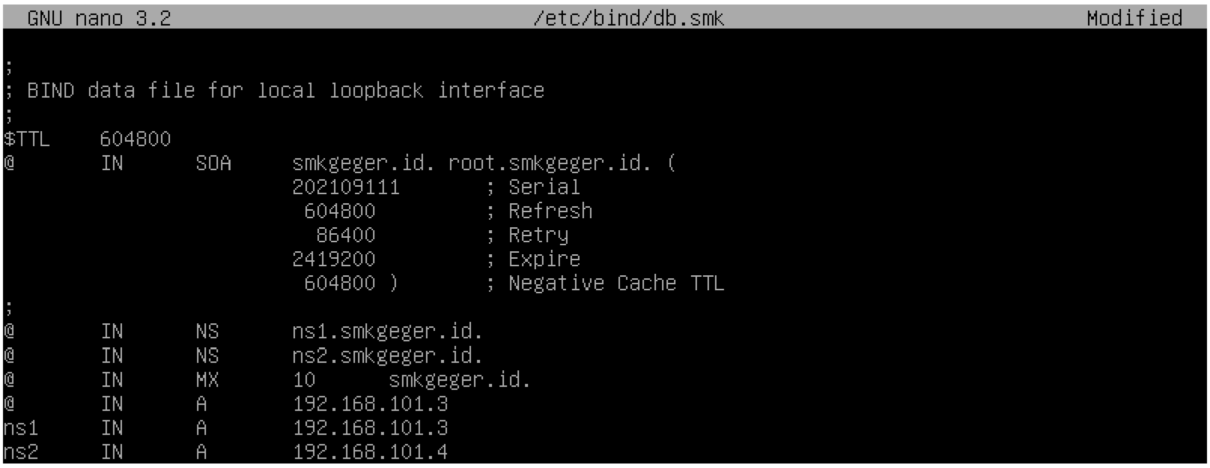
Selanjutnya kita konfigurasi record untuk forward zone terlebih dahulu

Salin file db.local menjadi db.smk

root@server1:/etc/bind# cp db.local db.smk

Lalu edit dengan nano, ubah localhost menjadi smkgeger.id

root@server1:/etc/bind# nano db.smk




```
$TTL    604800
@       IN      SOA     smkgeger.id. root.smkgeger.id. (
                        202109111       ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.smkgeger.id.
@       IN      NS      ns2.smkgeger.id.
@       IN      A       192.168.101.3
@       IN      MX 10 smkgeger.id.
ns1     IN      A       192.168.101.3
ns2     IN      A       192.168.101.4
```

@ menandakan record ini merujuk ke hostname yaitu smkgeger.id

IN menunjukkan jenis jaringan Internet

Jenis record DNS **A, NS, MX, CNAME, TXT dan SOA**. **A** menunjukkan Alamat IP domain, **NS** menunjukkan Alamat IP DNS Server, **MX** Alamat dari Mail Server, **CNAME** Alias (Canonical Name), **TXT** entri khusus, **SOA** yang menunjukkan authoritative nameserver untuk zona, dengan rincian tentang administrator, serial number, dan refresh rate.

Setiap kita melakukan perubahan pada record, kita harus mengganti juga serial number untuk standarnya yaitu format YYYYMMDDXX. Misal 202109111 berarti Tahun 2021 bulan 9 tanggal 11 perubahan ke 1

Lebih lengkap silakan baca di : https://wiki.debian.org/Bind9

Selanjutnya kita konfigurasi reverse zone

salin file db.172 menjadi db.192

```
root@server1:/etc/bind# cp db.127 db.192
```

lalu edit dengan nano, ubah localhost menjadi smkgeger.id

```
root@server1:/etc/bind# nano db.192
$TTL    604800
@       IN      SOA     smkgeger.id. root.smkgeger.id. (
                        202109111       ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.smkgeger.id.
@       IN      NS      ns2.smkgeger.id.
3       IN      PTR     smkgeger.id.
3       IN      PTR     ns1.smkgeger.id.
4       IN      PTR     ns2.smkgeger.id.
```

Lalu save dan restart service bind9

```
root@server1:/etc/bind# systemctl restart bind9
```

Setelah berhasil di konfigurasi, berikutnya ubah dns resolver menjadi IP kita 192.168.101.3

```
root@server1:/etc/bind# nano /etc/resolv.conf
nameserver 192.168.101.3
```

```
  GNU nano 3.2                    /etc/resolv.conf                    Modified

nameserver 192.168.101.3
```

Lalu gunakan perintah dig untuk mengecek dns

root@server1:/etc/bind# dig smkgeger.id

```
root@server1:~# dig smkgeger.id

; <<>> DiG 9.11.5-P4-5.1+deb10u5-Debian <<>> smkgeger.id
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 13636
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 2, ADDITIONAL: 3

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: 378ed011c824811e3adc347f613d3632328b296c06f6d6e3 (good)
;; QUESTION SECTION:
;smkgeger.id.                   IN      A

;; ANSWER SECTION:
smkgeger.id.            604800  IN      A       192.168.101.3

;; AUTHORITY SECTION:
smkgeger.id.            604800  IN      NS      ns2.smkgeger.id.
smkgeger.id.            604800  IN      NS      ns1.smkgeger.id.

;; ADDITIONAL SECTION:
ns1.smkgeger.id.        604800  IN      A       192.168.101.3
ns2.smkgeger.id.        604800  IN      A       192.168.101.4

;; Query time: 0 msec
;; SERVER: 192.168.101.3#53(192.168.101.3)
;; WHEN: Sun Sep 12 06:05:22 WIB 2021
;; MSG SIZE  rcvd: 152

root@server1:~#
```

Perhatikan record smkgeger.id sudah tertuju ke IP Server1, nanti kita gunakan dns ini di Windows Client agar bisa mengakses layanan web secara privat.

Setelah selesai melakukan pengecekan, ubah dns resolver kembali ke semula (jika kalian butuh mendownload paket dari internet) karena kalau tidak dirubah maka server kita tidak bisa mengakses domain umum seperti google.com

2.7.2.2 Konfigurasi DNS Slave (Secondary) di Server2

Install BIND9

root@server2:~# apt install bind9 dnsutils

```
root@server2:~# apt install bind9 dnsutils
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  bind9utils dns-root-data libirs161 net-tools python3-ply
Suggested packages:
  bind9-doc resolvconf ufw rblcheck python-ply-doc
The following NEW packages will be installed:
  bind9 bind9utils dns-root-data dnsutils libirs161 net-tools python3-ply
0 upgraded, 7 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/1,988 kB of archives.
After this operation, 6,150 kB of additional disk space will be used.
Do you want to continue? [Y/n] y_
```

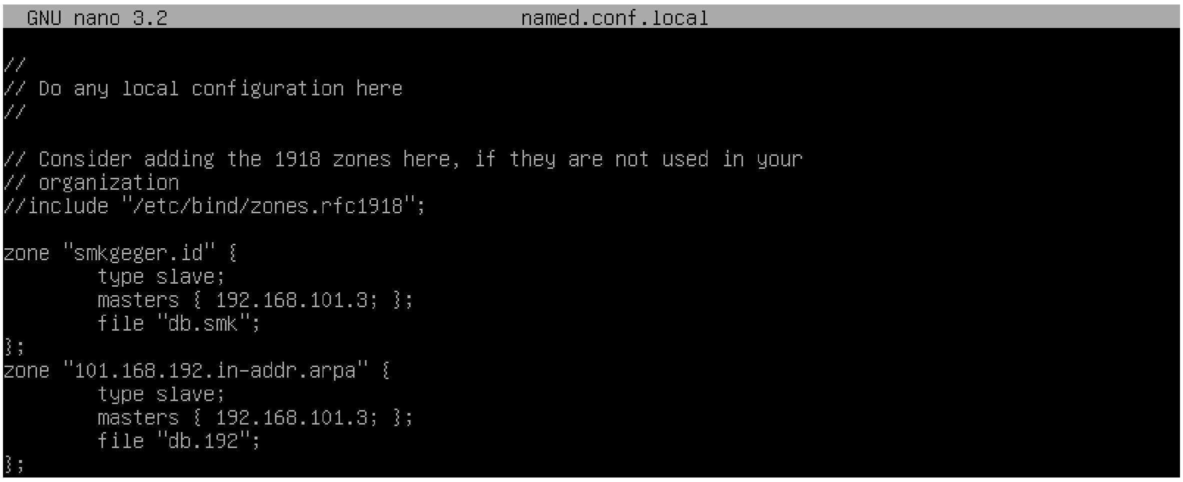
Pindah ke direktori bind

root@server2:~# cd /etc/bind

```
root@server2:~# cd /etc/bind
root@server2:/etc/bind#
```

Buat zone smkgeger.id

```
root@server2:/etc/bind# nano named.conf.local

zone "smkgeger.id" {

        type slave;

        masters { 192.168.101.3; };

        file "db.smk";

};

zone "101.168.192.in-addr.arpa" {

        type slave;

        masters { 192.168.101.3; };

        file "db.192";

};
```



Penjelasan syntax konfigurasi

zone “smkgeger.id” {	Baris ini menunjukkan bahwa kita membuat sebuah domain dengan nama smkgeger.id
type slave;	Baris ini menunjukkan bahwa zone yang kita buat sebagai dns cadangan
masters { 192.168.101.3; };	Baris ini menunjukkan bahwa master dns dari zone ini beralamat IP 192.168.101.3 (Server1)
file “db.smk”;	Baris ini menunjukkan bahwa file record untuk domain smkgeger.id terletak di db.smk, yang secara default berada di direktori /var/cache/bind.
};	Penutup konfigurasi forward zone
zone “101.168.192.in-addr.arpa”{	Baris ini menunjukkan bahwa kita membuat sebuah reverse zone dengan Alamat IP 192.168.101.x dengan format terbalik yaitu 101.168.192

type slave;	Sama seperti forward zone tadi, ini menunjukkan bahwa zone yang kita buat yaitu sebagai dns cadangan.
file “db.192”;	Sama seperti forward zone tadi, ini menunjukkan bahwa file record dari zone 101.168.192.in-addr.arpa terletak di /var/cache/bind/db.192
};	Penutup konfigurasi reverse zone

Lalu save dan restart service bind9

```
root@server2:/etc/bind# systemctl restart bind9
```

Kemudian lihat log dengan perintah

```
root@server2:/etc/bind# tail -f /var/log/syslog
```

```
root@server2:/etc/bind# tail -f /var/log/syslog
Sep 12 07:27:35 server2 named[1238]: transfer of 'smkgeger.id/IN' from 192.168.101.3#53: Transfer co
mpleted: 1 messages, 7 records, 190 bytes, 0.001 secs (190000 bytes/sec)
Sep 12 07:27:35 server2 named[1238]: zone smkgeger.id/IN: sending notifies (serial 202109111)
Sep 12 07:27:36 server2 named[1238]: managed-keys-zone: Key 20326 for zone . acceptance timer comple
te: key now trusted
Sep 12 07:27:36 server2 named[1238]: resolver priming query complete
Sep 12 07:27:36 server2 named[1238]: zone 101.168.192.in-addr.arpa/IN: Transfer started.
Sep 12 07:27:36 server2 named[1238]: transfer of '101.168.192.in-addr.arpa/IN' from 192.168.101.3#53
: connected using 192.168.101.4#40457
Sep 12 07:27:36 server2 named[1238]: zone 101.168.192.in-addr.arpa/IN: transferred serial 202109111
Sep 12 07:27:36 server2 named[1238]: transfer of '101.168.192.in-addr.arpa/IN' from 192.168.101.3#53
 Transfer status: success
Sep 12 07:27:36 server2 named[1238]: transfer of '101.168.192.in-addr.arpa/IN' from 192.168.101.3#53
 Transfer completed: 1 messages, 7 records, 212 bytes, 0.003 secs (70666 bytes/sec)
Sep 12 07:27:36 server2 named[1238]: zone 101.168.192.in-addr.arpa/IN: sending notifies (serial 2021
09111)
```

transfer sukses

Lalu cek di direktori /var/cache/bind

```
root@server2:/etc/bind# ls /var/cache/bind
```

```
root@server2:/etc/bind# ls /var/cache/bind/
db.192  db.smk  managed-keys.bind  managed-keys.bind.jnl
root@server2:/etc/bind# _
```

Uji coba dari Windows

Buka CMD Win+R , lalu jalankan nslookup dan setting server ke 192.168.101.4

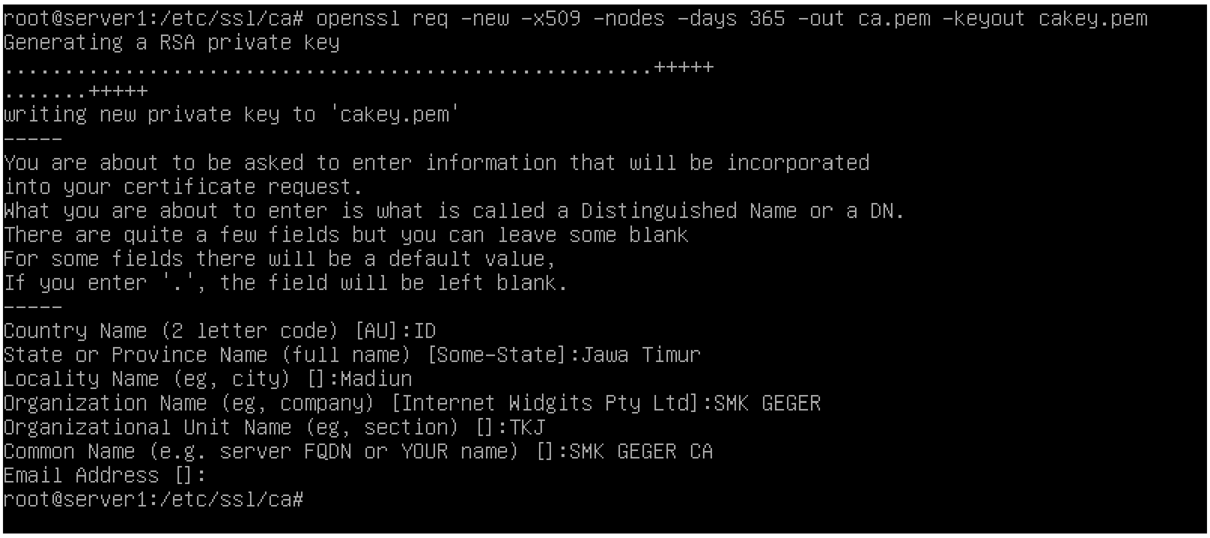
```
C:\Windows\system32\cmd.exe - nslookup
Microsoft Windows [Version 10.0.19043.1110]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dikac>nslookup
Default Server:  one.one.one.one
Address:  1.1.1.1

> server 192.168.101.4
Default Server:  [192.168.101.4]
Address:  192.168.101.4

> smkgeger.id
Server:  [192.168.101.4]
Address:  192.168.101.4

Name:    smkgeger.id
Address:  192.168.101.3

>
```

2.8 Membuat Certificate Authority (CA)

Apa itu Certificate Authority? CA adalah otoritas yang dipercaya untuk mengelola sertifikat SSL/TLS untuk web, alamat email dan perusahaan. Di server yang ada di internet kita bisa menggunakan Let's encrypt untuk mendapat sertifikat SSL/TLS secara gratis. Karena kita mengkonfigurasi server di lokal maka kita bisa membuat CA sendiri dengan openssl lalu di import ke Client.

Pertama kita pindah ke direktori /etc/ssl dan membuat folder baru yang bernama ca

```
root@server1:~# cd /etc/ssl

root@server1:/etc/ssl# mkdir ca

root@server1:/etc/ssl# cd ca/

root@server1:/etc/ssl/ca# openssl req -new -x509 -nodes -days 365 -out ca.pem -keyout cakey.pem
```



```
root@server1:/etc/ssl/ca# openssl req -new -x509 -nodes -days 365 -out ca.pem -keyout cakey.pem
Generating a RSA private key
......................................................+++++
.......+++++
writing new private key to 'cakey.pem'
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:ID
State or Province Name (full name) [Some-State]:Jawa Timur
Locality Name (eg, city) []:Madiun
Organization Name (eg, company) [Internet Widgits Pty Ltd]:SMK GEGER
Organizational Unit Name (eg, section) []:TKJ
Common Name (e.g. server FQDN or YOUR name) []:SMK GEGER CA
Email Address []:
root@server1:/etc/ssl/ca#
```

Generating a RSA private key

writing new private key to 'cakey.pem'

Country Name (2 letter code) [AU]:**ID**

State or Province Name (full name) [Some-State]:**Jawa Timur**

Locality Name (eg, city) []:**Madiun**

Organization Name (eg, company) [Internet Widgits Pty Ltd]:**SMK GEGER**

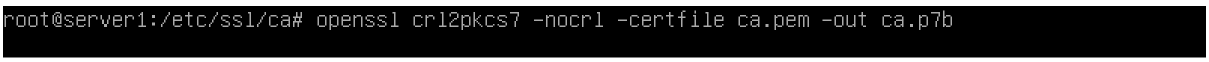
Organizational Unit Name (eg, section) []:**TKJ**

Common Name (e.g. server FQDN or YOUR name) []:**SMK GEGER CA**

Email Address []:

Mengkonversi CA agar bisa di import ke client Windows

```
root@server1:/etc/ssl/ca# openssl crl2pkcs7 -nocrl -certfile ca.pem -out ca.p7b
```



```
root@server1:/etc/ssl/ca# openssl crl2pkcs7 -nocrl -certfile ca.pem -out ca.p7b
```

Setelah membuat CA sekarang kita buat CSR(Certificate Signing Request) untuk domain smkgeger.id yang nanti akan di signing/tanda tangan ke CA yang tadi kita buat.

root@server1:/etc/ssl/ca# openssl req -new -nodes -out smk.csr -keyout smk.key

Generating a RSA private key

writing new private key to 'smk.key'

Country Name (2 letter code) [AU]:**ID**

State or Province Name (full name) [Some-State]:**East Java**

Locality Name (eg, city) []:**Madiun**

Organization Name (eg, company) [Internet Widgits Pty Ltd]:**SMK GEGER**

Organizational Unit Name (eg, section) []:**TKJ**

Common Name (e.g. server FQDN or YOUR name) []:**smkgeger.id**

Email Address []:**admin@smkgeger.id**

Please enter the following 'extra' attributes

to be sent with your certificate request

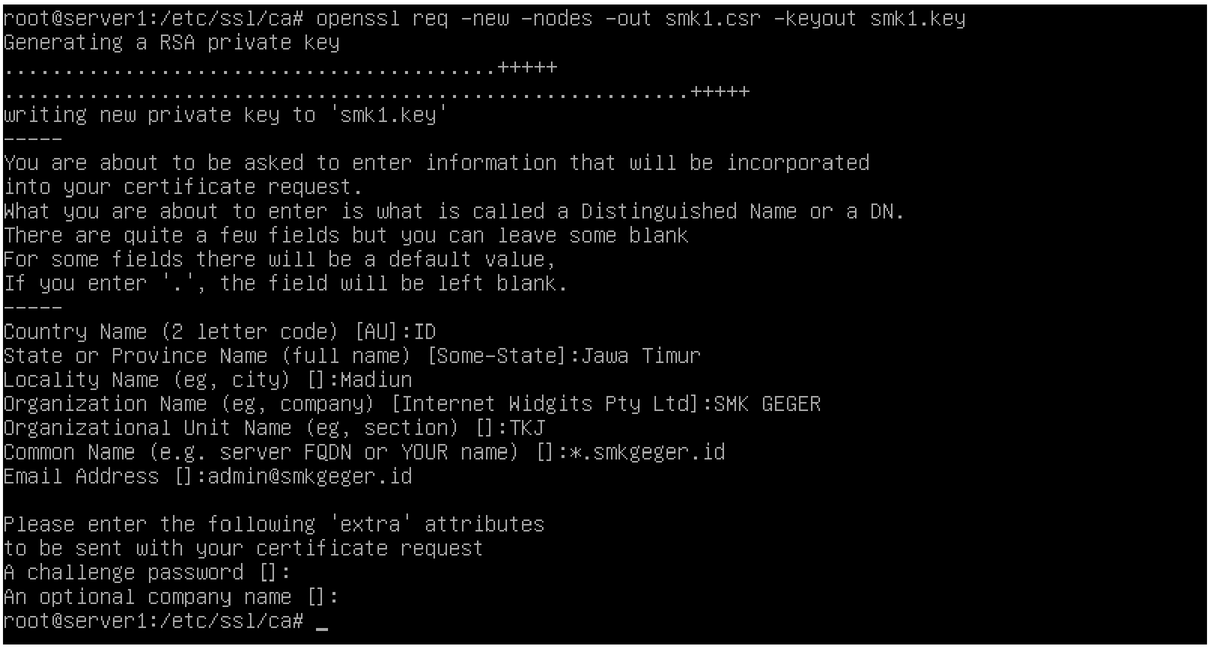
A challenge password []:

An optional company name []:

```
root@server1:/etc/ssl/ca# openssl req -new -nodes -out smk.csr -keyout smk.key
Generating a RSA private key
....+++++
.........+++++
writing new private key to 'smk.key'
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:ID
State or Province Name (full name) [Some-State]:Jawa Timur
Locality Name (eg, city) []:Madiun
Organization Name (eg, company) [Internet Widgits Pty Ltd]:SMK GEGER
Organizational Unit Name (eg, section) []:TKJ
Common Name (e.g. server FQDN or YOUR name) []:smkgeger.id
Email Address []:admin@smkgeger.id

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
root@server1:/etc/ssl/ca# _
```

Isi sama seperti membuat CA tadi namun pada FQDN kita isikan **smkgeger.id** yang berarti kita membuat certificate smkgeger.id, setelah itu generate CSR juga untuk subdomain smkgeger misal web2.smkgeger.id dengan cara yang sama namun FQDN nya di isi dengan *.smkgeger.id jadi nanti semua subdomain ada di 1 file sertifikat saja.

```
root@server1:/etc/ssl/ca# openssl req -new  -nodes -out smk1.csr -keyout smk1.key

Generating a RSA private key

writing new private key to 'smk1.key'

Country Name (2 letter code) [AU]:ID

State or Province Name (full name) [Some-State]:East Java

Locality Name (eg, city) []:Madiun

Organization Name (eg, company) [Internet Widgits Pty Ltd]:SMK GEGER

Organizational Unit Name (eg, section) []:TKJ

Common Name (e.g. server FQDN or YOUR name) []:*.smkgeger.id

Email Address []:admin@smkgeger.id

Please enter the following 'extra' attributes

to be sent with your certificate request

A challenge password []:

An optional company name []:
```



Setelah berhasil membuat CSR, selanjutnya kita tanda tangani dengan CA

```
root@server1:/etc/ssl/ca# openssl x509 -req -in smk.csr -out smk.crt -CA ca.pem -CAkey cakey.pem -days 365 -set_serial 01

Signature ok

subject=C = ID, ST = Jawa Timur, L = Madiun, O = SMK GEGER, OU = TKJ, CN = smkgeger.id, emailAddress = admin@smkgeger.id
```

```
root@server1:/etc/ssl/ca# openssl x509 -req -in smk.csr -out smk.crt -CA ca.pem -CAkey cakey.pem -da
ys 365 -set_serial 01
Signature ok
subject=C = ID, ST = Jawa Timur, L = Madiun, O = SMK GEGER, OU = TKJ, CN = smkgeger.id, emailAddress
 = admin@smkgeger.id
Getting CA Private Key
root@server1:/etc/ssl/ca#
```

Selanjutnya tanda tangani juga sertifikat untuk subdomain smkgeger

```
root@server1:/etc/ssl/ca# openssl x509 -req -in smk1.csr -out smk1.crt -CA ca.pem -CAkey cakey.pem -days 365 -set_serial 02

Signature ok

subject=C = ID, ST = Jawa Timur, L = Madiun, O = SMK GEGER, OU = TKJ, CN = *.smkgeger.id, emailAddress = admin@smkgeger.id
```

```
root@server1:/etc/ssl/ca# openssl x509 -req -in smk1.csr -out smk1.crt -CA ca.pem -CAkey cakey.pem -
days 365 -set_serial 01
Signature ok
subject=C = ID, ST = Jawa Timur, L = Madiun, O = SMK GEGER, OU = TKJ, CN = *.smkgeger.id, emailAddre
ss = admin@smkgeger.id
Getting CA Private Key
root@server1:/etc/ssl/ca# _
```

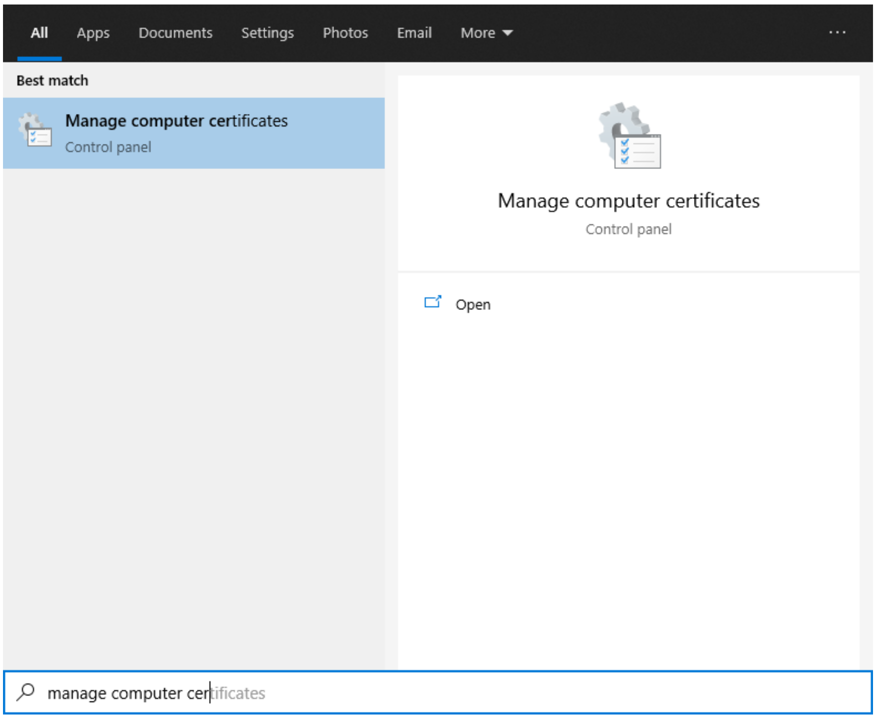
Coba kita lihat file CA dan Certificate yang baru kita buat, file smk.crt untuk smkgeger.id file smk1.crt untuk semua subdomain smkgeger.id

```
root@server1:/etc/ssl/ca# ls
```

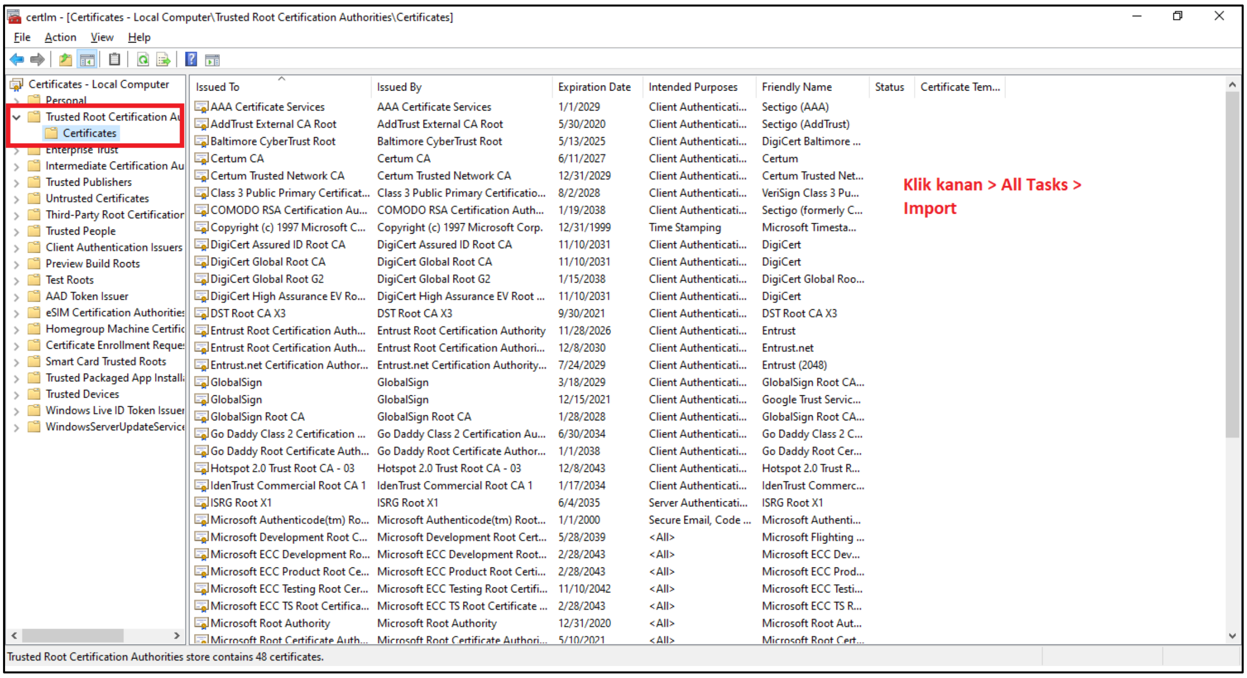
```
root@server1:/etc/ssl/ca# ls
cakey.pem  ca.p7b  ca.pem  smk1.crt  smk1.csr  smk1.key  smk.crt  smk.csr  smk.key
root@server1:/etc/ssl/ca#
```

Selanjutnya tinggal kita import file CA ke Windows Client

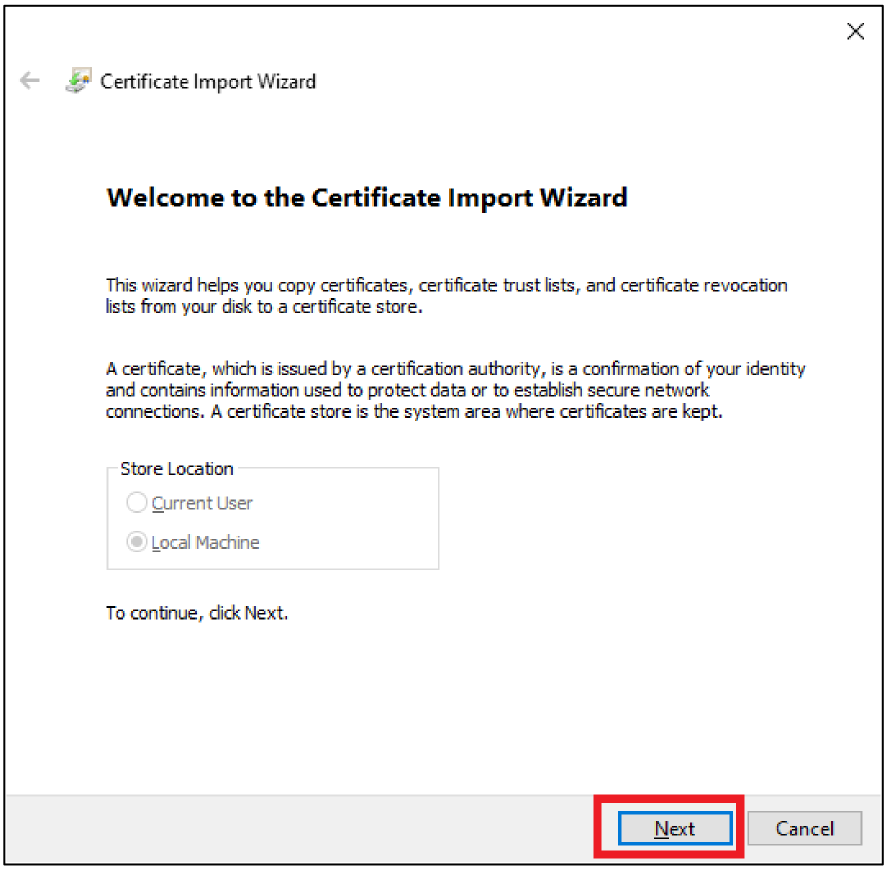
Transfer file ca.p7b ke Windows lalu Cari Manage Computer Certificates



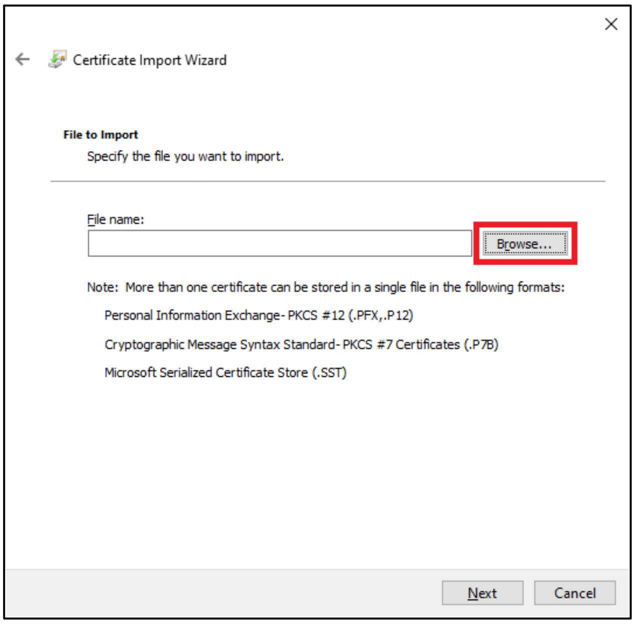
Klik **Trusted Root Certificate Authority** > **Certificate** > **Klik Kanan** > **All Tasks** > **Import**

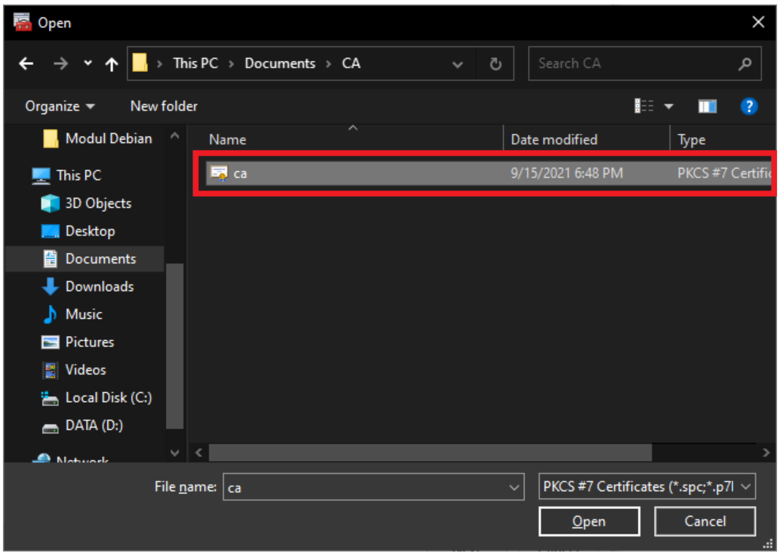


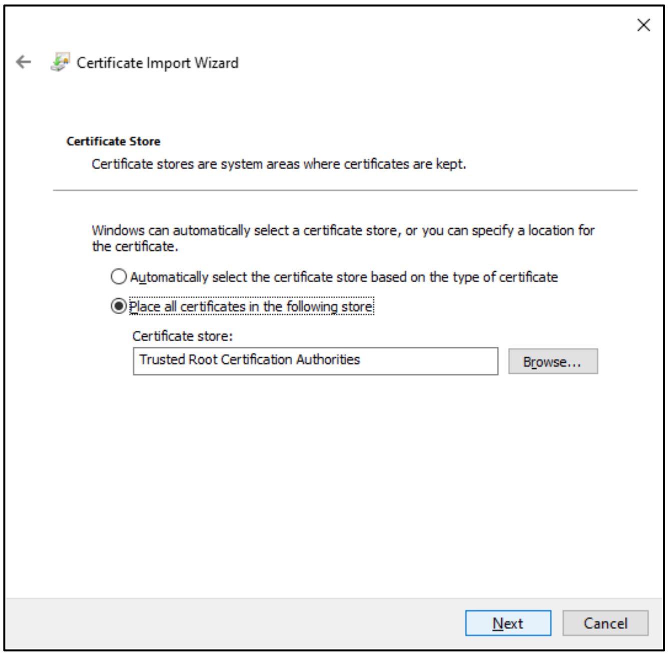
Kemudian klik **Next** > **Browse** > **Masukkan File CA** > klik **Next**

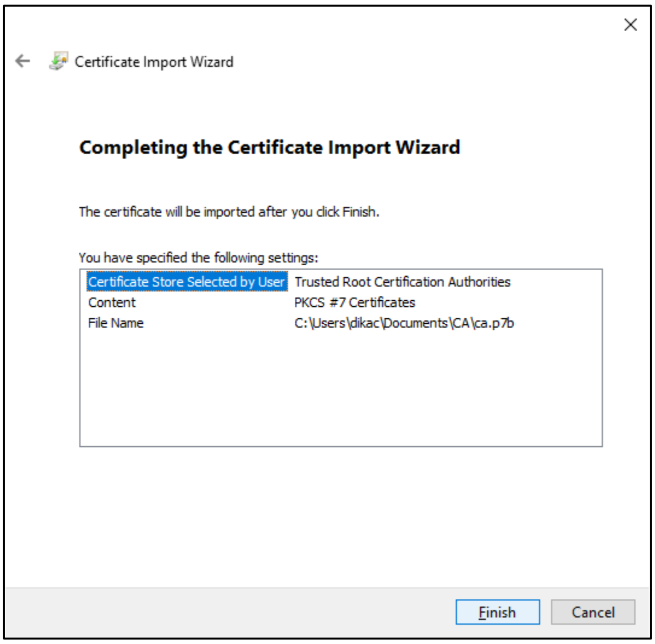


Setelah itu klik **Browse** > Pilih file CA **> Place All following store(Trusted Root Certification Authorities)** > Klik **Next >** Terakhir klik **Finish**









Berhasil!

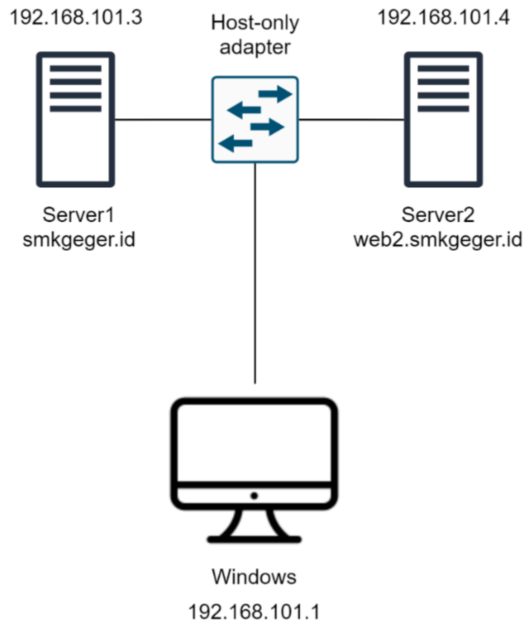
CA siap digunakan, nanti kita akan coba di konfigurasi Web Server

2.9 Konfigurasi Web dan Database Server

Apa itu web server? Web server adalah perangkat lunak yang menyimpan dan mengirimkan konten website seperti HTML,CSS kepada perangkat client seperti browser. Jadi client bisa mengakses konten yang dimiliki Web Server dengan menggunakan protokol HTTP atau HTTPS.

Apa itu Database Server? Database Server adalah perangkat lunak yang menyediakan layanan pengelolaan basis data dan melayani komputer atau program aplikasi basis data yang menggunakan model klien/server.

Topologi Web Server



Sebelum membuat Web Server, konfigurasikan domain sesuai dengan topologi.

Tambahkan subdomain web2 di DNS Server dan arahkan ke Server2. Jangan lupa untuk mengubah angka serialnya.

```
  GNU nano 3.2                                          db.smk

;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     smkgeger.id. root.smkgeger.id. (
                        202109112       ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.smkgeger.id.
@       IN      NS      ns2.smkgeger.id.
@       IN      A       192.168.101.3
ns1     IN      A       192.168.101.3
ns2     IN      A       192.168.101.4
web2    IN      A       192.168.101.4
```

2.9.1 Instalasi dan Konfigurasi LAMP Stack di Server1

LAMP adalah singkatan dari (Linux Apache MariaDB dan PHP), jadi ini adalah satu kesatuan lengkap yang biasanya di install ketika ingin membangun sebuah webserver. Jika ingin membuat web server dengan HTML saja, maka tidak perlu melakukan Instalasi php.

Install Apache2

```
root@server1:~# apt install apache2
```

```
root@server1:~# apt install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libbrotli1 libcurl4 libjansson4 liblua5.2-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser openssl-blacklist
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libbrotli1 libcurl4 libjansson4 liblua5.2-0 ssl-cert
0 upgraded, 13 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/2,959 kB of archives.
After this operation, 9,660 kB of additional disk space will be used.
Do you want to continue? [Y/n] y_
```

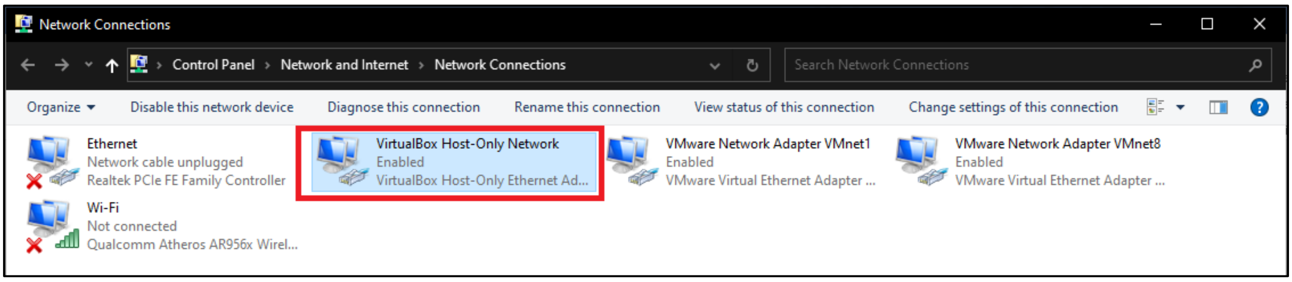
Cek status apache2

```
root@server1:~# systemctl status apache2
```

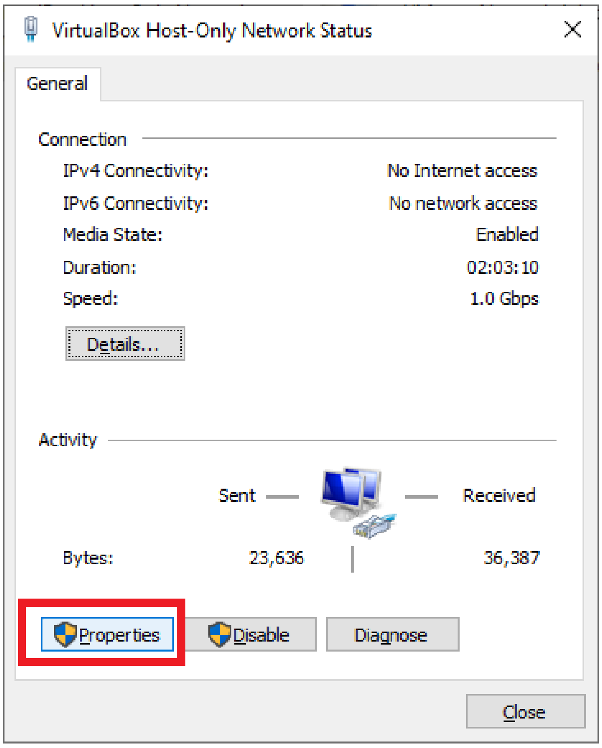
```
root@server1:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2021-09-12 13:21:16 WIB; 14s ago
     Docs: https://httpd.apache.org/docs/2.4/
 Main PID: 2379 (apache2)
    Tasks: 55 (limit: 545)
   Memory: 4.6M
   CGroup: /system.slice/apache2.service
           ├─2379 /usr/sbin/apache2 -k start
           ├─2381 /usr/sbin/apache2 -k start
           └─2382 /usr/sbin/apache2 -k start

Sep 12 13:21:16 server1 systemd[1]: Starting The Apache HTTP Server...
Sep 12 13:21:16 server1 systemd[1]: Started The Apache HTTP Server.
root@server1:~#
```

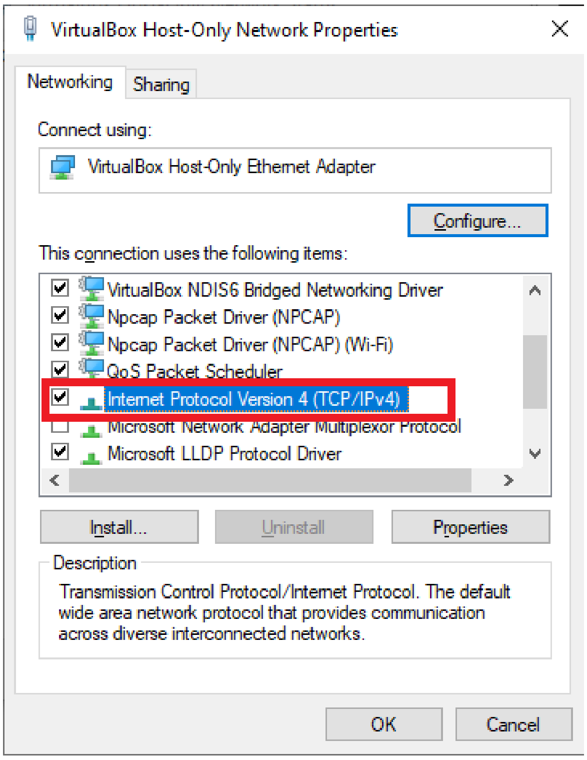
Cek di Web Browser Windows

Matikan Internet di Windows, Lalu pergi ke Control Panel > Network and Internet > Network Connections > Klik 2x Pada Adapter VirtualBox Host-only Network

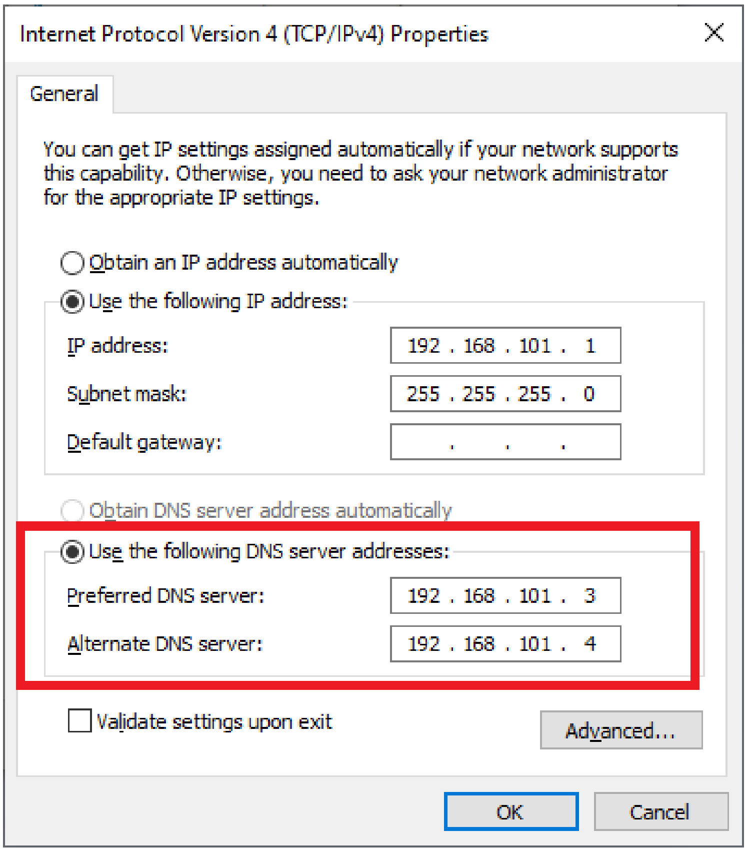


Klik Properties > Pilih Internet Protocol Version 4

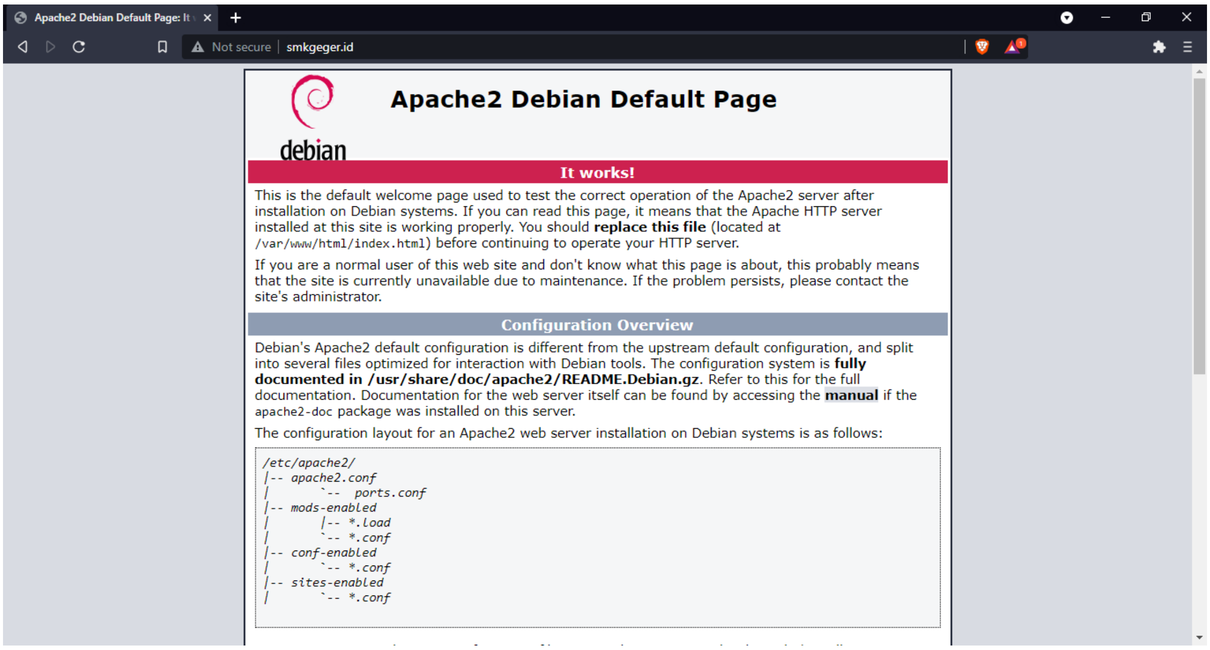




Ubah DNS ke 192.168.101.3 dan 192.168.101.4, Lalu Klik OK



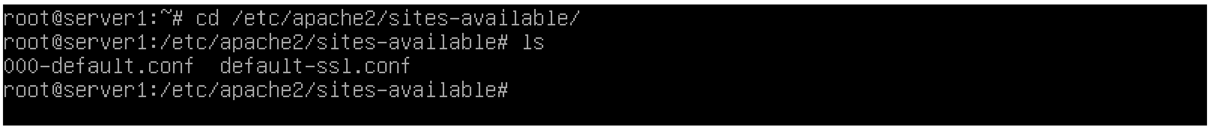
Buka Browser, kunjungi http://smkgeger.id dan terlihat tampilan default apache2



Kita coba ubah tampilan tersebut dengan file HTML Sederhana.

Pindah ke direktori konfigurasi Apache2, lalu coba kita lihat isinya

```
root@server1:~# cd /etc/apache2/sites-available/
root@server1:/etc/apache2/sites-available# ls
000-default.conf  default-ssl.conf
```



Salin file 000-default.conf misal menjadi smk.conf

```
root@server1:/etc/apache2/sites-available# cp 000-default.conf smk.conf
```

Lalu edit dengan teks editor nano dan ubah menjadi seperti berikut

```
root@server1:/etc/apache2/sites-available# nano smk.conf
<VirtualHost *:80>
    ServerName smkgeger.id
    ServerAdmin admin@smkgeger.id
    DocumentRoot /var/www/smk
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

Tampilan Konfigurasi Sebelum dirubah, abaikan tanda komentar (#)

```
  GNU nano 3.2                              smk.conf

<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/html

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet

^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Uncut Text ^T To Spell   ^_ Go To Line M-E Redo
```

Setelah dirubah

```
  GNU nano 3.2                              smk.conf                                  Modified

<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        ServerName smkgeger.id

        ServerAdmin admin@smkgeger.id
        DocumentRoot /var/www/smk

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet

^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Uncut Text ^T To Spell   ^_ Go To Line M-E Redo
```

Penjelasan Syntax Konfigurasi

<VirtualHost *:80>	Menunjukkan bahwa web server berjalan di semua IP dengan port 80
ServerName smkgeger.id	Menunjukkan web server ini berjalan dengan nama domain smkgeger.id
ServerAdmin admin@smkgeger.id	Menunjukkan alamat email yang harus dihubungi ketika web server mengalami gangguan
DocumentRoot /var/www/smk	Menunjukkan lokasi file web server
ErrorLog ${APACHE_LOG_DIR}/error.log	Menunjukkan lokasi file log error
CustomLog ${APACHE_LOG_DIR}/access.log combined	Menunjukkan lokasi file access log
</VirtualHost>	Penutup konfigurasi

Setelah itu save konfigurasi dan aktifkan konfigurasi smk.conf tersebut dengan perintah **a2ensite**

Dan disable default konfigurasi apache2 dengan perintah **a2dissite**

```
root@server1:/etc/apache2/sites-available# a2ensite smk.conf

root@server1:/etc/apache2/sites-available# a2dissite 000-default.conf
```

```
root@server1:/etc/apache2/sites-available# a2ensite smk.conf
Enabling site smk.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@server1:/etc/apache2/sites-available# a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@server1:/etc/apache2/sites-available#
```

Lalu buat direktori **/var/www/smk** dan file **index.html**

```
root@server1:/etc/apache2/sites-available# mkdir /var/www/smk

root@server1:/etc/apache2/sites-available# nano /var/www/smk/index.html
```

```
root@server1:/etc/apache2/sites-available# mkdir /var/www/smk
root@server1:/etc/apache2/sites-available# nano /var/www/smk/index.html_
```

Isi file index.html

```
  GNU nano 3.2                        /var/www/smk/index.html

<html>
<h1>Selamat Datang di Web SMKGEGER</h1>_
</html>
```

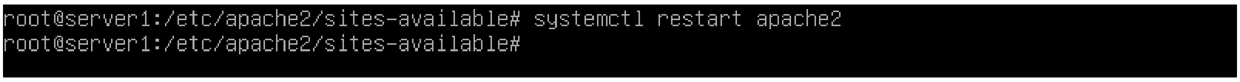
Simpan, lanjut ubah owner folder /var/www/smk menjadi user www-data dan group www-data

```
root@server1:/etc/apache2/sites-available# chown www-data:www-data /var/www/smk
```

```
root@server1:/etc/apache2/sites-available# chown www-data:www-data /var/www/smk
root@server1:/etc/apache2/sites-available# ls -l /var/www/
total 8
drwxr-xr-x 2 www-data www-data 4096 Sep 12 13:21 html
drwxr-xr-x 2 www-data www-data 4096 Sep 12 20:20 smk
root@server1:/etc/apache2/sites-available#
```

Restart service apache2

root@server1:/etc/apache2/sites-available# systemctl restart apache2



Coba kunjungi web lagi dari Windows Client

Konfigurasi HTTPS

Salin file contoh yaitu file default-ssl.conf menjadi smk-ssl.conf

root@server1:/etc/apache2/sites-available# cp defaults-ssl.conf smk-ssl.conf

Lalu edit dengan nano

root@server1:/etc/apache2/sites-available# nano smk-ssl.conf

<IfModule mod_ssl.c>

<VirtualHost _default_:443>

ServerAdmin admin@smkgeger.id

ServerName smkgeger.id

DocumentRoot /var/www/smk

ErrorLog ${APACHE_LOG_DIR}/error.log

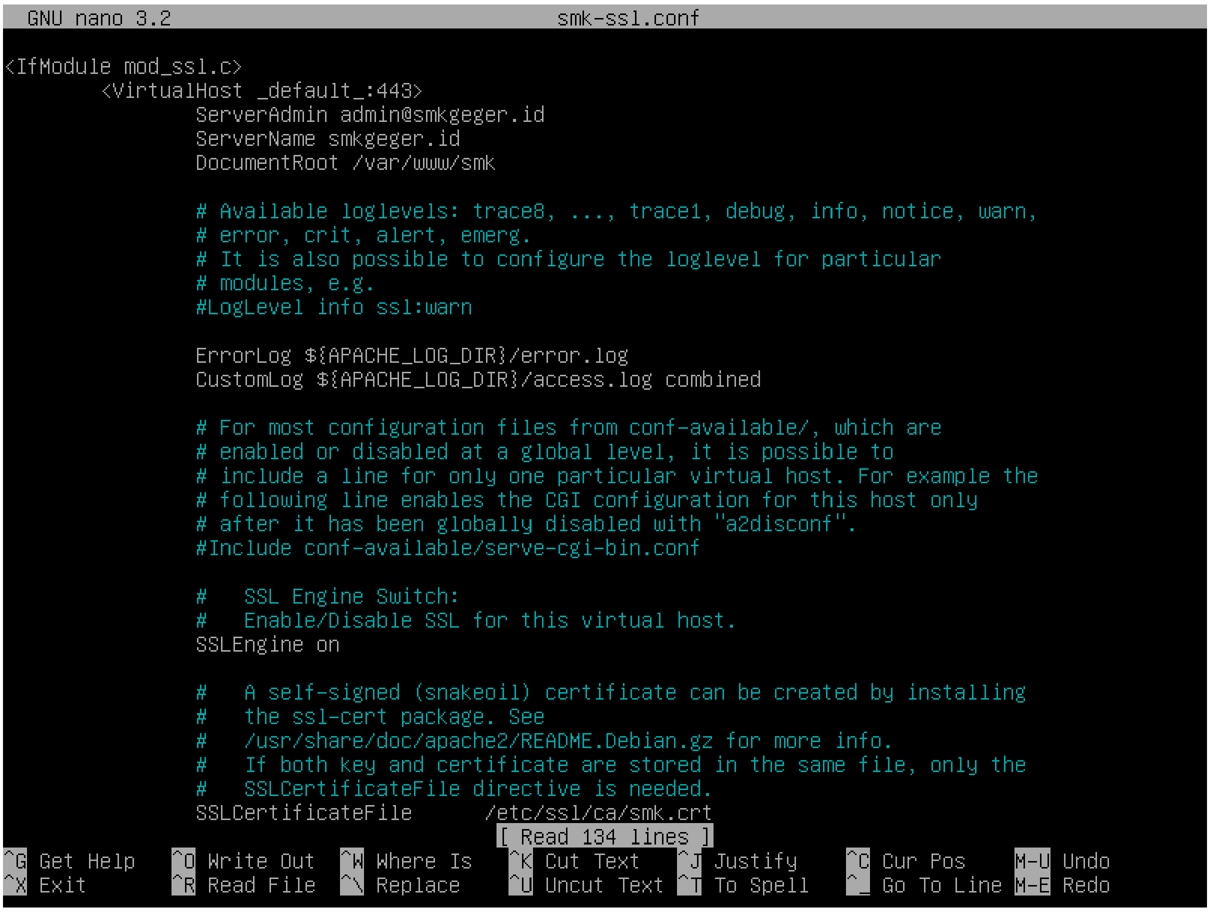
CustomLog ${APACHE_LOG_DIR}/access.log combined

SSLEngine on

SSLCertificateFile /etc/ssl/ca/smk.crt

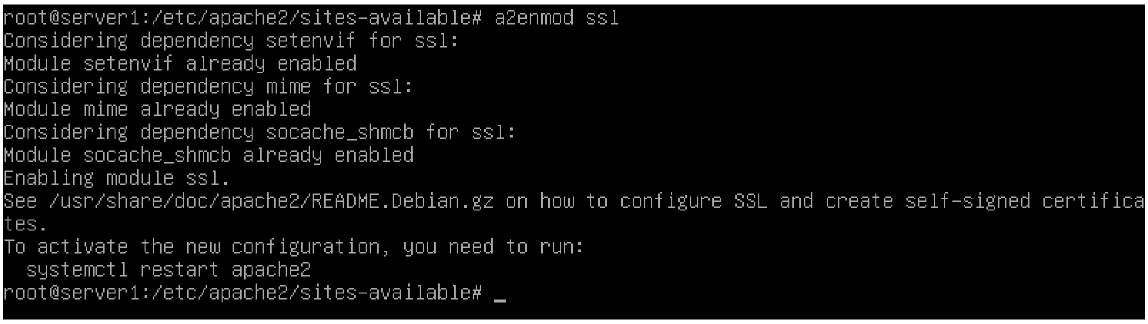
SSLCertificateKeyFile /etc/ssl/ca/smk.key

```
<FilesMatch "\.(cgi|shtml|phtml|php)$">
                        SSLOptions +StdEnvVars
            </FilesMatch>
            <Directory /usr/lib/cgi-bin>
                        SSLOptions +StdEnvVars
            </Directory>
  </VirtualHost>
</IfModule>
```



Kemudian save, dan enable ssl di apache2

```
root@server1:/etc/apache2/sites-available# a2enmod ssl
```



Lalu Enable smk-ssl dan restart Apache2

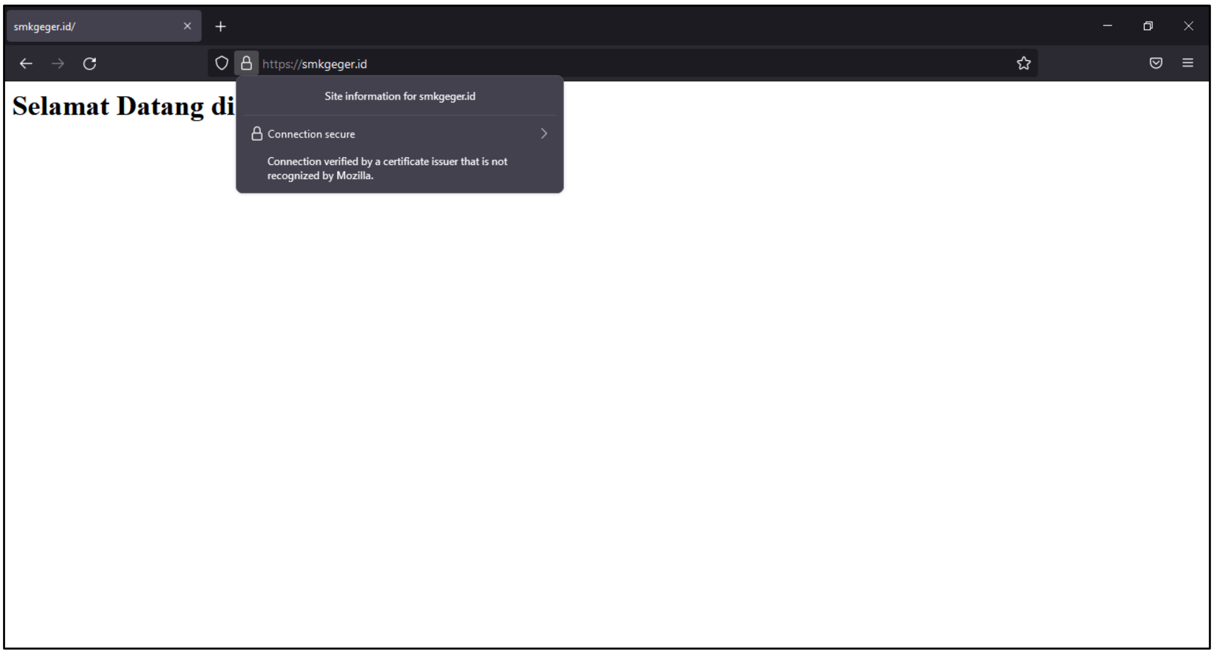
```
root@server1:/etc/apache2/sites-available# a2eniste smk-ssl.conf
```

```
root@server1:/etc/apache2/sites-available# a2ensite smk-ssl.conf
Enabling site smk-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@server1:/etc/apache2/sites-available#
```

```
root@server1:/etc/apache2/sites-available# systemctl restart apache2
```

```
root@server1:/etc/apache2/sites-available# systemctl restart apache2
```

Test di Browser Client



Terlihat kita berhasil mengakses smkgeger.id dengan https , Jika ada warning maka ada kesalahan saat membuat CA

Instalasi MariaDB

Disini kita akan melakukan Instalasi database server dengan MariaDB.

```
root@server1:~# apt  install mariadb-server mariadb-client
```

```
root@server1:~# apt install mariadb-server mariadb-client
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  galera-3 gawk libaio1 libcgi-fast-perl libcgi-pm-perl libconfig-inifiles-perl libdbd-mysql-perl
  libdbi-perl libencode-locale-perl libfcgi-perl libhtml-parser-perl libhtml-tagset-perl
  libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl
  liblwp-mediatypes-perl libmariadb3 libmpfr6 libreadline5 libsigsegv2 libsnappy1v5
  libterm-readkey-perl libtimedate-perl liburi-perl mariadb-client-10.3 mariadb-client-core-10.3
  mariadb-common mariadb-server-10.3 mariadb-server-core-10.3 mysql-common psmisc rsync socat
Suggested packages:
  gawk-doc libclone-perl libmldbm-perl libnet-daemon-perl libsql-statement-perl libdata-dump-perl
  libipc-sharedcache-perl libwww-perl mailx mariadb-test netcat-openbsd tinyca
The following NEW packages will be installed:
  galera-3 gawk libaio1 libcgi-fast-perl libcgi-pm-perl libconfig-inifiles-perl libdbd-mysql-perl
  libdbi-perl libencode-locale-perl libfcgi-perl libhtml-parser-perl libhtml-tagset-perl
  libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl
  liblwp-mediatypes-perl libmariadb3 libmpfr6 libreadline5 libsigsegv2 libsnappy1v5
  libterm-readkey-perl libtimedate-perl liburi-perl mariadb-client mariadb-client-10.3
  mariadb-client-core-10.3 mariadb-common mariadb-server mariadb-server-10.3
  mariadb-server-core-10.3 mysql-common psmisc rsync socat
0 upgraded, 36 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/22.5 MB of archives.
After this operation, 171 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Konfigurasi MariaDB

Secara default user root di mysql tidak memiliki password, agar lebih aman setelah Instalasi mariadb kita jalankan perintah mysql_secure_installation

```
root@server1:~# mysql_secure_installation
```

```
root@server1:~# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user.  If you've just installed MariaDB, and
you haven't set the root password yet, the password will be blank,
so you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password ensures that nobody can log into the MariaDB
root user without the proper authorisation.

Set root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
 ... Success!


By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
 ... Success!

Normally, root should only be allowed to connect from 'localhost'.  This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] _
```

Enter current password for root: klik Enter di keyboard

Set root password? [Y/n] pilih **y**

New password: Masukkan password root baru disini (tidak terlihat)

Re-enter password: Masukkan password lagi

Remove anonymous users? [Y/n] pilih **y**

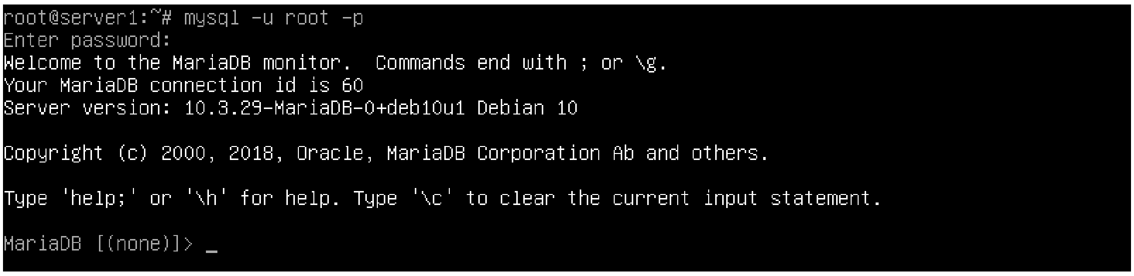
Disallow root login remotely? [Y/n] pilih **y**

Remove test database and access to it? [Y/n] pilih **y**

Reload privielge tables now? [Y/n] pilih **y**

Lalu coba login

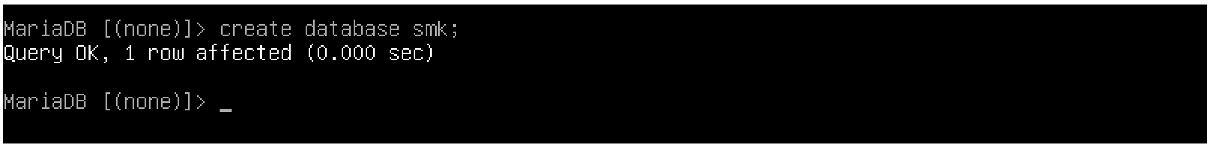
root@server1:~# mysql -u root -p



```
root@server1:~# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 60
Server version: 10.3.29-MariaDB-0+deb10u1 Debian 10

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> _
```

Membuat database baru

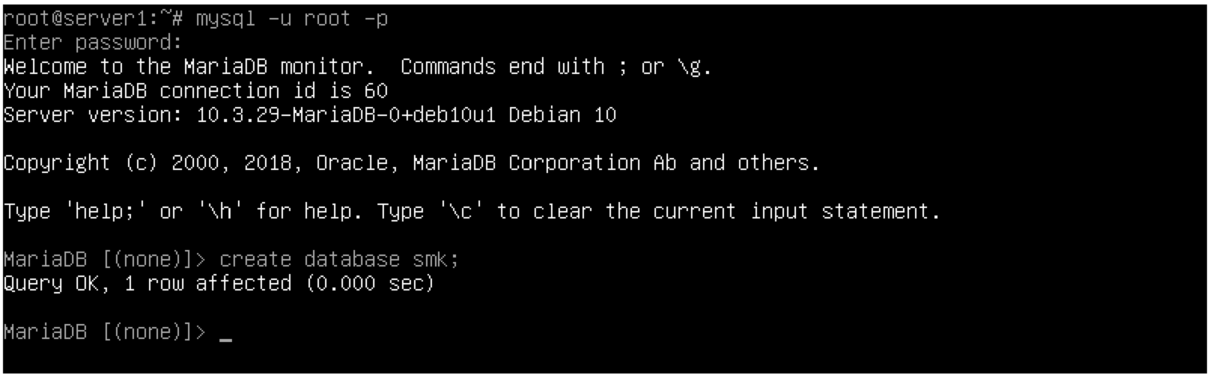
MariaDB [(none)]> create database smk;



```
MariaDB [(none)]> create database smk;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> _
```

Melihat database

MariaDB [(none)]> show databases;



```
root@server1:~# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 60
Server version: 10.3.29-MariaDB-0+deb10u1 Debian 10

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database smk;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> _
```

Membuat user baru

Contoh membuat user arya dengan password arya

MariaDB [(none)]> create user ‘arya’@localhost identified by ‘arya’;

```
MariaDB [(none)]> create user 'arya'@localhost identified by 'arya';
Query OK, 0 rows affected (0.000 sec)
```

Melihat semua user

```
MariaDB [(none)]> select User from mysql.user;
```

```
MariaDB [(none)]> select User from mysql.user;
+------+
| User |
+------+
| arya |
| root |
+------+
2 rows in set (0.000 sec)
```

Memberi akses hak user ke database

```
MariaDB [(none)]> grant all privileges on smk.* to ‘arya’@localhost
```

Contoh memberi akses user arya ke database smk

```
MariaDB [(none)]> grant all privileges on smk.* to 'arya'@localhost;
Query OK, 0 rows affected (0.000 sec)
```

Merefresh hak akses user

```
MariaDB [(none)]> flush privileges;
```

```
MariaDB [(none)]> flush privileges;
Query OK, 0 rows affected (0.000 sec)
```

Melihat hak akses user

```
MariaDB [(none)]> show grants for `arya`@localhost;
```

```
MariaDB [(none)]> show grants for 'arya'@localhost;
+-------------------------------------------------------------------------------------------
----------+
| Grants for arya@localhost
          |
+-------------------------------------------------------------------------------------------
----------+
| GRANT USAGE ON *.* TO `arya`@`localhost` IDENTIFIED BY PASSWORD '*4BE3E290EAF57E521F5D3CF6E5D9D67C
9E363470' |
| GRANT ALL PRIVILEGES ON `smk`.* TO `arya`@`localhost`
          |
+-------------------------------------------------------------------------------------------
----------+
2 rows in set (0.000 sec)
```

Menghapus user

Contoh menghapus user arya

```
MariaDB [(none)]> drop user ‘arya’@localhost
```

Install PHP

PHP(Hypertext Preprocessor)

```
root@server1:~# apt install php libapache2-mod-php php-mysqli
```

```
root@server1:~# apt install php libapache2-mod-php php-mysqli
Reading package lists... Done
Building dependency tree
Reading state information... Done
Note, selecting 'php7.3-mysql' instead of 'php-mysqli'
The following additional packages will be installed:
  libapache2-mod-php7.3 libsodium23 php-common php7.3 php7.3-cli php7.3-common php7.3-json
  php7.3-opcache php7.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php7.3 libsodium23 php php-common php7.3 php7.3-cli
  php7.3-common php7.3-json php7.3-mysql php7.3-opcache php7.3-readline
0 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/4,301 kB of archives.
After this operation, 18.4 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

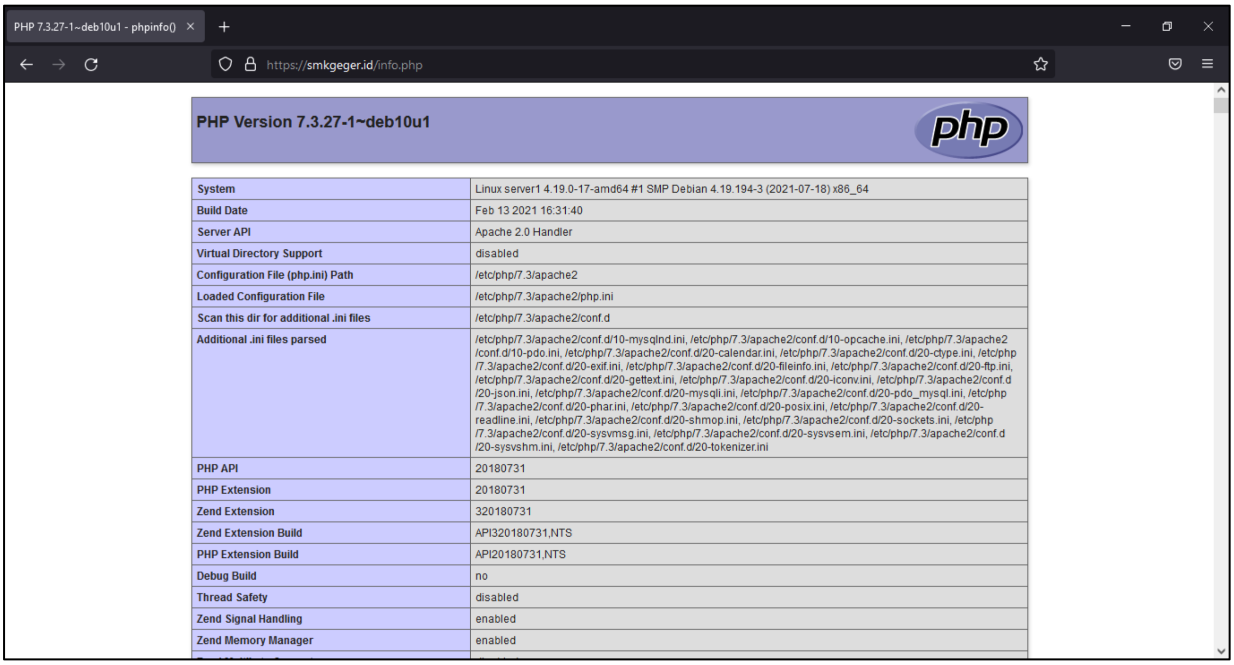
Lalu kita test dengan membuat file php info di **/var/www/smk**

```
root@server1:~# nano /var/www/smk/info.php

<?php phpinfo(); ?>
```

Save dan restart service apache2

```
root@server1:~# systemctl restart apache2
```

Coba kita kunjungi https://smkgeger.id/info.php



Terlihat file php kita berhasil di eksekusi, berarti konfigurasi PHP kita berhasil.

2.9.2 Instalasi dan Konfigurasi LEMP Stack di Server2

Perbedaan LAMP dengan LEMP yaitu LAMP Menggunakan Apache sedangkan LEMP menggunakan Nginx

Install Nginx, PHP, dan MariaDB

```
root@server2:~# apt install nginx php-fpm mariadb-server mariadb-client
```

Konfigurasi Nginx

Pindah ke direktori konfigurasi nginx di /etc/nginx/sites-available dan edit default konfigurasi dengan nano

```
root@server2:~# cd /etc/nginx/sites-available/

root@server2:/etc/nginx/sites-available# nano default
```

Kemudian sesuaikan konfigurasi seperti ini

```
server {
    listen 443 ssl;
    ssl_certificate /etc/ssl/ca/smk1.crt;
    ssl_certificate_key /etc/ssl/ca/smk1.key;
    root /var/www/smk;
    index index.php index.html index.htm index.nginx-debian.html;
    server_name web2.smkgeger.id;
    location / {
        try_files $uri $uri/ =404;
    }
    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/run/php/php7.3-fpm.sock;
    }
}
```

```
  GNU nano 3.2                    /etc/nginx/sites-available/default

server {
        listen 443 ssl;
        ssl_certificate /etc/ssl/ca/smk1.crt;
        ssl_certificate_key /etc/ssl/ca/smk1.key;
        root /var/www/smk;
        index index.php index.html index.htm index.nginx-debian.html;

        server_name web2.smkgeger.id;

        location / {
                try_files $uri $uri/ =404;
        }
        location ~ \.php$ {
                include snippets/fastcgi-php.conf;
                fastcgi_pass unix:/run/php/php7.3-fpm.sock;
        }
}
```

Transfer file sertifikat dari server1 ke server2 dengan menggunakan scp, kemudian buat konfigurasi php sederhana

```
root@server2:~# mkdir -p /var/www/smk

root@server2:~# nano /var/www/smk/index.php

<?php

echo "Welcome to Website\n";

echo gethostname();

?>
```

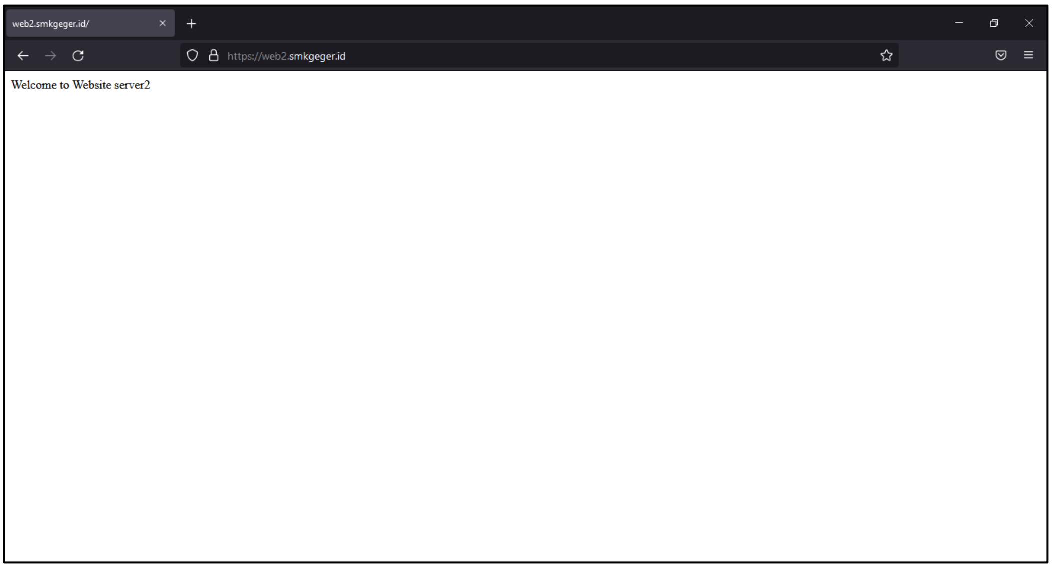
Save, selanjutnya ubah owner dan group folder /var/www/smk menjadi www-data

```
root@server2:~# chown -R www-data:www-data /var/www/smk
```

Restart service nginx

```
root@server2:~# systemctl restart nginx
```

Test dari Klien



2.10 Konfigurasi Mail Server dan WebMail

Apa itu Mail Server? Mail server adalah server yang digunakan untuk mengirim dan menerima email dari internet.

Komponen Mail Server

- MUA (Mail User Agent) komponen yang digunakan untuk mengirim dan menerima email contohnya Gmail, Yahoo, Outlook, Thunderbird dll.
- MTA (Mail Transfer Agent) komponen yang digunakan untuk mengirim dan menerima email dari satu komputer ke komputer lain maupun dari komputer ke server, memberi respons otomatis ketika email gagal dikirim. contohnya yaitu Postfix, Qmail, Sendmail, dll.
- MDA (Mail Delivery Agent) komponen yang digunakan untuk mengantarkan email dari MTA

Protokol yang digunakan di Mail Server

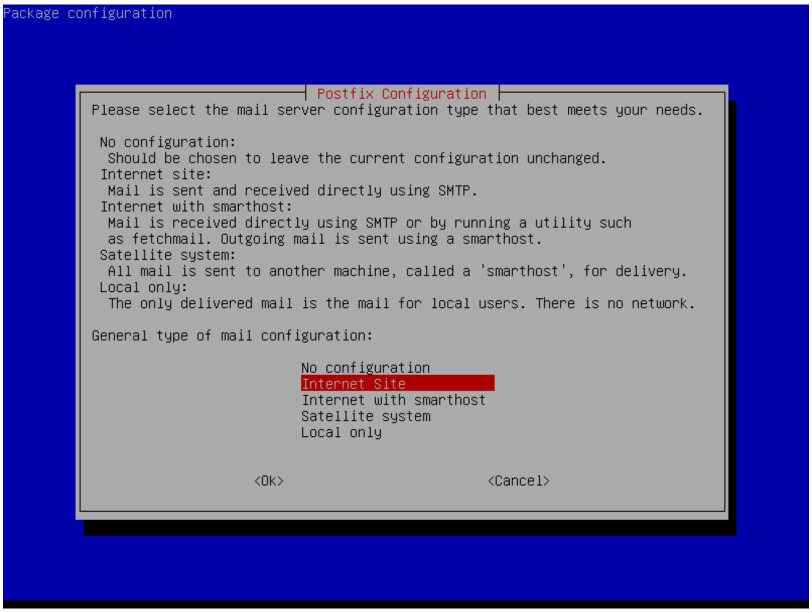
- SMTP (Simple Mail Transfer Protocol) protokol yang berfungsi untuk mengirim email dari lokal ke email server tujuan, proses komunikasi ini dilakukan di MTA
- POP3 (Post Office Protocol) protokol yang berfungsi untuk menerima email dari email server lain ke lokal.
- IMAP (Internet Message Access) protokol yang berfungsi untuk melakukan synchronize antara mail server dan client agar client bisa membaca email dari manapun asal terkoneksi ke jaringan.

Instalasi Postfix dan Dovecot(POP3/IMAP)

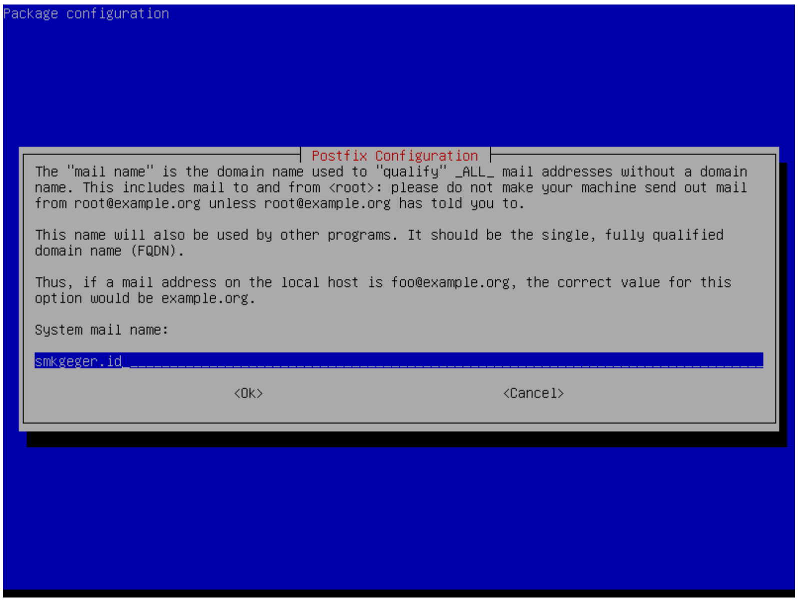
Install Postfix

```
root@server1:~# apt install postfix dovecot-pop3d dovecot-imapd
```

Nanti akan muncul menu dialog, pilih **Internet Site**



Lalu masukkan domain mail server **smkgeger.id**



Konfigurasi Postfix

```
root@server1:~# nano /etc/postfix/main.cf

mydomain = smkgeger.id

myorigin = $mydomain

home_mailbox = Maildir/
```

Tambahkan konfigurasi tersebut

Berikut perbandingan setelah di konfigurasi dan belum di konfgurasi

Sebelum

```
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = server1.smkgeger.id
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
myorigin = /etc/mailname
mydestination = $myhostname, smkgeger.id, server1.smkgeger.id, localhost.smkgeger.id, localhost
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all
```

Sesudah

```
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = server1.smkgeger.id
mydomain = smkgeger.id
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
myorigin = $mydomain
mydestination = $myhostname, smkgeger.id, server1.smkgeger.id, localhost.smkgeger.id, localhost
relayhost = _
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all
home_mailbox = Maildir/
```

Kemudian save dan restart service postfix

```
root@server1:~# systemctl restart postfix
```

Konfigurasi Dovecot

```
root@server1:~# nano /etc/dovecot/dovecot.conf
```

```
root@server1:~# nano /etc/dovecot/dovecot.conf
```

kita hilangkan komentar (#) pada listen, agar dovecot aktif di semua IP di Server

```
# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
#listen = *, ::
```

```
# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
listen = *, ::
```

Kemudian save, lanjut konfigurasi di

```
root@server1:~# nano /etc/dovecot/conf.d/10-auth.conf
```

```
root@server1:~# nano /etc/dovecot/conf.d/10-auth.conf _
```

Hilangkan komentar (#) pada disable plain text dan ubah menjadi no

```
# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGINDISABLED capability). Note that if the remote IP
# matches the local IP (ie. you're connecting from the same computer), the
# connection is considered secure and plaintext authentication is allowed.
# See also ssl=required setting.
#disable_plaintext_auth = yes
```

```
# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGINDISABLED capability). Note that if the remote IP
# matches the local IP (ie. you're connecting from the same computer), the
# connection is considered secure and plaintext authentication is allowed.
# See also ssl=required setting.
disable_plaintext_auth = no_
```

Kemudian geser ke bawah, cari **auth_mechanism = plain** ubah menjadi **auth_mechanism = plain login**

```
# Space separated list of wanted authentication mechanisms:
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp skey
#   gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = plain_
```

```
# Space separated list of wanted authentication mechanisms:
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp skey
#   gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = plain login
```

Setelah itu save, kemudian konfigurasi lagi di

root@server1:~# nano /etc/dovecot/conf.d/10-mail.conf

```
root@server1:~# nano /etc/dovecot/conf.d/10-mail.conf _
```

Cari konfigurasi mail_location, beri komentar (#) pada

mail_location = mbox:~/mail:INBOX=/var/mail/%u

dan hilangkan tanda komentar(#) pada **mail_location = maildir:~/Maildir**

```
# See doc/wiki/Variables.txt for full list. Some examples:
#
#   mail_location = maildir:~/Maildir
#   mail_location = mbox:~/mail:INBOX=/var/mail/%u
#   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
#
# <doc/wiki/MailLocation.txt>
#
mail_location = mbox:~/mail:INBOX=/var/mail/%u
```

```
# See doc/wiki/Variables.txt for full list. Some examples:
#
_   mail_location = maildir:~/Maildir
#   mail_location = mbox:~/mail:INBOX=/var/mail/%u
#   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
#
# <doc/wiki/MailLocation.txt>
#
#mail_location = mbox:~/mail:INBOX=/var/mail/%u
```

Save kemudian restart service dovecot

root@server1:~# systemctl restart postfix

```
root@server1:~# systemctl restart dovecot_
```

Selanjutnya buat maildir untuk setiap user yang dibuat

root@server1:~# maildirmake.dovecot /etc/skel/Maildir

```
root@server1:~# maildirmake.dovecot /etc/skel/Maildir_
```

Lalu kita test dengan mengirim email di localhost

Pertama kita tambahkan user baru, misal arya dan adit

root@server1:~# adduser arya

```
root@server1:~# adduser arya
Adding user `arya' ...
Adding new group `arya' (1001) ...
Adding new user `arya' (1001) with group `arya' ...
Creating home directory `/home/arya' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for arya
Enter the new value, or press ENTER for the default
        Full Name []: Arya Pramudika
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
root@server1:~# _
```

```
root@server1:~# adduser adit
```

```
root@server1:~# adduser adit
Adding user `adit' ...
Adding new group `adit' (1002) ...
Adding new user `adit' (1002) with group `adit' ...
Creating home directory `/home/adit' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for adit
Enter the new value, or press ENTER for the default
        Full Name []: Aditya Putra
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
root@server1:~# _
```

Lalu coba kirim email dengan telnet ke port smtp yaitu port 25

```
root@server1:~# telnet localhost 25

mail from: arya

rcpt to: adit

data

Test Pesan

.

quit
```

```
root@server1:~# telnet localhost 25
Trying ::1...
Connected to localhost.
Escape character is '^]'.
220 server1.smkgeger.id ESMTP Postfix (Debian/GNU)
mail from: arya
250 2.1.0 Ok
rcpt to: adit
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
Test Pesan
.
250 2.0.0 Ok: queued as EC7902B51
quit
221 2.0.0 Bye
Connection closed by foreign host.
root@server1:~# _
```

Setelah itu telnet lagi ke localhost dengan port 110

```
root@server1:~# telnet localhost 110

user adit

pass adit

retr 1
```

```
root@server1:~# telnet localhost 110
Trying ::1...
Connected to localhost.
Escape character is '^]'.
+OK Dovecot (Debian) ready.
user adit
+OK
pass adit
+OK Logged in.
retr 1
+OK 388 octets
Return-Path: <arya@smkgeger.id>
X-Original-To: adit
Delivered-To: adit@smkgeger.id
Received: from localhost (localhost [IPv6:::1])
        by server1.smkgeger.id (Postfix) with SMTP id EC7902B51
        for <adit>; Tue, 14 Sep 2021 09:06:48 +0700 (WIB)
Message-Id: <20210914020653.EC7902B51@server1.smkgeger.id>
Date: Tue, 14 Sep 2021 09:06:48 +0700 (WIB)
From: arya@smkgeger.id

Test Pesan
.
```

Terlihat pesan yang kita kirim tadi berhasil diterima oleh adit

2.10.1 Konfigurasi Webmail

Untuk webmail kita gunakan roundcube, dengan alamat di mail.smkgeger.id

Sebelum Instalasi roundcube tambahkan dulu subdomain mail.smkgeger.id mengarah ke IP Server1 dan jangan lupa untuk menambah serial numbernya

```
root@server1:~# nano /etc/bind/db.smk

mail    IN      A       192.168.1.103
```

```
$TTL    604800
@       IN      SOA     smkgeger.id. root.smkgeger.id. (
                        202109163       ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.smkgeger.id.
@       IN      NS      ns2.smkgeger.id.
@       IN      MX      10      smkgeger.id.
@       IN      A       192.168.101.3
ns1     IN      A       192.168.101.3
ns2     IN      A       192.168.101.4
web2    IN      A       192.168.101.4
mail    IN      A       192.168.101.3
```

Instalasi Roundcube

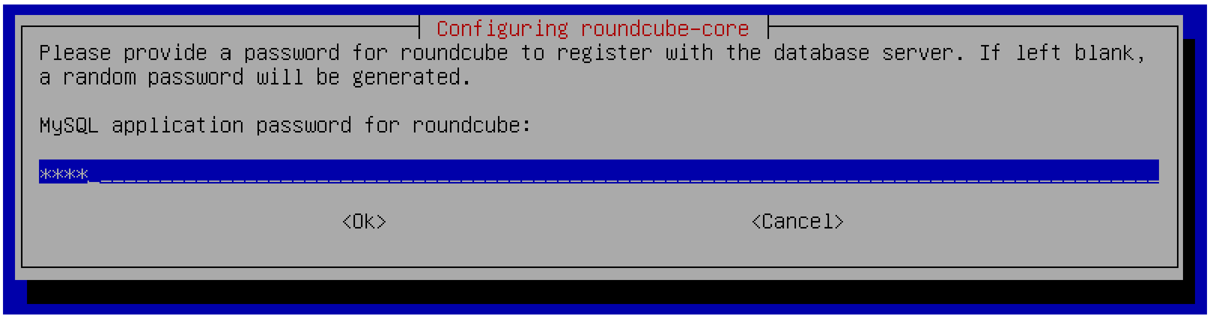
root@server1:~# apt install roundcube

```
root@server1:~# apt install roundcube
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  aspell aspell-en dbconfig-common fontconfig-config fonts-dejavu-core libaspell15 libfontconfig1
  libgd3 libjbig0 libjpeg62-turbo libtiff5 libwebp6 libxpm4 libxslt1.1 php-auth-sasl php-gd
  php-intl php-mail-mime php-mbstring php-net-sieve php-net-smtp php-net-socket php-pear
  php-pspell php-xml php7.3-gd php7.3-intl php7.3-mbstring php7.3-pspell php7.3-xml roundcube-core
  roundcube-mysql
Suggested packages:
  aspell-doc spellutils dbconfig-mysql | dbconfig-pgsql | dbconfig-sqlite | dbconfig-sqlite3
  | dbconfig-no-thanks libgd-tools roundcube-plugins php-net-ldap2 php-crypt-gpg php-net-ldap3
The following NEW packages will be installed:
  aspell aspell-en dbconfig-common fontconfig-config fonts-dejavu-core libaspell15 libfontconfig1
  libgd3 libjbig0 libjpeg62-turbo libtiff5 libwebp6 libxpm4 libxslt1.1 php-auth-sasl php-gd
  php-intl php-mail-mime php-mbstring php-net-sieve php-net-smtp php-net-socket php-pear
  php-pspell php-xml php7.3-gd php7.3-intl php7.3-mbstring php7.3-pspell php7.3-xml roundcube
  roundcube-core roundcube-mysql
0 upgraded, 33 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/7,557 kB of archives.
After this operation, 28.6 MB of additional disk space will be used.
Do you want to continue? [Y/n] y_
```

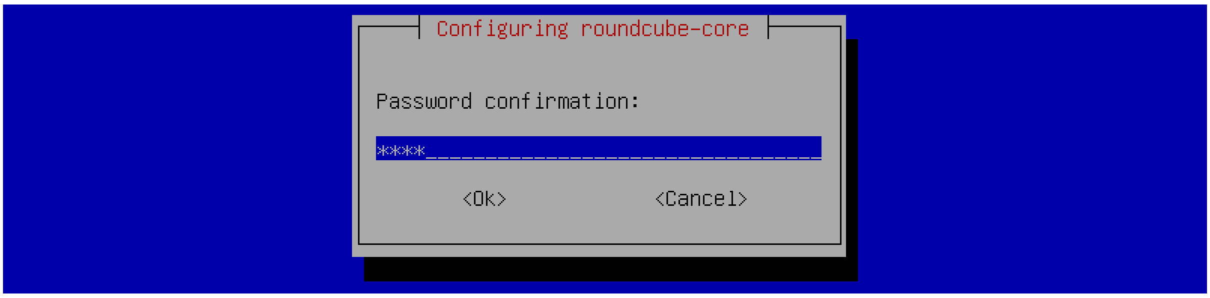
ketika meginstall roundcube akan muncul dialog**, Configure database for roundcube with dbconfig-common?** Pilih **Yes**

```
                    Configuring roundcube-core

 The roundcube package must have a database installed and configured before it can be used.
 This can be optionally handled with dbconfig-common.

 If you are an advanced database administrator and know that you want to perform this
 configuration manually, or if your database has already been installed and configured, you
 should refuse this option. Details on what needs to be done should most likely be provided
 in /usr/share/doc/roundcube.

 Otherwise, you should probably choose this option.

 Configure database for roundcube with dbconfig-common?

                  <Yes>                                    <No>
```

Masukkan password untuk database roundcube



Masukkan password sekali lagi

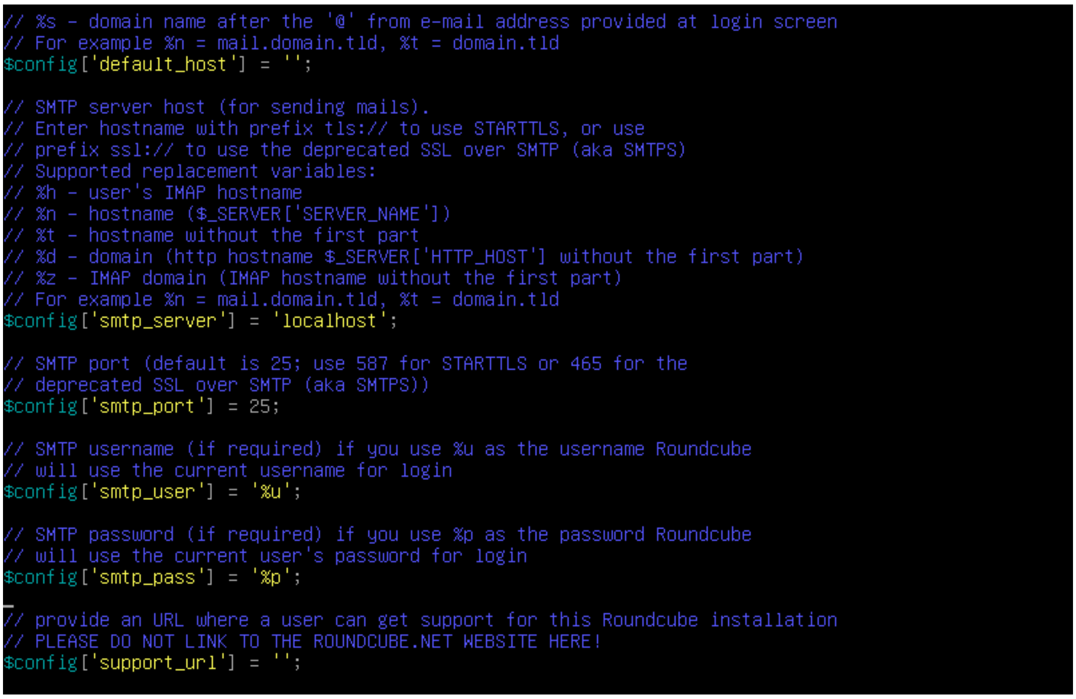


Setelah roundcube berhasil di install selanjutnya ke tahap konfigurasi

Konfigurasi Roundcube

root@server1:~# nano /var/lib/roundcube/config/config.inc.php

Geser ke bawah, ubah default host menjadi smkgeger.id smtp_server ke smkgeger.id, smtp_user dan smtp_pass kosongkan



```
// %s - domain name after the '@' from e-mail address provided at login screen
// For example %n = mail.domain.tld, %t = domain.tld
$config['default_host'] = 'smkgeger.id';

// SMTP server host (for sending mails).
// Enter hostname with prefix tls:// to use STARTTLS, or use
// prefix ssl:// to use the deprecated SSL over SMTP (aka SMTPS)
// Supported replacement variables:
// %h - user's IMAP hostname
// %n - hostname ($_SERVER['SERVER_NAME'])
// %t - hostname without the first part
// %d - domain (http hostname $_SERVER['HTTP_HOST'] without the first part)
// %z - IMAP domain (IMAP hostname without the first part)
// For example %n = mail.domain.tld, %t = domain.tld
$config['smtp_server'] = 'smkgeger.id';

// SMTP port (default is 25; use 587 for STARTTLS or 465 for the
// deprecated SSL over SMTP (aka SMTPS))
$config['smtp_port'] = 25;

// SMTP username (if required) if you use %u as the username Roundcube
// will use the current username for login
$config['smtp_user'] = '';

// SMTP password (if required) if you use %p as the password Roundcube
// will use the current user's password for login
$config['smtp_pass'] = '';

// provide an URL where a user can get support for this Roundcube installation
// PLEASE DO NOT LINK TO THE ROUNDCUBE.NET WEBSITE HERE!
$config['support_url'] = '';
```

Kemudian buat konfigurasi apache untuk webmail

```
root@server1:~# cd /etc/apache2/sites-available

root@server1:/etc/apache2/sites-available# cp smk-ssl.conf mail.conf
```

```
root@server1:~# cd /etc/apache2/sites-available/
root@server1:/etc/apache2/sites-available# cp smk-ssl.conf mail.conf
```

Lalu edit dengan nano

```
root@server1:/etc/apache2/sites-available# nano mail.conf

<IfModule mod_ssl.c>

    <VirtualHost _default_:443>

        ServerAdmin admin@smkgeger.id

        ServerName mail.smkgeger.id

        DocumentRoot /var/lib/roundcube

        ErrorLog ${APACHE_LOG_DIR}/error.log

        CustomLog ${APACHE_LOG_DIR}/access.log combined

        SSLEngine on

        SSLCertificateFile      /etc/ssl/ca/smk1.crt

        SSLCertificateKeyFile /etc/ssl/ca/smk1.key

        <FilesMatch "\.(cgi|shtml|phtml|php)$">

                SSLOptions +StdEnvVars
```

```
		</FilesMatch>
		<Directory /usr/lib/cgi-bin>
				SSLOptions +StdEnvVars
		</Directory>
	</VirtualHost>
</IfModule>
```

ServerName ubah menjadi mail.smkgeger.id kemudain SSLCertificate ubah ke /etc/ssl/ca/smk1.crt dan SSLCertificatekey ke /etc/ssl/ca/smk1.key

```
  GNU nano 3.2                          mail.conf                          Modified

<IfModule mod_ssl.c>
        <VirtualHost _default_:443>
                ServerAdmin admin@smkgeger.id
                ServerName mail.smkgeger.id
                DocumentRoot /var/lib/roundcube

                ErrorLog ${APACHE_LOG_DIR}/error.log
                CustomLog ${APACHE_LOG_DIR}/access.log combined

                SSLEngine on
                SSLCertificateFile      /etc/ssl/ca/smk1.crt
                SSLCertificateKeyFile /etc/ssl/ca/smk1.key

                <FilesMatch "\.(cgi|shtml|phtml|php)$">
                                SSLOptions +StdEnvVars
                </FilesMatch>
                <Directory /usr/lib/cgi-bin>
                                SSLOptions +StdEnvVars
                </Directory>

        </VirtualHost>
</IfModule>
```

Save dan enable site mail.conf

```
root@server1:/etc/apache2/sites-available# a2ensite mail.conf
```

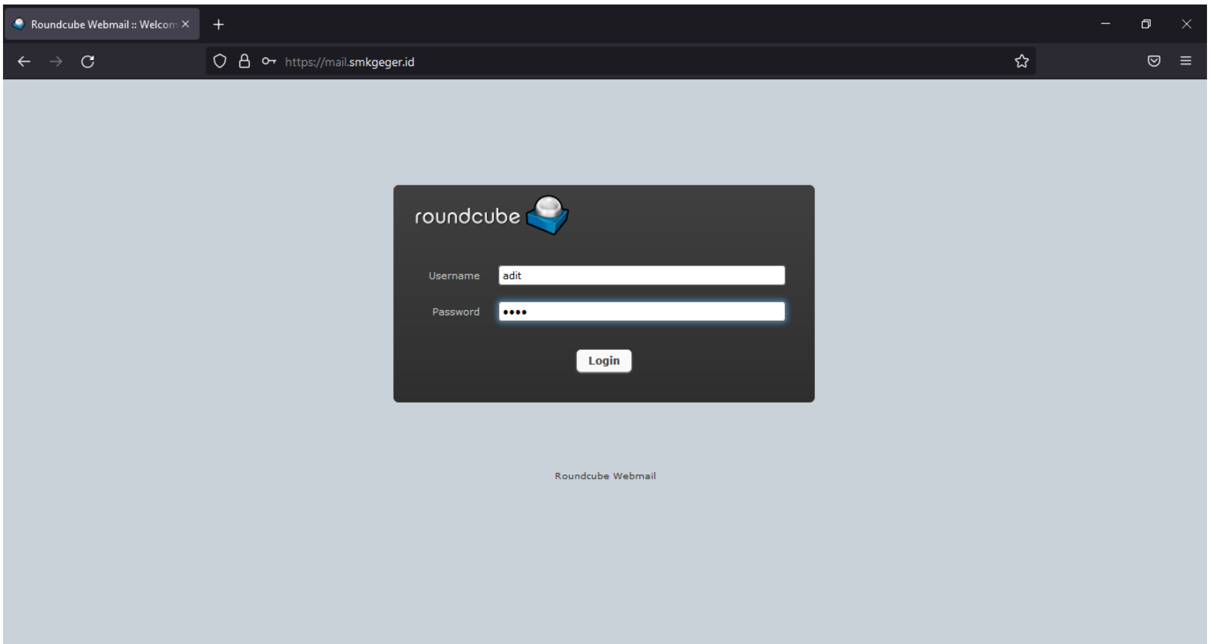
```
root@server1:/etc/apache2/sites-available# a2ensite mail.conf
Enabling site mail.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@server1:/etc/apache2/sites-available# _
```

Restart apache2

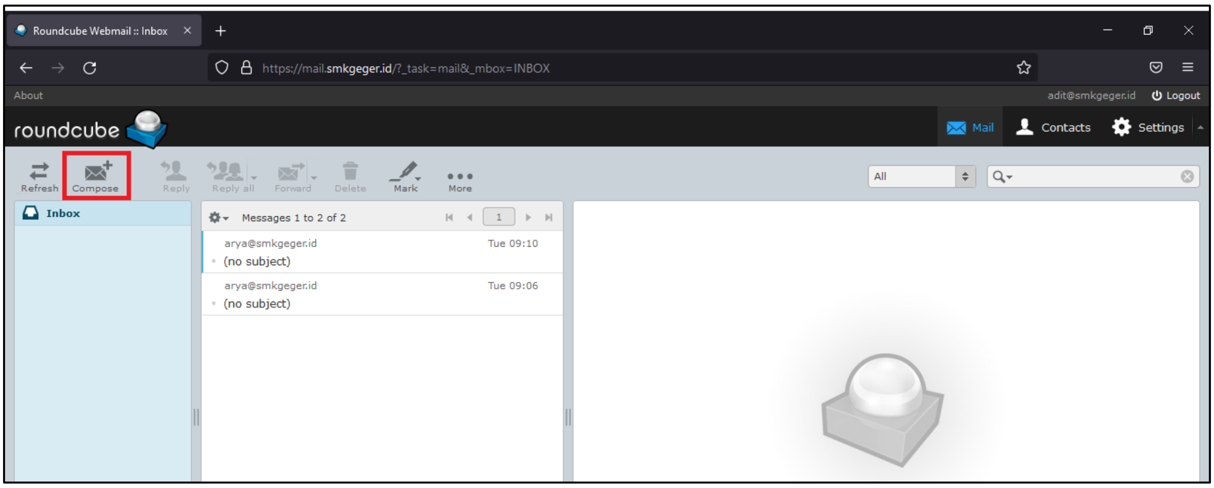
```
root@server1:/etc/apache2/sites-available# systemctl restart apache2
```

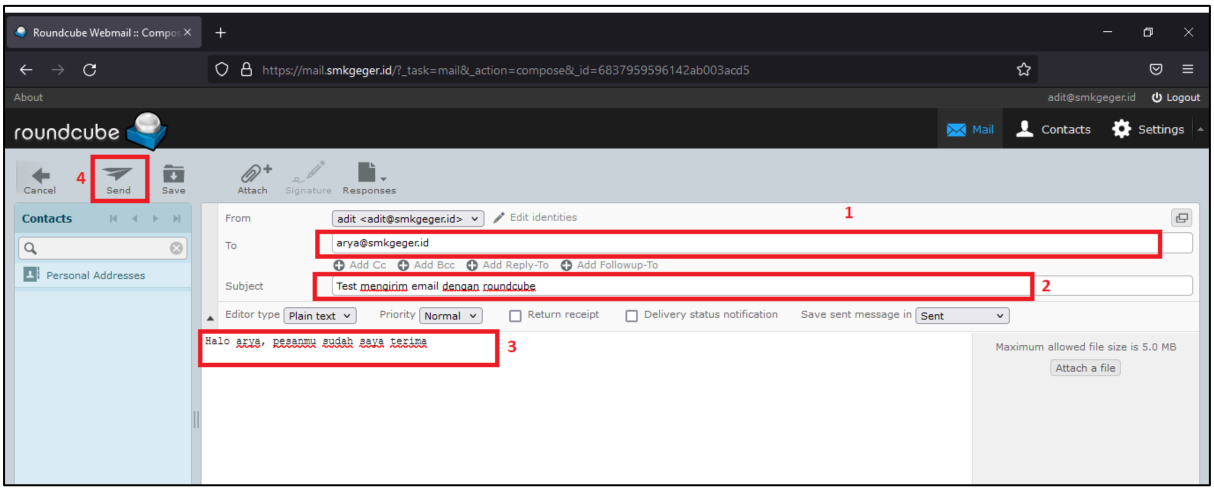
Lalu buka browser di client dan kunjungi https://mail.smkgeger.id

Login misal dengan user adit

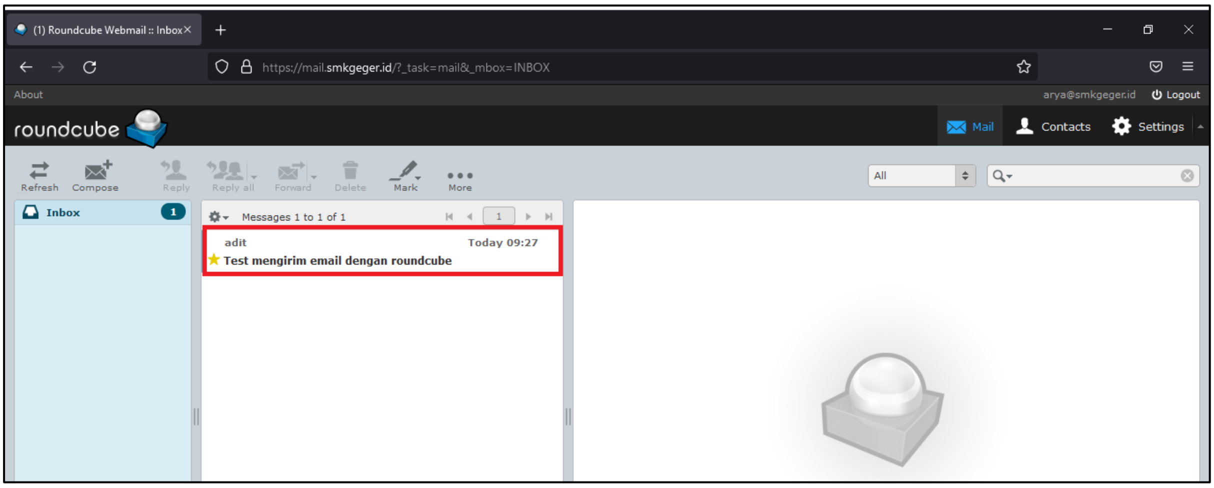


Lalu kirim pesan ke arya@smkgeger.id

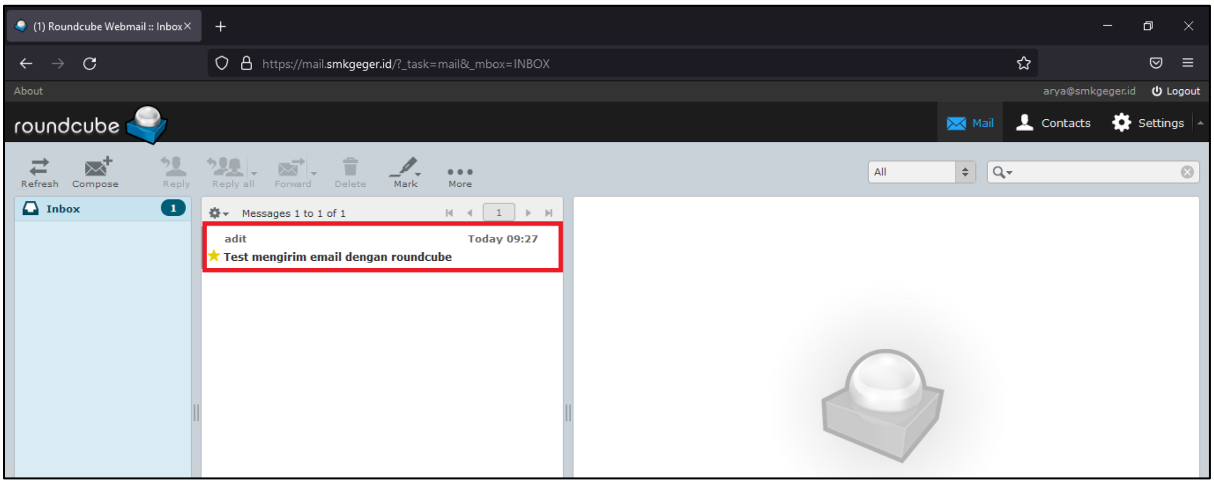




Pesan berhasil dikirim



Logout dari user adit dan login dengan user arya



Terlihat pesan dari adit berhasil di terima oleh arya

2.11 Konfigurasi File Sharing

Ada berbagai macam file sharing yang bisa digunakan di Debian, kali ini kita hanya akan berfokus pada Samba dan NFS.

Apa itu samba? Secara singkat samba merupakan software file sharing yang didasarkan pada protokol SMB yang biasa digunakan di sistem Operasi Windows. Samba adalah file sharing yang tepat jika client kita terdiri dari berbagai macam sistem operasi.

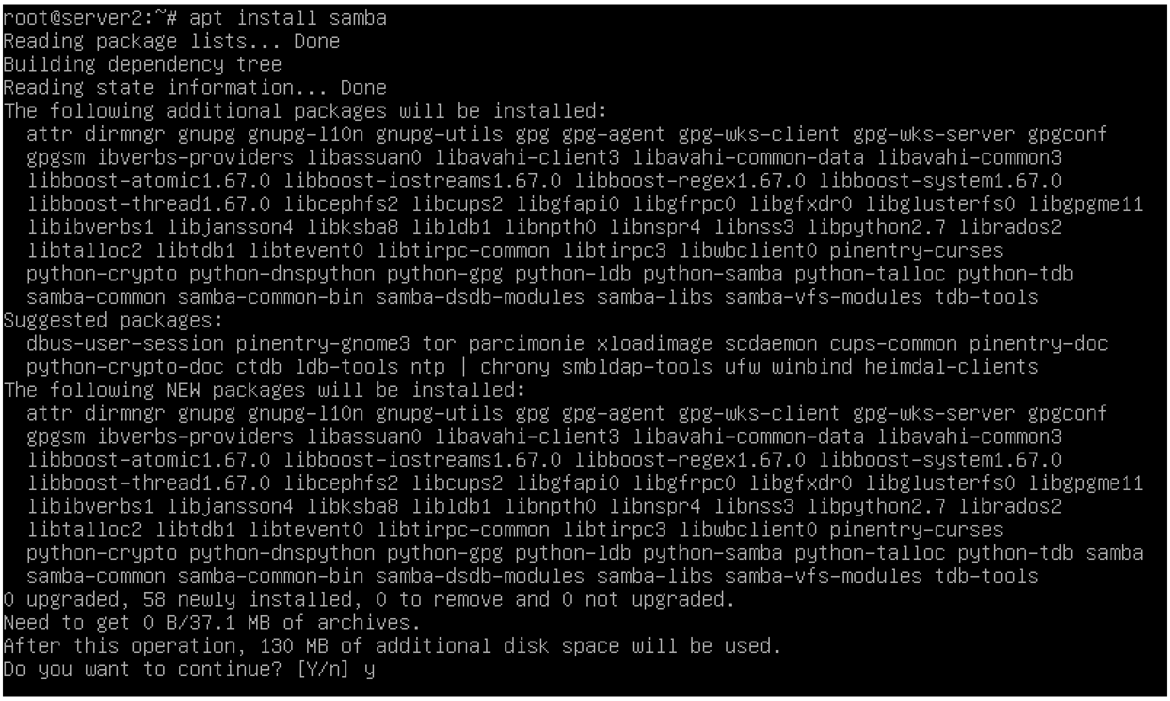
Apa itu NFS? NFS adalah file sharing yang umumnya digunakan di sistem operasi unix-like, jadi kebanyakan digunakan untuk file sharing antar sistem operasi Linux.

2.11.1 Samba

Kali ini kita gunakan server2 sebagai server samba, dan untuk client kita gunakan Windows

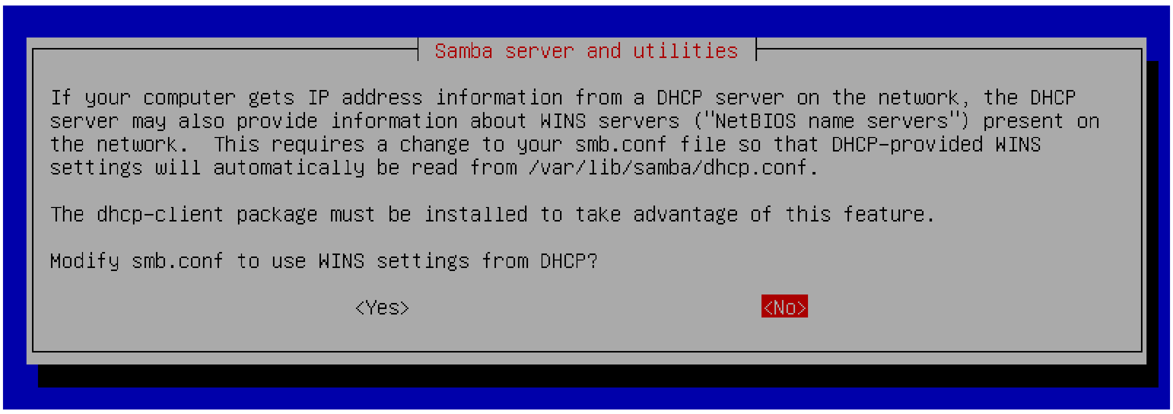
Instalasi Samba

```
root@server2:~# apt install samba
```



```
root@server2:~# apt install samba
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  attr dirmngr gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf
  gpgsm ibverbs-providers libassuan0 libavahi-client3 libavahi-common-data libavahi-common3
  libboost-atomic1.67.0 libboost-iostreams1.67.0 libboost-regex1.67.0 libboost-system1.67.0
  libboost-thread1.67.0 libcephfs2 libcups2 libgfapi0 libgfrpc0 libgfxdr0 libglusterfs0 libgpgme11
  libibverbs1 libjansson4 libksba8 libldb1 libnpth0 libnspr4 libnss3 libpython2.7 librados2
  libtalloc2 libtdb1 libtevent0 libtirpc-common libtirpc3 libwbclient0 pinentry-curses
  python-crypto python-dnspython python-gpg python-ldb python-samba python-talloc python-tdb
  samba-common samba-common-bin samba-dsdb-modules samba-libs samba-vfs-modules tdb-tools
Suggested packages:
  dbus-user-session pinentry-gnome3 tor parcimonie xloadimage scdaemon cups-common pinentry-doc
  python-crypto-doc ctdb ldb-tools ntp | chrony smbldap-tools ufw winbind heimdal-clients
The following NEW packages will be installed:
  attr dirmngr gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf
  gpgsm ibverbs-providers libassuan0 libavahi-client3 libavahi-common-data libavahi-common3
  libboost-atomic1.67.0 libboost-iostreams1.67.0 libboost-regex1.67.0 libboost-system1.67.0
  libboost-thread1.67.0 libcephfs2 libcups2 libgfapi0 libgfrpc0 libgfxdr0 libglusterfs0 libgpgme11
  libibverbs1 libjansson4 libksba8 libldb1 libnpth0 libnspr4 libnss3 libpython2.7 librados2
  libtalloc2 libtdb1 libtevent0 libtirpc-common libtirpc3 libwbclient0 pinentry-curses
  python-crypto python-dnspython python-gpg python-ldb python-samba python-talloc python-tdb samba
  samba-common samba-common-bin samba-dsdb-modules samba-libs samba-vfs-modules tdb-tools
0 upgraded, 58 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/37.1 MB of archives.
After this operation, 130 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Modify smb.conf to use WINS settings from DHCP? Pilih **No**



```
Samba server and utilities

If your computer gets IP address information from a DHCP server on the network, the DHCP
server may also provide information about WINS servers ("NetBIOS name servers") present on
the network.  This requires a change to your smb.conf file so that DHCP-provided WINS
settings will automatically be read from /var/lib/samba/dhcp.conf.

The dhcp-client package must be installed to take advantage of this feature.

Modify smb.conf to use WINS settings from DHCP?

              <Yes>                                   <No>
```

Konfigurasi Samba

Edit file dengan nano , lalu geser ke paling bawah dan tambahkan konfigurasi

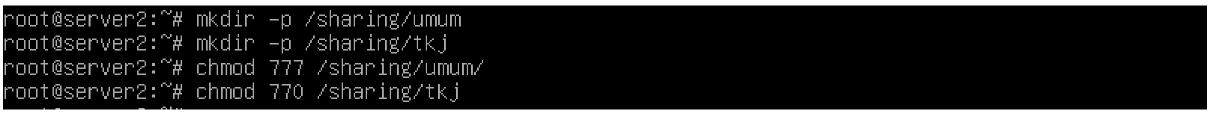
Membuat 2 file sharing 1 untuk umum, 1 khusus TKJ

```
root@server2:~# nano /etc/samba/smb.conf
[smkgeger]
  comment = File Sharing Umum
  path = /sharing/umum
  browseable = yes
  guest ok = yes
[tkj]
  comment = File Sharing TKJ
  path = /sharing/tkj
  writeable = yes
  browseable = yes
  guest ok = no
 valid users = @tkj
```

```
[smkgeger]
   comment = File Sharing Umum
   path = /sharing/umum
   browseable = yes
   guest ok = yes
   guest only = yes

[tkj]
   comment = File Sharing TKJ
   path = /sharing/tkj
   writeable = yes
   browseable = yes
   guest ok = no
   valid users = @tkj
```

Buat folder dan setting permission

```
root@server2:~# mkdir -p /sharing/umum
root@server2:~# mkdir -p /sharing/tkj
root@server2:~# chmod 777 /sharing/umum
root@server2:~# chmod 770 /sharing/tkj
```



```
root@server2:~# mkdir -p /sharing/umum
root@server2:~# mkdir -p /sharing/tkj
root@server2:~# chmod 777 /sharing/umum/
root@server2:~# chmod 770 /sharing/tkj
```

Membuat user baru dan memasukkan ke group tkj

```
root@server2:~# useradd andri

root@server2:~# smbpasswd -a andri

New SMB password:

Retype new SMB password:

Added user andri.

root@server2:~# usermod -aG tkj andri
```

```
root@server2:~# useradd andri
root@server2:~# smbpasswd -a andri
New SMB password:
Retype new SMB password:
root@server2:~# usermod -aG tkj andri
root@server2:~#
```

Kemudian restart service samba

```
root@server2:~# systemctl restart smbd
```

```
root@server2:~# systemctl restart smbd
```

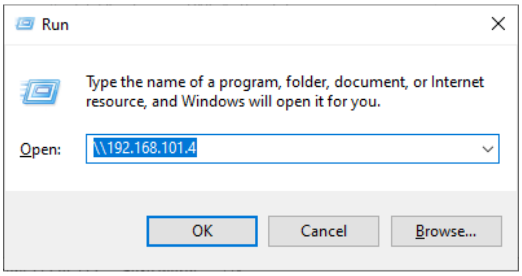
Buat file txt sederhana di sharing umum dan sharing tkj

```
root@server2:~# echo "Haloo" > /sharing/umum/halo.txt
root@server2:~#
```

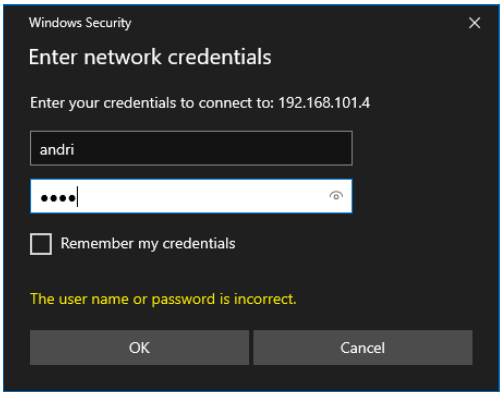
```
root@server2:~# echo "Teknik-Komputer" > /sharing/tkj/tkj.txt
root@server2:~# _
```

Test dari Windows Client

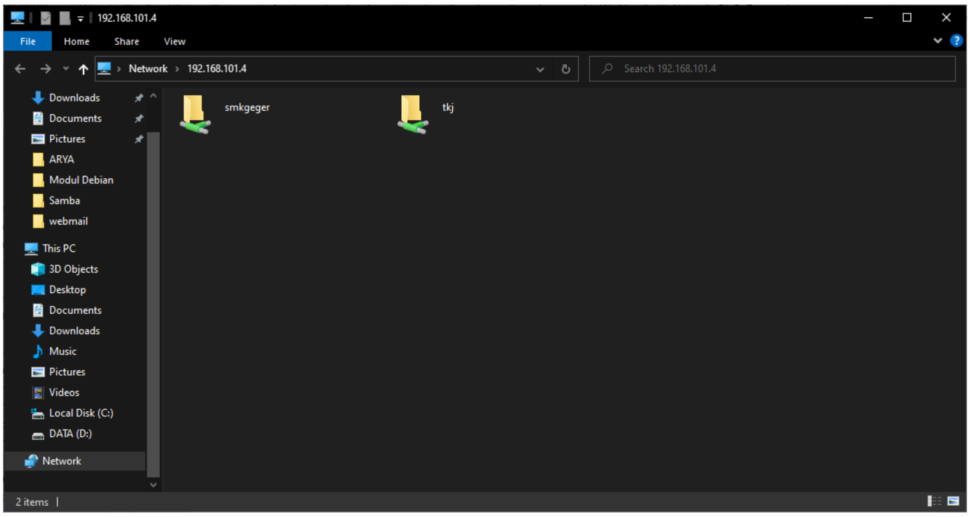
Tekan Win+R, masukkan alamat \\192.168.101.4



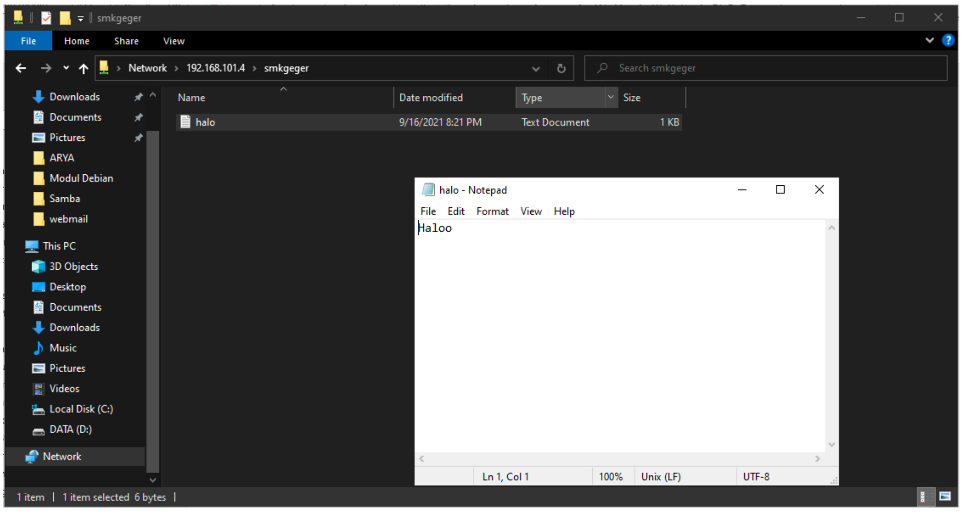
Nanti akan muncul dialog untuk memasukkan user, masukkan user andri



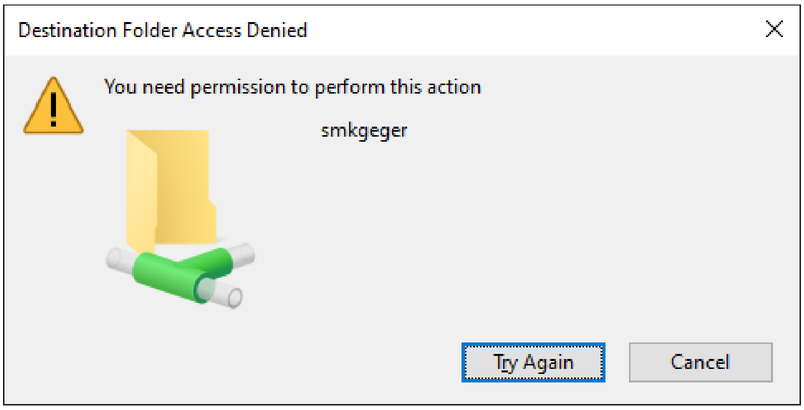
Pertama kita masuk di smkgeger



File txt yang kita buat tadi bisa dibaca



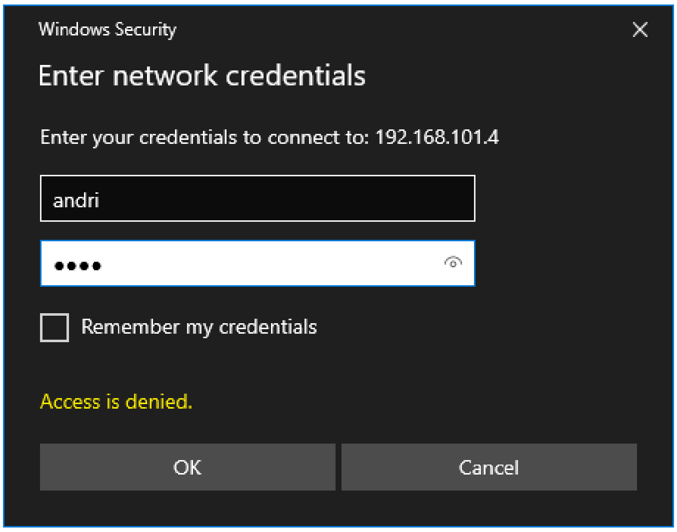
Coba kita buat folder baru apakah bisa?



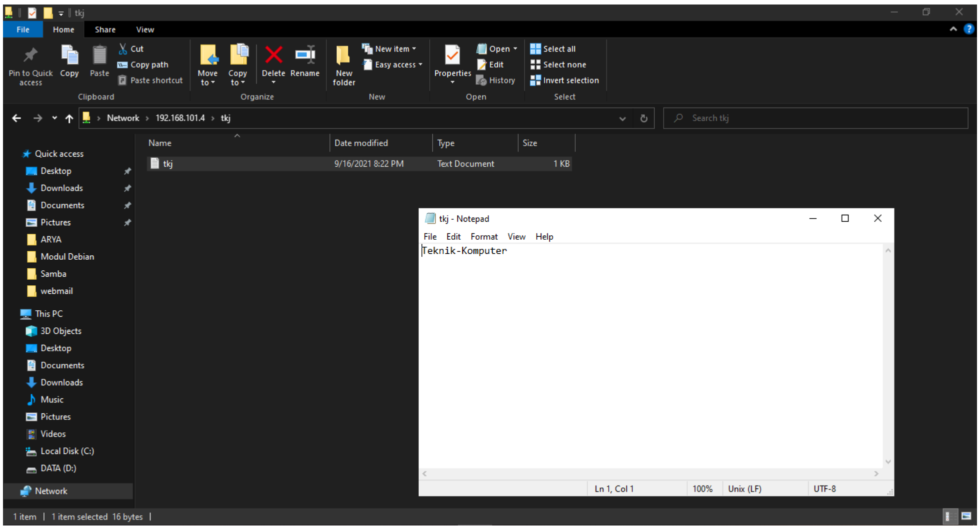
Kenapa tidak bisa? Karena di konfigurasi samba tadi kita tidak menambahkan konfigurasi writeable = yes jadi secara default membuat/memasukkan file dan folder di blokir.

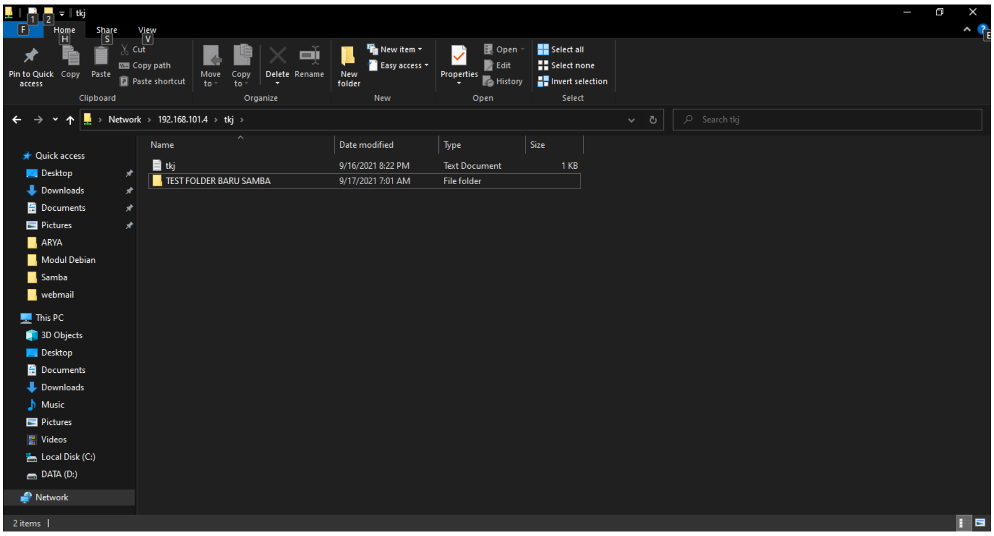
Coba kita masuk ke tkj

Pertama akan muncul dialog autentikasi, masukkan user andri lagi



Coba buka file dan buat folder baru





2.11.2 NFS

Kita gunakan Server 2 untuk NFS Server dan Server 1 sebagai client

Install NFS Server

```
root@server2:~# apt install nfs-kernel-server nfs-common
```

```
root@server2:~# apt install nfs-kernel-server nfs-common
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  keyutils libevent-2.1-6 libnfsidmap2 rpcbind
Suggested packages:
  open-iscsi watchdog
The following NEW packages will be installed:
  keyutils libevent-2.1-6 libnfsidmap2 nfs-common nfs-kernel-server rpcbind
0 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/663 kB of archives.
After this operation, 1,977 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Kemudian buat folder untuk nfs dan ubah permission

```
root@server2:~# mkdir -p /nfs

root@server2:~# chmod 777 /nfs
```

```
root@server2:~# mkdir /nfs
root@server2:~# chmod 777 /nfs
root@server2:~# _
```

Konfigurasi NFS

```
root@server2:~# nano /etc/exports

/nfs     192.168.101.0/24(rw, no_root_squash)
```

```
  GNU nano 3.2                                /etc/exports

# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/nfs     192.168.101.0/24(rw,no_root_squash,no_subtree_check)
```

/nfs adalah lokasi direktori yang di share

192.168.101.0/24 Alamat IP yang di ijinkan untuk mengakses

(rw, no_root_squash) Option untuk NFS agar bisa read write

Baca lebih lengkap disini : https://linux.die.net/man/5/exports

Setelah itu save lalu export dan restart service nfs

```
root@server2:~# exportfs -a

root@server2:~# systemctl restart nfs-kernel-server
```

Konfigurasi Client

Install nfs-common

```
root@server1:~# apt install nfs-common
```

```
root@server1:~# apt install nfs-common
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  keyutils libevent-2.1-6 libnfsidmap2 libtirpc-common libtirpc3 rpcbind
Suggested packages:
  open-iscsi watchdog
The following NEW packages will be installed:
  keyutils libevent-2.1-6 libnfsidmap2 libtirpc-common libtirpc3 nfs-common rpcbind
0 upgraded, 7 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/649 kB of archives.
After this operation, 1,930 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Buat folder untuk nfs client dan mount nfs server di client

```
root@server1:~# mkdir /nfs

root@server1:~# mount 192.168.101.4:/nfs /nfs

root@server1:~# df -h
```

```
root@server1:~# mount 192.168.101.4:/nfs /nfs
root@server1:~# df -h
Filesystem          Size  Used Avail Use% Mounted on
udev                228M     0  228M   0% /dev
tmpfs                49M  2.3M   47M   5% /run
/dev/sda1           6.9G  1.5G  5.1G  23% /
tmpfs               242M     0  242M   0% /sys/fs/cgroup
tmpfs                49M     0   49M   0% /run/user/0
192.168.101.4:/nfs  6.9G  929M  5.6G  14% /nfs
root@server1:~# _
```

Coba buat file baru di /nfs

```
root@server1:~# nano /nfs/test.txt

Ini adalah file test
```

```
root@server1:~# nano /nfs/test.txt
```

```
  GNU nano 3.2                                  /nfs/test.txt

Ini adalah file test
```

Kemudian save dan cek di server2

```
root@server2:~# cat /nfs/test.txt
```

```
root@server2:~# cat /nfs/test.txt
Ini adalah file test
root@server2:~# _
```

Mount permanen NFS

Edit file /etc/fstab lalu tambahkan

```
192.168.101.4:/nfs    /nfs    nfs    defaults,_netdev    0    0
```

```
  GNU nano 3.2                                    /etc/fstab

# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point>   <type>  <options>       <dump>  <pass>
# / was on /dev/sda1 during installation
UUID=00ac9205-0e13-4854-a345-2d980f895a67 /               ext4    errors=remount-ro 0       1
# swap was on /dev/sda5 during installation
UUID=d41a5bcc-cd73-47a0-84f4-f8171b5a7c48 none            swap    sw              0       0
/dev/sr0        /media/cdrom0   udf,iso9660 user,noauto     0       0
#NFS
192.168.101.4:/nfs      /nfs    nfs     defaults,_netdev        0       0
```

Lalu coba reboot, dan terlihat nfs sudah otomatis termount

```
Debian GNU/Linux 10 server1 tty1

server1 login: root
Password:
Last login: Fri Sep 17 15:44:37 WIB 2021 on tty1
Linux server1 4.19.0-17-amd64 #1 SMP Debian 4.19.194-3 (2021-07-18) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@server1:~# df -h
Filesystem          Size  Used Avail Use% Mounted on
udev                228M     0  228M   0% /dev
tmpfs                49M  1.7M   47M   4% /run
/dev/sda1           6.9G  1.5G  5.1G  23% /
tmpfs               242M     0  242M   0% /dev/shm
tmpfs               5.0M     0  5.0M   0% /run/lock
tmpfs               242M     0  242M   0% /sys/fs/cgroup
192.168.101.4:/nfs  6.9G  929M  5.6G  14% /nfs
tmpfs                49M     0   49M   0% /run/user/0
root@server1:~# _
```

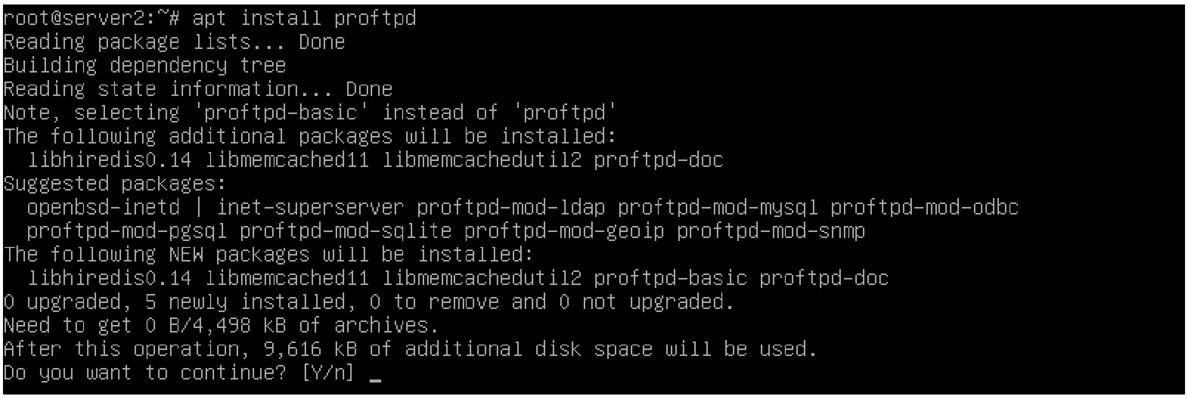
2.12 Konfigurasi FTP Server

FTP adalah singkatan dari File Transfer Protocol, sesuai dengan namanya FTP digunakan untuk transfer file dari server ke client maupun sebaliknya. Di debian terdapat berbagai macam software untuk ftp server, kali ini kita akan melakukan konfigurasi FTP server dengan software ProFTPD.

Kita gunakan Server2 untuk konfigurasi kali ini

Install ProFTPD

root@server2:~# apt install proftpd



```
root@server2:~# apt install proftpd
Reading package lists... Done
Building dependency tree
Reading state information... Done
Note, selecting 'proftpd-basic' instead of 'proftpd'
The following additional packages will be installed:
  libhiredis0.14 libmemcached11 libmemcachedutil2 proftpd-doc
Suggested packages:
  openbsd-inetd | inet-superserver proftpd-mod-ldap proftpd-mod-mysql proftpd-mod-odbc
  proftpd-mod-pgsql proftpd-mod-sqlite proftpd-mod-geoip proftpd-mod-snmp
The following NEW packages will be installed:
  libhiredis0.14 libmemcached11 libmemcachedutil2 proftpd-basic proftpd-doc
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/4,498 kB of archives.
After this operation, 9,616 kB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Setelah itu konfigurasi di /etc/proftpd/proftpd.conf

root@server2:~# nano /etc/proftpd/proftpd.conf

```
root@server2:~# nano /etc/proftpd/proftpd.conf
```

```
UseIPv6             off
ServerName          "FTP-SMKGEGER"
DefaultRoot         /home/ftp
Port                21
AuthOrder           mod_auth_pam.c* mod_auth_unix.c
```

Ubah seperti di atas, yang lainnya biarkan default

Kemudian save dan restart service proftpd

root@server2:~# systemctl restart proftpd

```
root@server2:~# systemctl restart proftpd_
```

Kemudian buat folder /home/ftp dan set permission ke 777

```
root@server2:~# mkdir /home/ftp
root@server2:~# chmod 777 /home/ftp
```

```
root@server2:~# mkdir /home/ftp
root@server2:~# chmod 777 /home/ftp
root@server2:~# _
```

Lalu buat user baru untuk ftp, misal user narji

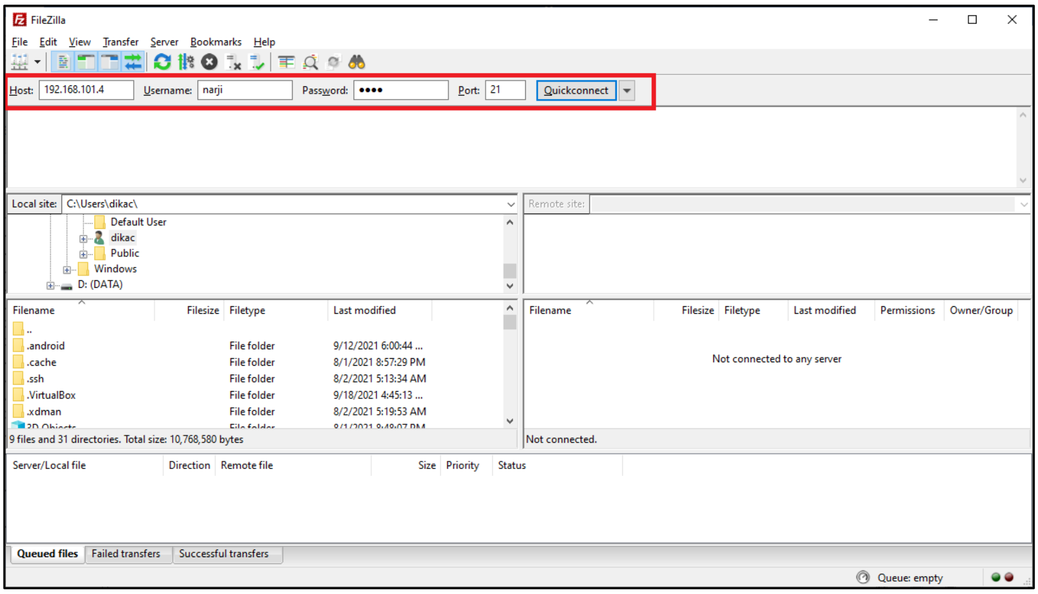
root@server2:~# adduser narji

```
root@server2:~# adduser narji
Adding user `narji' ...
Adding new group `narji' (1002) ...
Adding new user `narji' (1002) with group `narji' ...
Creating home directory `/home/narji' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for narji
Enter the new value, or press ENTER for the default
        Full Name []: Narji TKJ
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
root@server2:~#
```

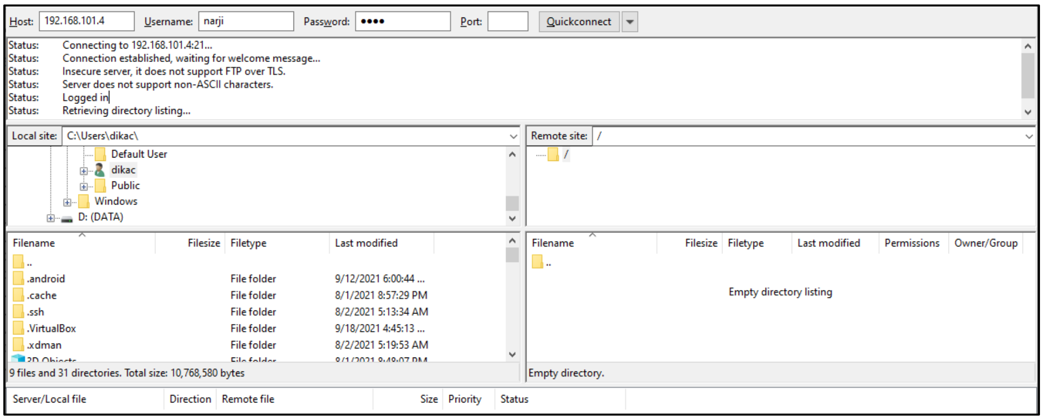
Tes dari Client

Untuk client bisa menggunakan ftp client seperti filezilla atau winscp. Disini saya menggunakan FileZilla

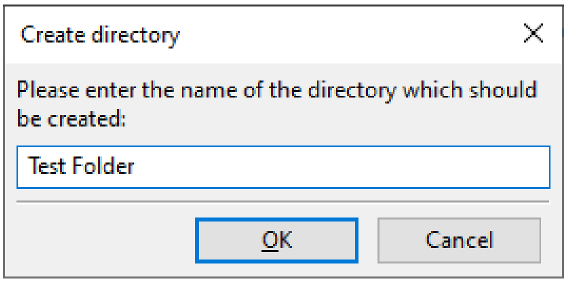
Buka Filezilla > Lalu Isikan Host, Username , Password dan Port 21 > Klik Quickconnect

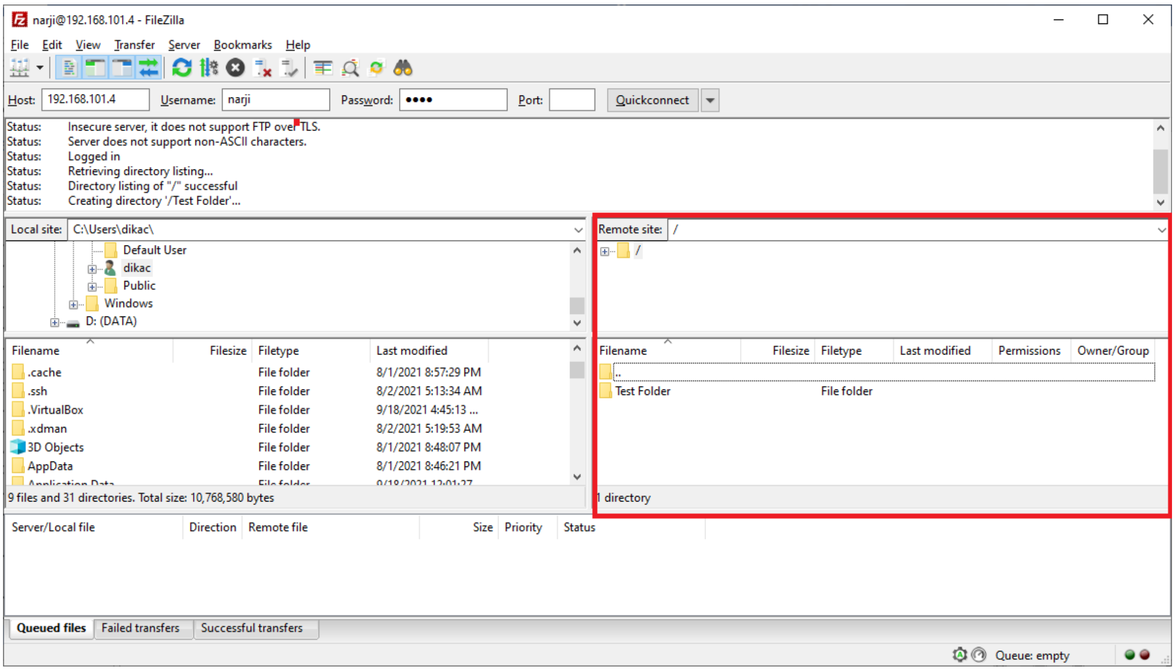


Berhasil Login



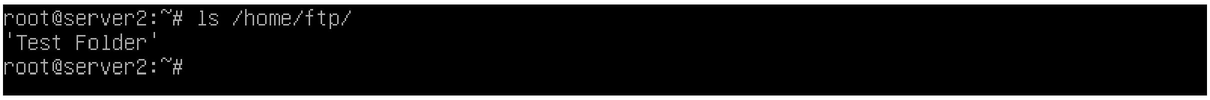
Coba buat direktori baru, klik kanan pada server kemudian pilih create directory dan isikan nama folder misal Test Folder





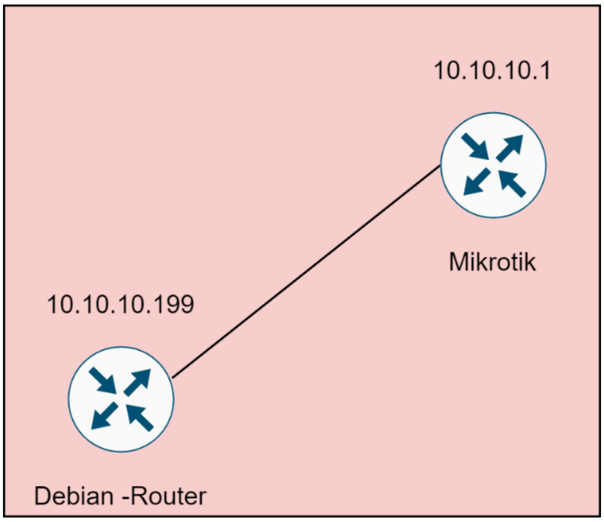
Coba cek di server

```
root@server2:~# ls /home/ftp
```



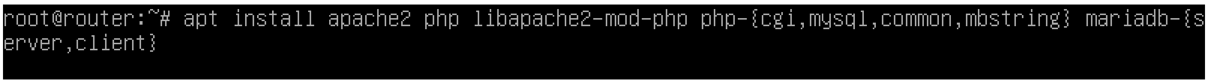
2.13 Konfigurasi Monitoring Server

Kali ini kita akan menggunakan Cacti untuk melakukan Monitoring Router Mikrotik, berarti kita menggunakan Debian-Router untuk monitoring



Install LAMP Stack

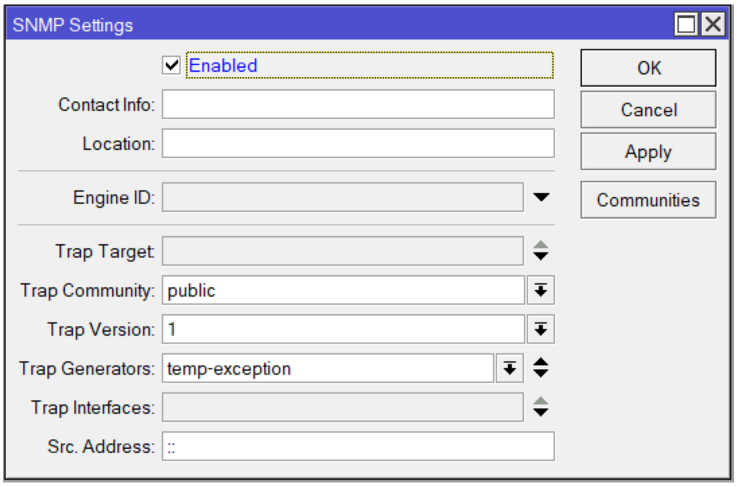
```
root@router:~# apt install apache2 php libapache2-mod-php php-{cgi,mysql,common,mbstring} mariadb-{server,client}
```



Konfigurasi Mikrotik

Untuk konfigurasi mikrotik agar bisa di monitoring lewat cacti aktifkan SNMP pada menu

IP > SNMP > Centang Opsi Enabled

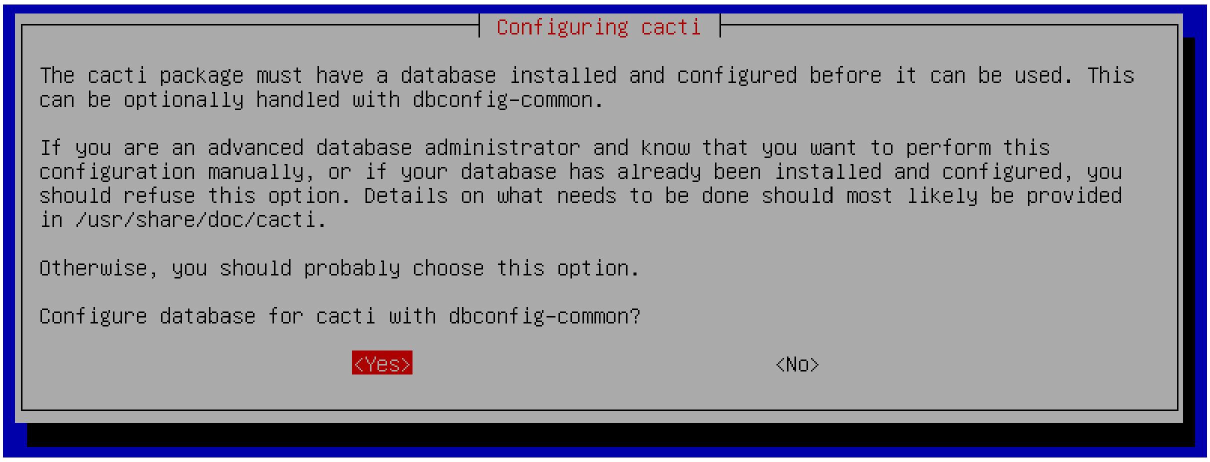


Install Cacti

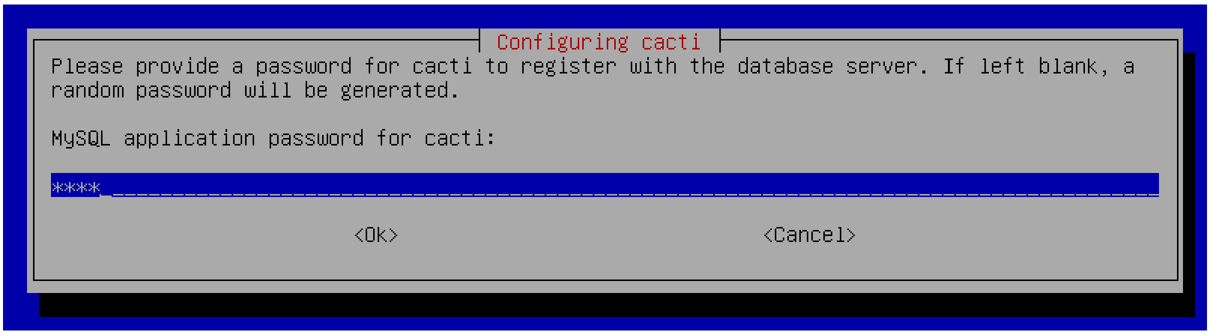
```
root@router:~# apt install cacti cacti-spine
```

Nanti akan muncul dialog, Web Server pilih **apache2**

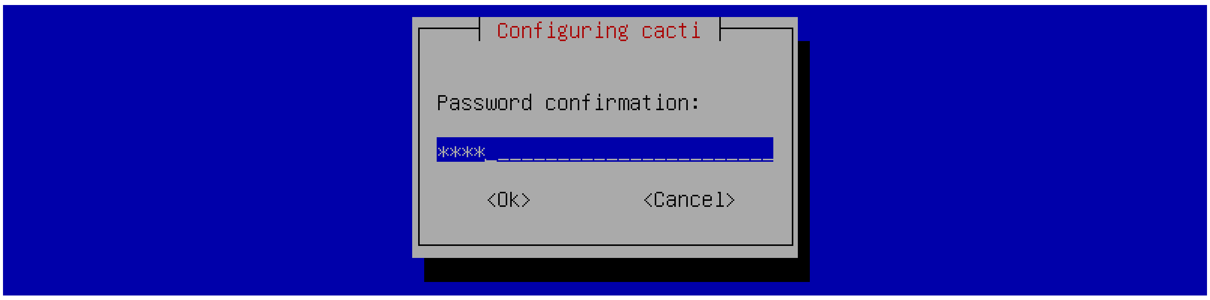
Configure database for cacti with dbconfig-common? Pilih **Yes**



Masukkan password untuk database cacti



Masukkan password lagi



Kemudian sebelum konfigurasi cacti, kita tambahkan subdomain untuk cacti di DNS Server jadi nanti alamat untuk mengakses cacti yaitu cacti.smkgeger.id

```
root@server1:~# nano /etc/bind/db.smk
```

```
$TTL    604800
@       IN      SOA     smkgeger.id. root.smkgeger.id. (
                        202109181       ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.smkgeger.id.
@       IN      NS      ns2.smkgeger.id.
@       IN      MX      10      smkgeger.id.
@       IN      A       192.168.101.3
ns1     IN      A       192.168.101.3
ns2     IN      A       192.168.101.4
web2    IN      A       192.168.101.4
mail    IN      A       192.168.101.3
cacti   IN      A       192.168.101.2_
```

Konfigurasi Cacti

```
root@router:~# cd /etc/apache2/sites-available/

root@router:/etc/apache2/sites-available# cp default-ssl.conf cacti.conf

root@router:/etc/apache2/sites-available# nano cacti.conf
```

```
root@router:~# cd /etc/apache2/sites-available/
root@router:/etc/apache2/sites-available# cp default-ssl.conf cacti.conf
root@router:/etc/apache2/sites-available# nano cacti.conf _
```

```
ServerAdmin admin@smkgeger.id

ServerName cacti.smkgeger.id

DocumentRoot /usr/share/cacti/site

ErrorLog ${APACHE_LOG_DIR}/error.log

CustomLog ${APACHE_LOG_DIR}/access.log combined

SSLEngine on

SSLCertificateFile      /etc/ssl/smk1.crt

SSLCertificateKeyFile /etc/ssl/smk1.key

<FilesMatch "\.(cgi|shtml|phtml|php)$">

        SSLOptions +StdEnvVars

</FilesMatch>

<Directory /usr/lib/cgi-bin>

        SSLOptions +StdEnvVars

</Directory>
```

```
<IfModule mod_ssl.c>
        <VirtualHost _default_:443>
                ServerAdmin admin@smkgeger.id
                ServerName cacti.smkgeger.id
                DocumentRoot /usr/share/cacti/site

                ErrorLog ${APACHE_LOG_DIR}/error.log
                CustomLog ${APACHE_LOG_DIR}/access.log combined

                SSLEngine on
                SSLCertificateFile      /etc/ssl/smk1.crt
                SSLCertificateKeyFile /etc/ssl/smk1.key

                <FilesMatch "\.(cgi|shtml|phtml|php)$">
                                SSLOptions +StdEnvVars
                </FilesMatch>
                <Directory /usr/lib/cgi-bin>
                                SSLOptions +StdEnvVars
                </Directory>

        </VirtualHost>
</IfModule>
```

Transfer sertifikat ssl dari server1 ke router kemudian letakkan di /etc/ssl

```
root@server1:/etc/ssl/ca# scp smk1* root@192.168.101.2:/etc/ssl
```

```
root@server1:/etc/ssl/ca# scp smk1* root@192.168.101.2:/etc/ssl
The authenticity of host '192.168.101.2 (192.168.101.2)' can't be established.
ECDSA key fingerprint is SHA256:oOEOu5jvCxP9zDLyJr3I7+LoGIOMH/OnFlj8YQdux2c.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.168.101.2' (ECDSA) to the list of known hosts.
root@192.168.101.2's password:
smk1.crt                                          100% 1253   783.7KB/s   00:00
smk1.csr                                          100% 1058    93.7KB/s   00:00
smk1.key                                          100% 1704   879.3KB/s   00:00
root@server1:/etc/ssl/ca#
```

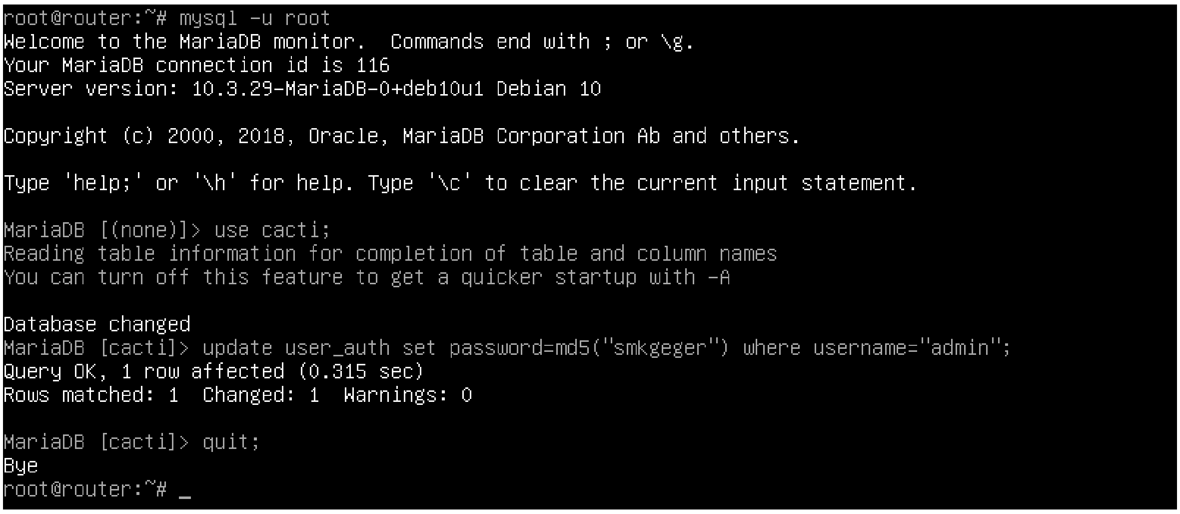
Enable ssl, situs cacti dan restart apache2

```
root@router:/etc/apache2/sites-available# a2enmod ssl

root@router:/etc/apache2/sites-available# a2ensite cacti

root@router:/etc/apache2/sites-available# systemctl restart apache2
```

```
root@router:/etc/apache2/sites-available# a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certifica
tes.
To activate the new configuration, you need to run:
  systemctl restart apache2
root@router:/etc/apache2/sites-available# a2ensite cacti.conf
Enabling site cacti.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@router:/etc/apache2/sites-available# systemctl restart apache2
```

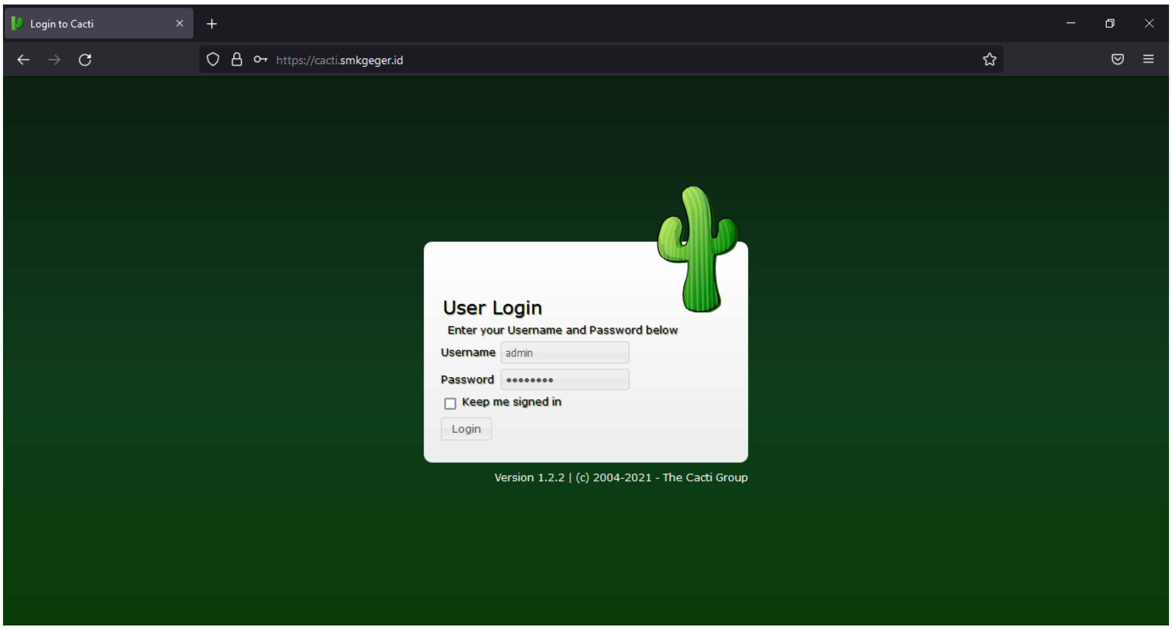
Setelah itu reset password default admin cacti dengan mysql

```
root@router:~# mysql -u root

MariaDB [(none)]> use cacti

MariaDB [cacti]> update user_auth set password=md5("smkgeger") where username="admin" ;

MariaDB [cacti]> quit;
```

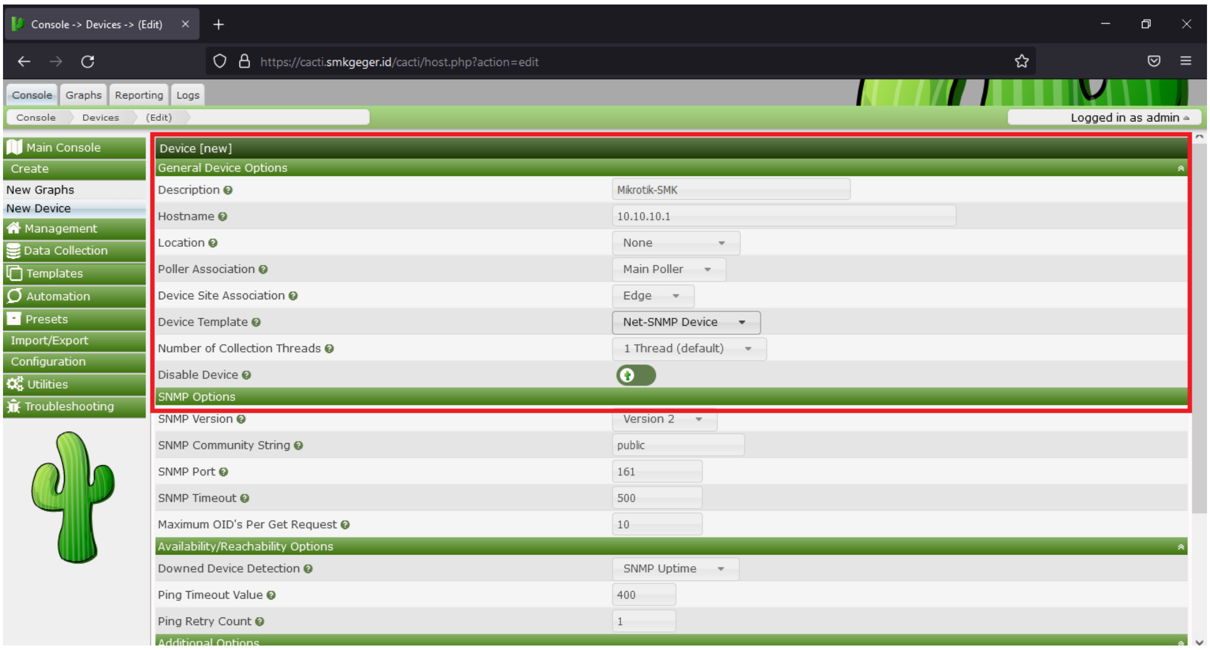


Kemudian kunjungi di browser Windows client, lalu login dengan user admin dan password smkgeger

https://cacti.smkgeger.id



Kemudian tambahkan mikrotik, Klik Create > New Device

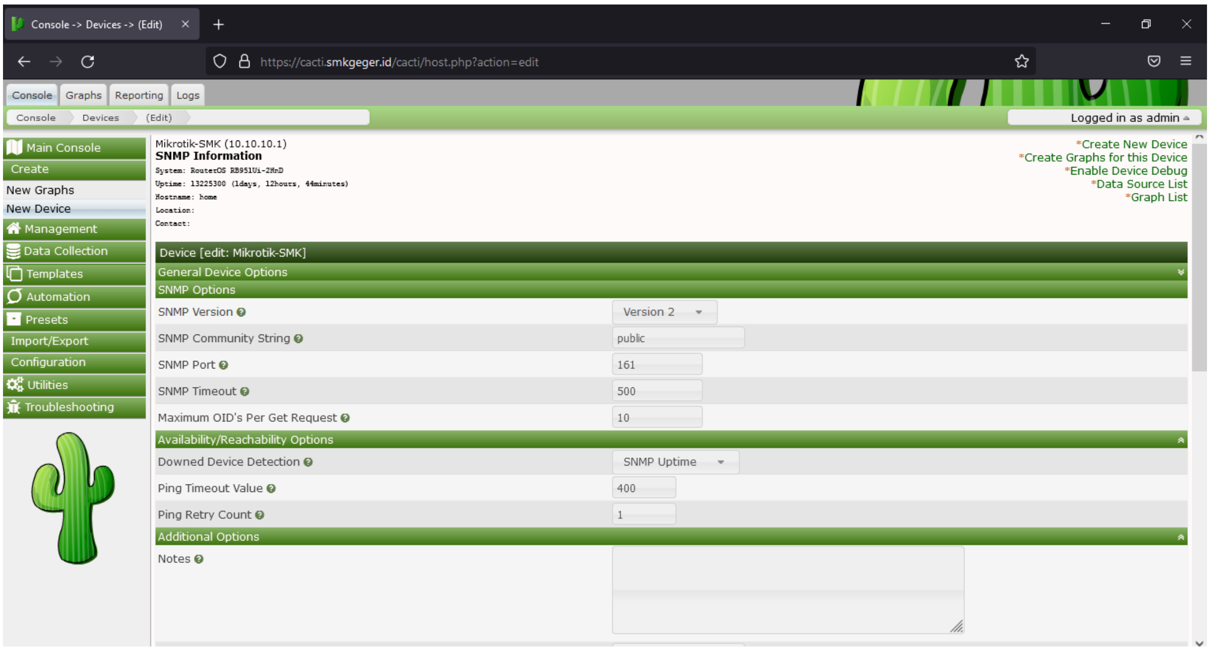


Description = Masukkan Nama Perangkat

Hostname = Masukkan Alamat IP Mikrotik

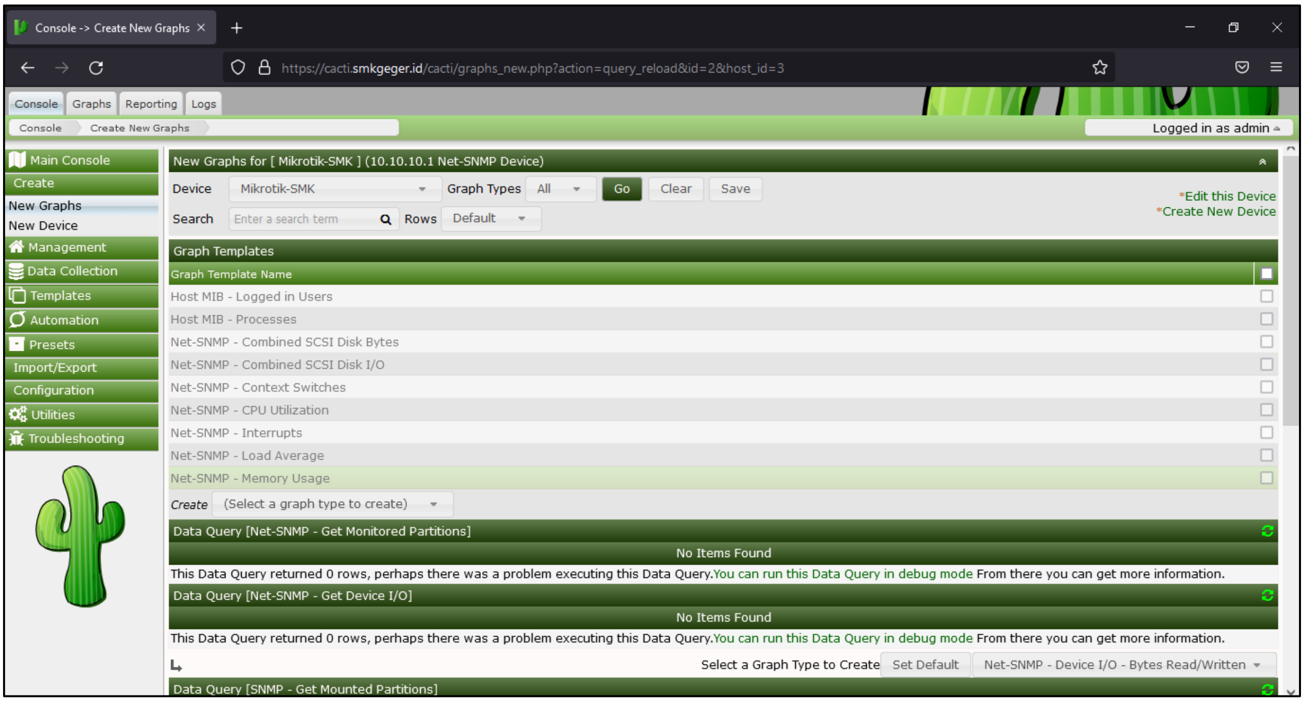
Device Template = Net-SNMP Device

Kemudian geser kebawah dan klik Create



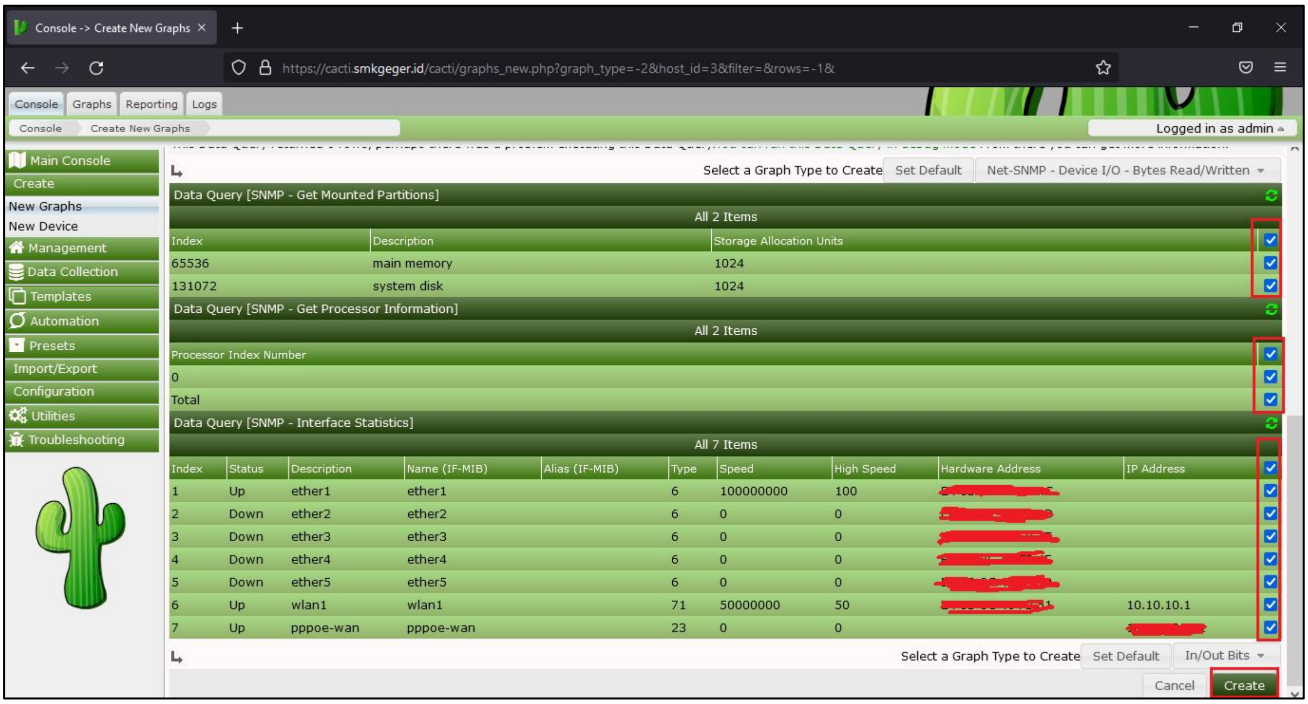
Mikrotik berhasil ditambahkan

Selanjutnya buat grafik untuk monitoring, Klik Create > New Graphs

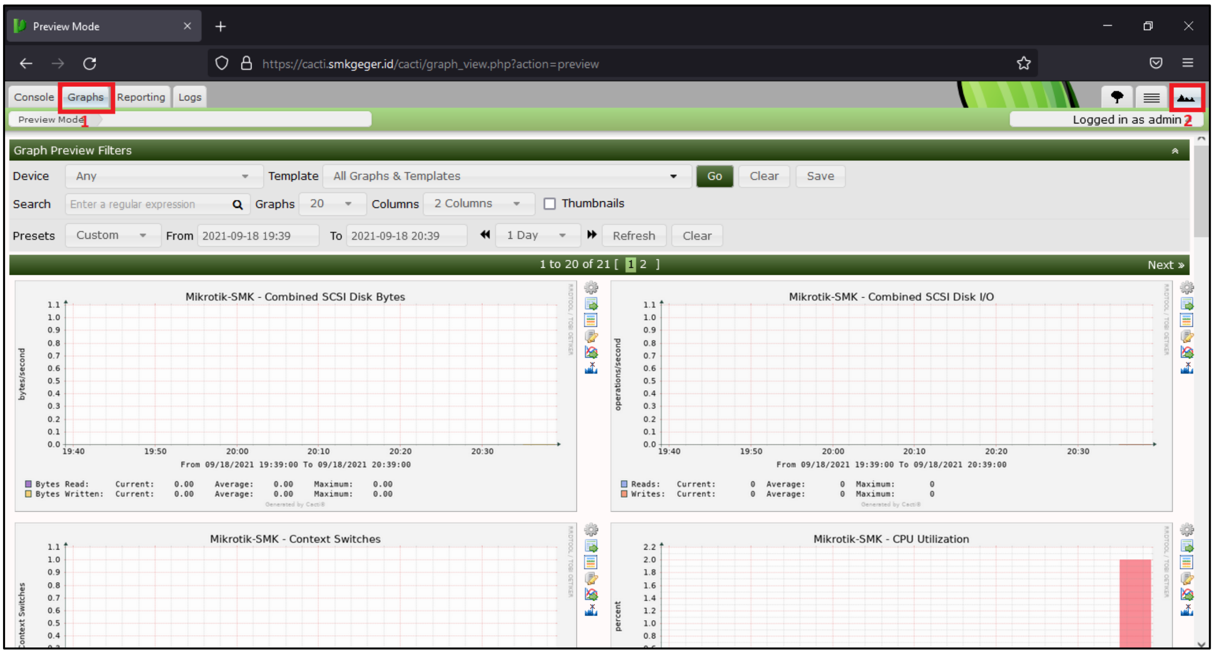


Device = Pilih Mikrotik yang tadi ditambahkan

Centang semua opsi grafik



Lalu pergi ke menu Graphics di menu sebelah kanan pilih Preview



Jika grafik belum muncul maka tunggu kurang lebih 5 Menit.

Monitoring Mikrotik Berhasil!!

BAB 3 Penutupan

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